

Chapter5

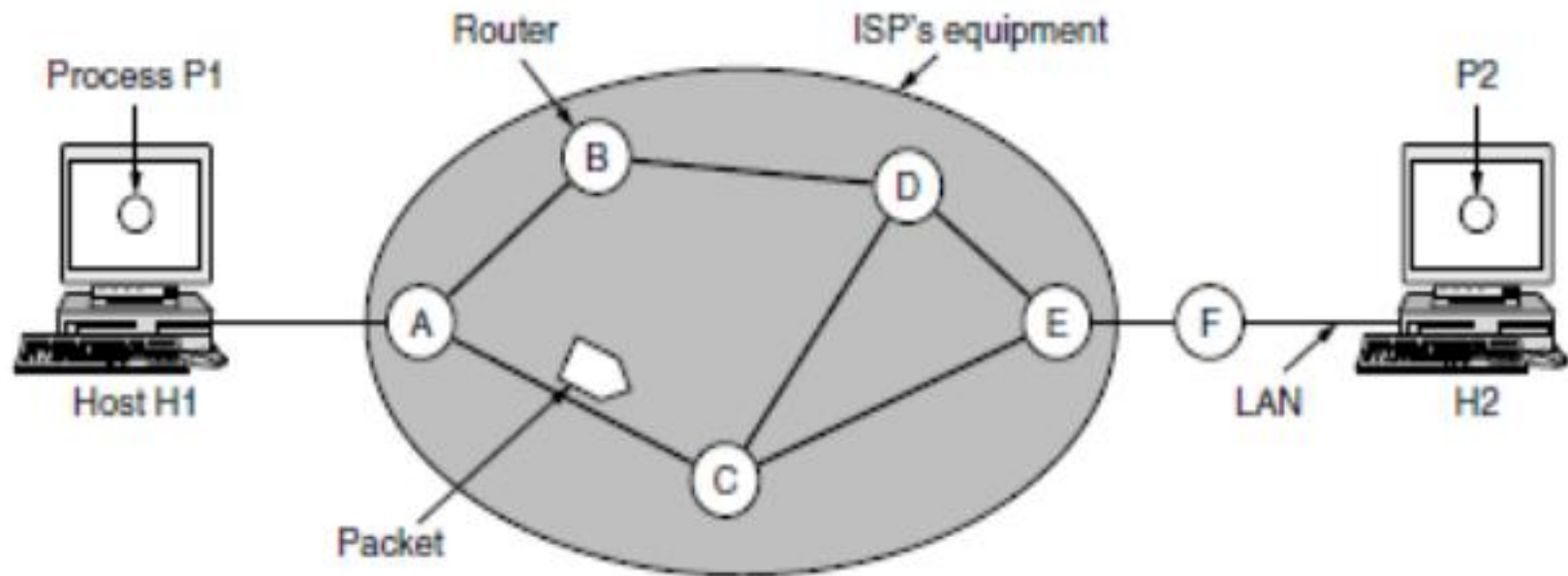
The Network Layer

(around 14 hours)

Outline

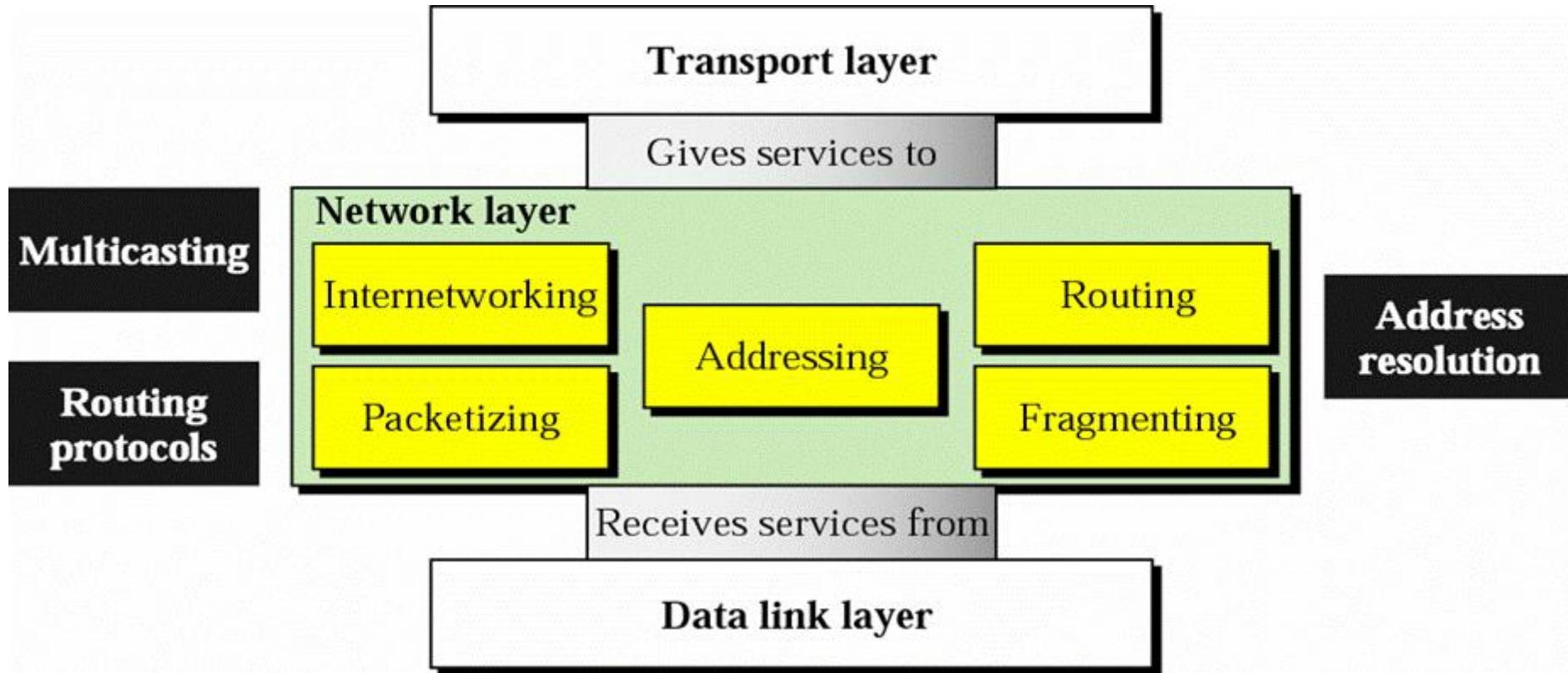
- 5.1 Introduction to Network Layer
- 5.2 Routing Algorithms
- 5.3 Congestion Control
- 5.4 Quality of Service
- 5.5 Internetworking
- 5.6 The network layer in Internet

5.1.1 Environment of Network Layer Protocols



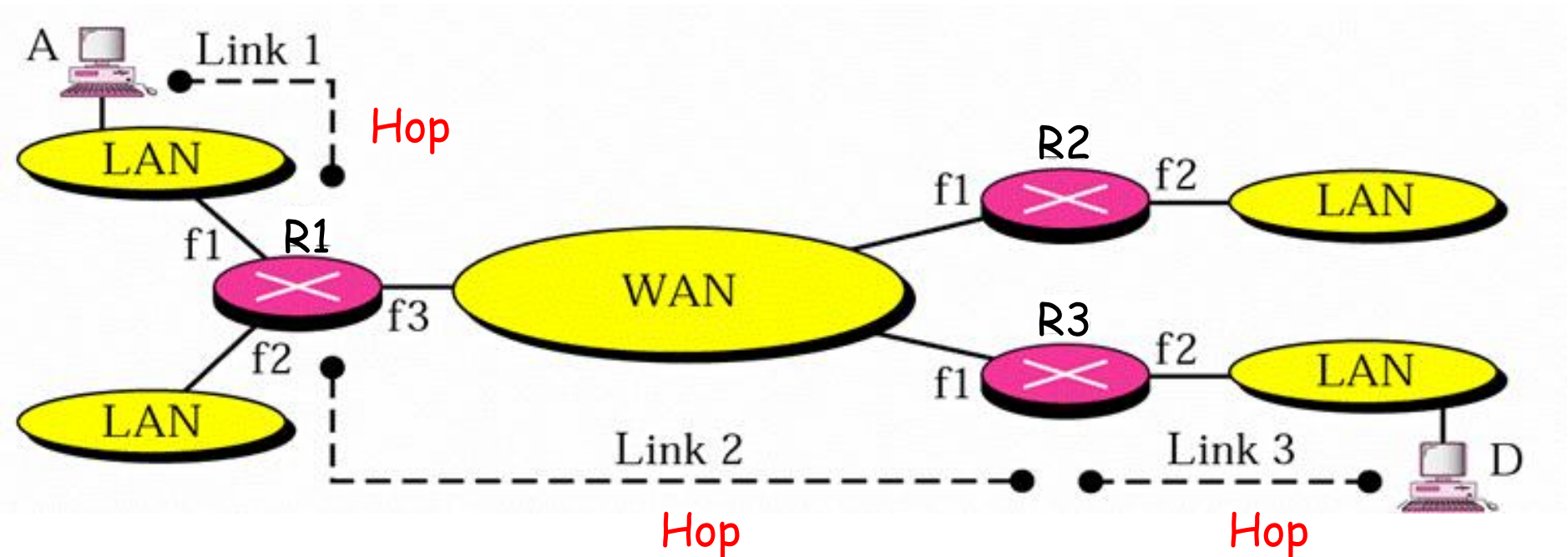
Store-and-Forward Packet Switching
(存储转发 分组交换)

(1) Positions and Functions of Network Layer



Network layer is the lowest layer that deals with **end-to-end (host-to-host)** transmission (端到端传输)

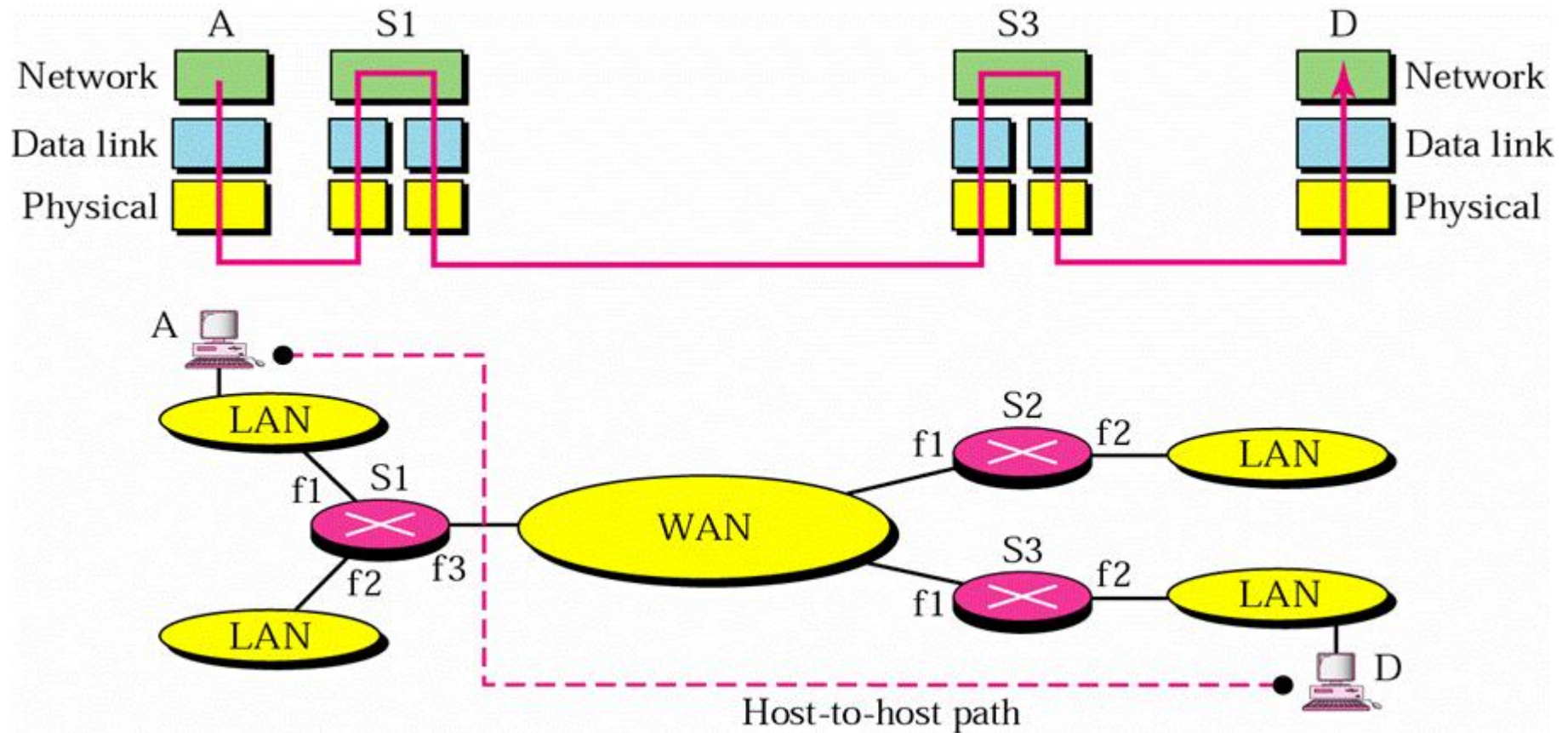
(2) The concept of Hop(跳)



经过一个路由器就是1跳

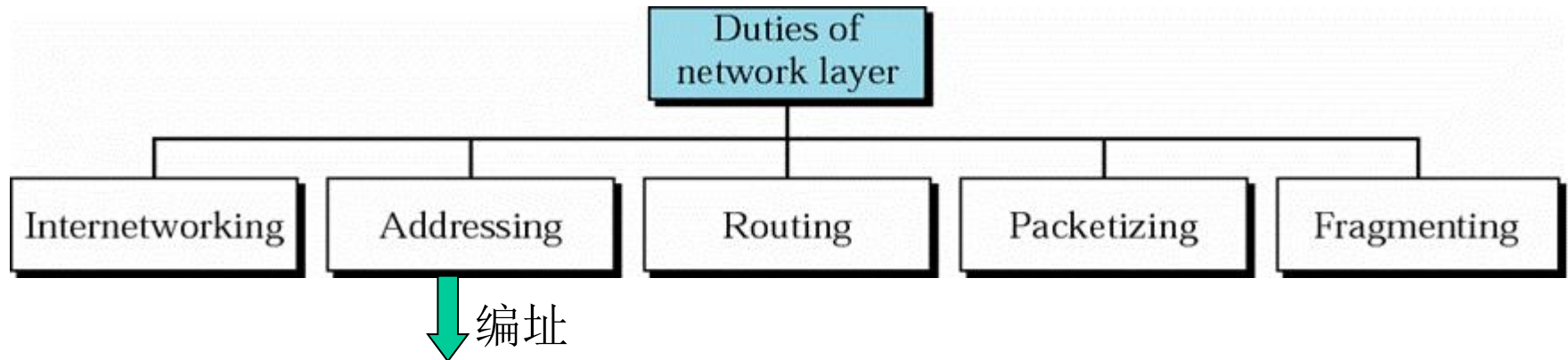
Next Hop: 下一跳

Network Layer in an Internetwork: Host-to-Host Delivery(主机-主机传输)



[Forouzan]

(3) Network Duties(网络的功能)



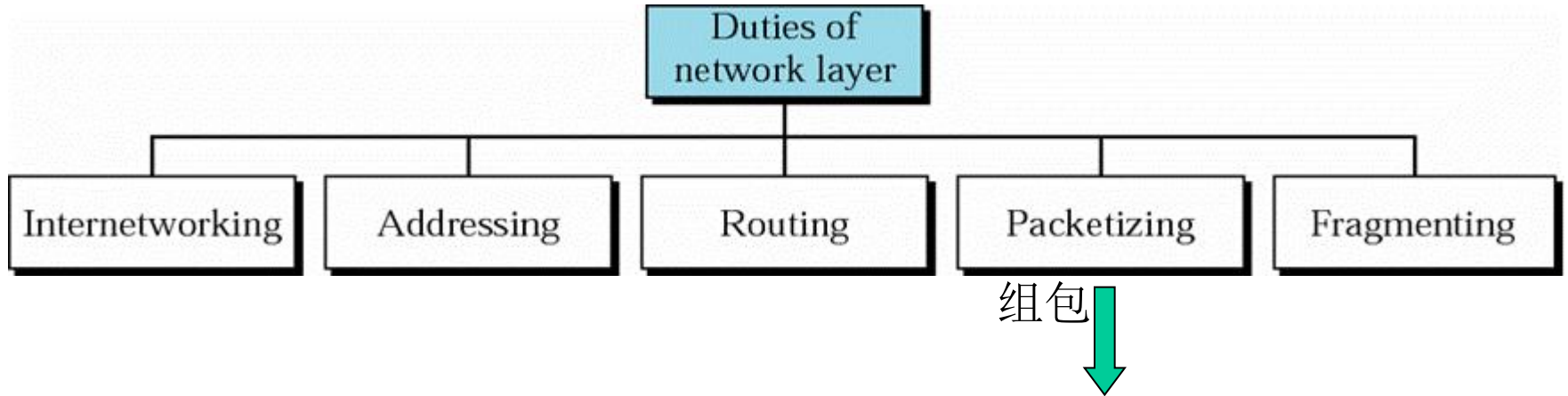
The addresses of a host/router must be **uniquely**(数值唯一) and **universally**(格式统一)

Two devices on the internet can never have the same address.
(Address per interface to network)

- A **MAC address** is responsible for **local identification**(本地标识)
- An **IP address** for **global identification**(全局标识)

[Forouzan]

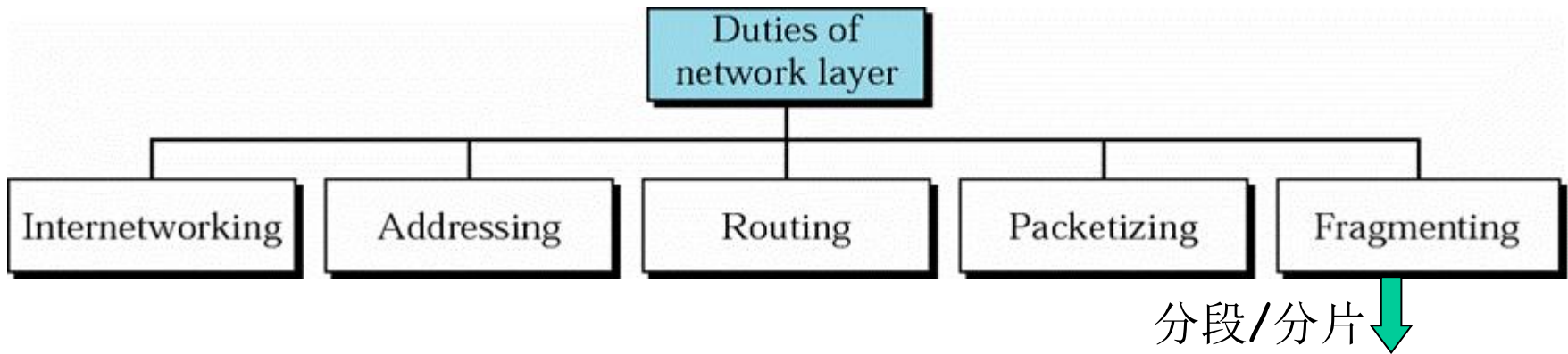
(3) Network Duties(cont.)



Encapsulated(封装的) message received from upper layer and forming **packets**(分组), a task common to all layers. In Internet, packetizing is done by **IP protocol**.

[Forouzan]

(3) Network Duties(cont.)



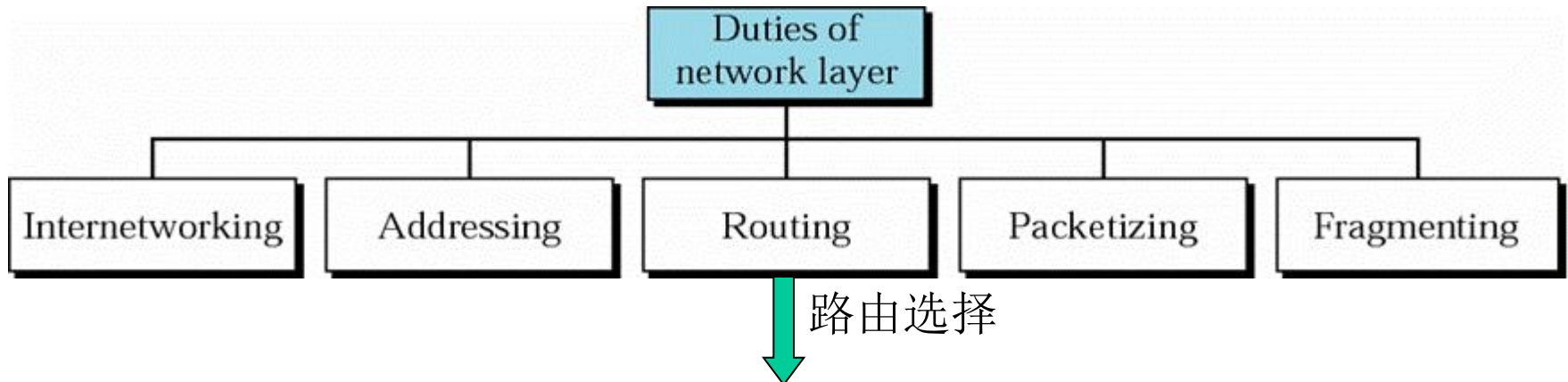
The network layer must be able to operate on top of **any data-link layer technology**. All these technologies can handle a different packet length.

MTU(Maximum Transmission Unit, 最大传输单元) is the largest size packet that can be delivered in bytes without fragmentation.

The network layer must be able to **fragment(分段)** transport layer PDUs into smaller units so that they can be transferred over various data-link layer technologies.

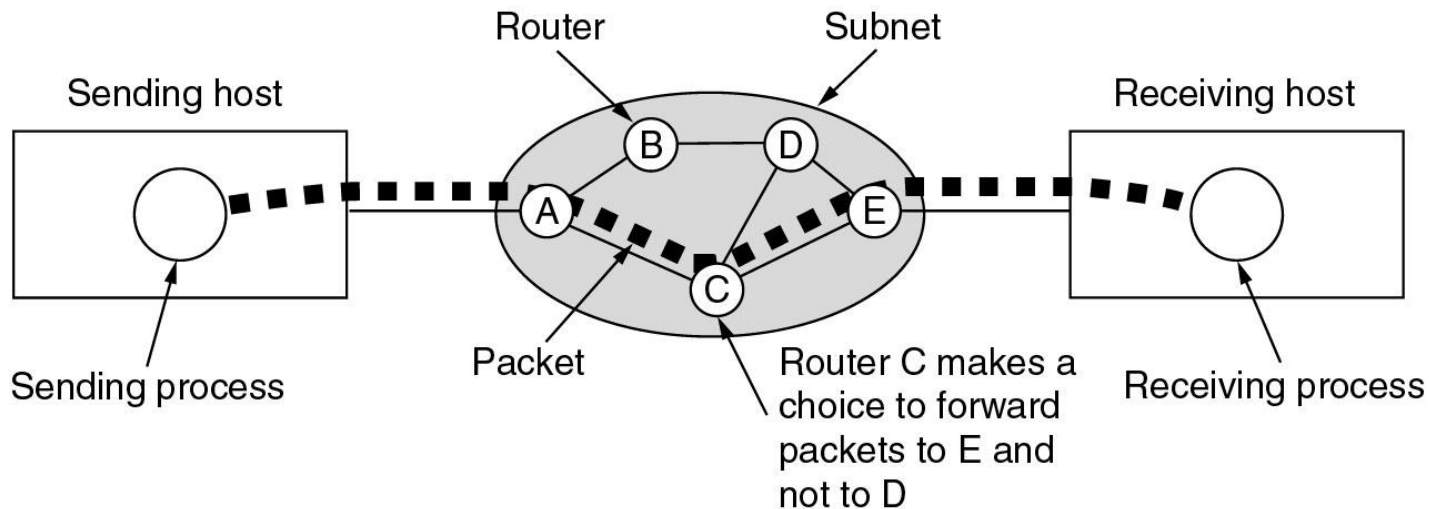
- IEEE802.3(Ethernet) MTU is **1500 bytes**
- IEEE802.11(WiFi) MTU is **2312 bytes**
- IEEE802.15.4 MTU is **127 bytes**
- BLE (blue tooth low energy) MTU ranges from **23 to 251 bytes**.

(3) Network Duties(cont.)



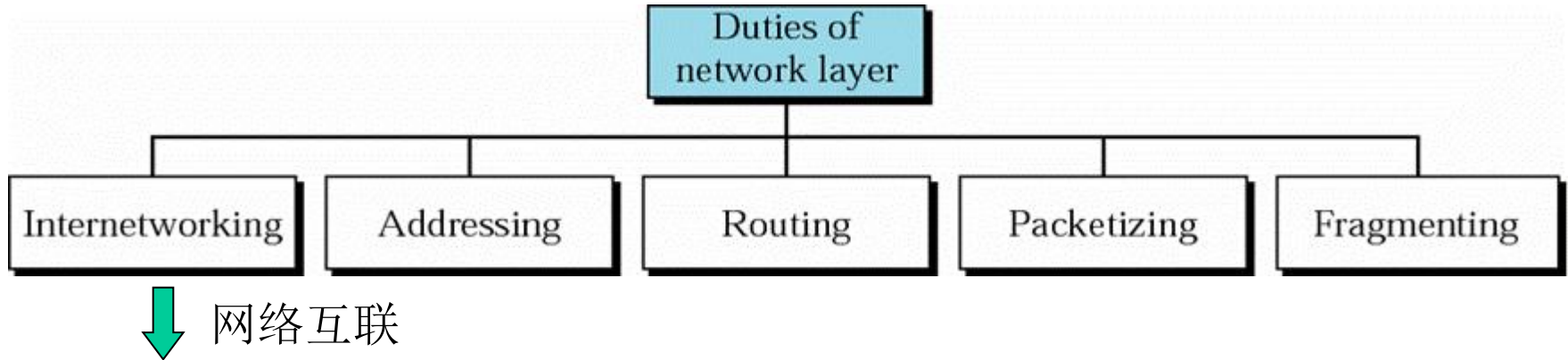
To send the packet to destination, who is the **next hop**?

Packet cannot choose the route; **the routers(路由器)** connecting the LANs/WANs makes this decision.



[Forouzan]

(3) Network Duties(cont.)



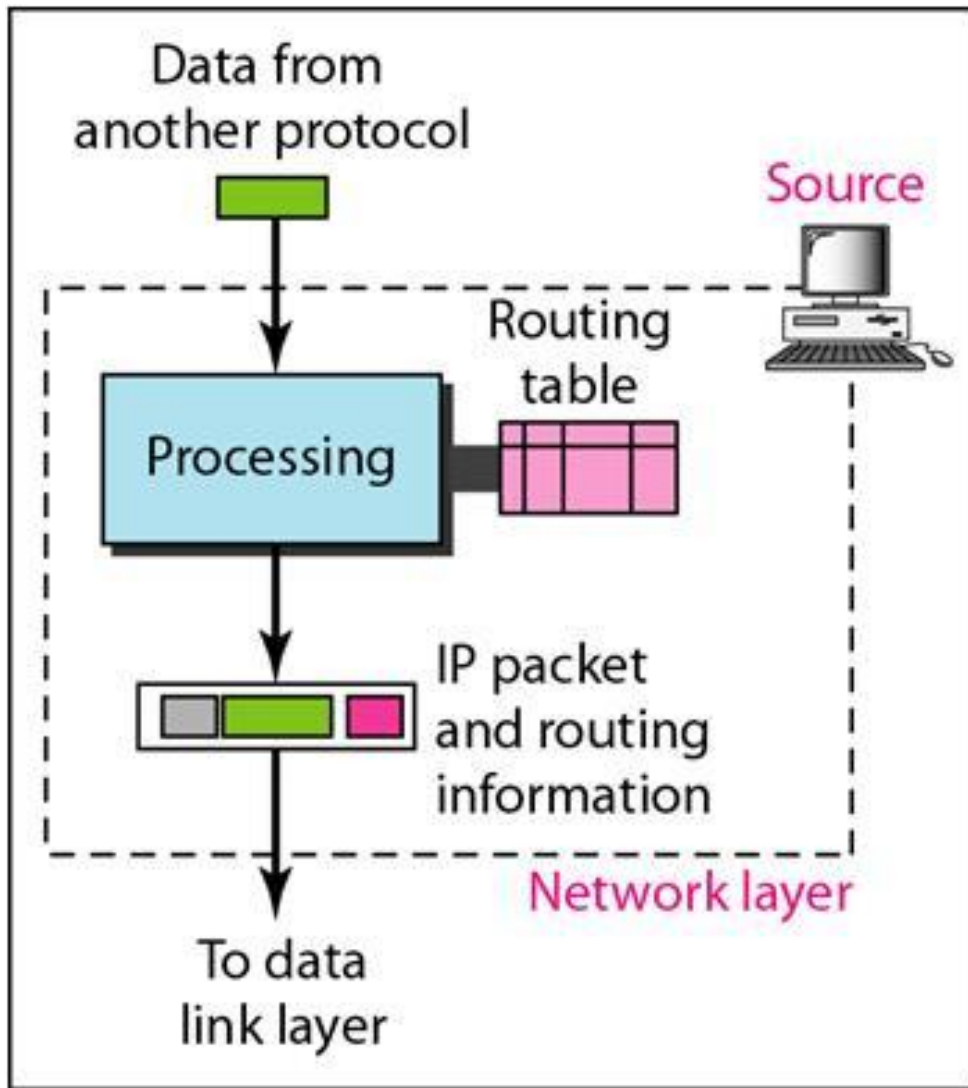
Interconnecting **different networks** (LAN or WAN) and making them look the **same** to the transport layer.

不同的网络：网络地址不同、包格式不同、实现技术不同

The transport layer should not be worried about the underlying physical network! (网络的个数及其具体技术对于传输层透明)

[Forouzan]

(4) Network Layer at the Source



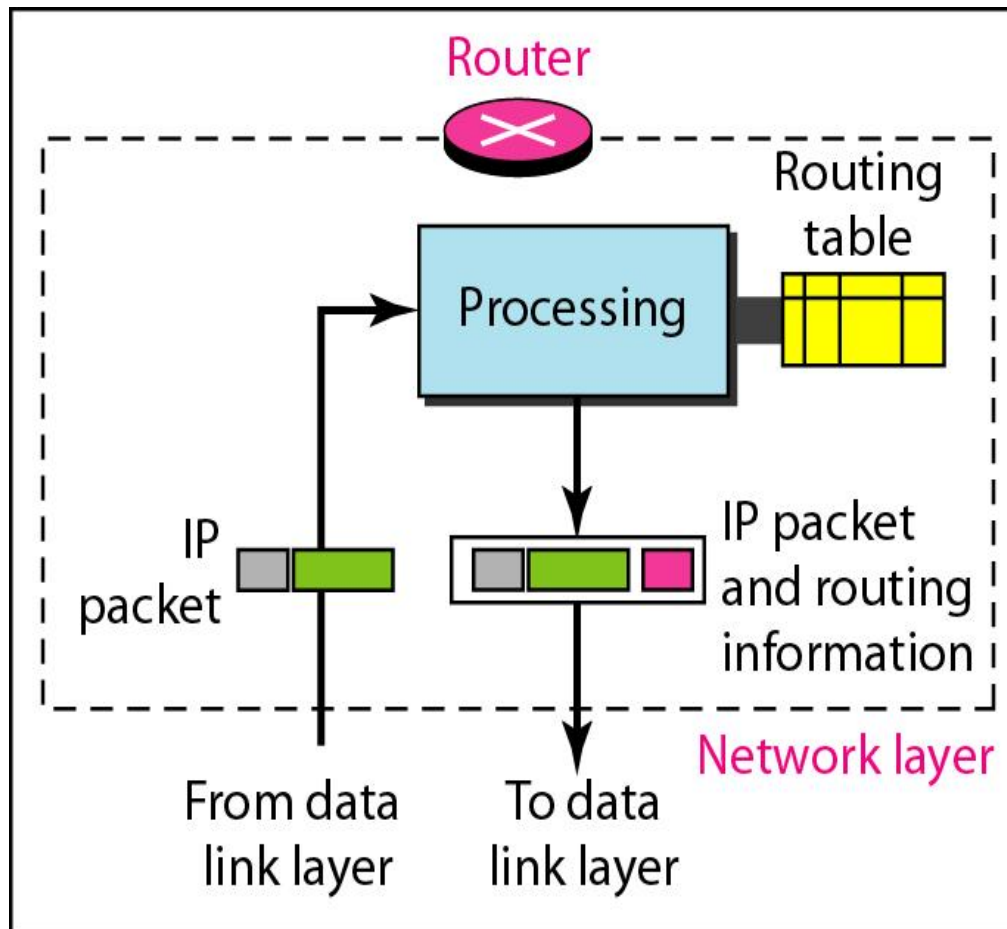
The source network layer **create a packet** carrying **addresses** of source host and destination host.

The source also checks routing table for find the **routing information** (e.g. outgoing interface, next hop, Ethernet or Wifi).

[Forouzan]

(4) Network Layer at the Router

Responsible for **routing** the packet



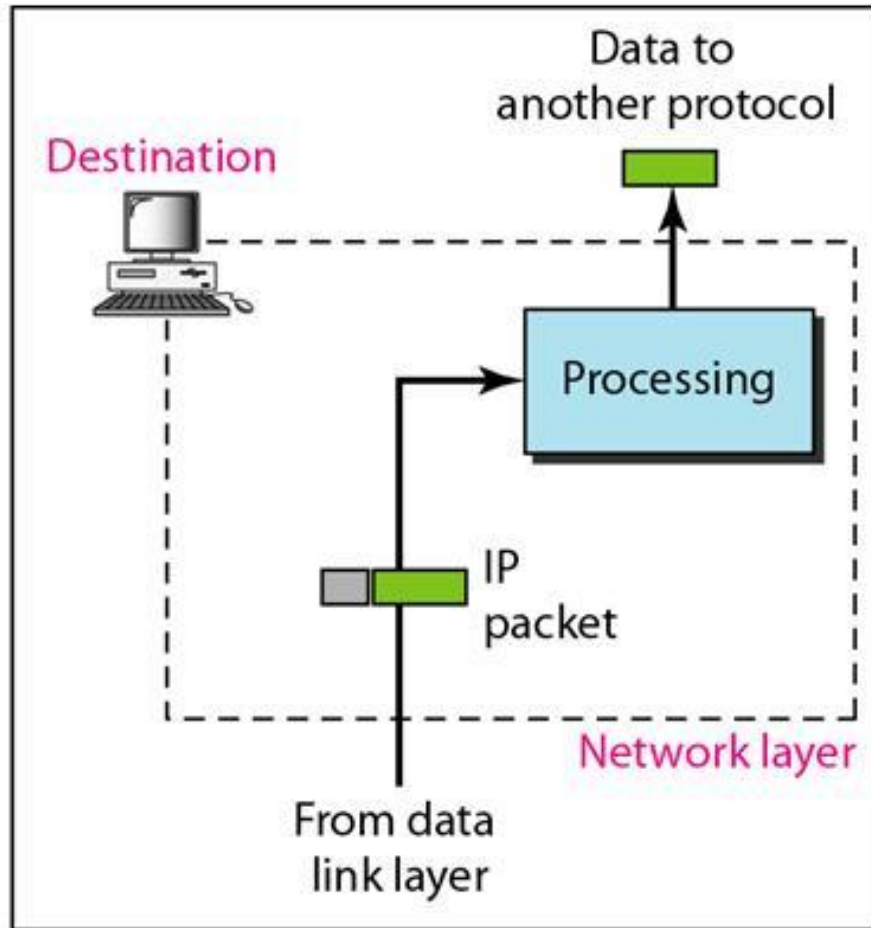
When a packet arrives, the router checks its routing table and finds the **interface** from which the packet must be sent using **routing table**.

c. Network layer at a router

[Forouzan]

(4) Network Layer at the Destination

Responsible for **address verification**



b. Network layer at destination

The destination makes sure that the **destination address** on the packet is the same as the address of the host.

[Forouzan]

下列关于网络层的描述正确的是()

- ☒ A 网络层地址在网络范围内是唯一的
- ☐ B 路由器router和交换机switch都是网络层的设备
- ☒ C 网络层具有数据路由 (Routing) 能力, 能够根据目的地址进行路由转发
- ☒ D 网络层数据处理单位是分组packet

5.1.2 Services Provided to Transport Layer(传输层)

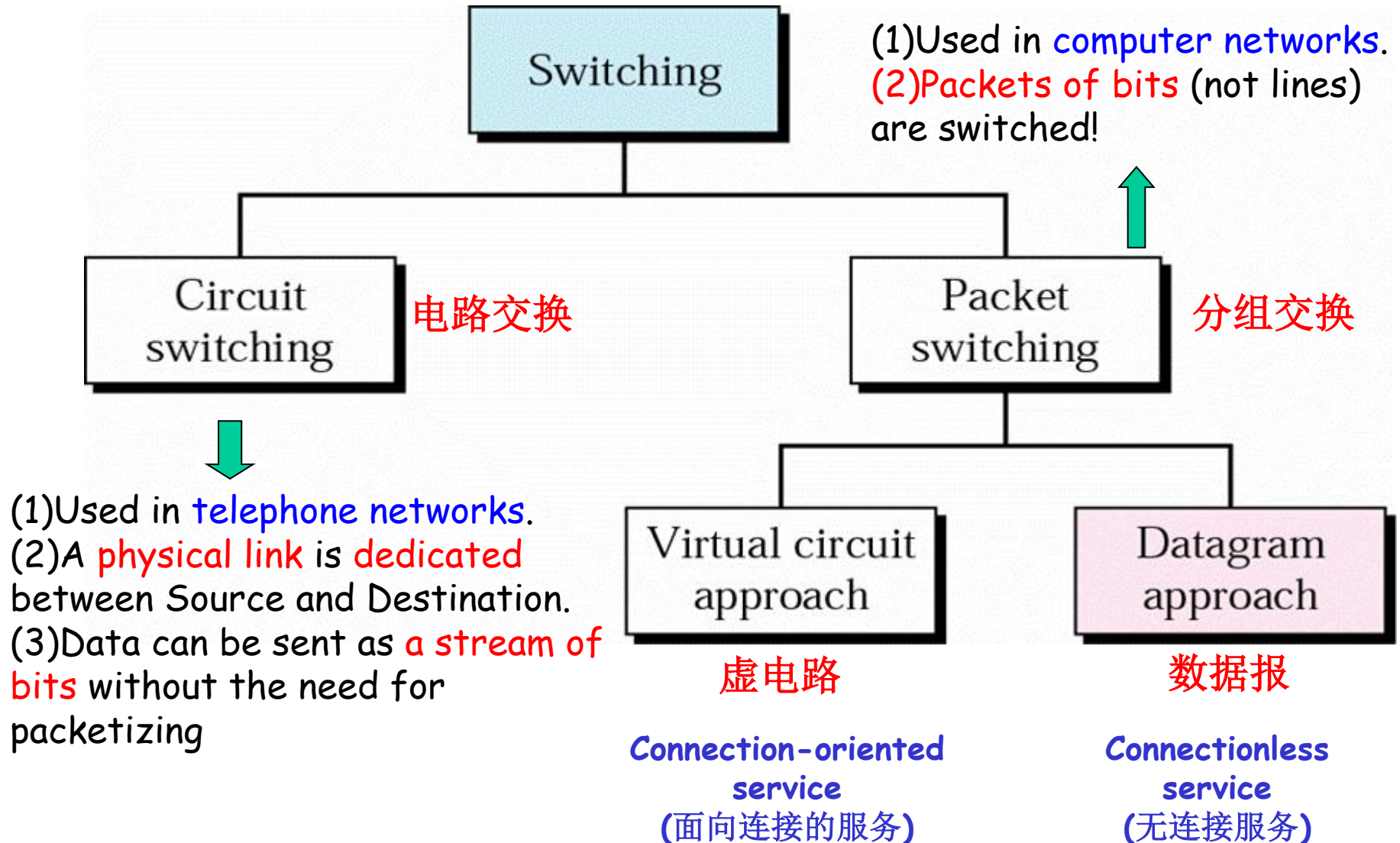
■ Design Goal

- ◆ The services should be independent of the router technology.
- ◆ The transport layer should be shielded from the number, type, and topology of the routers present
- ◆ The network addresses made available to the transport layer should use a uniform(统一)numbering plan, even across LANs and WANs

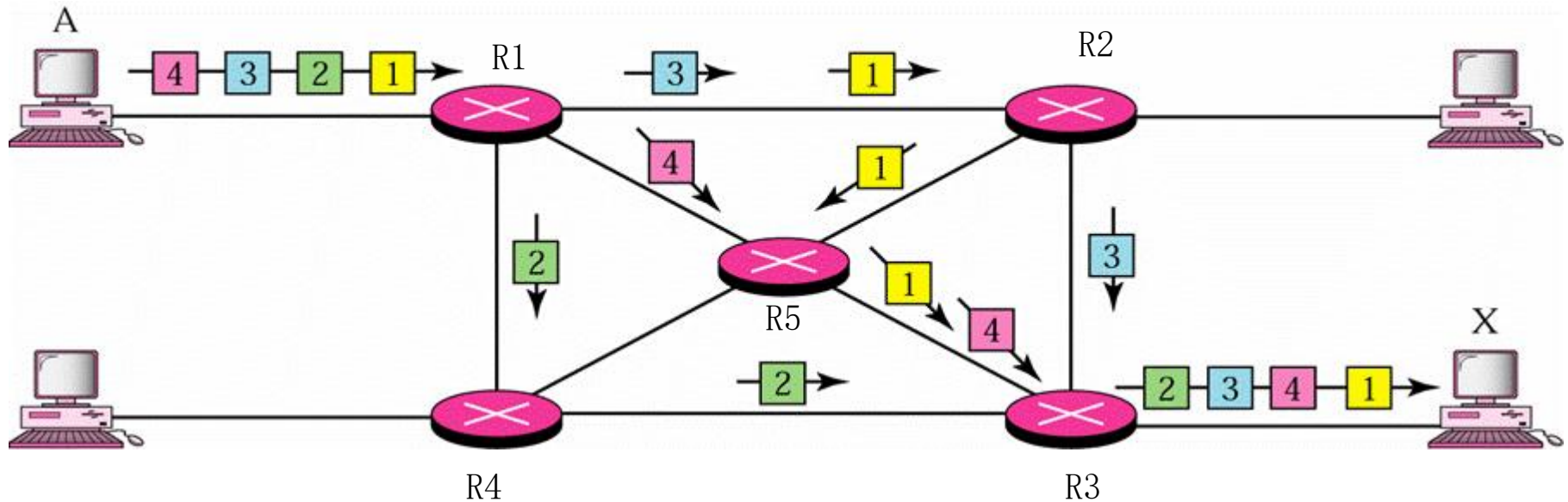
■ Two classes of service

- ◆ Connectionless Service (数据报--Datagram)
- ◆ Connection-Oriented Service (虚电路—Virtual Circuit)

Switching (交换)



5.1.3 Implementation of Connectionless Service: Datagram(数据报) Network



Routing: where to forward the packet?

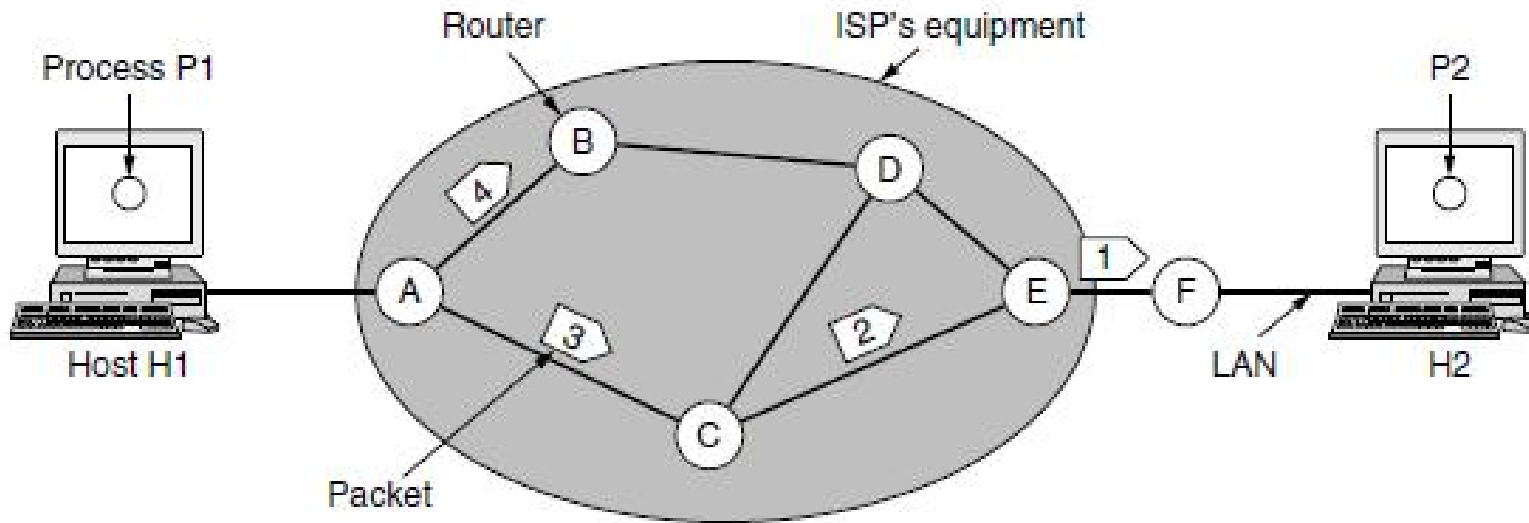
Routing Table of R1

Dest.	Line(Next Hop , 下一跳)
R1	---
R2	R2
R3	R5
R4	R4
R5	R5

The router forwards packets according to the **routing table**.

[Forouzan]

Routing within a Datagram(数据报) Subnet



A's table (initially)

A	-
B	B
C	C
D	B
E	C
F	C

Dest. Line

A's table (later)

A	-
B	B
C	C
D	B
E	B
F	B

C's table

A	A
B	A
C	-
D	E
E	E
F	E

E's table

A	C
B	D
C	C
D	D
E	-
F	F

The routing tables are **dynamic** and are **updated** periodically.

Review(1)

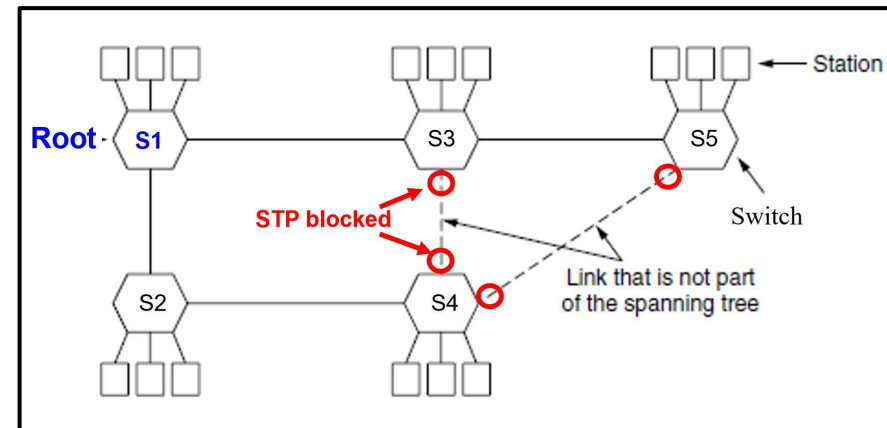
■ Data Link Layer Switching

◆ Backward Learning Bridges

- ◆ **Source MAC:** building forward(station) table
- ◆ **Destination MAC**
 - ◆ **Unicast addr:** forward, discard, flooding
 - ◆ **Broadcast addr:** flooding

◆ Spanning Tree Bridges

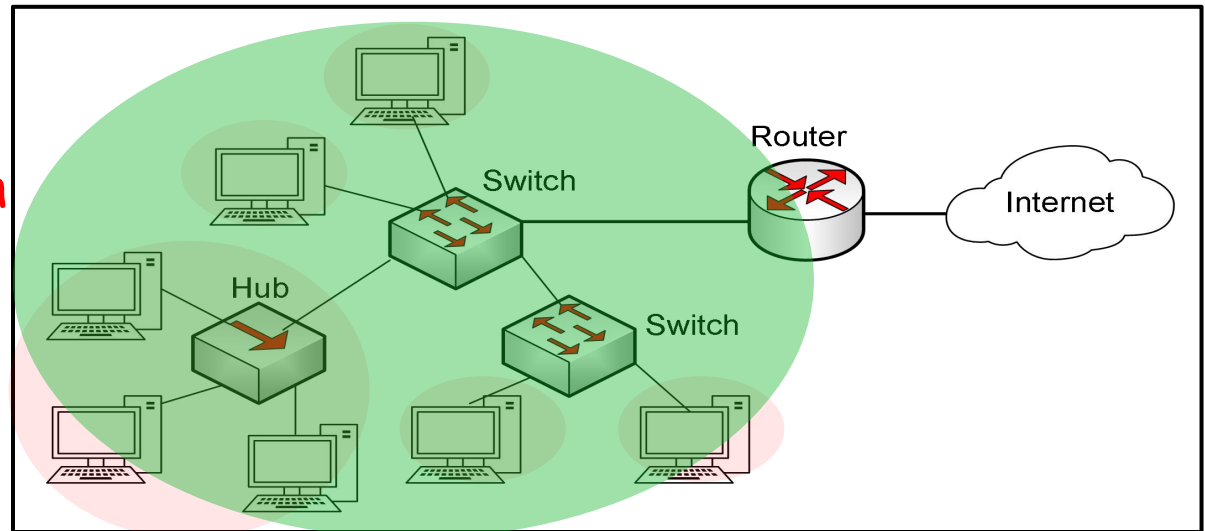
- ◆ **Redundant topology, loops**
- ◆ **Idea: loop-free topology**
 - ◆ electing root
 - ◆ shortest path to the root
 - block ports
 - detect topology changes and update the tree



Review(2)

■ Repeaters, Hubs, Bridges, Switches, Routers

- ◆ Layer
- ◆ Collision domain
- ◆ Broadcast domain



■ Virtual LANs

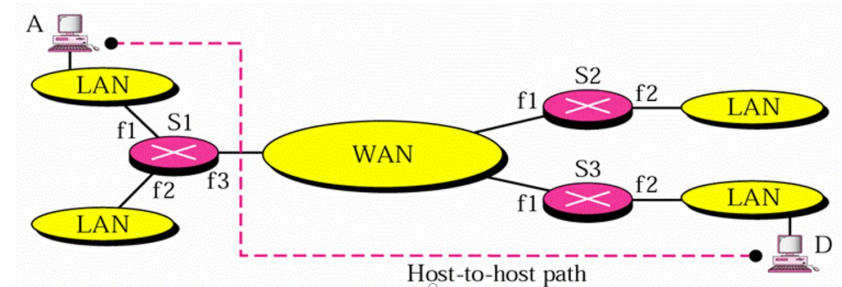
- ◆ Decoupling the logical topology from the physical topology
- ◆ Prohibit
 - packets from traveling to destinations on different VLANs
 - multicast and broadcast across colors
- ◆ Protocol: IEEE802.1Q
- ◆ Devices: switches and bridges

Review(3)

■ Network layer

◆ Positions

- Network layer
- Host-to-host transmission

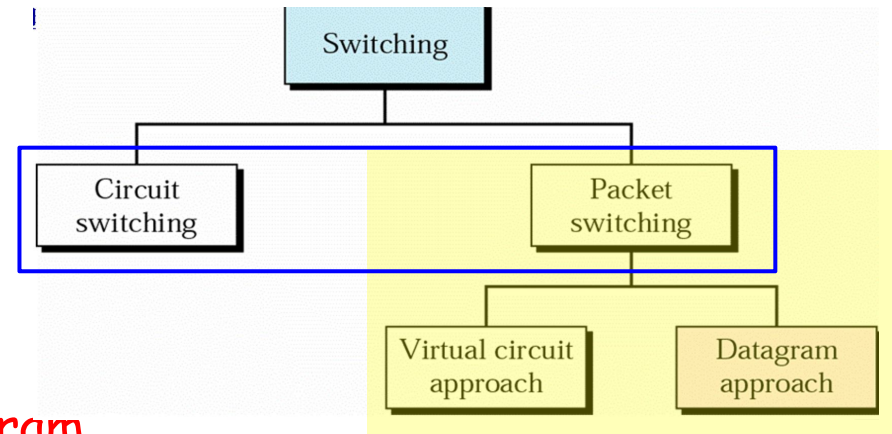


◆ Functions

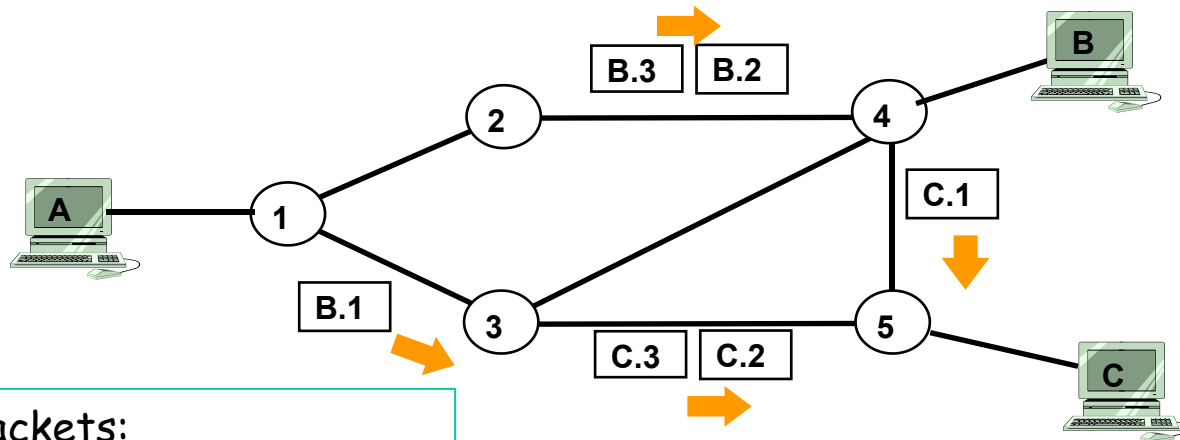
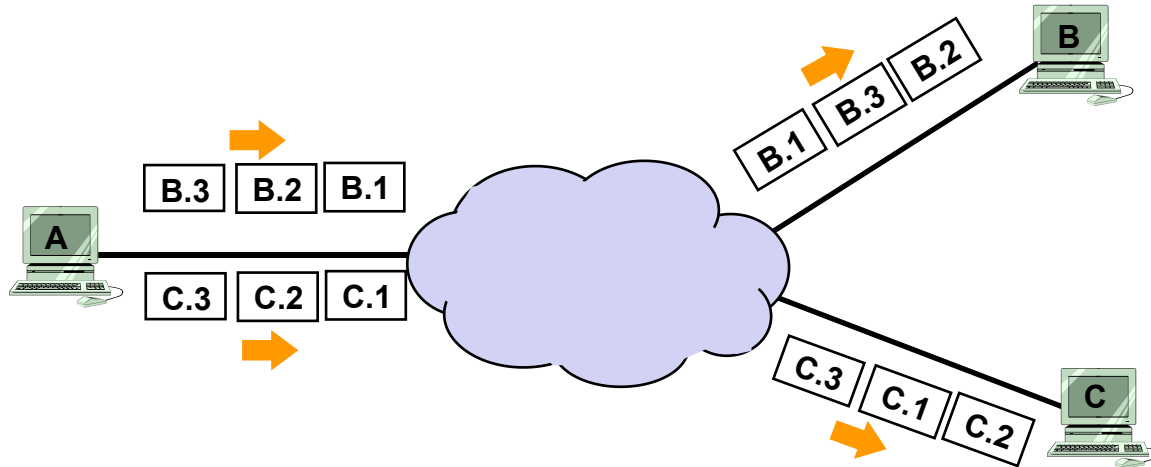
- packetizing (transmission unit---packet)
- fragmenting
- addressing
- routing
- internetworking
- QoS, congestion control

◆ Services

- Connectionless Service--Datagram
- Connection-Oriented Service--Virtual Circuit



Routing within a Datagram Network

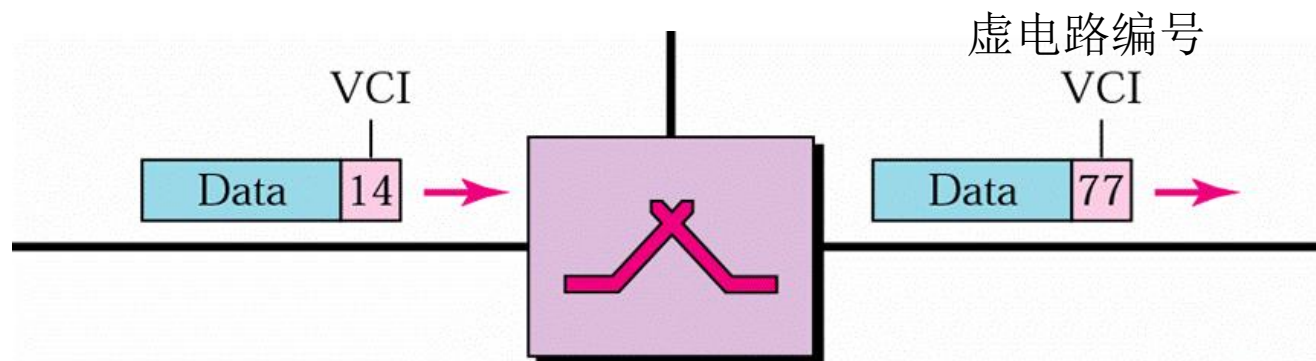


The packets:

- carry the **full address**
- may arrive **out of order**

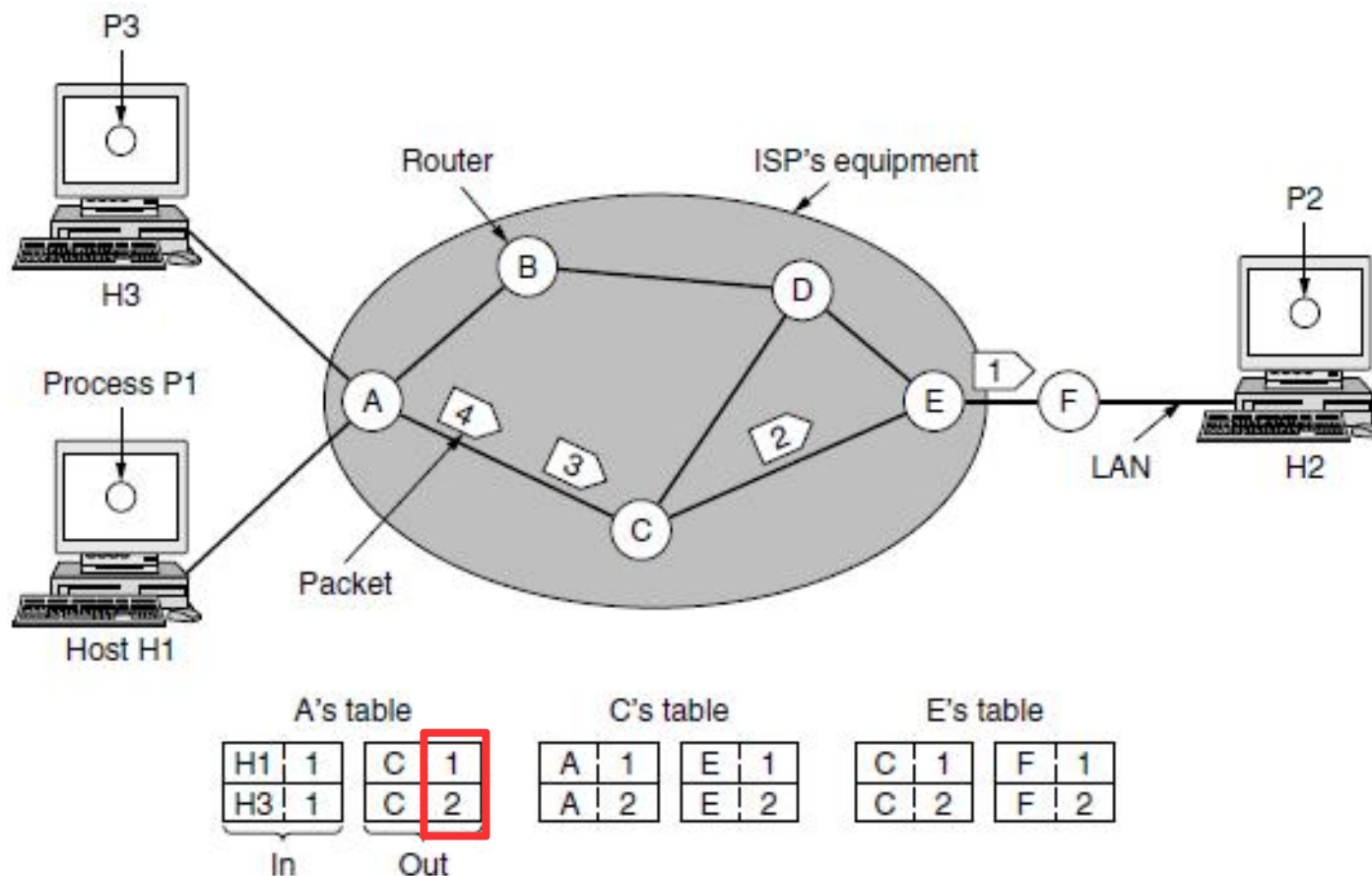
5.1.4 Implementation of Connection-oriented Service: Virtual Circuit(虚电路) Network

- Connection-oriented service
- Virtual circuit is established and end-to-end path is decided before data transmission
 - ◆ When a connection is established, a route(virtual circuit) from source to destination is chosen and stored in routers' tables
 - ◆ When the connection is released, the virtual circuit is terminated
- Virtual Circuit Identifier(VCI): local significant
 - ◆ Much shorter than address



[Forouzan]

Routing within a Virtual-circuit Subnet

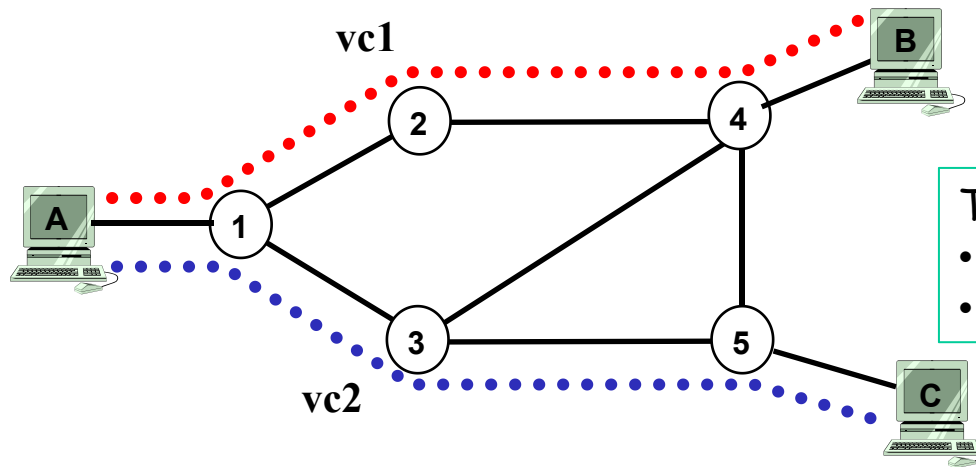
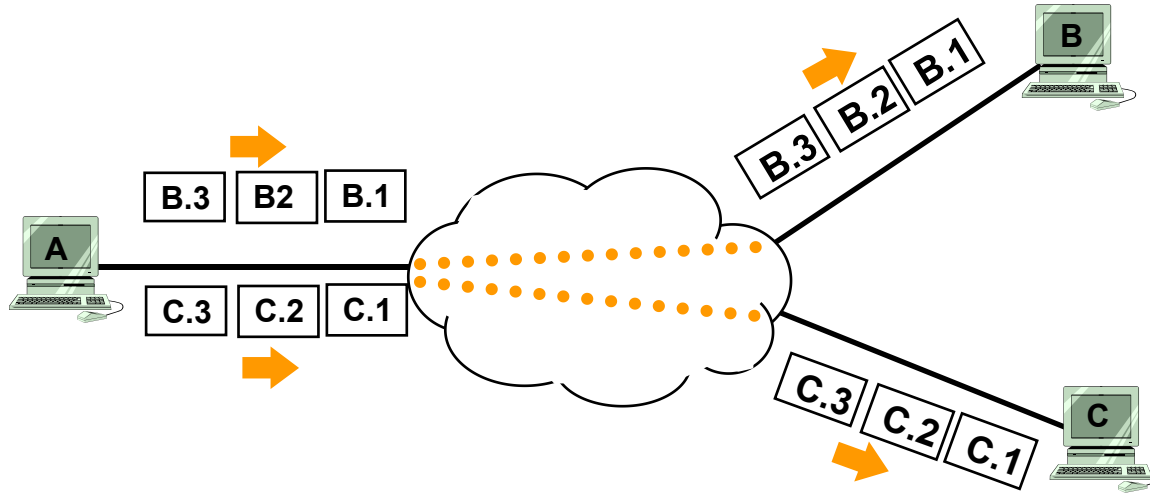


label switching(标记交换)

VC1: H1-A-C-E-F-H2

VC2: H3-A-C-E-F-H2

Routing within a Virtual-circuit Subnet(2)



● **vc1: A--1--2--4--B**

● **vc2: A--1--3--5--C**

The packets:

- carry **short and fixed-length vci**
- arrive **in order**

5.1.5 Virtual Circuit vs. Datagram

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service 服务质量	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control 拥塞控制	Difficult	Easy if enough resources can be allocated in advance for each VC

补: End-to-end argument

- Internet community: routers' job is moving packets around and nothing else.

- ◆ the network is inherently unreliable
- ◆ the hosts should do error control, flow control ...
- ◆ The network service is connectionless (datagram), best effort service(尽力而为的服务)

- End-to-end argument: 网络的智能放在端节点

END-TO-END ARGUMENTS IN SYSTEM DESIGN

J.H. Saltzer, D.P. Reed and D.D. Clark*

M.I.T. Laboratory for Computer Science

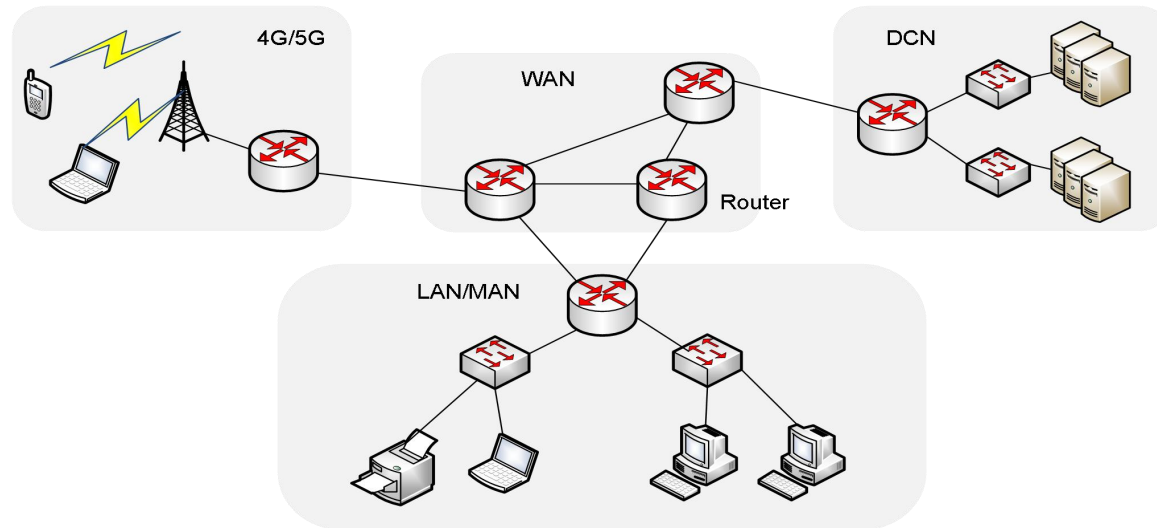
This paper presents a design principle that helps guide placement of functions among the modules of a distributed computer system. The principle, called the end-to-end argument, suggests that functions placed at low levels of a system may be redundant or of little value when compared with the cost of providing them at that low level. Examples discussed in the paper include bit error recovery, security using encryption, duplicate message suppression, recovery from system crashes, and delivery acknowledgement. Low level mechanisms to support these functions are justified only as performance enhancements.

Outline

- 5.1 Introduction to Network Layer
- 5.2 Routing Algorithms
 - ◆ (補) *Concept of Routing and Routing Table*
 - ◆ 5.2.1 Optimality principle
 - ◆ 5.2.2 Dijkstra's "Shortest Path" Algorithm
 - ◆ 5.2.3 Flooding
 - ◆ 5.2.4 Distance Vector Routing (DVR)
 - ◆ 5.2.5 Link State Routing (LSR)
 - ◆ 5.2.6 Hierarchical Routing
- 5.3 Congestion Control
- 5.4 Quality of Service
- 5.5 Internetworking
- 5.6 The network layer in Internet

(1)Routing(路由): Overview

- An internet(互联网) is a **combination of networks** connected by **routers**. When a packet goes from a source to a destination, it may pass through many routers.



- A router consults a **routing table** when a packet is ready to be forwarded. The routing table specifies the **optimum path (最佳路径)** for the packet.

(1)Routing: Overview(cont.)

- **Routing Algorithms:** algorithms used to construct and update routing tables
- **Routing Protocols(路由协议):** a combination of rules/procedures that lets routers inform one another when changes occur; mostly based on sharing/combining information between routers at different networks.
 - ◆ RIP, OSPF, BGP

路由表 = 路由算法 + 路由协议

(2) Responsibility of Routing Algorithms

- Responsible for deciding which output line an incoming packet should be transmitted on
 - ◆ If the subnet uses datagrams internally, this decision must be made for every arriving data packet
 - ◆ If the subnet uses virtual circuits internally, routing decisions are made only when a new virtual circuit is being set up.
- Routing(路由) & Forwarding(转发)
 - ◆ Routing: Generate and update routing tables using routing algorithms and protocols
 - ◆ Forwarding: Research the outbound route of the packet according to the routing table

(3)How to evaluate routing algorithms?

■ Goals

- ◆ Correctness
- ◆ Simplicity
- ◆ Robustness
- ◆ Stability: 算法可快速收敛达到稳定平衡状态
- ◆ Fairness
- ◆ Optimality
 - Minimizing mean packet delay
 - Maximizing total network throughput

■ Performance Criteria

- ◆ Least cost, such as minimum hops

(4) Classification of Routing Algorithms

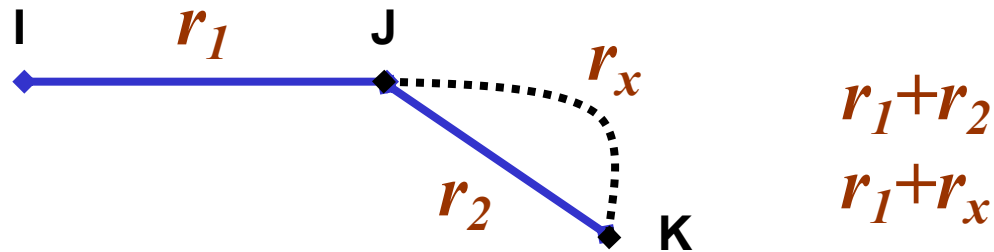
- **Nonadaptive(非自适应) algorithms (Static Routing)**
 - ◆ The choice of the route to use is computed in advance, off-line, and downloaded to the routers when the network is booted
 - ◆ The routing decisions are not based on measurements or estimates of the current traffic and topology
- **Adaptive(自适应) algorithms**
 - ◆ Changing their routing decisions to reflect changes in the topology, and usually the traffic as well 适应网络拓扑和业务量的变化

Routing Algorithms

- (補)Concept of Routing and Routing Table
- *5.2.1 Optimality principle*
- 5.2.2 Dijkstra's "Shortest Path" Algorithm
- 5.2.3 Flooding
- 5.2.4 Distance Vector Routing (DVR)
- 5.2.5 Link State Routing (LSR)
- 5.2.6 Hierarchical Routing

The Optimality Principle(最优化原则)

- If router J is on the optimal path from router I to router K, then the optimal path from J to K also falls along the same route
 - ◆ If a route r_x better than r_2 existed from J to K, it could be concatenated with r_1 to improve the route from I to K
 - ◆ Contradicting our statement that r_1r_2 is optimal



最优路径上任意两点之间的路径也是最优的

Routing Algorithms

- (補)Concept of Routing and Routing Table
- 5.2.1 Optimality principle
- *5.2.2 Dijkstra's "Shortest Path Tree" Algorithm(白魔)*
- 5.2.3 Flooding
- 5.2.4 Distance Vector Routing (DVR)
- 5.2.5 Link State Routing (LSR)
- 5.2.6 Hierarchical Routing

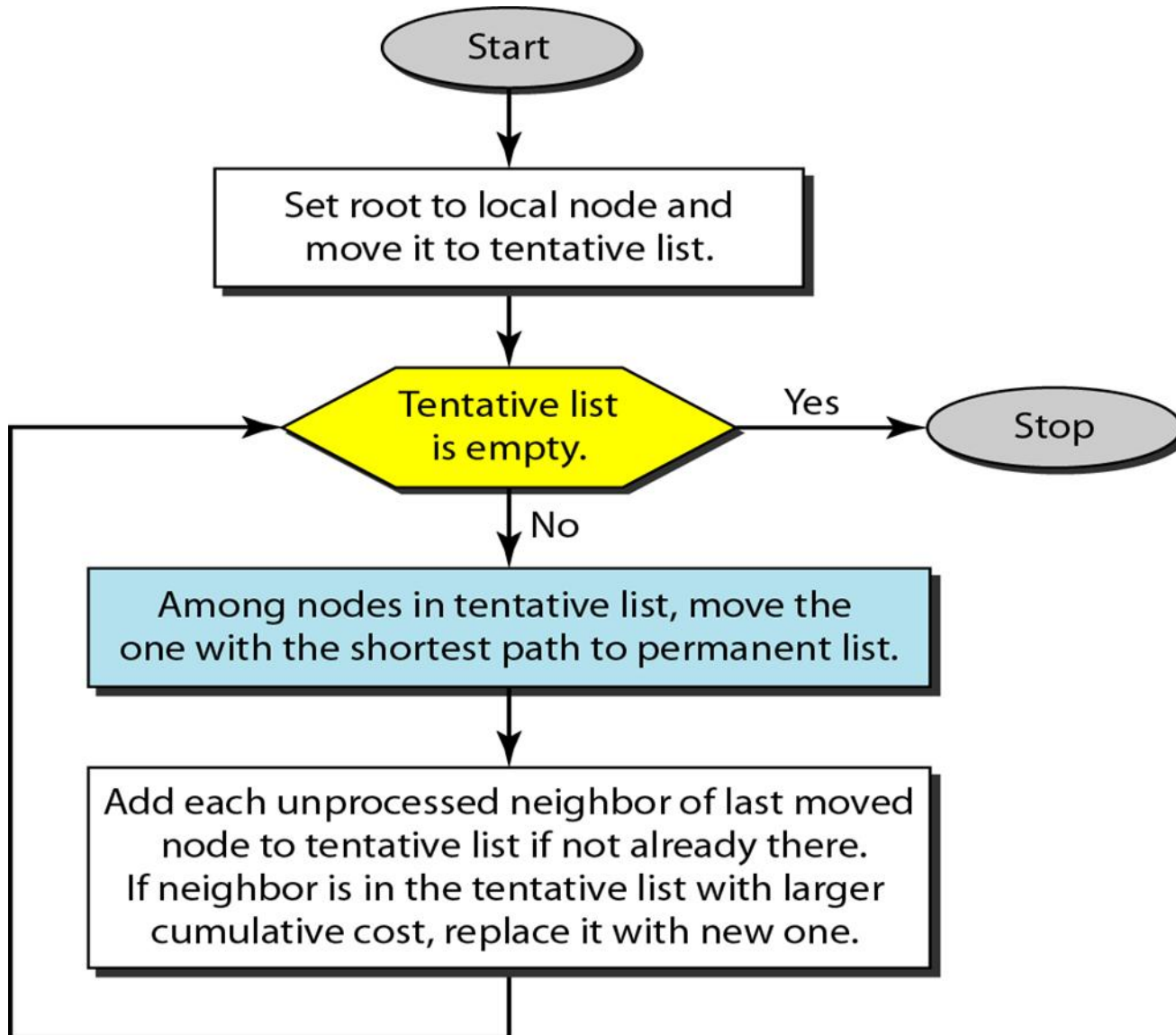
(1) Shortest Path Routing(最短路径选路)

- Build a weighted and directed graph of subnet(有向加权图)
 - ◆ Nodes represent **routers**
 - ◆ Arcs(边) represents communication lines
- Weight
 - ◆ **Hops**
 - ◆ Geographic **distance** in kilometers
 - ◆ Mean queuing and transmission **delay**
 - ◆ **A function of** the distance, bandwidth, average traffic, communication cost, mean queue length, measured delay, and other factors

(2) Dijkstra

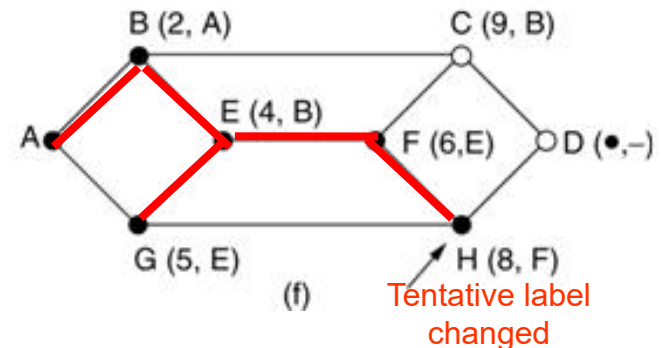
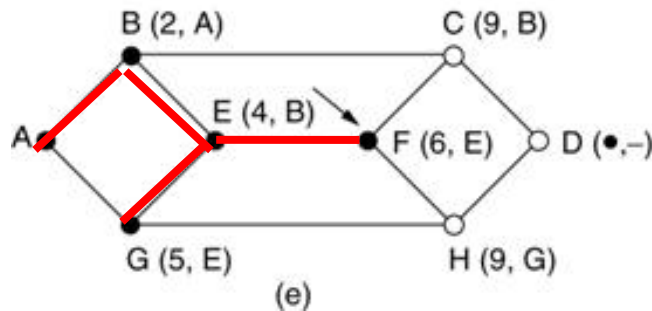
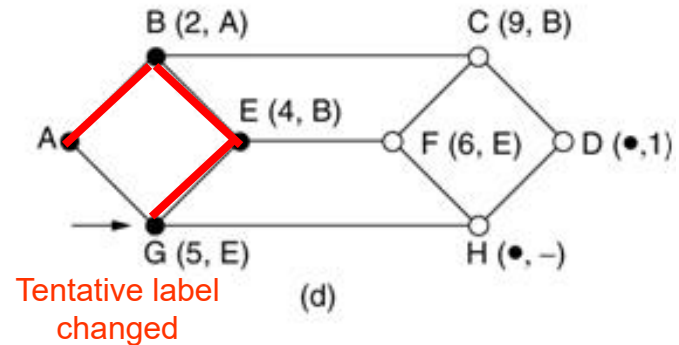
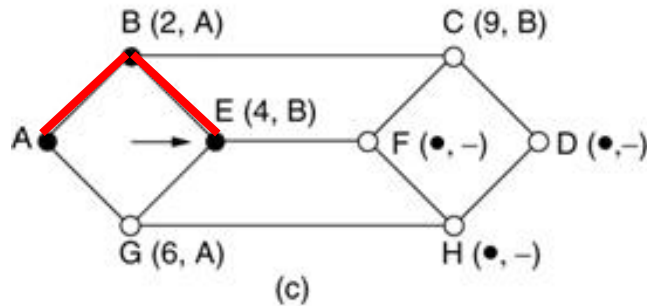
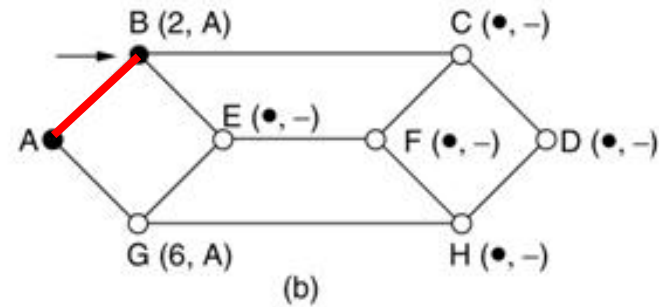
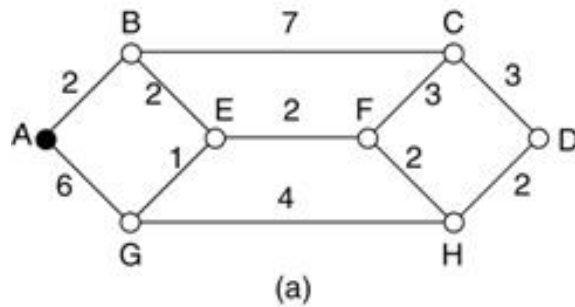
- Find **shortest paths tree**(SPT最短路径树) from given source node to all other nodes
- Routing Algorithm
 - ◆ 1. Starting with the **local node s** (router): the **root** of the tree.
 - ◆ 2. Assign a cost of 0 to this node and make it the first **permanent node**(永久节点).
 - ◆ 3. Examine each neighbor node of the node that was the last permanent node **P**.
 - ◆ 4. Assign a **cumulative cost** (累计开销) to each node and make it **tentative** (临时节点) .
 - ◆ 5. Among the list of tentative nodes
 - Find the node with the **smallest** cumulative cost and make it permanent.
 - If a node can be reached from more than one direction, Select the direction with the shortest cumulative cost.
 - ◆ 6. Repeat steps 3 to 5 until **every** node becomes permanent.

Dijkstra算法的流程图

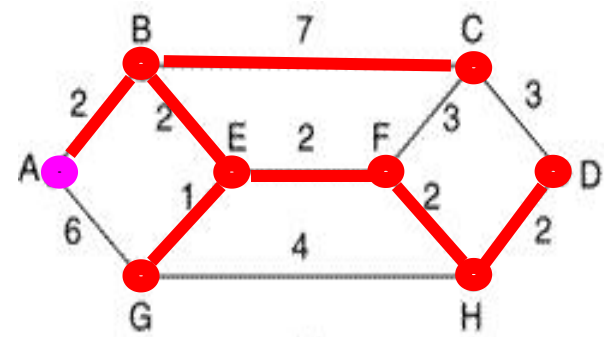


[Forouzan]

Example of Dijkstra Algorithm



Example of Dijkstra

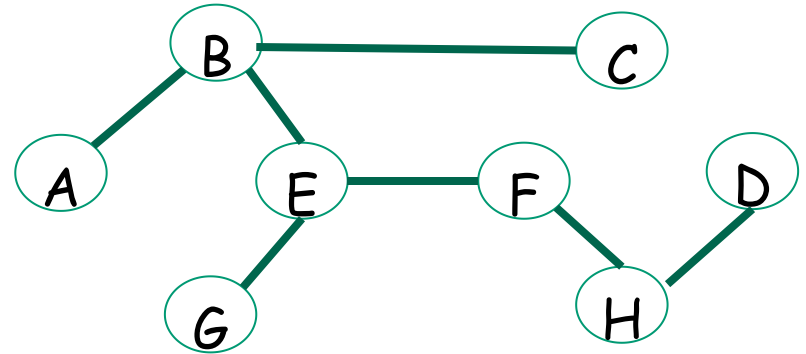
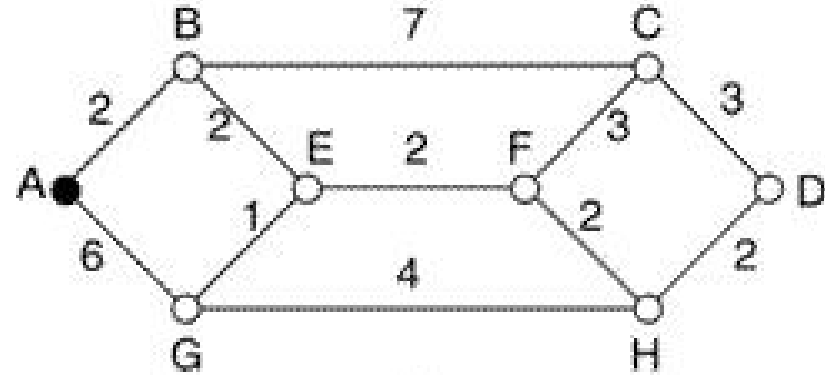


A	B	C	D	E	F	G	H	Permanent Node
∞	∞	∞	∞	∞	∞	∞	∞	
0	∞	∞	∞	∞	∞	∞	∞	A
0	2, A	∞	∞	∞	∞	6, A	∞	B
0	---	9, B	∞	4, E	∞	6, A	∞	E
0	---	9, B	∞	---	6, E	5, E	∞	G
0	---	9, B	∞	---	6, E	---	9, G	F
0	---	9, B	∞	---	---	---	8, F	H
0	---	9, B	10, H	---	---	---	---	C
0	---	---	10, H	---	---	---	---	D

What we get by using Dijkstra?

A's routing table

Dest.	Cost	Next Hop
A	0	--
B	2	B
C	9	B
D	10	B
E	4	B
F	6	B
G	5	B
H	9	B



A's shortest path tree

几个问题(课下思考)

- **Dijkstra**算法工作的条件是什么？
- 为什么每个路由器的路由表只负责找到下一跳？而不需要保存整个最优路径？

Routing Algorithms

- (補)Concept of Routing and Routing Table
- 5.2.1 Optimality principle
- 5.2.2 Dijkstra's "Shortest Path" Algorithm
- *5.2.3 Flooding*
- 5.2.4 Distance Vector Routing (DVR)
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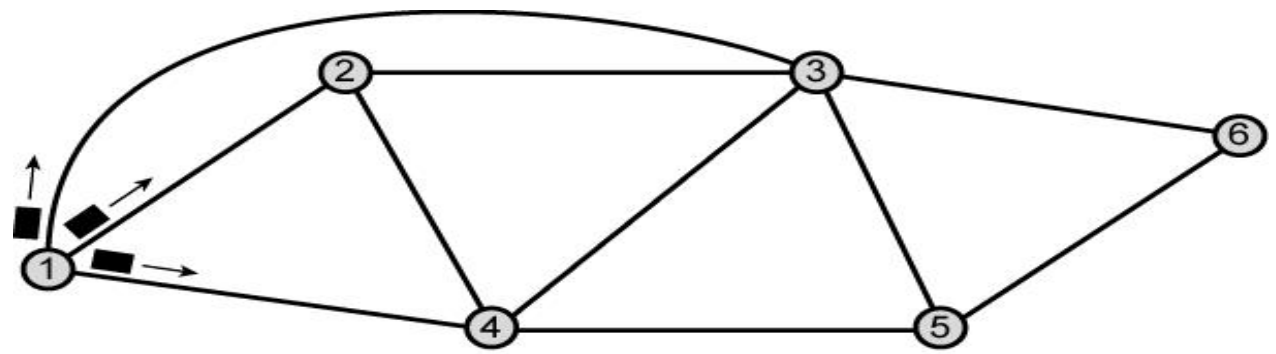
(1) Major Routing Strategies

- Fixed Routing(固定查表选路)
 - ◆ Single permanent route for each source to destination pair
- Flooding(洪泛/泛洪)
 - ◆ Packet sent to every neighbor
 - ◆ Incoming packets forwarded to every link except incoming link
- Random Routing(随机选路)
 - ◆ Selects one outgoing path to forward incoming packet
 - ◆ Selection can be random or round robin
- Adaptive Routing(自适应选路)
 - ◆ Routing decisions **change** as conditions on the network change
 - ◆ Requires **information about network**
 - ◆ Used by almost all packet switching networks

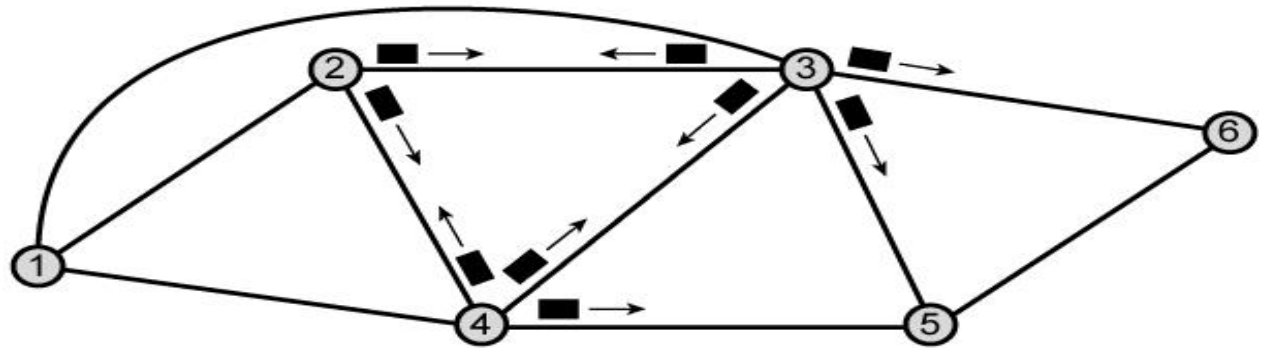
(2) Flooding(洪泛法): Ideas

- 无需路由表
- No network information required
- Incoming packets retransmitted on every link except incoming link
- Eventually a number of copies will arrive at destination

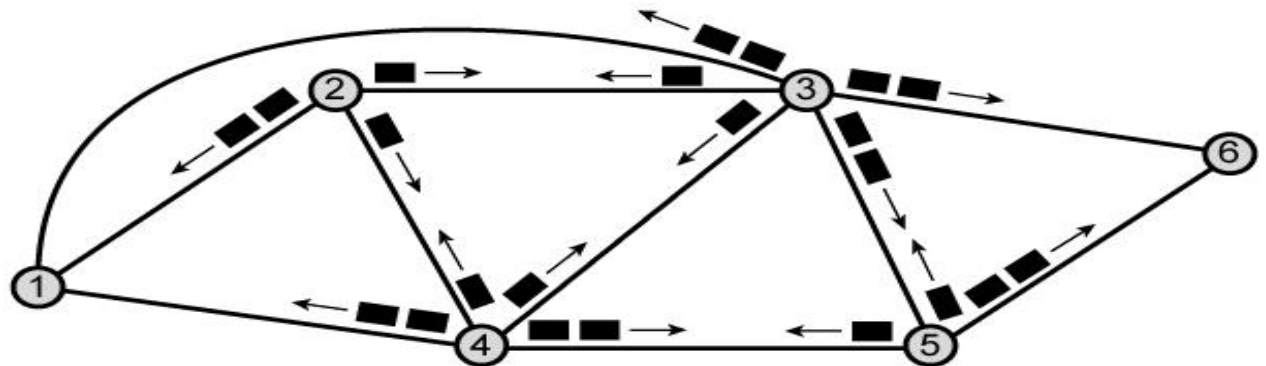
Example of Flooding



(a) First hop



(b) Second hop



(c) Third hop

(3) Flooding: Problems and Solutions

■ Problem

- ◆ Generates vast numbers of duplicate packets

■ Solutions

- ◆ The header of each packet contains a hop counter(跳计数器)
 - Counter is decremented at each hop
 - When the counter reaches zero, discard the packet
 - Sender initializes the hop counter as the length of the path from source to destination, or the subnet diameter
- ◆ Source puts a sequence number in each packet(序列号)
 - Each router records maximal seq per source telling which seq have already been seen

(4) Properties of Flooding

- All possible routes are tried
 - ◆ Very robust 健壮
- At least one packet will have taken minimum hop count route
- All nodes are visited
 - ◆ Useful to **distribute routing information**

Routing Algorithms

- (補)Concept of Routing and Routing Table
- 5.2.1 Optimality principle
- 5.2.2 Dijkstra's "Shortest Path" Algorithm
- 5.2.3 Flooding
- *5.2.4 Distance Vector Routing (DVR)*
- 5.2.5 Link State Routing (LSR)
- 5.2.6 Hierarchical Routing

(1) Distance Vector Routing(距离矢量选路)

- Based on Bellman-Ford algorithm

- Each router maintains a **routing table** →

- ◆ The best known **distance** to each destination

- The distance might be number of hops, time delay in milliseconds, ...

- ◆ Which line to use (下一跳/接口)(方向)

Vector (distance, direction)

Node C		
Dest.	Dist.	NextHop
A	7	D
B	1	B
D	1	D

- The table is updated by exchanging information with **neighbors** 路由器周期向所有邻居广播整个路由表

- ◆ Routing information sent: **(Destination, Distance)**--矢量

- Periodically (**RIP:30s**)

- ◆ When routing information received

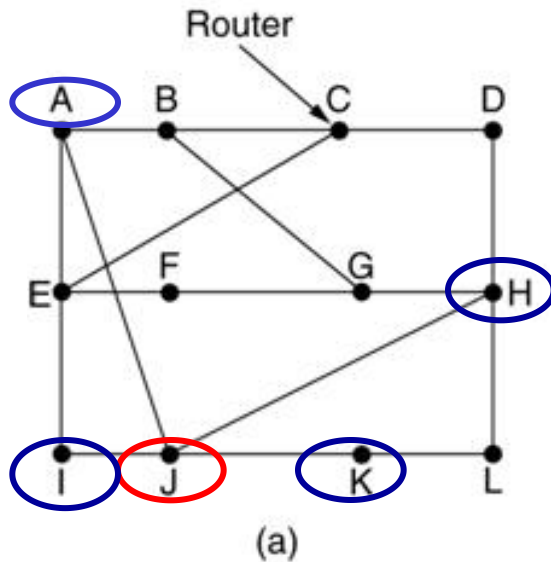
- Router **updates local table** if receive a **better** route

- Refreshes existing routes

路由器根据收到的邻居路由表更新自己的路由表

- ◆ **Delete routing table** items if they time out (**RIP: 180s**)

DVR: Example in textbook



+

To	A	I	H	K	New estimated delay from J	
					↓	Line
A	0	24	20	21	8	A
B	12	36	31	28	20	A
C	25	18	19	36	28	I
D	40	27	8	24	20	H
E	14	7	30	22	17	I
F	23	20	19	40	30	I
G	18	31	6	31	18	H
H	17	20	0	19	12	H
I	21	0	14	22	10	I
J	9	11	7	10	0	-
K	24	22	22	0	6	K
L	29	33	9	9	15	K

JA delay is 8 JI delay is 10 JH delay is 12 JK delay is 6

Vectors received from J's four neighbors

New routing table for J

(b)

Taking destination B as an example,

$$JB = JA + AB = 8 + 12 = 20$$

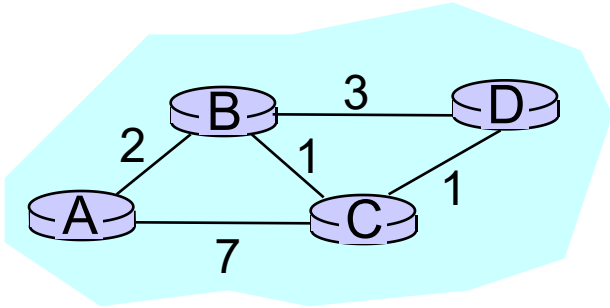
$$JB = JI + IB = 10 + 36 = 46$$

$$JB = JH + HB = 12 + 31 = 43$$

$$JB = JK + KB = 6 + 28 = 34$$

Thus, A is selected as next hop.

DVR Example 2: Initial Status



Node A

Dest.	Cost	NextHop
B	2	B
C	7	C
D	∞	-

Node B

Dest.	Cost	NextHop
A	2	A
C	1	C
D	3	D

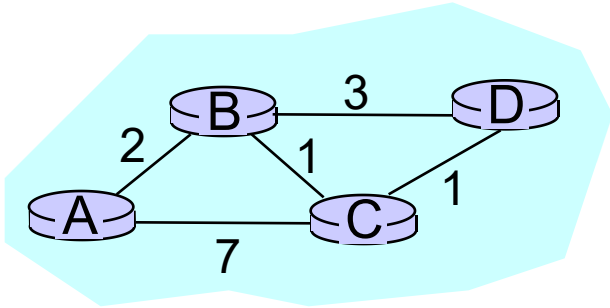
Node C

Dest.	Cost	NextHop
A	7	A
B	1	B
D	1	D

Node D

Dest.	Cost	NextHop
A	∞	-
B	3	B
C	1	C

Example 2: 1st Iteration (C → A)



Node A

Dest.	Cost	NextHop
B	2	B
C	7	C
D	8	C

Node B

Dest.	Cost	NextHop
A	2	A
C	1	C
D	3	D

$$D(A, D) = D(A, C) + D(C, D) = 7 + 1 = 8$$

(D(C,A), D(C,B), D(C,D))

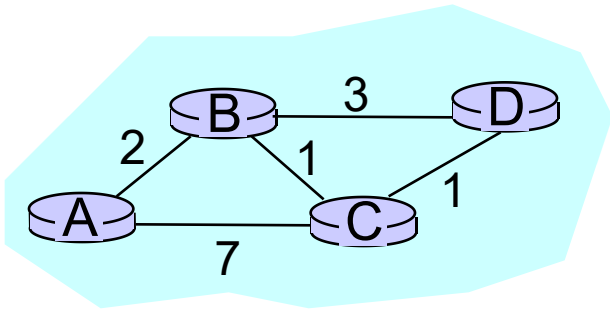
Node C

Dest.	Cost	NextHop
A	7	A
B	1	B
D	1	D

Node D

Dest.	Cost	NextHop
A	∞	-
B	3	B
C	1	C

Example 2: 1st Iteration ($B \rightarrow A$, $C \rightarrow A$)



Node A

Dest.	Cost	NextHop
B	2	B
C	3	B
D	5	B

Node B

Dest.	Cost	NextHop
A	2	A
C	1	C
D	3	D

$$D(A,D) = D(A,B) + D(B,D) = 2 + 3 = 5$$

$$D(A,C) = D(A,B) + D(B,C) = 2 + 1 = 3$$
$$D(A,C) = 7$$

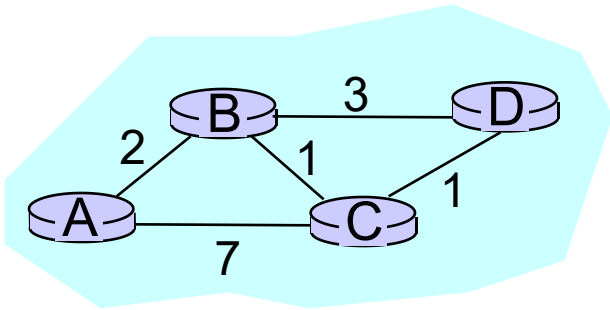
Node C

Dest.	Cost	NextHop
A	7	A
B	1	B
D	1	D

Node D

Dest.	Cost	NextHop
A	∞	-
B	3	B
C	1	C

Example 2: End of 1st Iteration



Node A

Dest.	Cost	NextHop
B	2	B
C	3	B
D	5	B

Node B

Dest.	Cost	NextHop
A	2	A
C	1	C
D	2	C

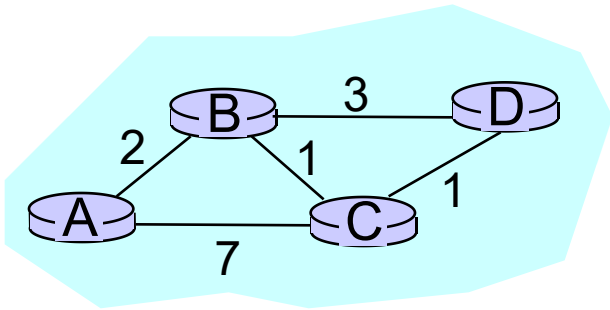
Node C

Dest.	Cost	NextHop
A	3	B
B	1	B
D	1	D

Node D

Dest.	Cost	NextHop
A	5	B
B	2	C
C	1	C

Example 2: End of 2nd Iteration



Node A

Dest.	Cost	NextHop
B	2	B
C	3	B
D	4	B

Node B

Dest.	Cost	NextHop
A	2	A
C	1	C
D	2	C

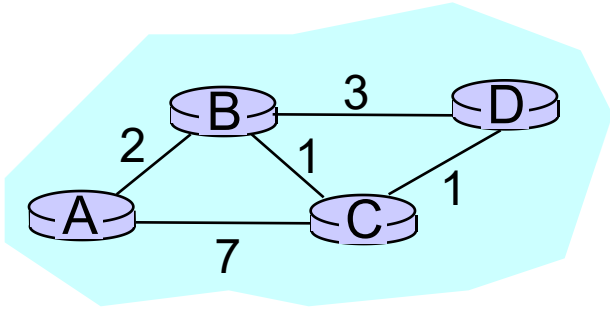
Node C

Dest.	Cost	NextHop
A	3	B
B	1	B
D	1	D

Node D

Dest.	Cost	NextHop
A	4	C
B	2	C
C	1	C

Example 2: End of 3rd Iteration



Node A

Dest.	Cost	NextHop
B	2	B
C	3	B
D	4	B

Node B

Dest.	Cost	NextHop
A	2	A
C	1	C
D	2	C

Node C

Dest.	Cost	NextHop
A	3	B
B	1	B
D	1	D

Node D

Dest.	Cost	NextHop
A	4	C
B	2	C
C	1	C

Nothing changes → algorithm terminates

DVR补充练习

■ 408-2021年

某网络中所有路由器均采用DVR计算路由。若路由器E与邻居路由器A,B,C和D之间的直接链路距离分别是8, 10, 12和6, 且E收到邻居路由器A,B,C,和D到目的网络net1的距离分别是1, 23, 20和22, 则路由器E更新后的到达目的网络net1的距离是多少?

EA=8, EB=10, EC=12, ED=6

Net1: EA+A-Net1=**8** + **1** =9

Net1: EB+B-Net1=**10** + **23** =33

Net1: EC+C-Net1=**12** + **20** =32

Net1: ED+D-Net1=**6** + **22** =28

(2)Features and Problems of DVR

■ 4 key features

- ◆ Sharing only with immediate **neighbours**.
- ◆ Sharing **knowledge about the entire network(routing table)** with neighbours.
- ◆ Sending knowledge about the entire network through all its interface.**(Broadcast Updates)**
- ◆ Sharing at **regular intervals (Periodic Updates)**.

路由器周期地向所有邻居 广播整个路由表

■ Issues

- ◆ 收敛速度可能会比较慢
- ◆ react rapidly to good news, but leisurely to bad news
(好消息反应迅速, 坏消息反应慢)

Review(1)

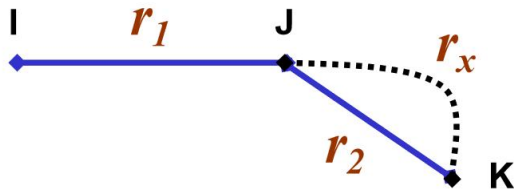
■ Routing algorithms

◆ Overview

- Goals: discovery the **optimum path** (最优路径)
- Concepts:
 - **Router** (layer 3): **Routing vs. Forwarding**
 - **Routing table** = routing algorithms + routing protocol
- Classification: **Nonadaptive algorithms** vs. **adaptive algorithms** (reflect changes in the current topology and traffic)

◆ Optimality principle

- 最优路径上的任意两点之间的路径也是最优的



Routing Table

Destination	Next hop	

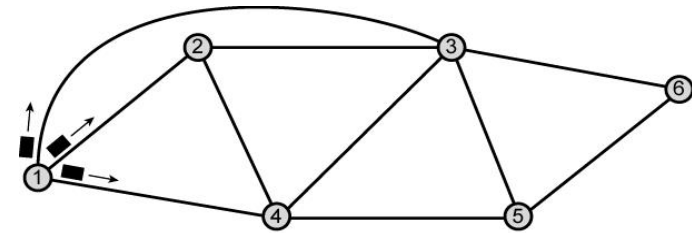
Review (2)

■ Shortest Path Algorithms

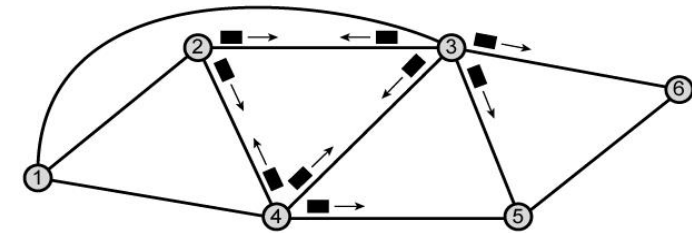
- ◆ **Dijkstra**: 单源, 非负权值有向图
- ◆ **Bellman-ford**: 允许负权值有向图(不要求)

■ Flooding

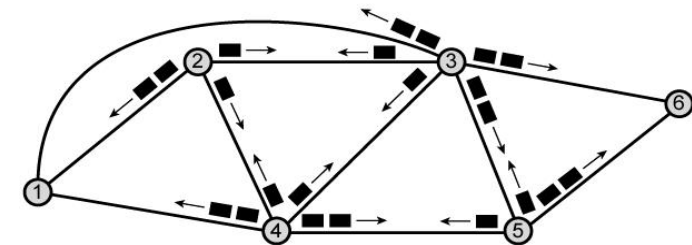
- ◆ forwarded to every link except incoming link
- ◆ Problem: generates **vast** numbers of **duplicate packets**
- ◆ Solutions: **hop counter**, **sequence number**
- ◆ Properties
 - packet is **delivered to every node**: useful to **distribute routing information**
 - tremendously **robust**: **all possible routes** are tried
 - at least one packet will have taken **minimum hop** count route



(a) First hop



(b) Second hop



(c) Third hop

Review (3)

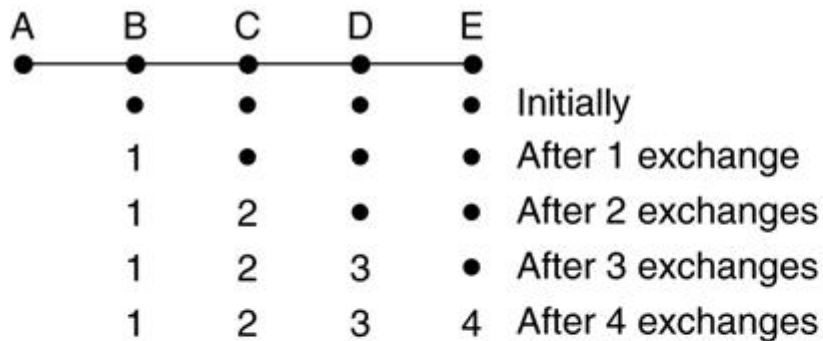
■ DVR

	DVR
Algorithm	Bellman-ford
What information to share	routing table(destination and distance)
Whom to share with	neighbors
When to share	periodically
What does router knows	distance (not path)
Issues	react leisurely to bad news

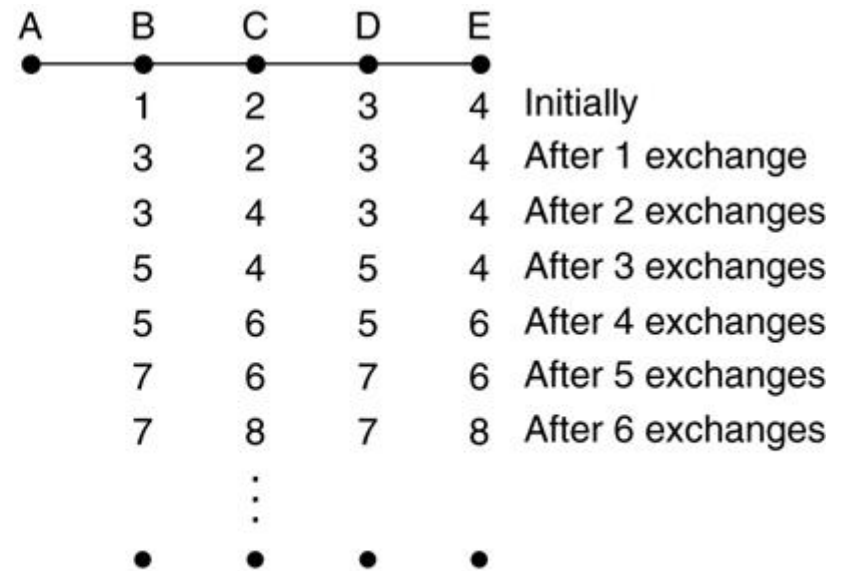
Count-to-Infinity Problem

(3) Count-to-Infinity(无穷计数) Problem

- Problems: Tedious comparing/updating process, slow response to **infinite loop (count-to-infinity, 无穷计数)** problem, huge list to be maintained!!



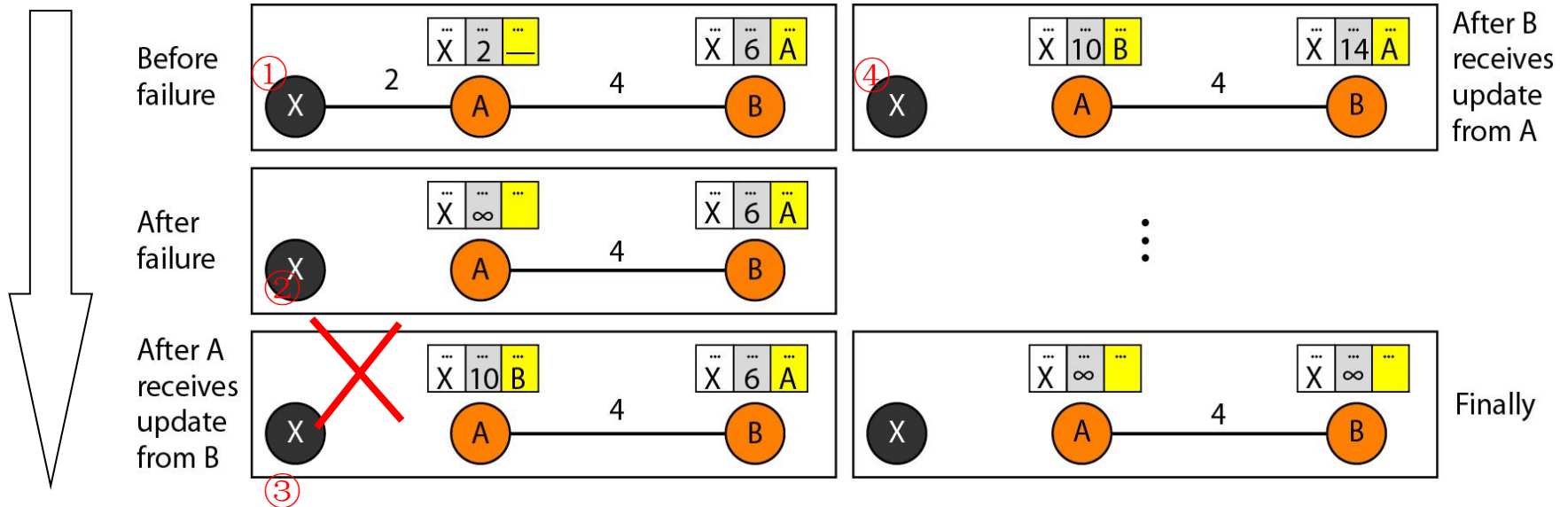
(a)



(b)

Bad news travels slowly

DVR: Infinite Loop Example



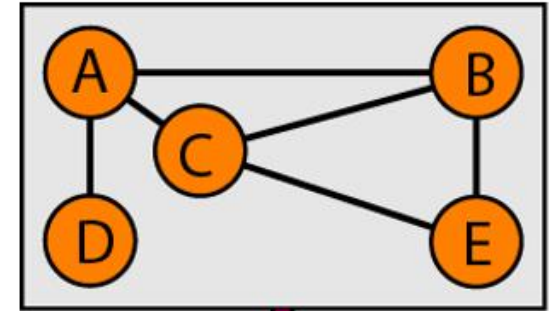
For a destination, router only knows distance, no idea of the path
(只知距离，不知路径)

Routing Algorithms

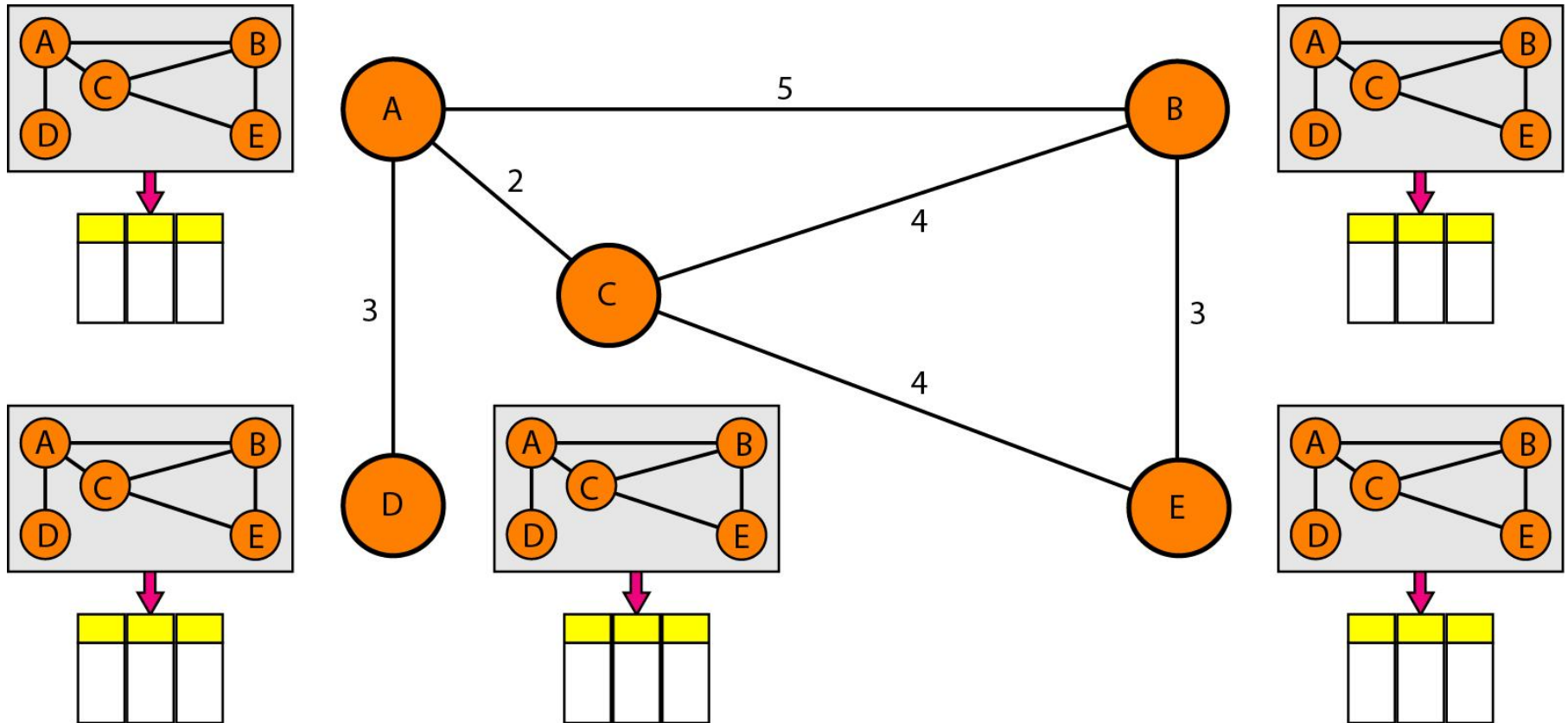
- (補)Concept of Routing and Routing Table
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- 5.2.4 Distance Vector Routing (DVR)
- *5.2.5 Link State Routing (LSR)*
- 5.2.6 Hierarchical Routing

(1) Link State Routing(链路状态选路)

- In LSR, each router shares its knowledge about the its neighbourhood with **every other routers** in the network.
- The Link-State Packet (LSP) defines the best known network topology is sent to every routers after it is constructed locally. (**flooding**)
- 3 keys to understand how LSR works:
 - ◆ Sharing knowledge about **network topology**.
 - ◆ Sharing **with every other routers** in the network.
 - ◆ Sharing **when there is a change**.
- From the received LSPs and knowledge of **entire topology**, a router can then calculate the **shortest path** (Dijkstra algorithm) from itself to every other routers in the area.



LSR: What routers know?



The routers have knowledge of **entire topology**

(2) Link State Routing: Operations

■ Each router must do the following

1. Discover its neighbors, learn their network address
2. Measure the delay or cost to each of its neighbors
3. Construct a packet (**LSP**: Link-state Packet, 链路状态包) telling all it has just learned
4. Send LSP to **all other routers** (not only neighbors) reliably (**flooding**)
5. Compute **the shortest path** tree to every other router (**Dijkstra's algorithm**)
6. Generate the **routing table** based on SPT

Step1: Learning about the Neighbors

■ HELLO packet

- ◆ For a broadcast network, broadcasting a special HELLO packet
- ◆ For a point-to-point network, unicast HELLO to neighbors

■ Neighbors send back a reply telling who it is

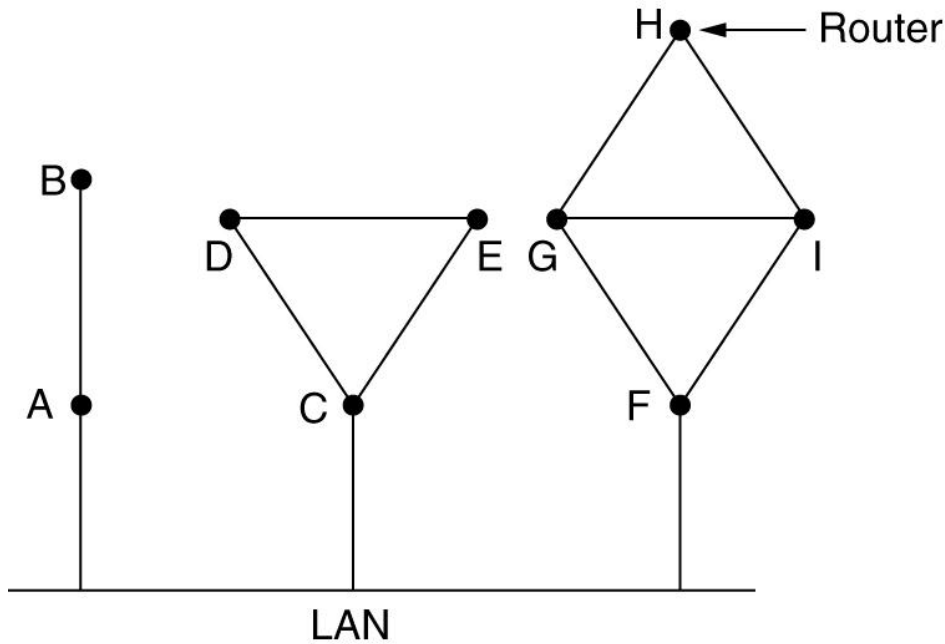
- ◆ Names (Route IDs) must be globally unique

■ Simplify the topology(自阅)

- ◆ When two or more routers are connected by a LAN or other multi-access network

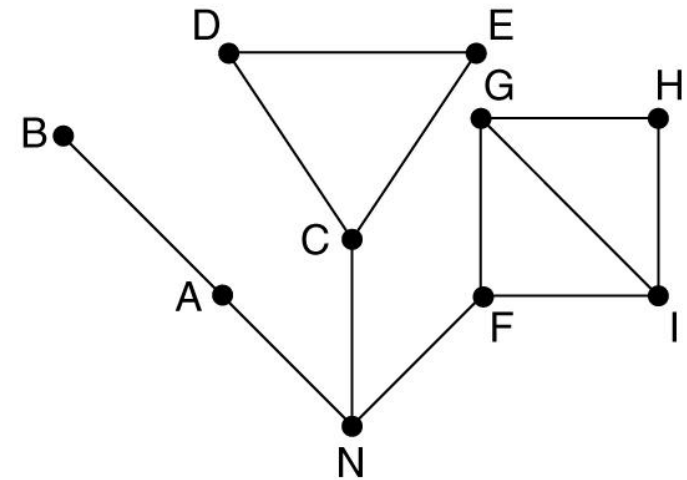
How to model the LAN

A LAN is considered as a Node(虚拟节点)



(a)

(a) Nine routers and a LAN.

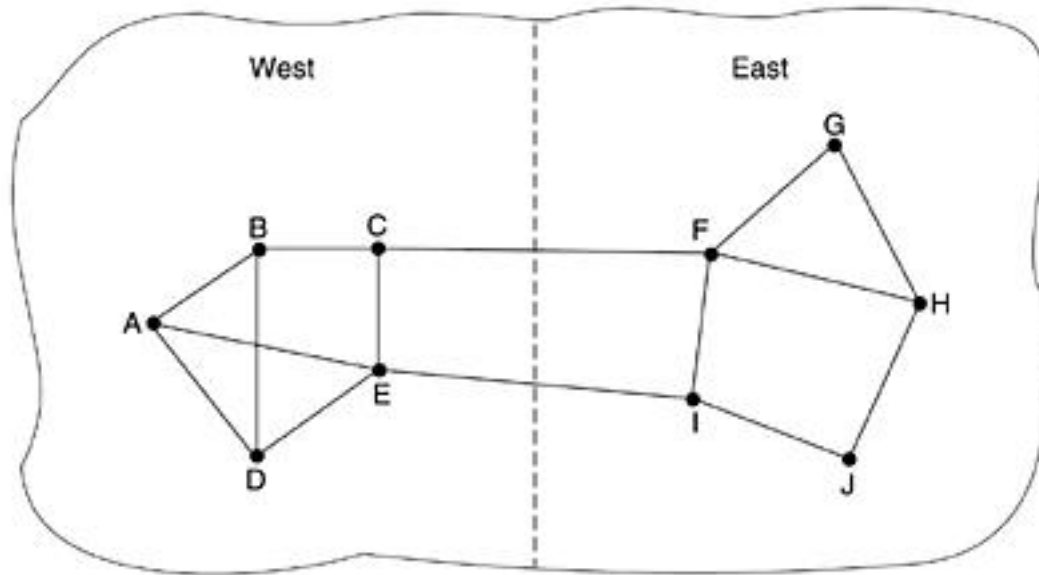


(b)

(b) A graph model of (a)

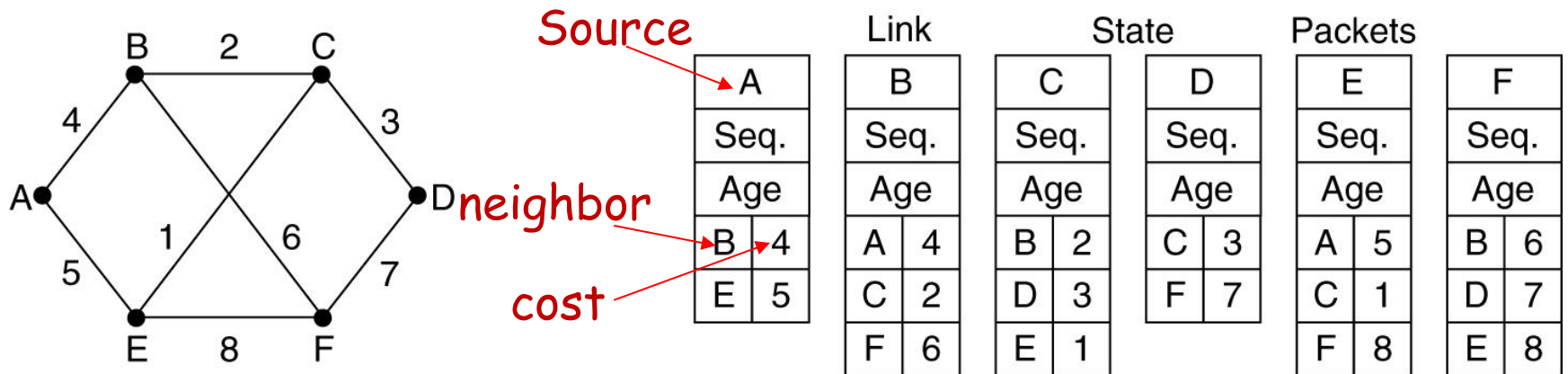
Step 2: Measuring Line Cost

- The delay to each of its neighbors
 - ◆ Measuring the **round-trip time** (ECHO Packet)
 - ◆ **Bandwidth**
 - ◆ An issue is whether to take the **load** into account when measuring the delay? (routing tables may oscillate, leading to erratic routing and many potential problems)



路由振荡，
导致不稳定

Step 3: Building Link State Packets



(a) A subnet. (b) The link state packets for this subnet

■ When to build LSP

- ◆ Build LSP periodically
- ◆ Build LSP when some **significant event** occurs
 - such as a line or neighbor going down or coming back up again or changing its properties appreciably

About Link State Packets

- **Sequence number:** used to check whether the received LSP is new
 - ◆ If LSP is new, forward with **flooding**
 - ◆ If LSP is a duplicate, it is **discarded**
 - ◆ If lower than the highest one seen so far ever arrives, it is rejected as being obsolete
 - ◆ If a router is down, and then restart, ...
- **Age:** used to delete looped LSP
 - ◆ If a router is down
 - ◆ If a router changed its routerID
 - ◆ If a router need to delete an LSP

Step 4: Distributing the Link State Packets

■ Responsibility

- ◆ Distributing the LSP **reliably**

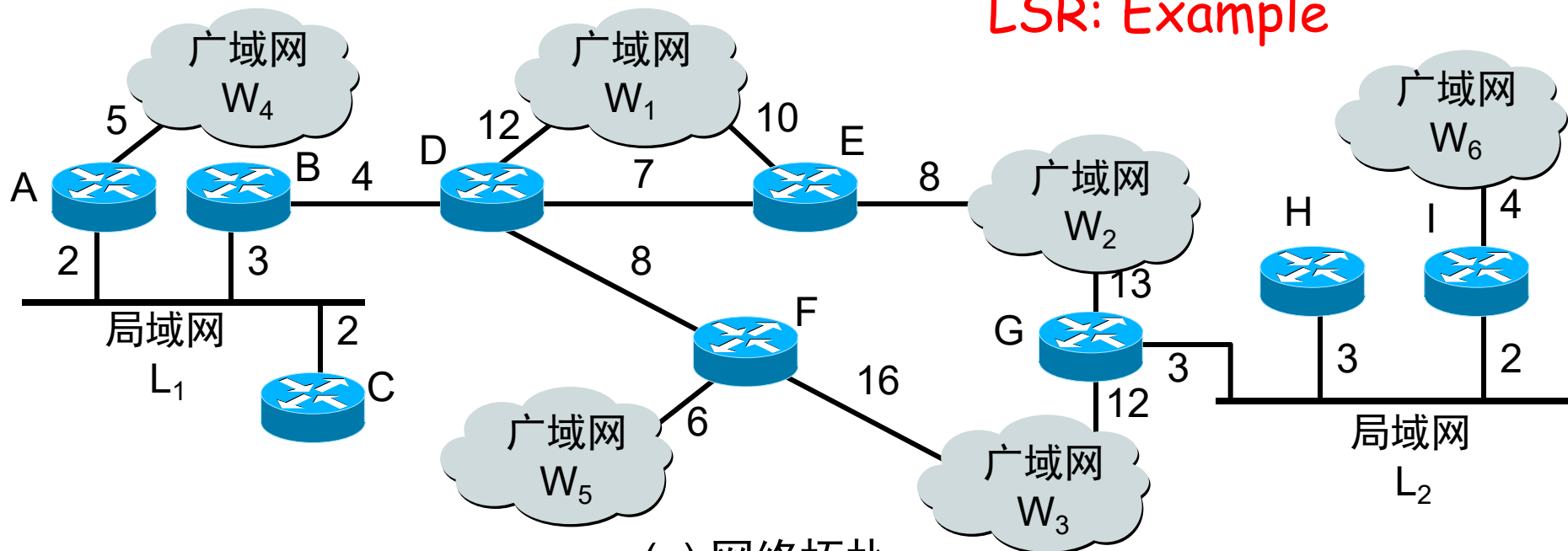
■ The basic distribution algorithm

- ◆ Use **flooding** to distribute LSPs
- ◆ Guard against errors on the router-router lines, all LSP are **acknowledged**

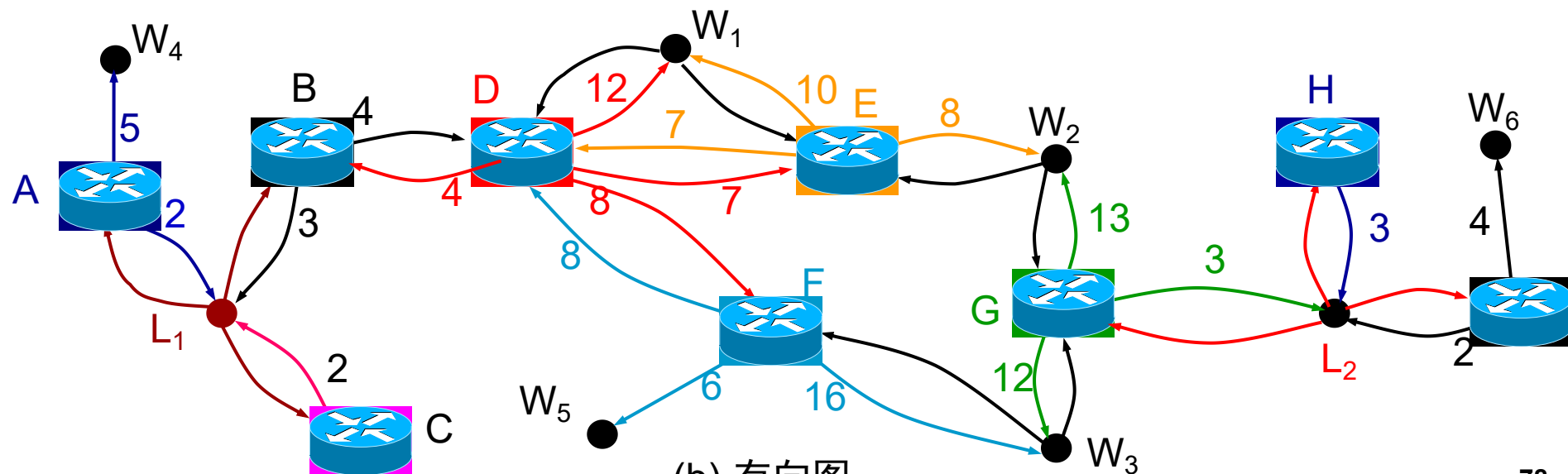
Step 5: Computing the New Route

- Once a router has accumulated a full set of LSPs, it can **construct the entire graph**
 - ◆ Every link is represented twice, once for each direction. The two values can be averaged or used separately.
- Run **Dijkstra's algorithm** locally to construct the shortest path to all possible destinations
- The results of this algorithm can be **installed in the routing tables**

LSR: Example

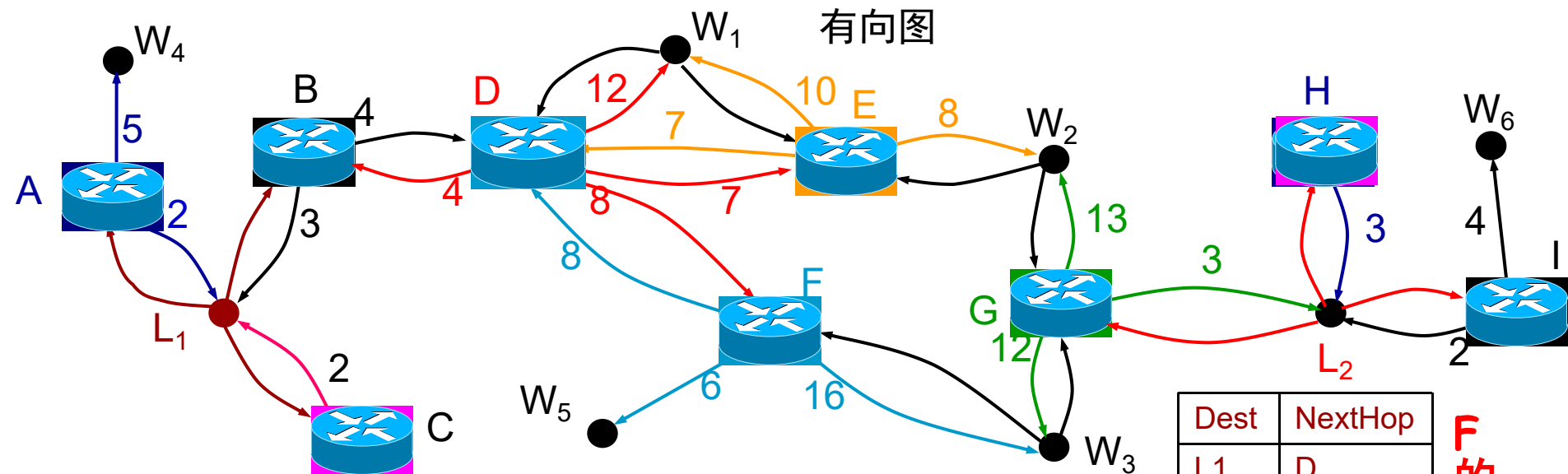


(a) 网络拓扑



(b) 有向图

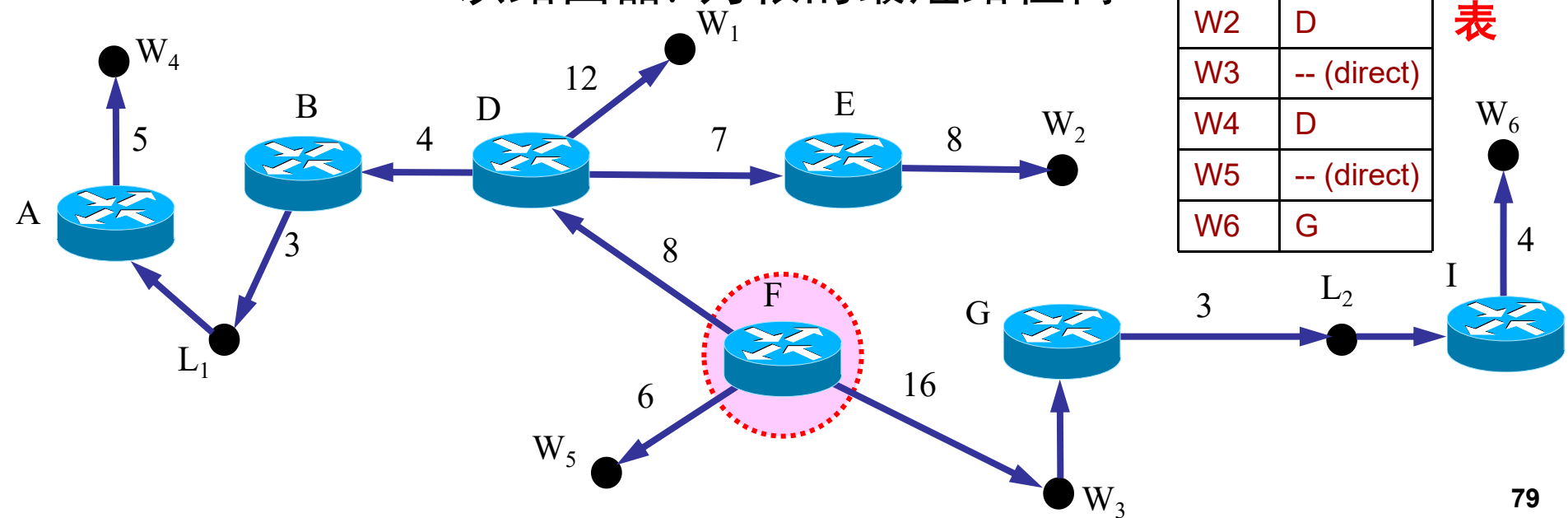
有向图



Dest	NextHop
L1	D
L2	G
W1	D
W2	D
W3	-- (direct)
W4	D
W5	-- (direct)
W6	G

F 的路由表

以路由器F为根的最短路径树



(3) Adoptions of Link State Routing

■ OSPF (Open Shortest Path First)

- ◆ designed several years after IS-IS
- ◆ Only for IP

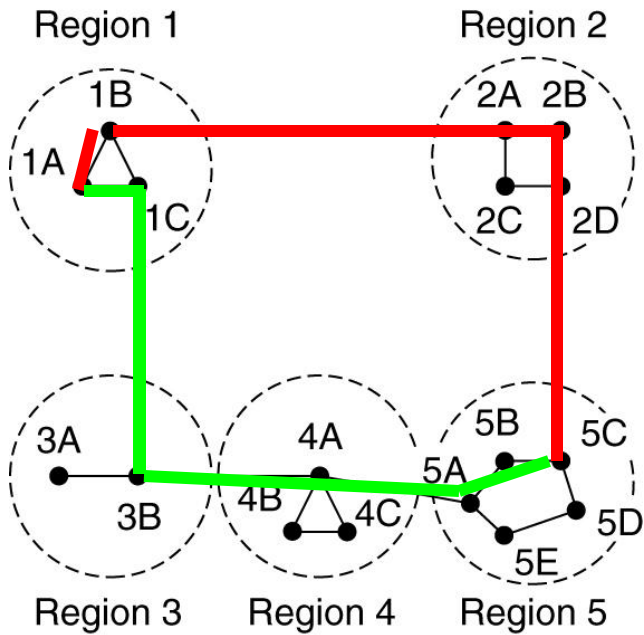
■ IS-IS (Intermediate System-Intermediate System):

- ◆ support multiple network layer protocols at the same time
- ◆ DECnet, ISO CLNP, IP, AppleTalk, Novell NLSP (IPX)

Routing Algorithms

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- 5.2.5 Link State Routing (LSR)
- *5.2.6 Hierarchical Routing*

Hierarchical Routing(层次选路/分级选路) (1)



1A→5C

(a)

Full table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

May take
sub-optimal route
(次优路由)

(c)

Hierarchical Routing (2)

- Levels : **regions**区域 -> **clusters**簇 -> **zones**区 -> **groups**群
- The router **knows** the **detailed topology** within its own region, but has **no idea** about the internal structure of other regions.
- Consider a subnet with 720 routers.
 - ◆ If there is no hierarchy: **720** routing table entries
 - ◆ If the subnet is partitioned into 24 regions of 30 routers each (Total: **53**)
 - 30 local entries
 - 23 remote entries
 - ◆ If a three-level hierarchy is chosen, with 8 clusters, each containing 9 regions of 10 routers (Total: **25**)
 - 10 entries for local routers
 - 8 entries for routing to other regions within its own cluster
 - 7 entries for distant clusters

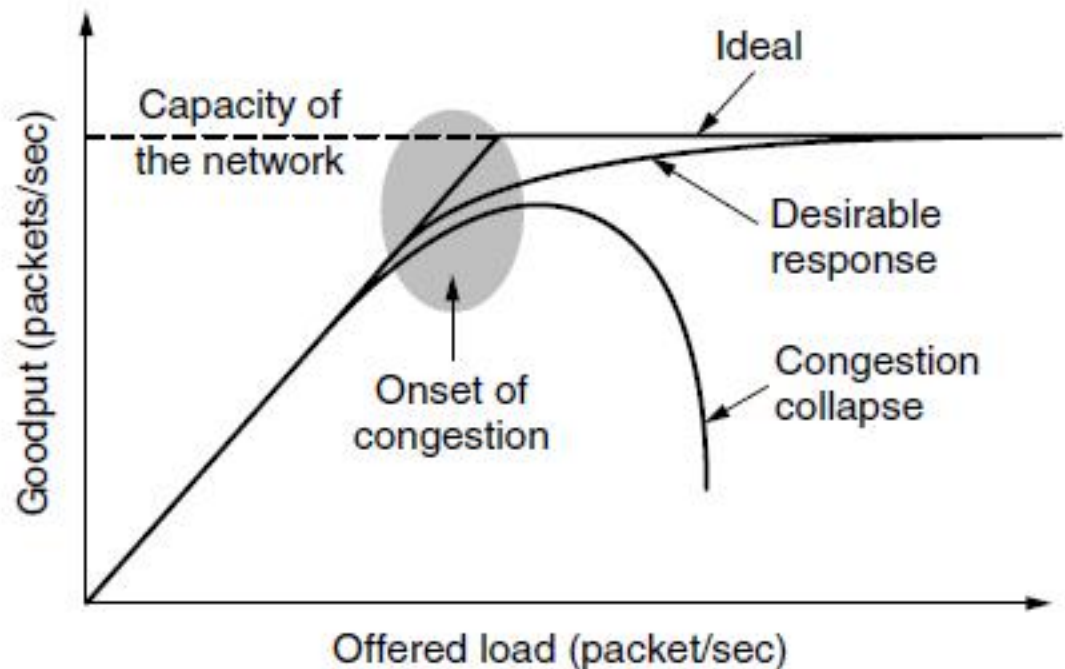
Outline

- 5.1 Introduction to Network Layer
- 5.2 Routing Algorithms
- 5.3 Congestion Control
- 5.4 Quality of Service
- 5.5 Internetworking
- 5.6 The network layer in Internet

(1) What is Congestion(拥塞)?

- Informally, “too many sources sending too much data too fast for **network** to handle” (资源不足)
- Congestion may occur if the **load** on the network (the number of packets sent to the network) is **greater than** the **capacity** of the network (the number of packets a network can handle)

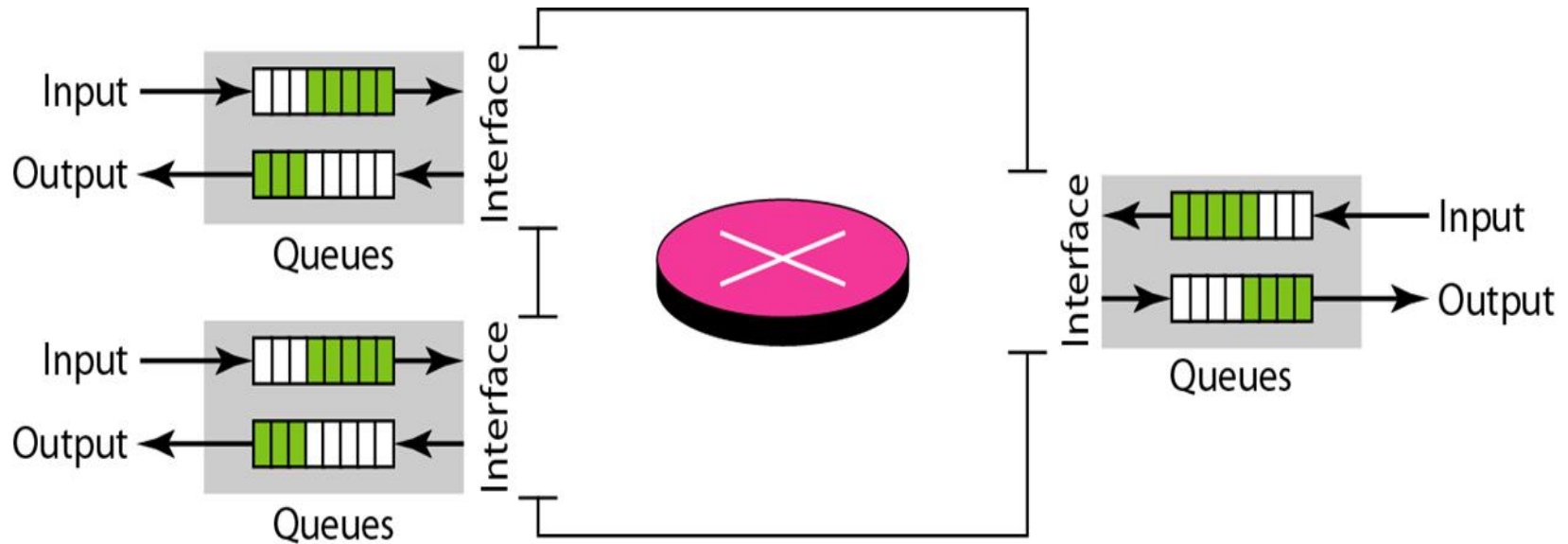
When **too much traffic** is offered, congestion sets in and performance degrades sharply



(2) Symptoms of Congestion

- Congestion happens in any system that involves waiting or **queuing** 
- 网络中导致拥塞的可能因素：
 - ◆ 计算资源：CPU处理速度慢
 - ◆ 带宽资源：链路带宽低
 - ◆ 存储资源：缓冲队列短
- Symptoms
 - ◆ **Long delays** (Queuing in router buffers)
 - ◆ **Lost packets** (Buffer overflow at routers)

Queues in a Router

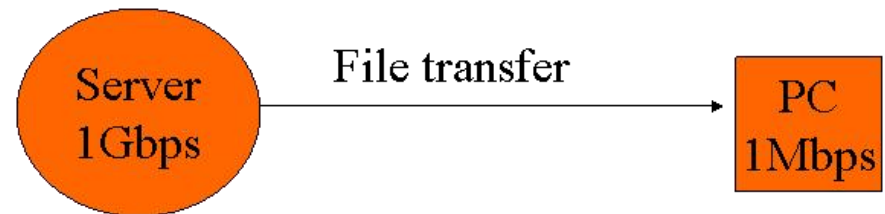


- If the rate of packet arrival is higher than the packet processing rate, the input queues become longer and longer
- If the packet departure rate is lesser than the packet processing, the output queues become longer and longer.

(3)Flow control vs. Congestion control

■ Flow Control(流量控制)

- ◆ To avoid overflowing **the receiver**
- ◆ **Local issue**, involves **only 2 endpoints**, the sender and the receiver



■ Congestion Control(拥塞控制)

- ◆ To avoid congesting **the network(routers)**
- ◆ **Global issue**, concerns all nodes/hosts along the path

5.3.1 Approaches to Congestion Control

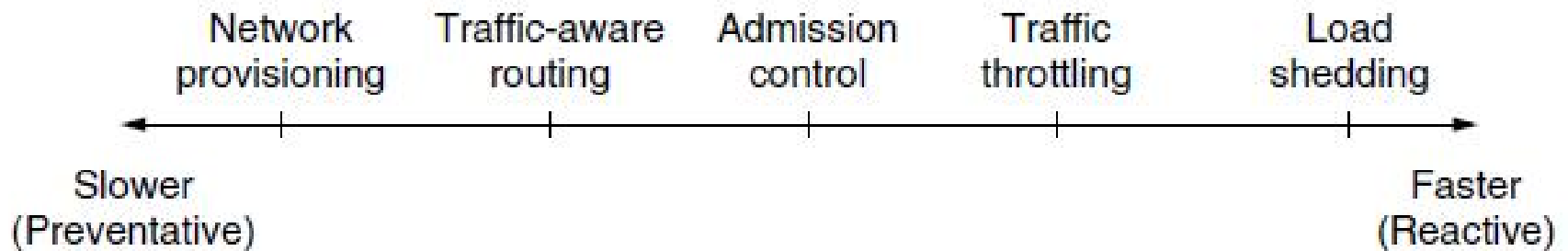
■ Two types of solutions

◆ Increasing the resources

- Network provisioning(网络供给)
- Traffic aware-routing(业务量感知路由)

◆ Decreasing the load

- Admission control(准入控制)
- Traffic throttling(流量限制)
- Load shedding(负载掉落)



5.3.4 Traffic Throttling(业务量减速/流量限制)

- Feedback based solution
- 基本过程：拥塞检测->拥塞通知->业务量限制/调节
- Step 1: Router--detects congestion, by
 - ◆ utilization of the output links(输出链路利用率：不够精确)
 - ◆ buffering of queued packets inside the router（排队分组）
 - ◆ number of packets that are lost due to insufficient buffering（分组丢包数量：太迟）

Estimate of the queuing delay:

EWMA (Exponentially Weighted Moving Average, 指数加权移动均值)

$$d_{\text{new}} = \alpha d_{\text{old}} + (1 - \alpha)s$$

- ◆ d: recent queue length
- ◆ s: sample of the instantaneous queue length
- ◆ $0 < \alpha < 1$ 平衡因子，体现了历史值对当前估计值的影响程度

Traffic Throttling(流量限制)

■ Step 2: Transfer Information about Congestion

(通知发送方: 抑制分组、显式explicit/隐式implicit congestion notification拥塞通知)

- ◆ Sending a packet(**Choke Packet**, 抑制分组) to the **source host**, giving it the destination found in the packet
- ◆ The original packet is tagged so that it will not generate any more choke packets
- ◆ To avoid increasing load on the network, the router may only send choke packets at a low rate

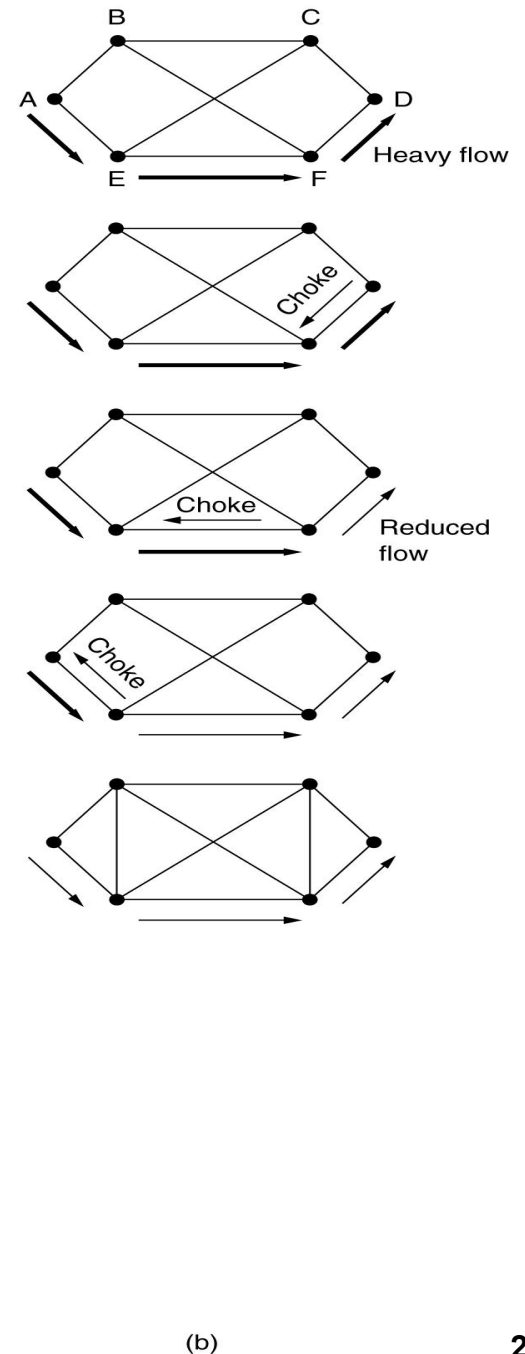
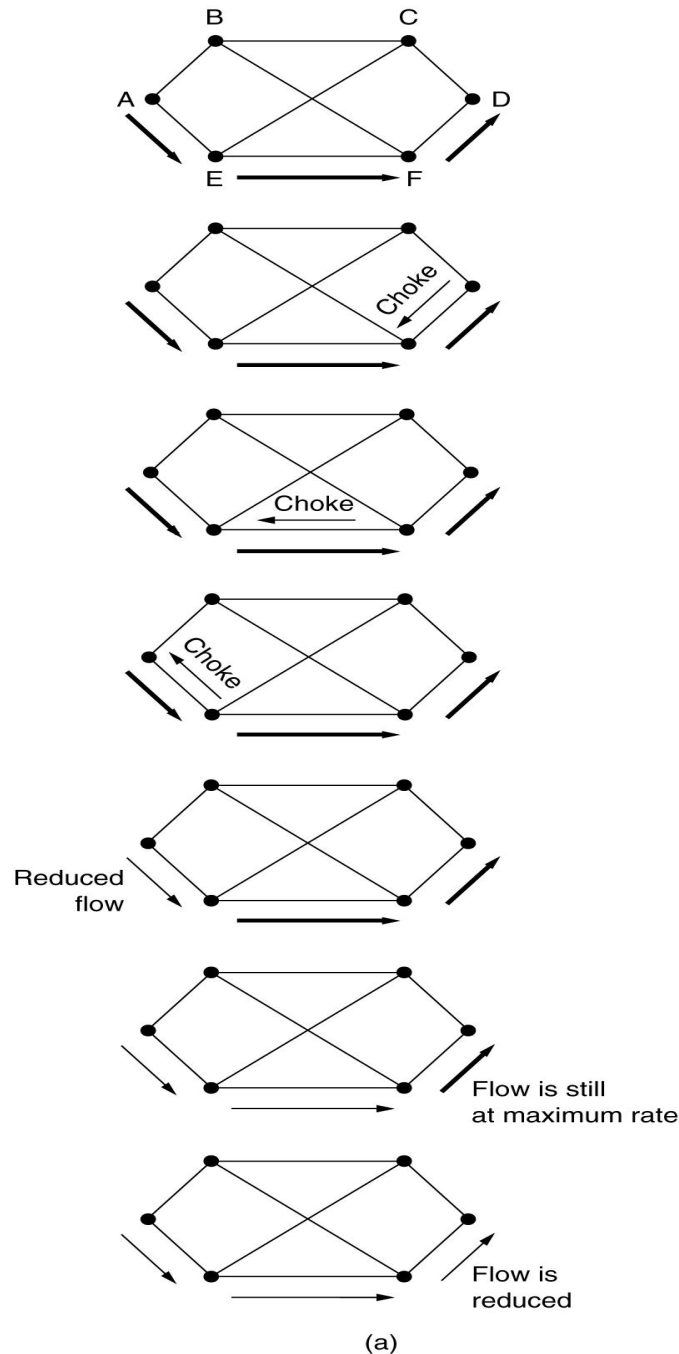
■ Step 3: Source--adjust System Operation

- ◆ Receiving choke packet, the source may **reduce the traffic** sent to the specified destination by x percent (e.g. 50%)

Other Feedback Mechanisms: Hop-by-Hop Backpressure

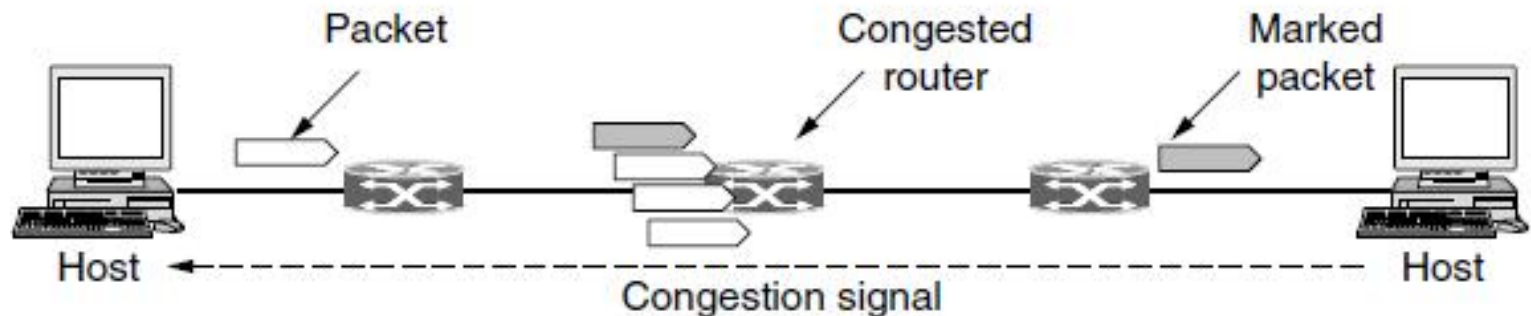
(a) A choke packet that affects **only the source**.

(b) A choke packet that affects **each hop** it passes through.



Other Feedback Mechanisms: ECN

- **Explicit Congestion Notification**, 显式拥塞通知
 - ◆ Used by IP and TCP
 - ◆ Two bits in the IP packet header are used to record whether the packet has experienced congestion.
 - ◆ Router detecting congestion marks the packet as it is forwarded.
 - ◆ The destination will then echo any marks back to the sender as an explicit congestion signal in its next reply packet.



Review (1)

■ Distance Vector Routing vs Link State Routing

	DVR	LSR
Algorithm	Bellman-ford	Dijkstra
What information to share	routing table(destination and distance)	network topology (LSP)
Whom to share with	neighbors	All routers in the network
When to share	periodically	When there is change
What does router knows	distance, not path	Entire network topology
Count-to-infinity problem	Yes	No
Information utilized in routing	Rumor based routing	Neighbor link status based routing

Review (2)

■ Hierarchical routing

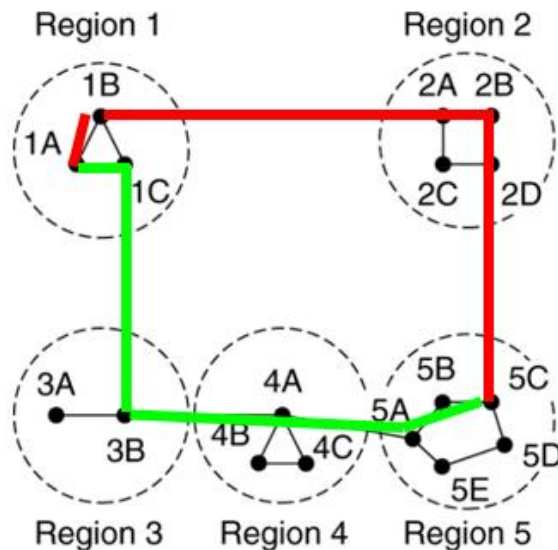
- ◆ Arranging routers in a **hierarchical manner**

Levels : **regions** -> **clusters** -> **zones** -> **groups**

- ◆ **Minimize** the size of the routing table

- **knows** the **detailed topology** within its **own region**
- has **no idea** about the internal structure of **other regions**.

- ◆ May take **sub-optimal** route



Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

Review(3)

■ Congestion Control

- ◆ The **load** is **greater than** the **capacity** of the network

$\Sigma \text{对资源需求} > \text{可用资源}$

resources= {CUP, buffers, bandwidth}

- ◆ Symptoms: **Long delays, lost packets**

- ◆ Congestion control approaches

slower

- Network provisioning
(网络供给)
- Traffic aware-routing
(流量感知路由)

Increasing
the
resources

preventative 预防性措施
(open-loop control)

Decreasing
the **load**

reactive 反应性措施
(feedback control)

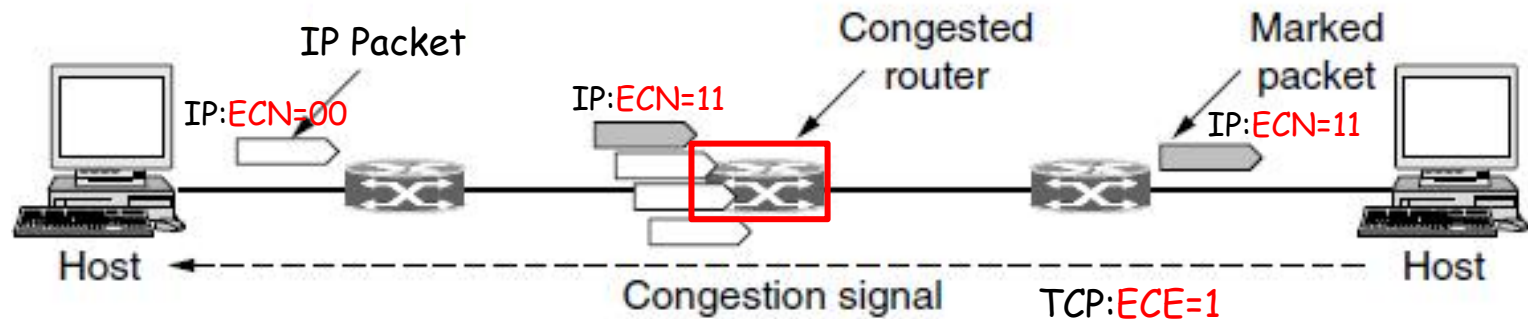
faster

- Admission control
(准入控制)
- Traffic throttling
(流量限制)
- Load shedding
(负载脱落)

Review (4)

■ Traffic throttling

- 1) Router detects congestion
- 2) Congestion notification: choke packets, ECN



- 3) Source reduces traffic

5.3.5 Loading Shedding(负载掉落)

■ Drop packets

- ◆ When routers are being inundated by packets, just **throw them away**

■ 关键问题：选择丢弃哪些数据分组

■ Policies: depend on the type of applications

- ◆ Pick packets at **random** to drop
- ◆ Wine(old is better than new) & Milk(new is better than old)
 - E.g. file transfer, multimedia
- ◆ 智能丢弃: Requires cooperation from the senders
 - E.g. algorithms for compressing video, ...
 - The low-priority packets being cheaper to send than the high-priority ones
 - Allow hosts to exceed the limits specified in agreement, but all excess traffic must be marked as low priority

Load shedding in Internet: RED

- Random Early Detection (随机早期检测)
- Drop packets before the situation has become hopeless
 - ◆ When the **average queue length** on some line **exceeds a threshold**, the line is said to be congested, and a small fraction of the **packets are dropped at random** (随机丢包)
- How should the router tell the source about the problem?
 - ◆ Just **discard** the selected packet
 - ◆ The affected sender will **notice the loss** when there is no acknowledgement, **implicit notification** (隐式通知)
- Sources respond to **lost packets** by slowing down their transmission rate
 - ◆ **In wireless networks**, where most losses are due to noise on the air link, **this approach cannot be used**.

Outline

- 5.1 Introduction to Network Layer
- 5.2 Routing Algorithms
- 5.3 Congestion Control
- 5.4 Quality of Service
- 5.5 Internetworking
- 5.6 The network layer in Internet

5.4.1 Primary QoS Parameters(服务质量参数)

■ Reliability

- ◆ Packet error rate
- ◆ Packet loss rate
- ◆ Bit error rate(BER)

■ Delay: the time period from 1st bit of packet sent to last bit received

■ Jitter(抖动): the variation in delay of packets belonging to the same flow

■ Bandwidth: maximum data rate

QoS Requirements of Various Applications

Application	Reliability	Delay	Jitter	Bandwidth
E-mail	High	Low	Low	Low
File transfer	High	Low	Low	Medium
Web access	High	Medium	Low	Medium
Remote login	High	Medium	Medium	Low
Audio on demand	Low	Low	High	Medium
Video on demand	Low	Low	High	High
Telephony	Low	High	High	Low
Videoconferencing	Low	High	High	High

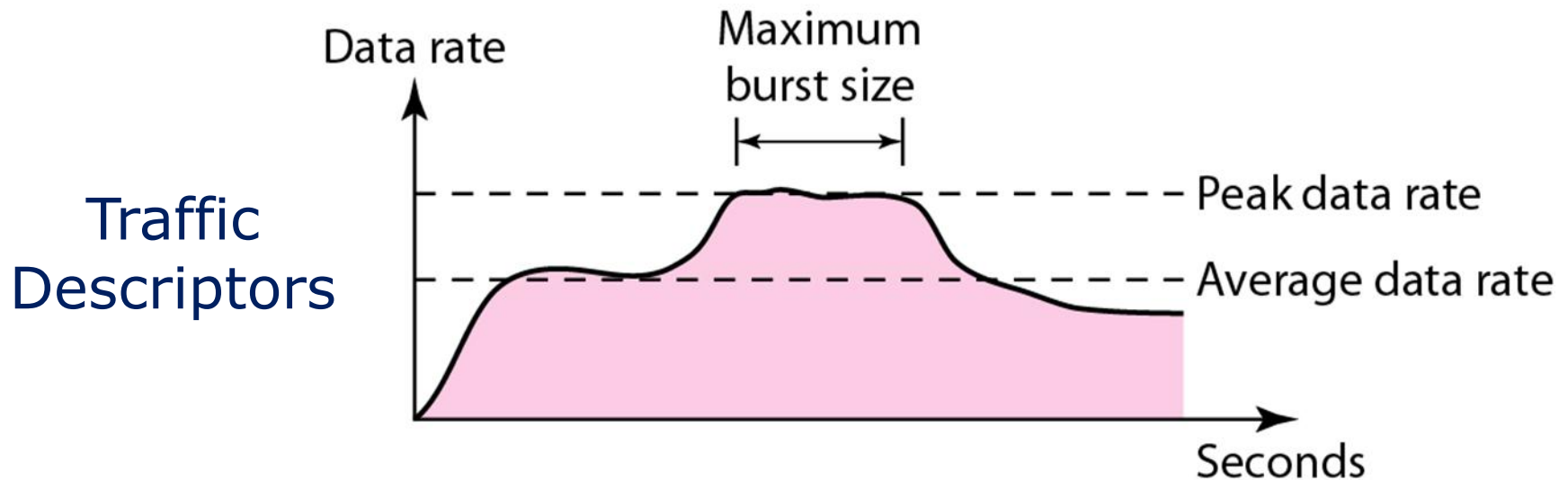
Techniques to improve QoS

- *5.4.2 Traffic shaping* 流量整形
- 5.4.3 Packet Scheduling 分组调度
- 5.4.4 Admission control(自阅)
- 5.4.5 Integrated Services(不讲)
- 5.4.6 Differentiated Services(不讲)

(1)Traffic Shaping(业务量/流量整形)

■ Problem

- ◆ **Traffic burst(流量突发)**: Host emits the packets irregularly, which may cause congestion to occur in network



Traffic Shaping流量整形

■ Service Level Agreement (SLA服务等级合约)

- ◆ Between the user and the carrier(网络-服务质量提供者)
- ◆ The user: traffic shaping reduces congestion and helps carrier live up to its promise (流量整形)
- ◆ The carrier网络运营商: traffic policing(流量监管)
(**CAR**: Committed Access Rate) 承诺的接入速度

■ Traffic shaping(流量整形--平滑流量)

- ◆ Regulate the average rate and burstiness of data transmission, **smooth out the traffic**

■ Traffic Policing(流量监管)

- ◆ Monitoring a traffic flow
- ◆ Packets in **excess of the agreed pattern** might be **dropped or marked** as having lower priority

(2) Leaky Buckets and Token Buckets

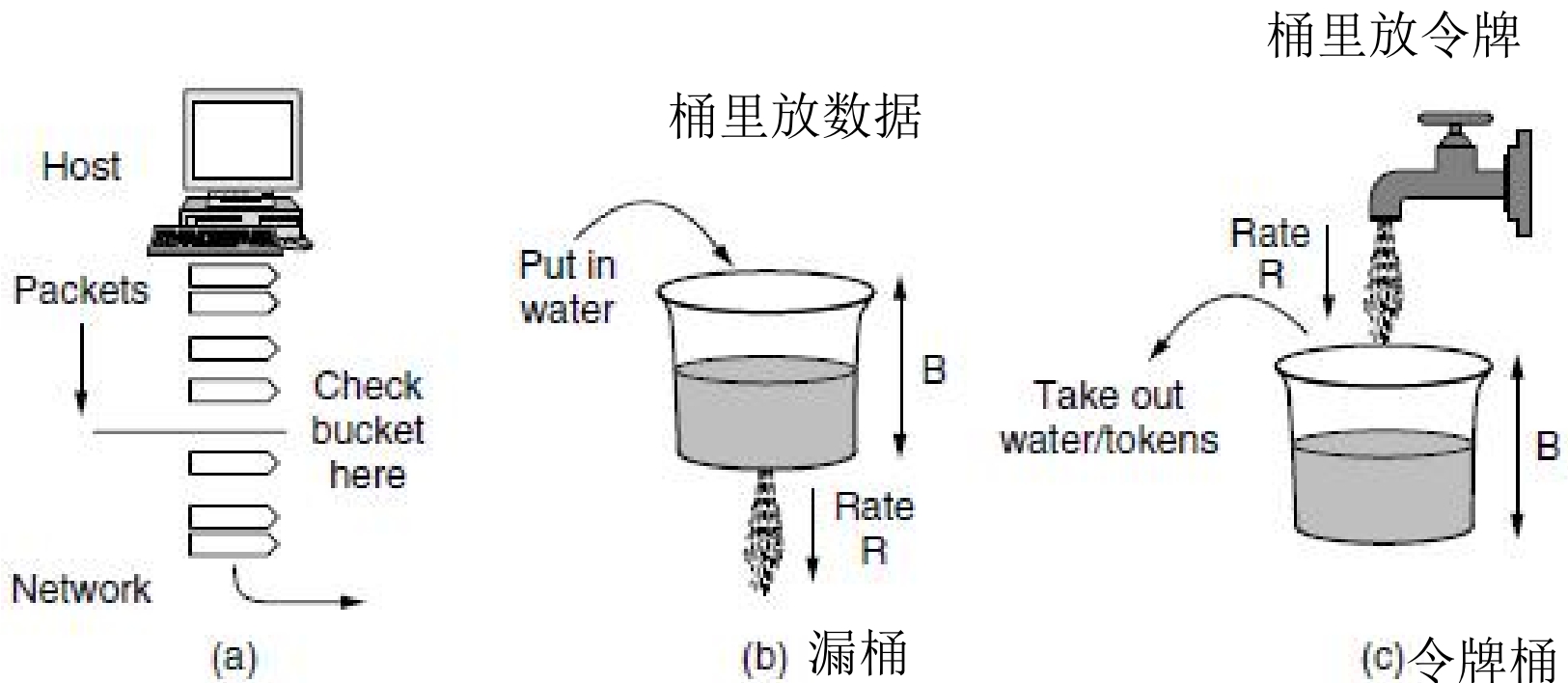
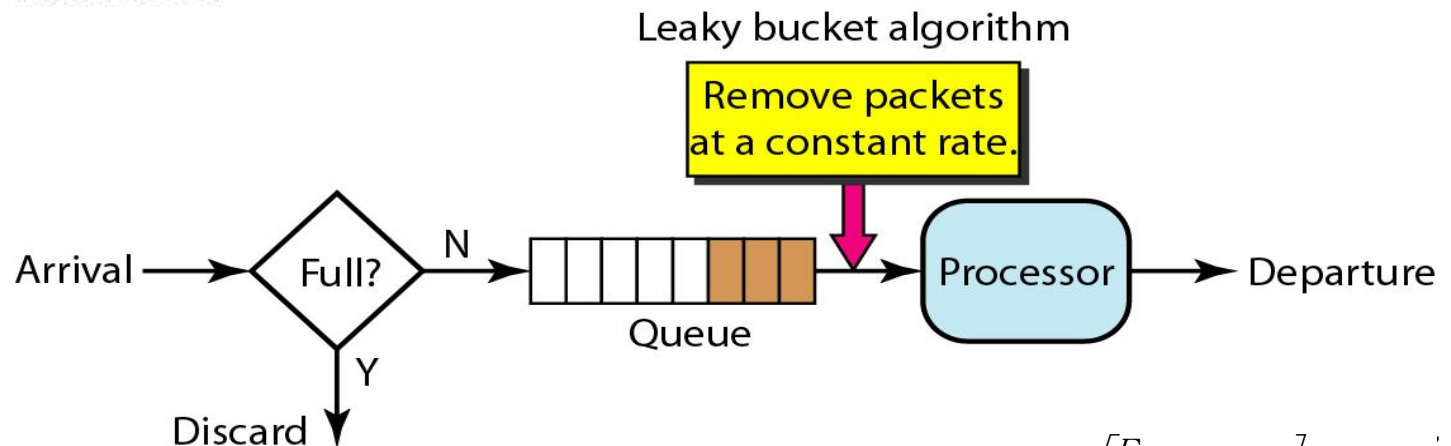
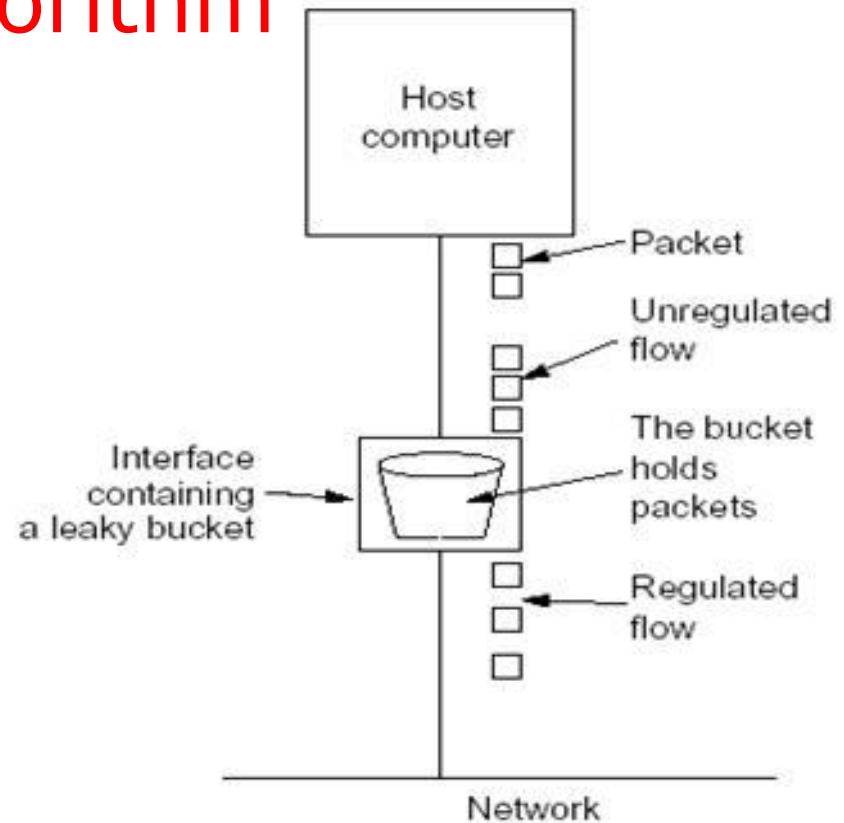
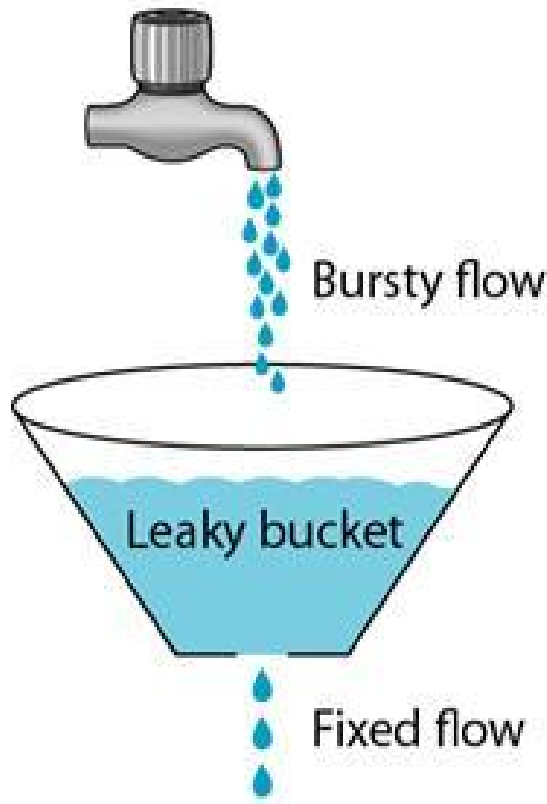


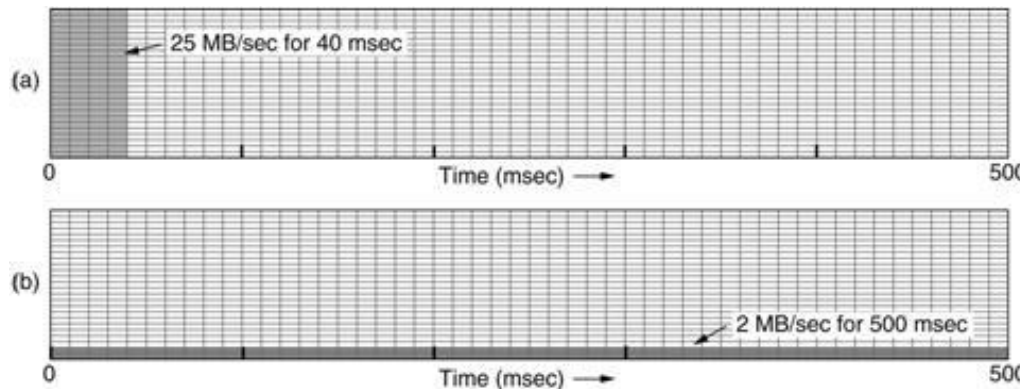
Figure 5-28. (a) Shaping packets. (b) A leaky bucket. (c) A token bucket.

Leaky Bucket (漏桶) Algorithm



Leaky Bucket: Example

- Turns an uneven flow of packets into an even flow
 - ◆ Smoothing out bursts (平滑突发流量)
 - ◆ Reducing the chances of congestion
- The byte-counting leaky bucket (桶满则数据丢失)



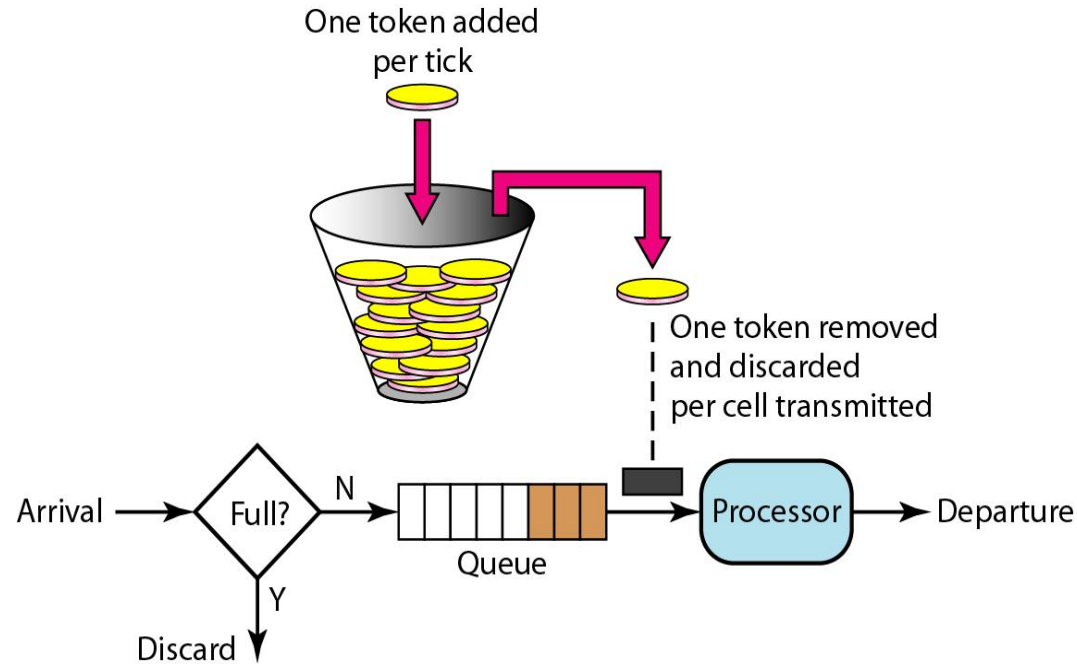
Bursting data

Output from a leaky bucket.

- Bucket can be used to **shape**整形 or **police**监督 packets entering the network
If a packet arrives when the bucket is full, the packet must be queued until enough water leaks out to hold it (**host shapes its traffic**) or be discarded (**network polices traffic**)

(3) Token Bucket (令牌桶) Algorithm

- Token bucket allows idle hosts to **accumulate credit** for the future in the forms of token.
- Token bucket **allows bursts**, but up to a regulated maximum length



Token to be collected based on idle time and can be used later to **sends bursty packets** as long as it is allowed by the bucket.

Leaky Bucket vs. Token Bucket

■ Burstiness

- ◆ **Leaky bucket** algorithm enforces a rigid output pattern at the **average rate**, no matter how bursty the traffic is
- ◆ **Token Bucket** allowing **some burstiness** in the output stream

■ When the bucket fills up

- ◆ Token bucket algorithm **throws away tokens** but **never discards packets**
- ◆ Leaky bucket algorithm **discards packets**

Token Bucket: Example

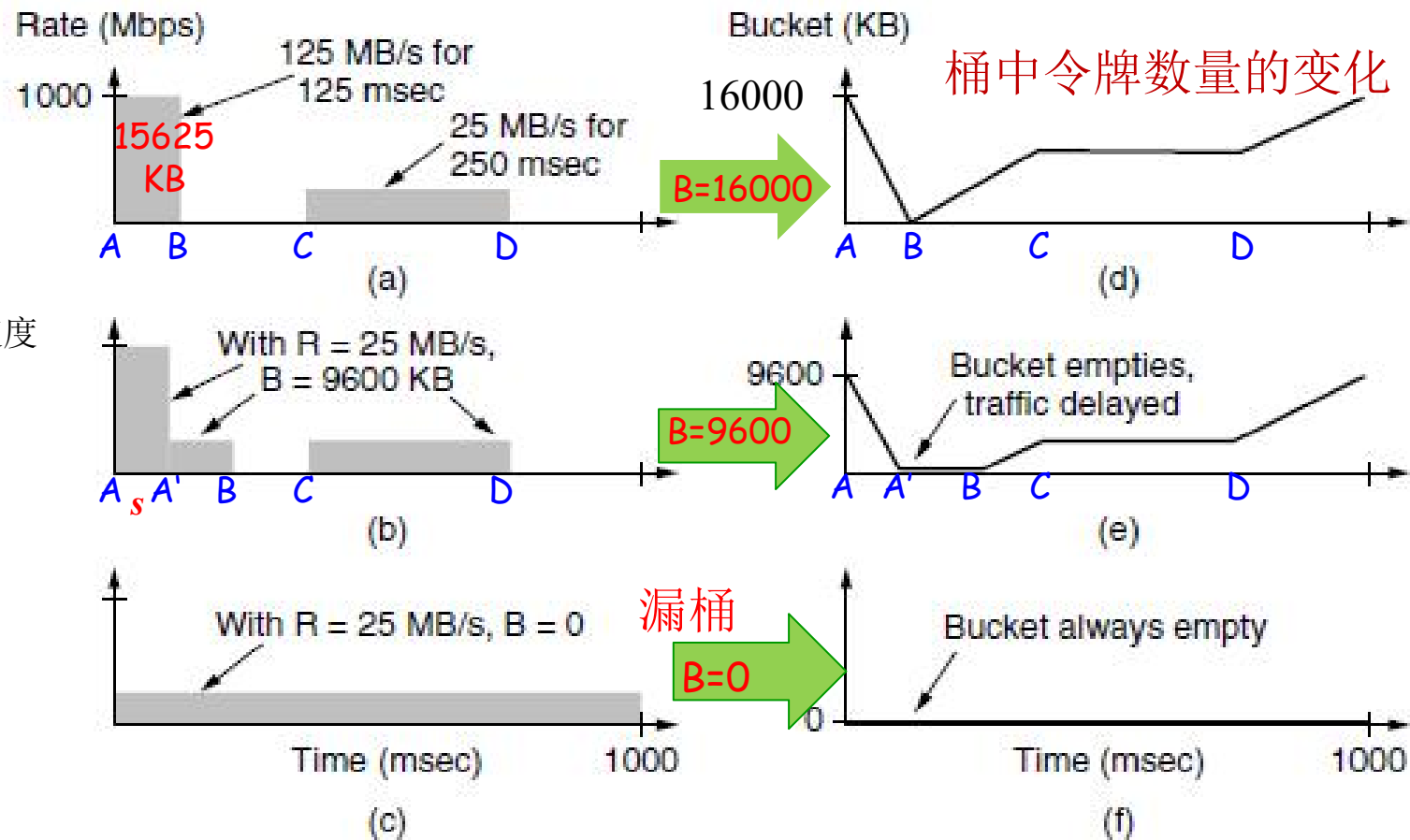


Figure 5-29. (a) Traffic from a host. Output shaped by a token bucket of rate 200 Mbps and capacity (b) 9600 KB and (c) 0 KB. Token bucket level for shaping with rate 200 Mbps and capacity (d) 16,000 KB, (e) 9600 KB, and (f) 0 KB.

The Length of the Maximum Rate Burst

经过令牌桶之后的突发数据持续时间:

burst length **S** sec

token bucket capacity **B** bytes

token arrival rate **R** bytes/sec

maximum output rate **M** bytes/sec

An output burst contains a maximum of **B + RS** bytes

the number of bytes in a maximum-speed burst of length **S** seconds is **MS**

$$B + RS = MS$$

$$S = B / (M - R)$$

会计算

If $B = 9600\text{KB}$, $M = 125 \text{ MB/sec}$, and $R = 25 \text{ MB/sec}$
then $S = 96 \text{ ms}$

Techniques to improve QoS

- 5.4.2 Traffic shaping
- *5.4.3 Packet Scheduling*
- 5.4.4 Admission control(自阅)
- 5.4.5 Integrated Services(不讲)
- 5.4.6 Differentiated Services(不讲)

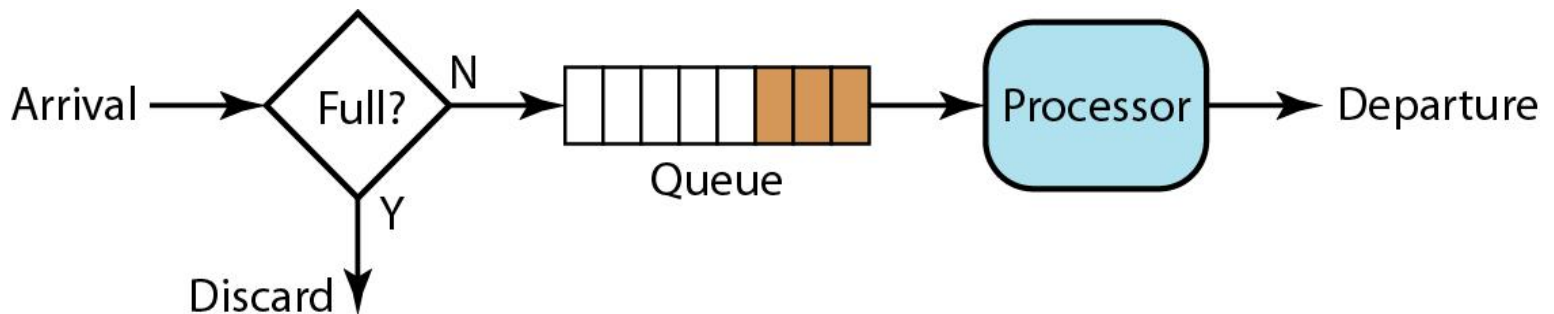
Packet Scheduling 分组调度

- Resources can be reserved(预留) for different flows
 - ◆ Bandwidth
 - ◆ Buffer space
 - ◆ CPU cycles

- Algorithms
 - ◆ FIFO (FCFS), tail drop
 - ◆ RED Algorithm
 - ◆ Fair queueing algorithm

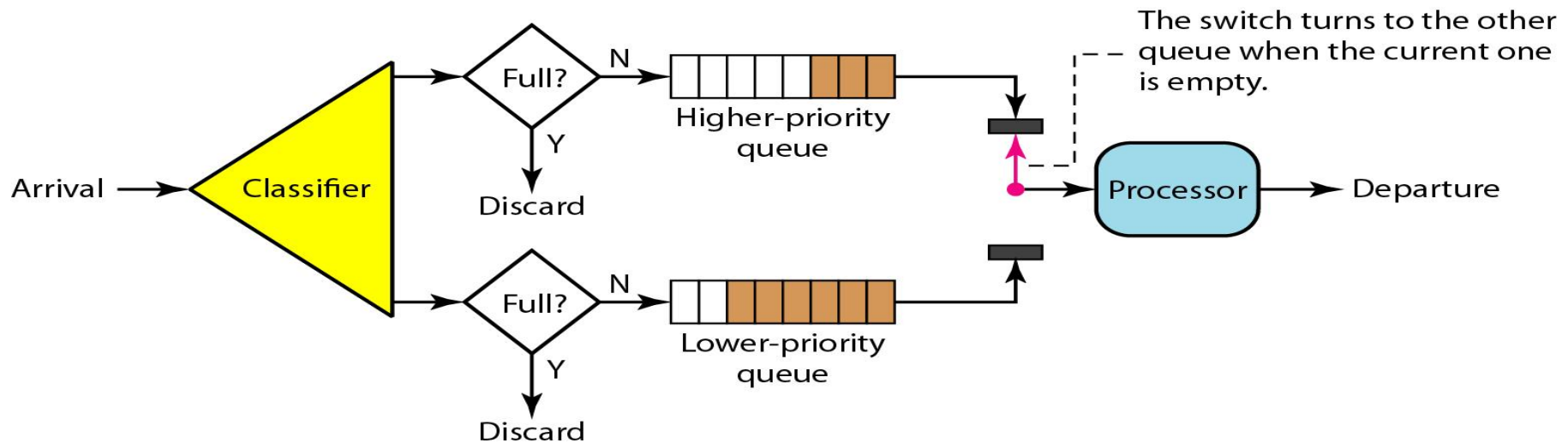
Packet Scheduling 分组调度: FIFO Queuing(先进先出队列)

- Packets from different flows arrive at a router for processing. A good scheduler treats different flows in a **fair** and **appropriate** manner to improve the **QoS**.
- Methods
 - ◆ FIFO queuing, Priority queuing, Weighted fair queue



FIFO: First In, First Out

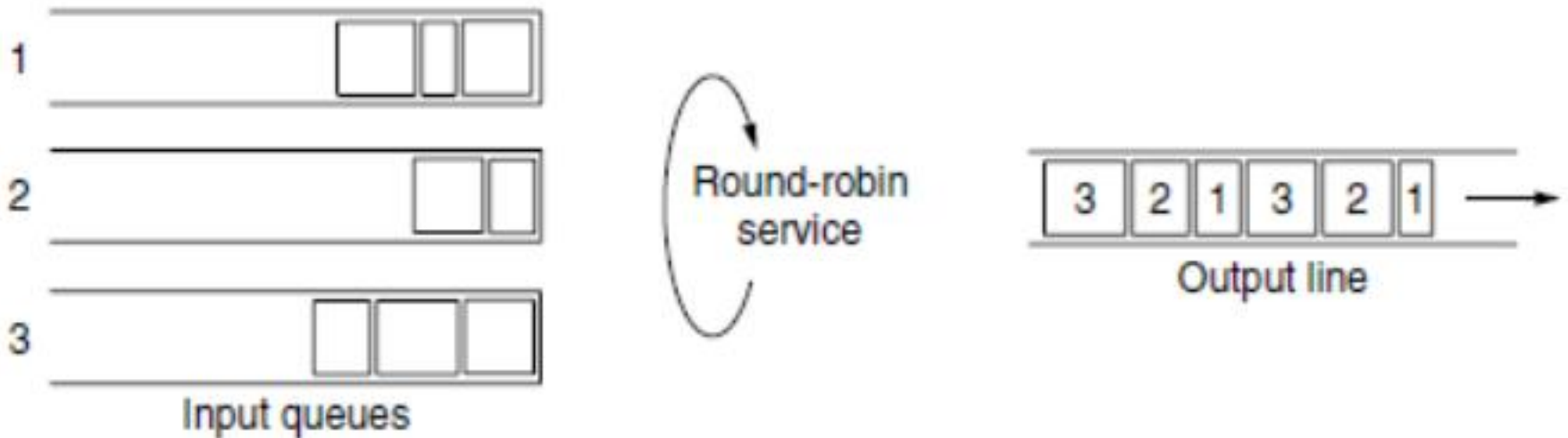
Packet Scheduling: Priority Queuing (优先级队列)



- **Packets** are first assigned a **priority** class. Each priority class has its **own queue**.
- When a line is idle, packets in the highest priority queue are processed first. 高优先级数据先发送
- **Better QoS** because **higher priority** traffic such as multimedia can reach the destination **with less delay**.
- Potential drawback: starvation of lower priority traffic, not fair. 对低优先级数据不公平

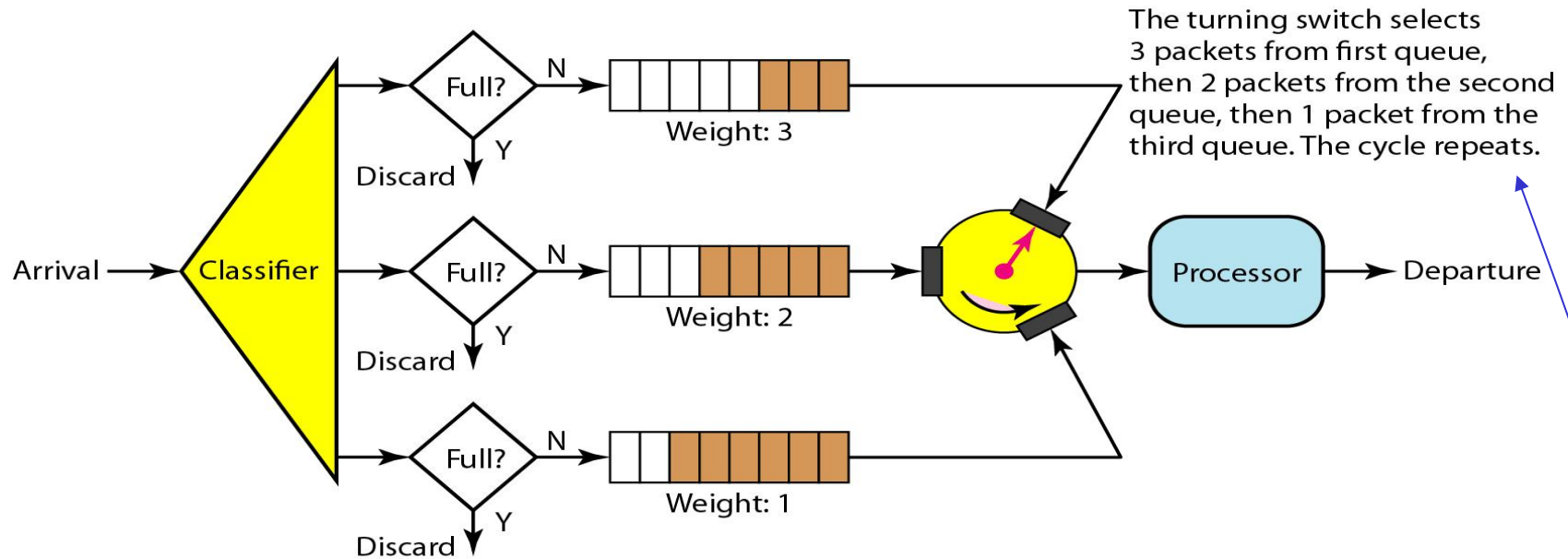
Packet Scheduling: Fair Queue (公平队列)

- Routers have separate **queues** for each flow for a given putput line
- Packet are processed in a **round robin**(循环) way.



- It gives **more bandwidth** to hosts that use **large packets** than to hosts that use **small packets**(对小包不公平)

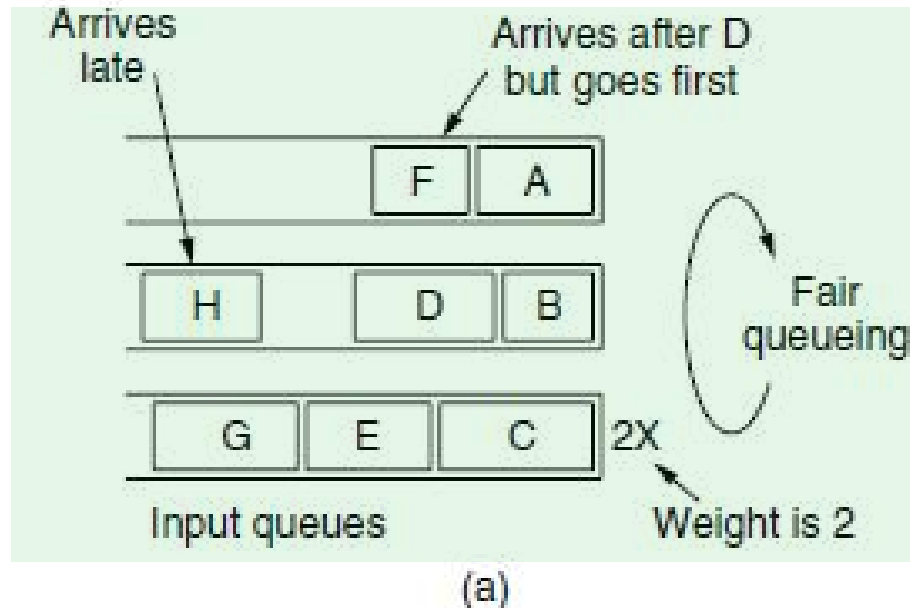
Packet Scheduling: Weighted Fair Queuing(公平加权队列)



- **Priority Queue + Fair Queue** (按权值轮流)
- Queues are weighted based on priority; **higher priority means higher weights**
- The system **processes packets in each queue in a round-robin** fashion with the number of packets selected from each queue based on corresponding weight.

WFQ Example

Byte-by-byte scheme: 按字节轮流处理，计算每个包的发送结束时间(以字节为单位虚拟时间)，然后按照结束时间来排序，按该顺序真正发送数据包



Packet	Arrival time	Length	Finish time	Output order
A	0	8	8	1
B	5	6	11	3
C	5	10	10	2
D	8	9	20	7
E	8 (10)	8	14	4
F	10 (8)	6	16	5
G	11	10	19	6
H	20	8	28	8

(b)

既考虑了高优先级数据的特权，又兼顾了小包和低优先级数据的公平性

Finishing times for the packets.

$$F_i = \max(A_i, F_{i-1}) + \frac{L_i}{W}$$

F: finish time, A: Arrival time, L: length of packet, W: weight

Outline

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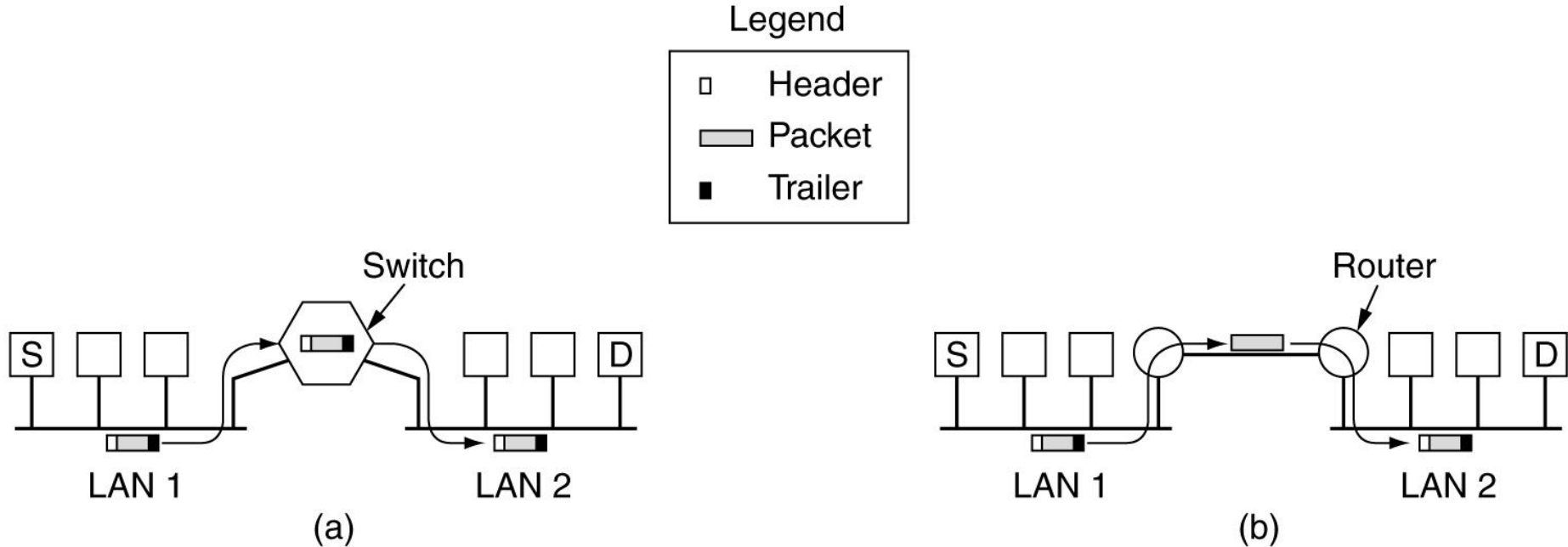
Connecting Networks

- Different networks & different protocols
 - ◆ Ethernet, Internet over cable, Fixed and mobile telephone networks, 802.11, 802.16, ATM, ...
- Purpose of interconnecting
 - ◆ Allow users on any network to communicate with users on all the other ones
- Internetwork(internet, 互联网): two or more networks are connected

Interconnection at different layers

- Layer 1: Repeater, HUB
- Layer 2: Bridges, Switches
 - ◆ Protocol translation: Ethernets/802.11
- Layer 3: Router (Packet translation)
- Layer 4: Transport gateway
 - ◆ TCP/Wireless TCP
 - ◆ TCP/SNA
- Above layer 4: Application gateway

Bridge vs. Router



Bridge/Switch: the entire frame is transported on the basis of its **MAC address**

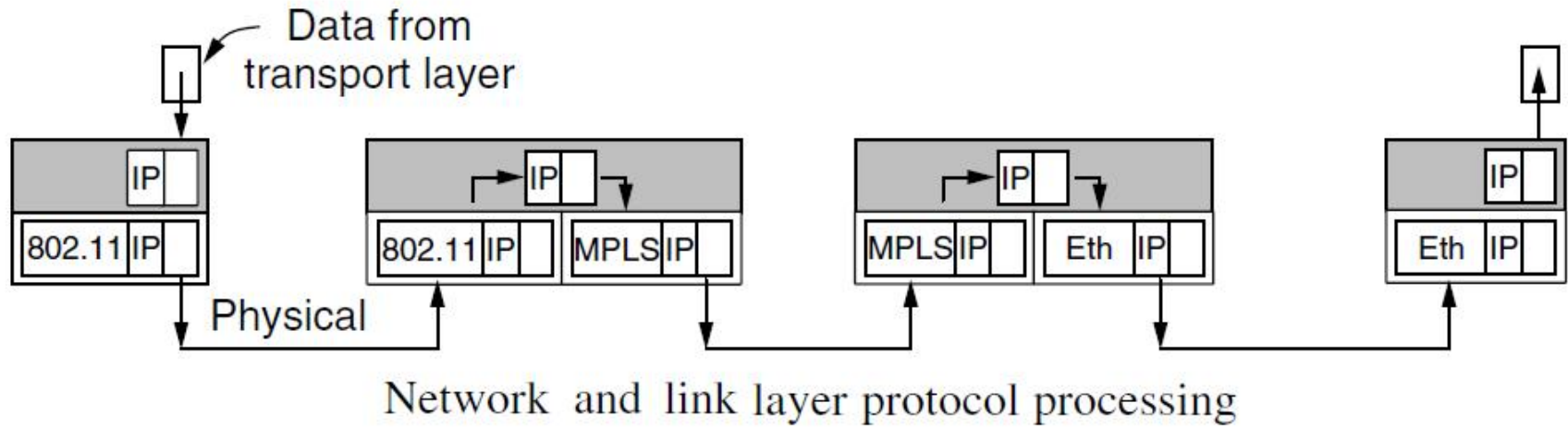
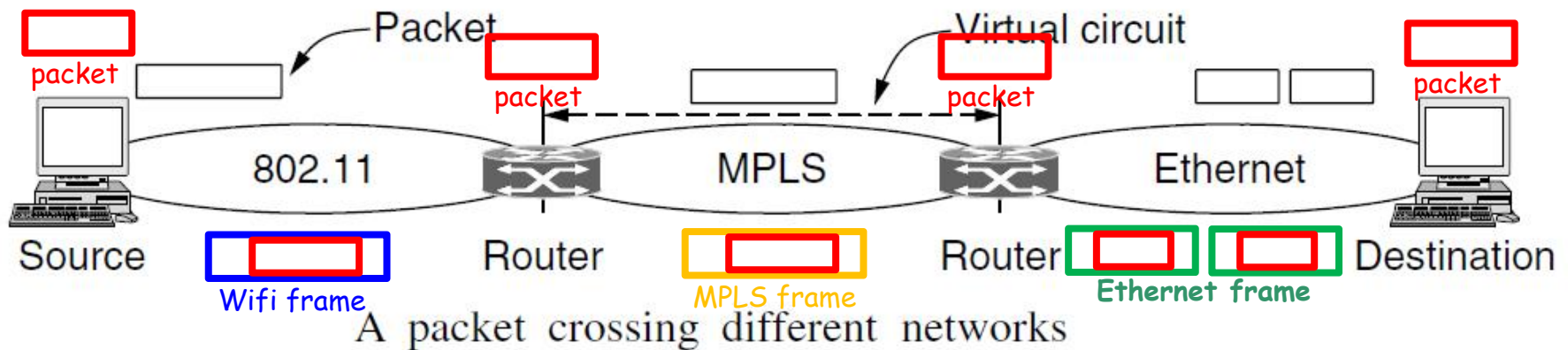
Router: packet is extracted from the frame and the **Layer-3 address**(eg.IP address) is used for deciding where to send it

5.5.1 How Networks Differ

Item	Some Possibilities
Service offered	Connectionless versus connection oriented
Addressing	Different sizes, flat or hierarchical
Broadcasting	Present or absent (also multicast)
Packet size	Every network has its own maximum
Ordering	Ordered and unordered delivery
Quality of service	Present or absent; many different kinds
Reliability	Different levels of loss
Security	Privacy rules, encryption, etc.
Parameters	Different timeouts, flow specifications, etc.
Accounting	By connect time, packet, byte, or not at all

5.5.2 How Networks Can Be Connected

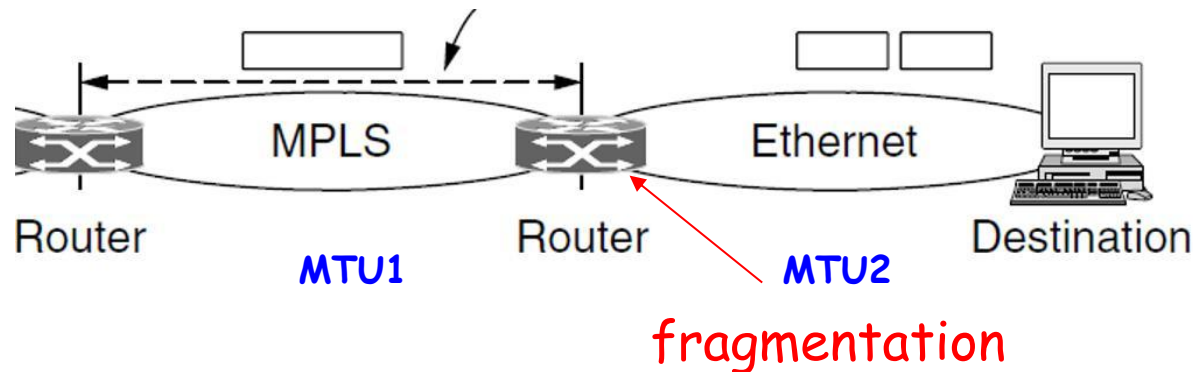
◆ Packet(IP datagram): Independent of Hardware



Encapsulation of IP Datagram in Ethernet Frame

5.5.5 Fragmentation(分段/分片)

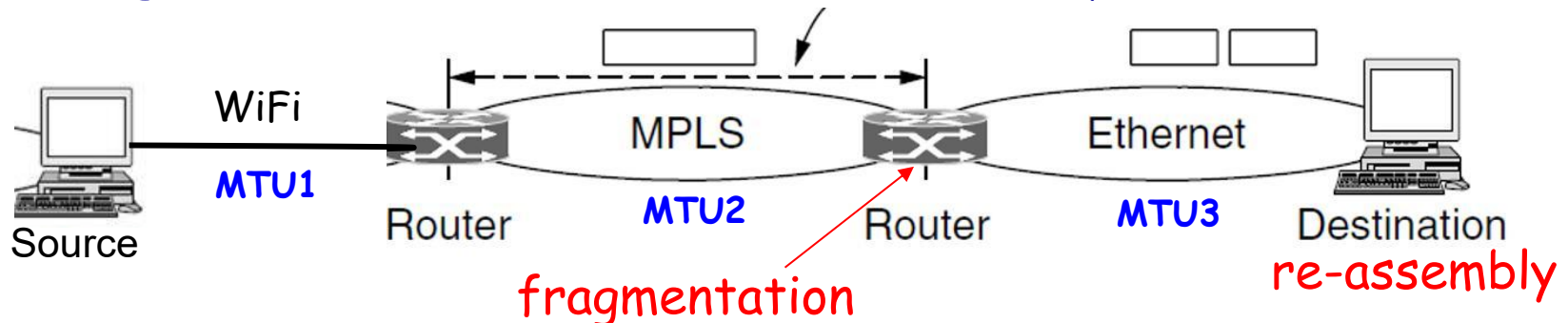
- **MTU: Maximum Transmission Unit (最大传输单元)**
- MTU is determined by
 - ◆ Hardware (e.g., the size of an Ethernet frame)
 - ◆ Operating system (e.g., all buffers are 512 bytes).
 - ◆ Protocols (e.g., the number of bits in the packet length field)
 - ◆ Compliance with some (inter)national standard.
 - ◆ Desire to reduce error-induced retransmissions to some level
 - ◆ Desire to prevent one packet from occupying the channel too long



- How to do fragmentation?
 - ◆ See **how IP does**

Fragmentation and Re-assembly(分段/分片和重装)

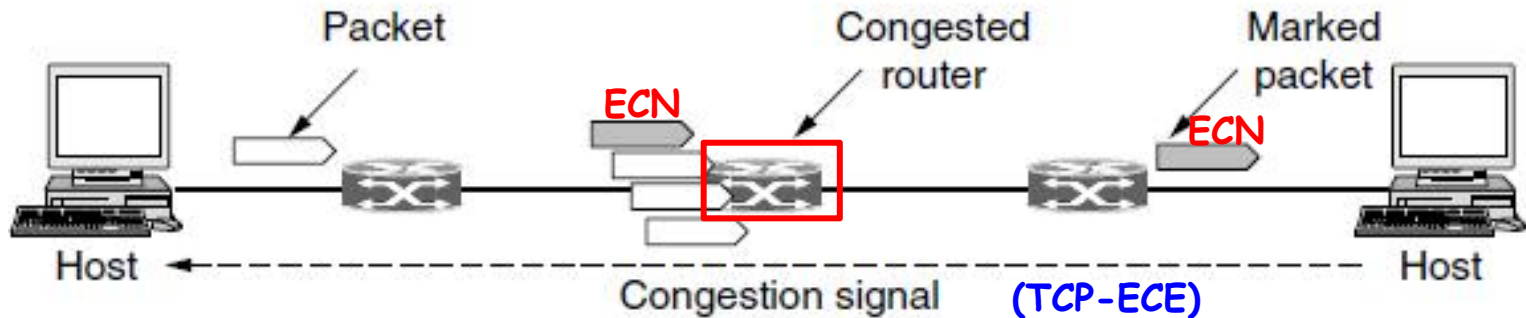
- Problem: A datagram can travel through different networks. different networks have different maximum frame sizes.
 - ◆ **MTU**: maximum transmission unit(最大传输单元)
 - Ethernet: MTU = 1500 bytes.
 - WiFi(IEEE802.11): MTU = 2312 bytes.
 - ◆ MTU may be as small as 128 bytes or less.
- Consequence: IP should adapt to different MTUs.
 - ◆ Choose convenient initial datagram size.
 - ◆ Divide datagrams when they traverse network with smaller MTU
 - ◆ Size of fragments determined by network with **minimum MTU**
- IP datagram is **fragmented** by the **source host** or any **router** in the path.
- IP datagram is **re-assembles** at **destination** only



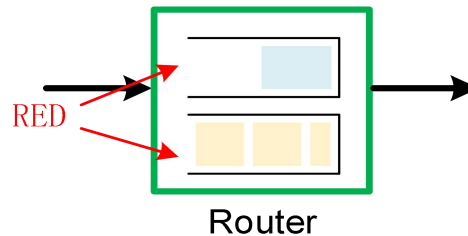
Review(1)

■ Congestion control

◆ Traffic throttling ECN



◆ Load shedding RED



Review(2)

QoS

◆ Parameters: reliability, delay, jitter, bandwidth

◆ Techniques to improve QoS

➤ traffic shaping

▣ Leaky buckets

▣ Token buckets

▣ $S = B / (M - R)$ (会计算)

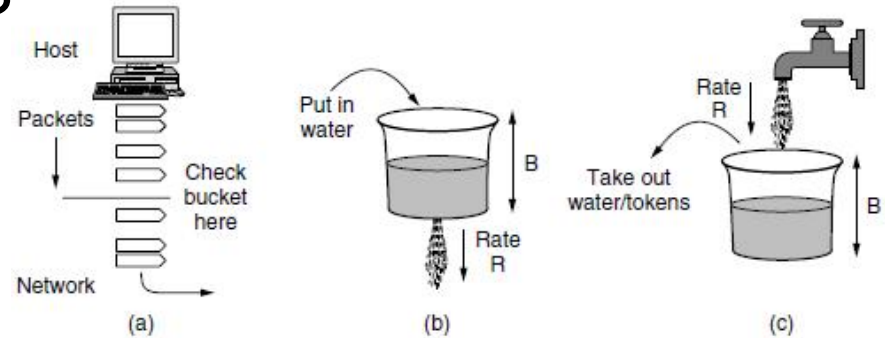


Figure 5-28. (a) Shaping packets. (b) A leaky bucket. (c) A token bucket.

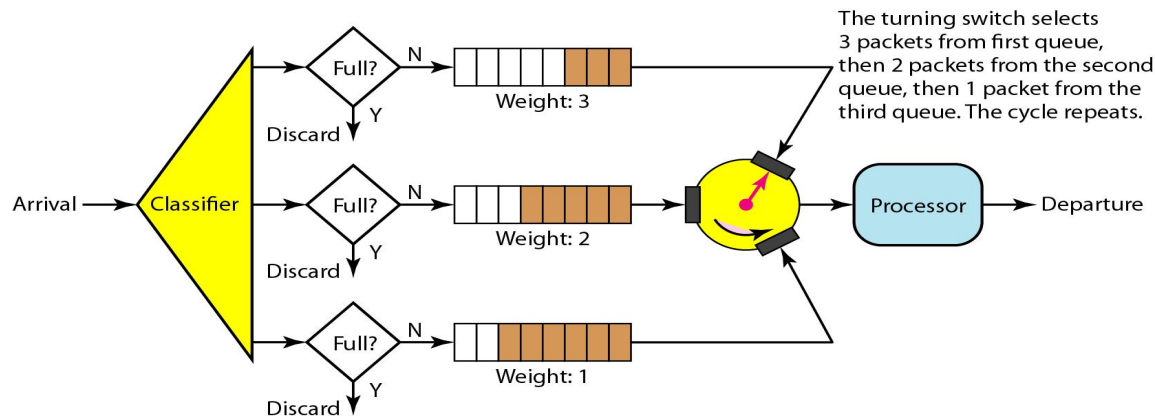
➤ Packet Scheduling

▣ FIFO(先进先出)

▣ PQ(优先级队列)

▣ FQ(公平队列)

▣ WFQ

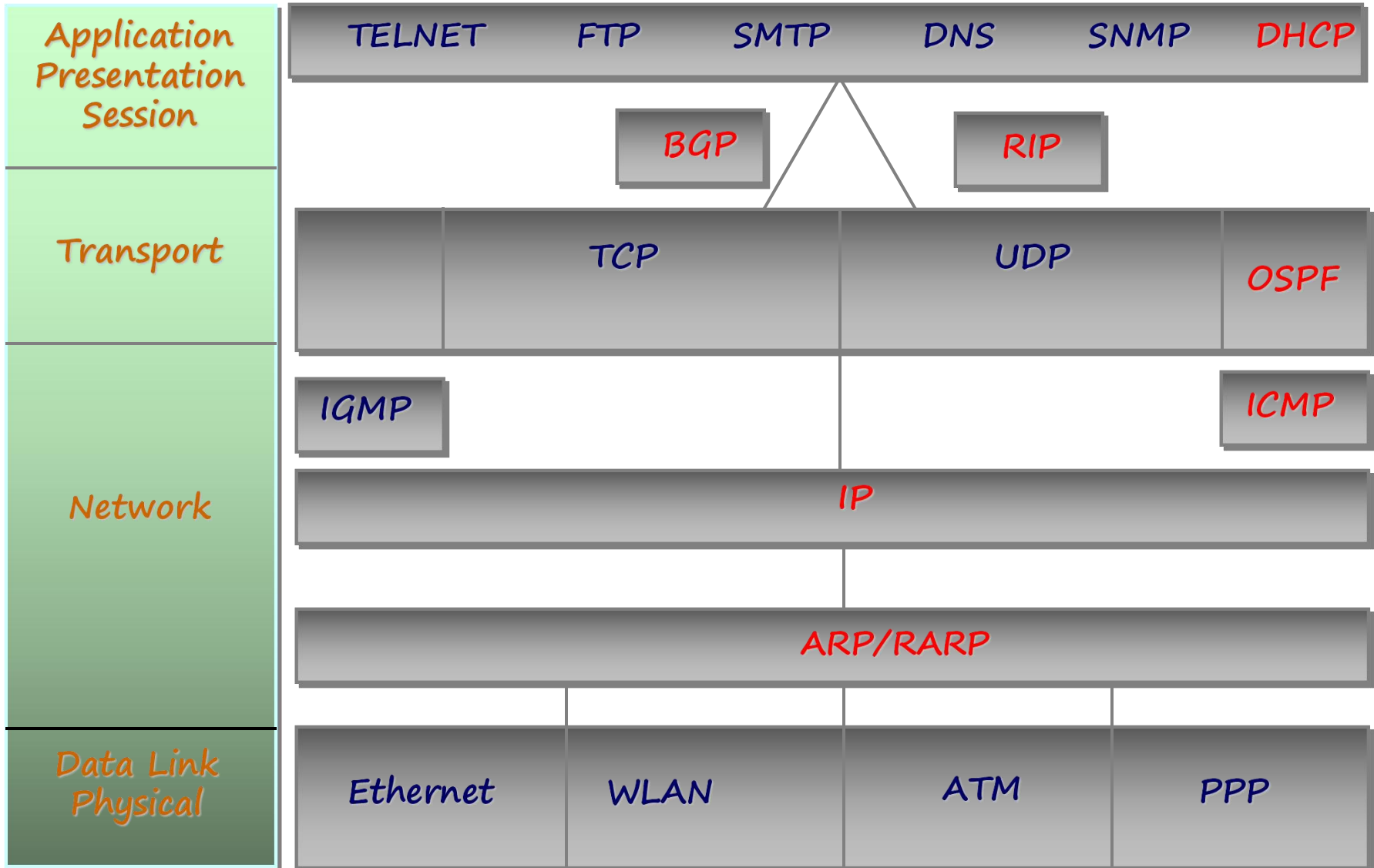


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 - ◆ 5.6.1 IPv4 Packet format
 - ◆ 5.6.2 IP Addressing
 - ◆ 5.6.3 IPv6
 - ◆ 5.6.4 ARP, DHCP and ICMP
 - ◆ 5.6.6&5.6.7 Routing Protocols: OSPF and BGP

OSI

TCP/IP

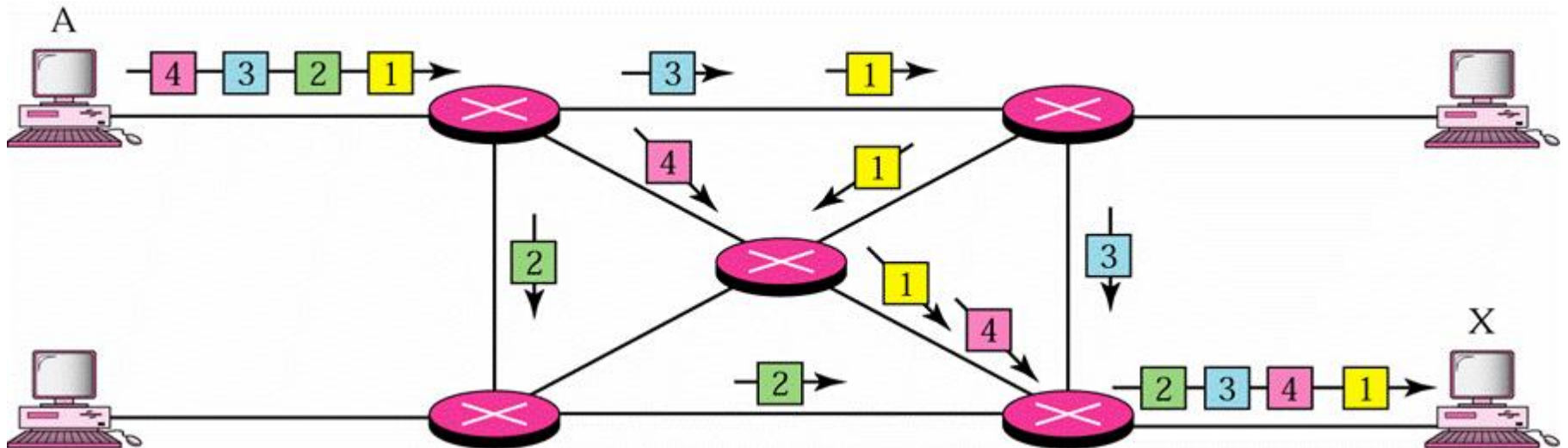


Design principles of IP

- **Make sure it works**-保证工作
- **Keep it simple**-保持简单
- **Make clear choices**-明确选择
- **Exploit modularity**-模块开发
- **Expect heterogeneity**-期望异构性
- **Avoid static options and parameters**-避免静态选项与参数
- **Look for a good design; it need not be perfect**-寻找好的而不是完美的设计
- **Be strict when sending and tolerant when receiving**-严格发送，宽容接收
- **Think about scalability**-考虑可扩展性
- **Consider performance and cost**-考虑性能与成本

Internet Protocol (IP)

- The **most-widely used** Internetworking protocol
- A **connectionless** network-layer protocol.
- Based on **datagram** routing.
- IP provides **best-effort** service, unreliable !!
- Routing in IP is a mixture of **hierarchical**, **distance vector** and **link state** routing algorithms.



[Forouzan]

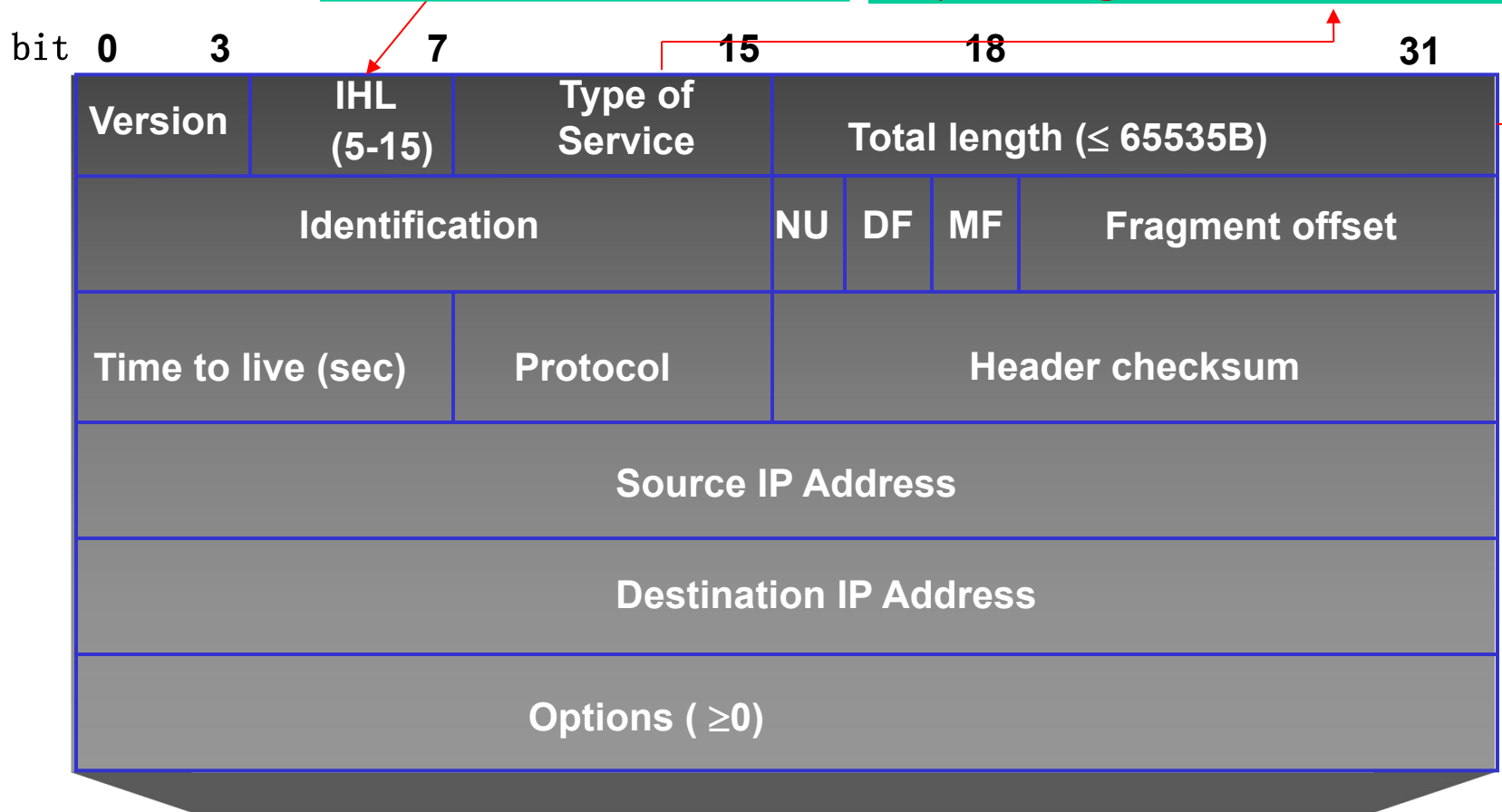
The network layer in the Internet

- 5.6.1 IPv4 Packet format
- 5.6.2 IP Addressing
- 5.6.3 IPv6
- 5.6.4 ARP, DHCP and ICMP
- 5.6.6&5.6.7 Routing Protocols: OSPF and BGP

IP Header

IP Header Length in terms of 32-bit words

top 6 bits are used to Service Class, the bottom 2 bits are used to carry explicit congestion notification (ECN)



Header length + Data length of the IP packet in terms of number of bytes

The IPv4 Header (1)

■ Version (4 bits)

■ IHL(4 bits)

◆ How long the header is, in 32-bit words, Val:5~15, 20~60 bytes

■ The Differentiated services field

◆ Originally, it was called the Type of service field (TOS)

➤ ~~a 3-bit Precedence field: (Priority 0~7)~~

➤ ~~3 flags: Delay, Throughput, Reliability~~

◆ **Now**, the top 6 bits are used to mark the packet with its **service class** the bottom 2 bits are used to carry explicit congestion notification (**ECN**) information

■ Total Length(16 bits)

◆ Includes **both header and data**, in bytes

◆ The maximum length is 65,535 bytes

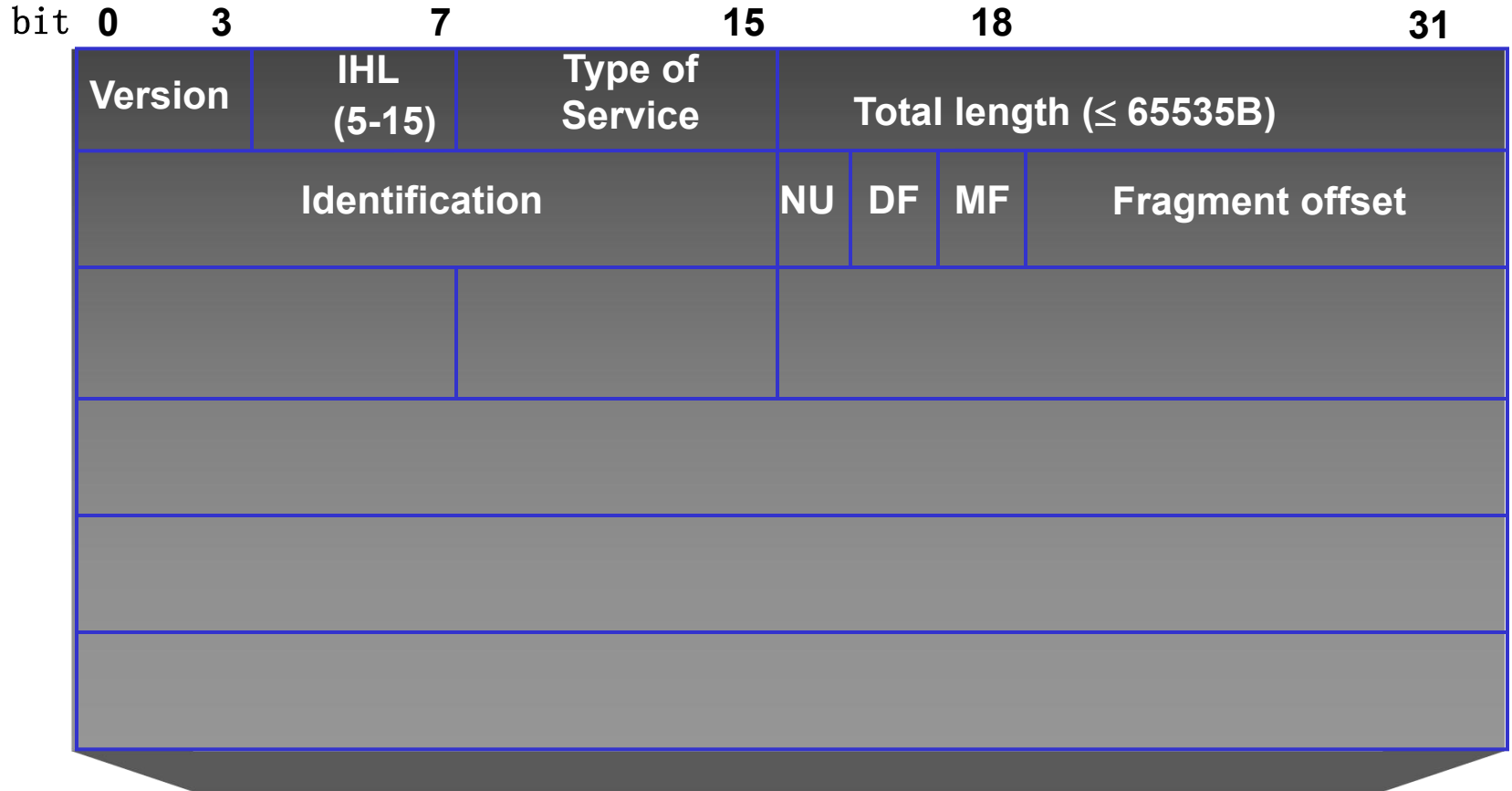
eg 4**5** 00 **00** 30

->0000 0000 0011 0000

-> $2^5 + 2^4 = 48$

-> $3 \times 2^4 = 48$

IP Header



The IPv4 Header (2)

■ Identification (16 bits)

- ◆ Allow the destination host to determine which datagram a newly arrived **fragment** belongs to
- ◆ All the fragments of a datagram contain the **same Identification** value

■ NU(1 bit): Null(未定义)

■ DF (1 bit)

- ◆ **Don't Fragment**. It is an order to the routers not to fragment the datagram (It is used as part of the process to discover the path MTU)

■ MF (1 bit)

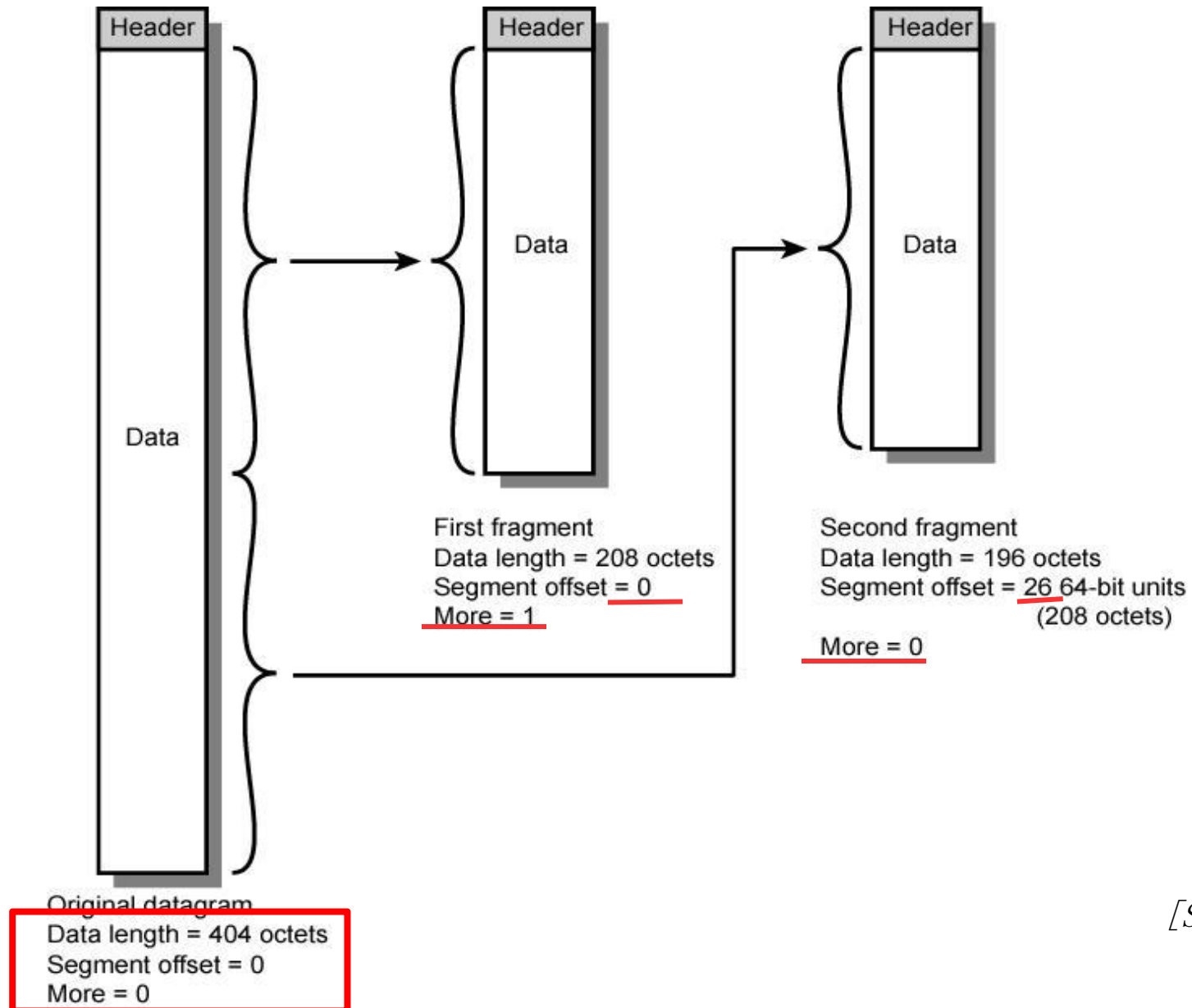
- ◆ **More** Fragments

■ Fragment offset (13 bits)

- ◆ Tell where in the current datagram this fragment belongs, **in 8 bytes** (0~8191)

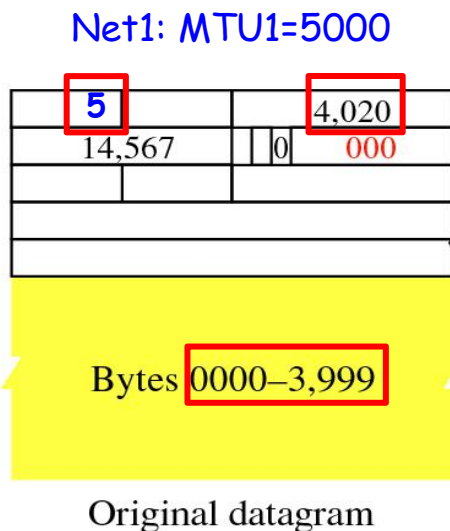
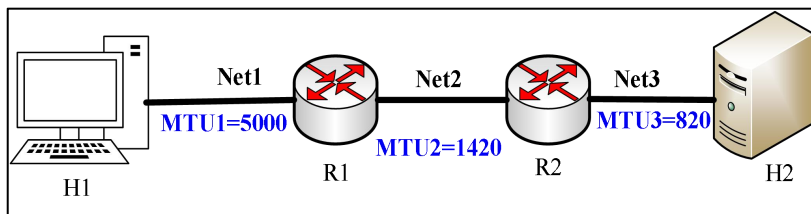
82 fc **40 00**

Fragmentation Example (1)



[Stallings]

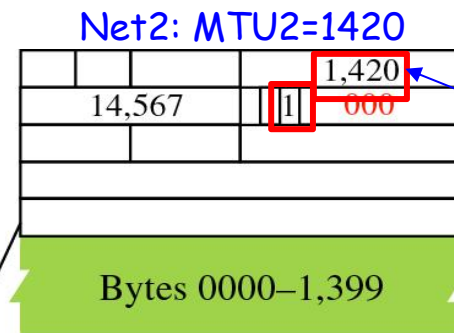
Fragmentation Example (2)



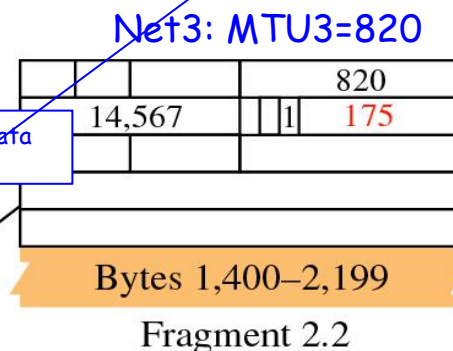
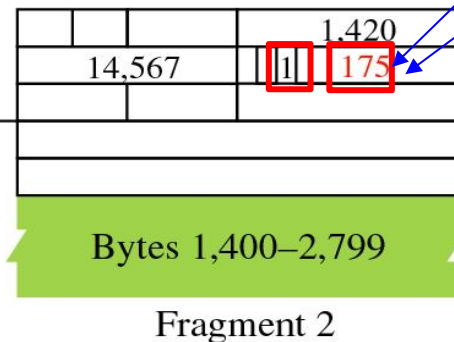
total length=4020Byte
 IHL=5*4=20Byte
 data length=4020-20=4000

Question: MTU2=1400

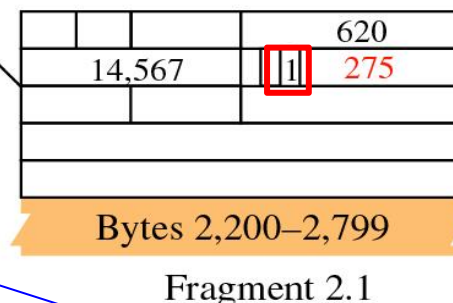
first-fragment data length=1376
 offset= 172



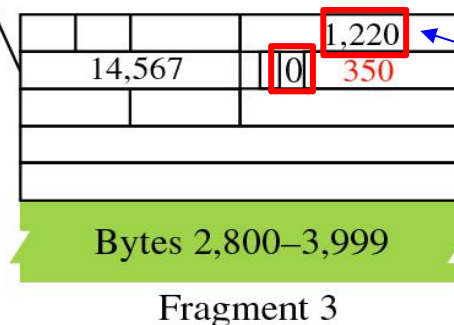
header length=20Byte
 first-fragment data length= $[(MTU2-20)//8]*8$



offset=first-fragment data length/8 =1400/8=175



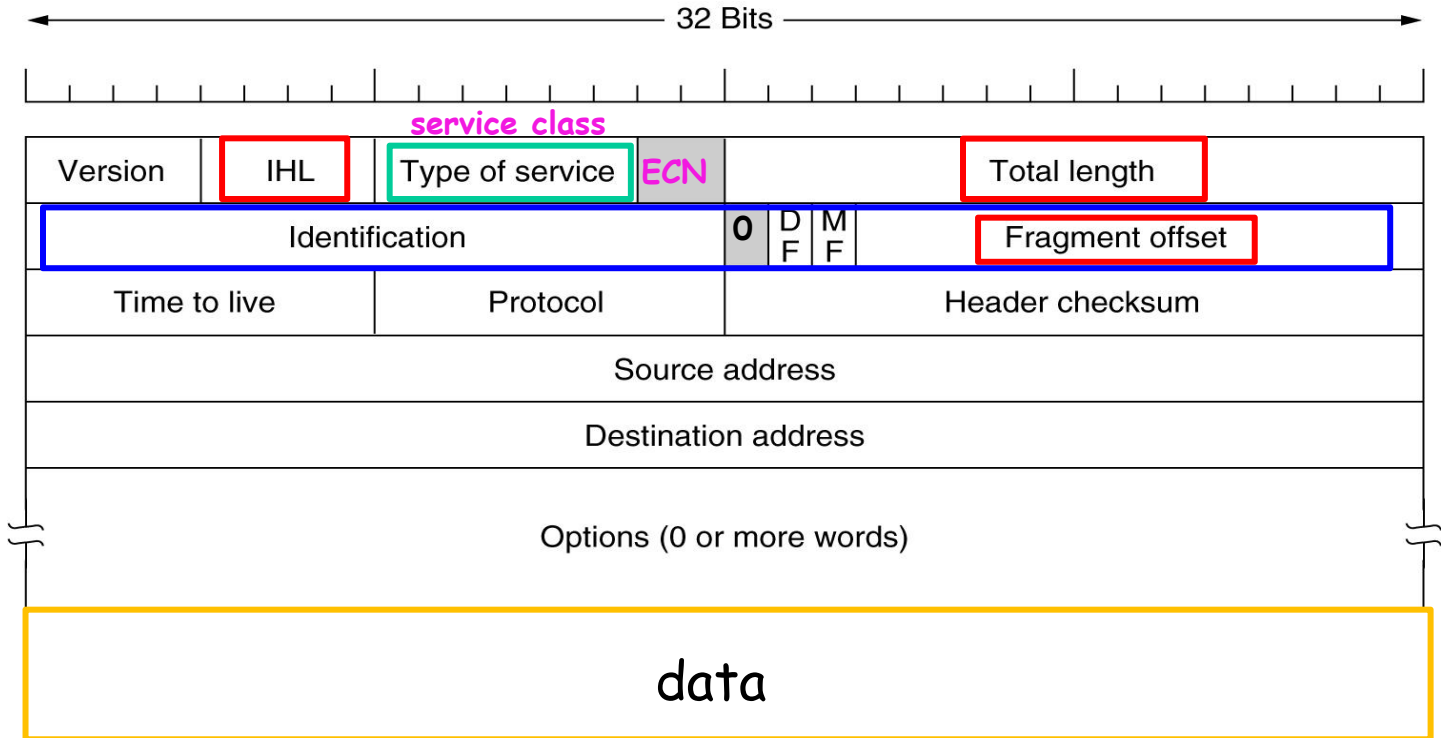
last fragment data length
 =4020-350(offset)*8=1220



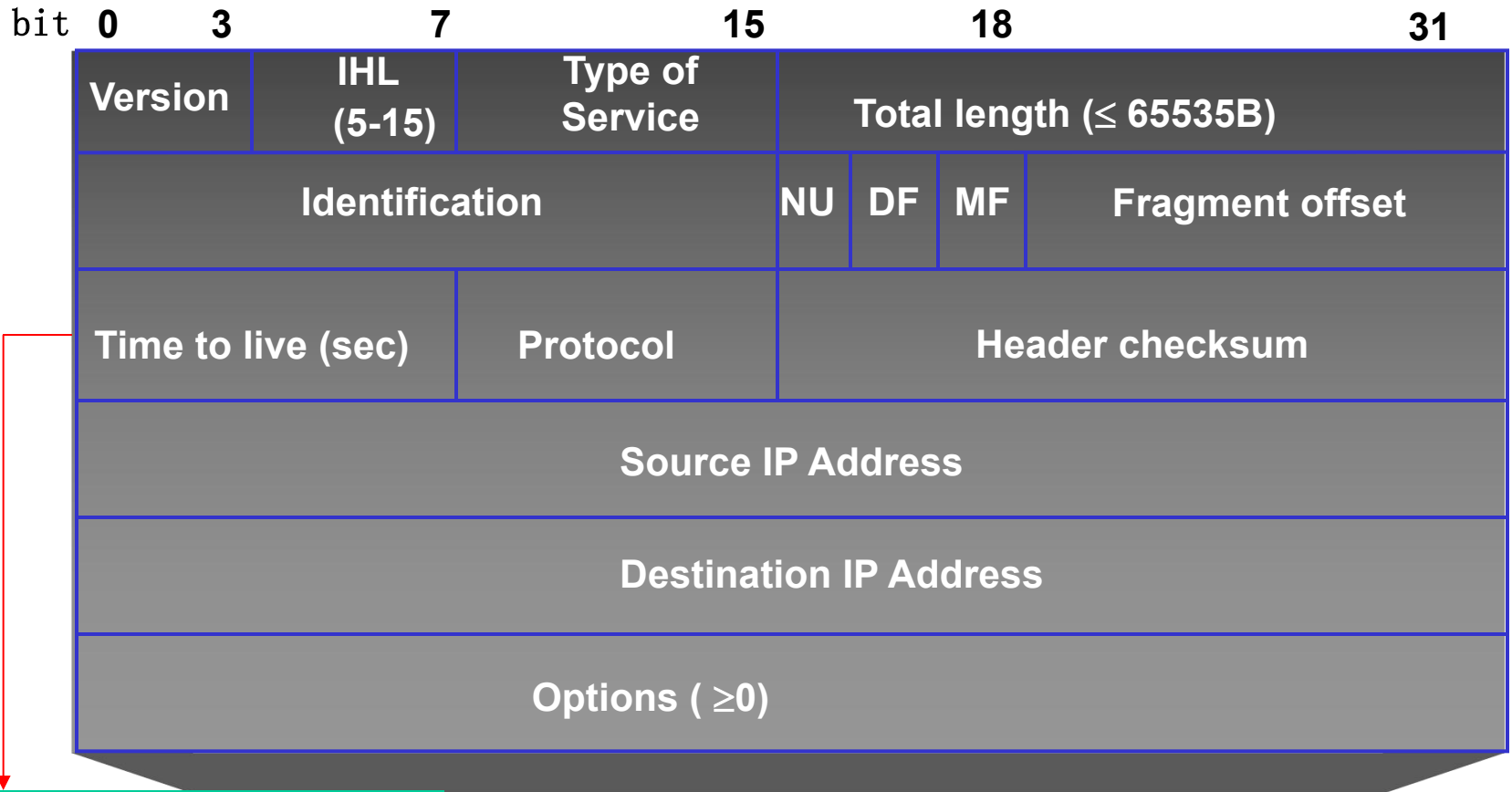
原始IP包总长度计算
 (1) 找最后一片 (即MF=0), 该片的段偏移量*8+该片的总长度
 (2) 各数据片的长度域-20, 相加+20

Review

■ IPv4 PDU



IP Header



to control max number
of hops visited by a
datagram by count
decrement.

The IPv4 Header (3)

■ TTL: Time to live (8 bits)

- ◆ A counter used to limit packet **lifetimes**. It is supposed to count time in **seconds**
- ◆ It must be **decremented on each hop**
- ◆ When it hits **zero**, the packet is discarded and a warning packet is sent back to the source host

■ Protocol field (8 bits)

- ◆ Which transport process to give it to (TCP/UDP,...)
1-ICMP, 6-TCP, 17-UDP, 89-OSPF, ...

■ The Header checksum (16 bits)

- ◆ Verifies the header only
- ◆ Must be **recomputed** at each hop

80 06 f5 a5

■ Source & Destination address(32+32 bits)

- ◆ 32-bit, Dotted-decimal notation(点分十进制)

10000000 00001011 00000011 00011111

80 0b 03 1f

128.11.3.31

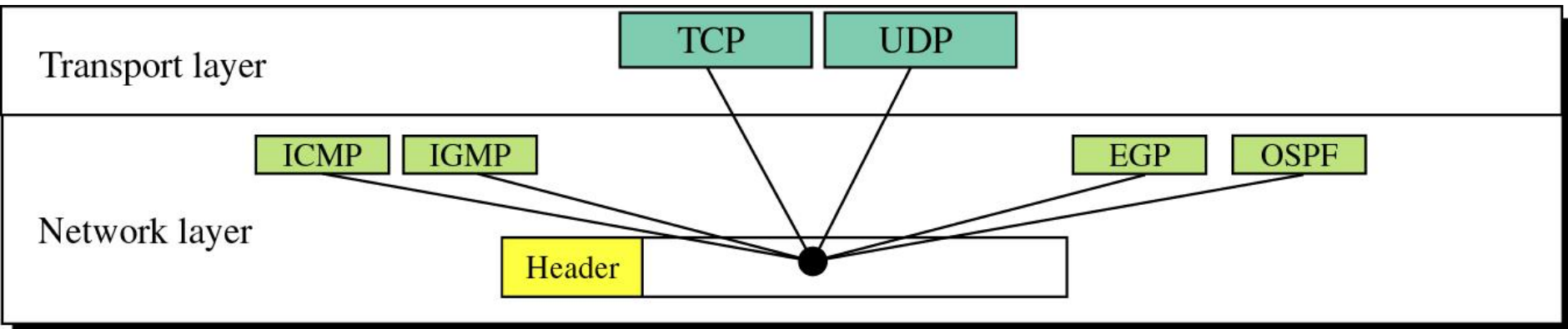
80 a8 b1 99

128.168.177.153

3b 40 6e d5

128.168.177.153

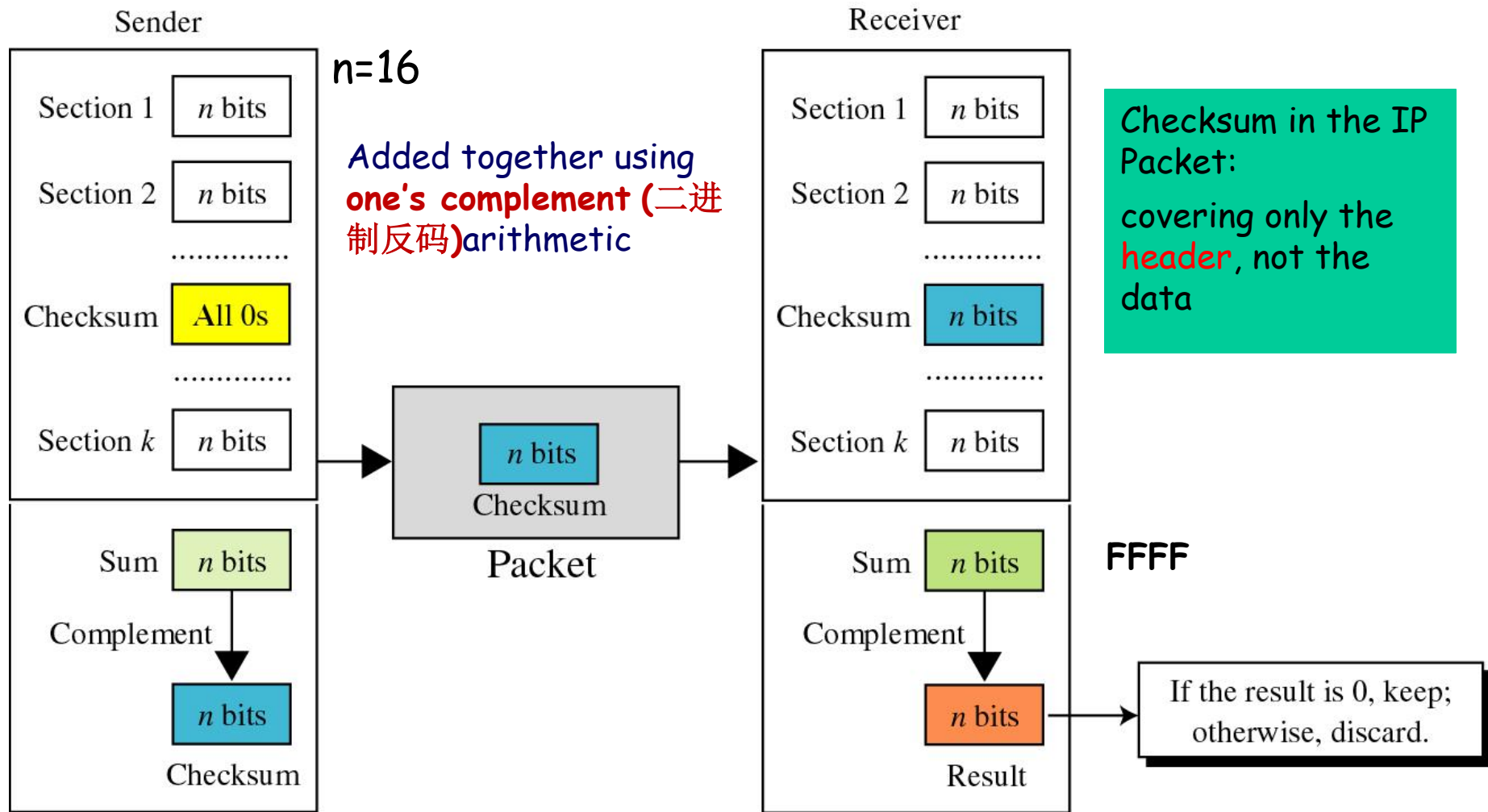
Protocol Field



- Defines the **higher level protocol** (transport-layer) that uses the services of IP-layer.
- An IP datagram can encapsulate data from several higher-level protocols such as: TCP=6, UDP=17, ICMP=1, IGMP=2, OSPF=89
- Specifies final destination protocol to which IP datagram should be delivered

[Forouzan]

Header Checksum(校验和) Concept(RFC 1071)



问题：如果使用二进制异或运算检错效果如何？

[Forouzan]

Example of Checksum Calculation

4	5	0	28	
1			0	0
4	17		0	
10.12.14.5				
12.6.7.9				



4, 5, and 0	➔	01000101	00000000
28	➔	00000000	00011100
1	➔	00000000	00000001
0 and 0	➔	00000000	00000000
4 and 17	➔	00000100	00010001
0	➔	00000000	00000000
10.12	➔	00001010	00001100
14.5	➔	00001110	00000101
12.6	➔	00001100	00000110
7.9	➔	00000111	00001001
Sum	➔	01110100	01001110
Checksum	➔	10001011	10110001

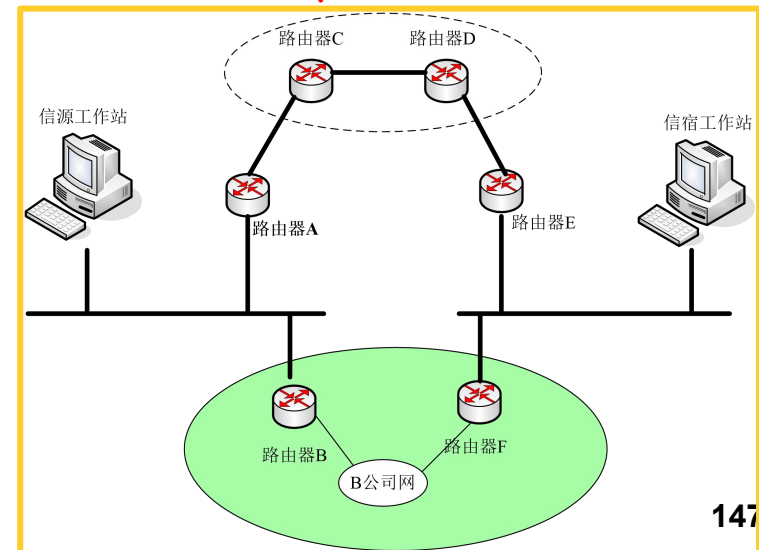
11111111 11111111



Some of the IP options

- Security option
- Strict source routing option 严格源选路(A→C→D→E→H2)
 - ◆ Gives the **complete path from source to destination** as a sequence of IP addresses
- Loose source routing option 宽松源选路(E→H2)
 - ◆ Requires the packet to **traverse the list of routers specified**, and in the order specified, but it is **allowed to pass through other routers** on the way
- Record route option 记录路由
 - ◆ Tells the **routers along the path** to append their IP address to the option field
- The Timestamp option 时间戳
 - ◆ Like the Record route option, except that in addition to **recording its 32-bit IP address**, each router **also records a 32-bit timestamp**

Option	Description
Security	Specifies how secret the datagram is
Strict source routing	Gives the complete path to be followed
Loose source routing	Gives a list of routers not to be missed
Record route	Makes each router append its IP address
Timestamp	Makes each router append its address and timestamp



The network layer in the Internet

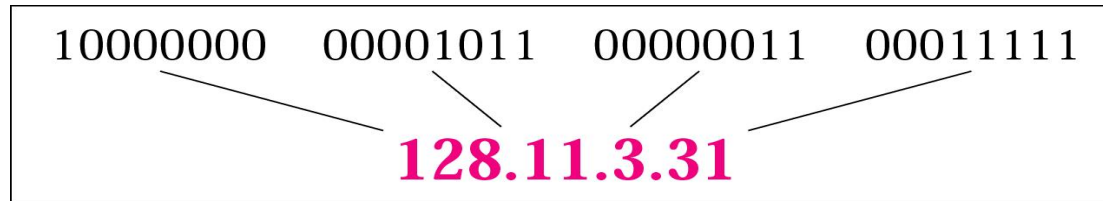
- 5.6.1 IPv4 Packet format
- 5.6.2 IP Addressing
- 5.6.3 IPv6
- 5.6.4 ARP, DHCP and ICMP
- 5.6.6&5.6.7 Routing Protocols: OSPF and BGP

IPv4 Addressing

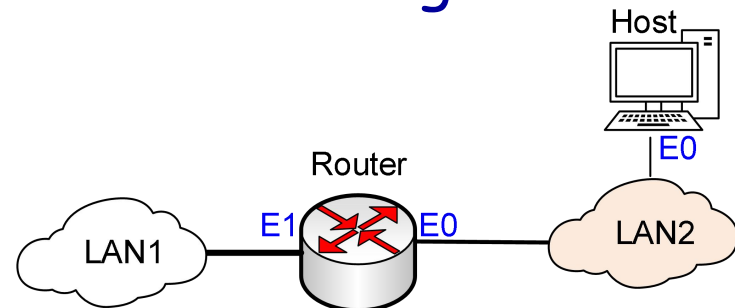
- **Classful Addressing**(分类地址): 1981
- Subnetting(子网): 1985
- Supernetting(超网): 1993
- **Classless Addressing**(无类别地址): 1995

(1) IPv4 Address

- Identifier used in network layer to identify each device connected to the Internet is called the **Internet address** or **IP address**.
- 32-bit, Dotted-decimal notation(点分十进制)



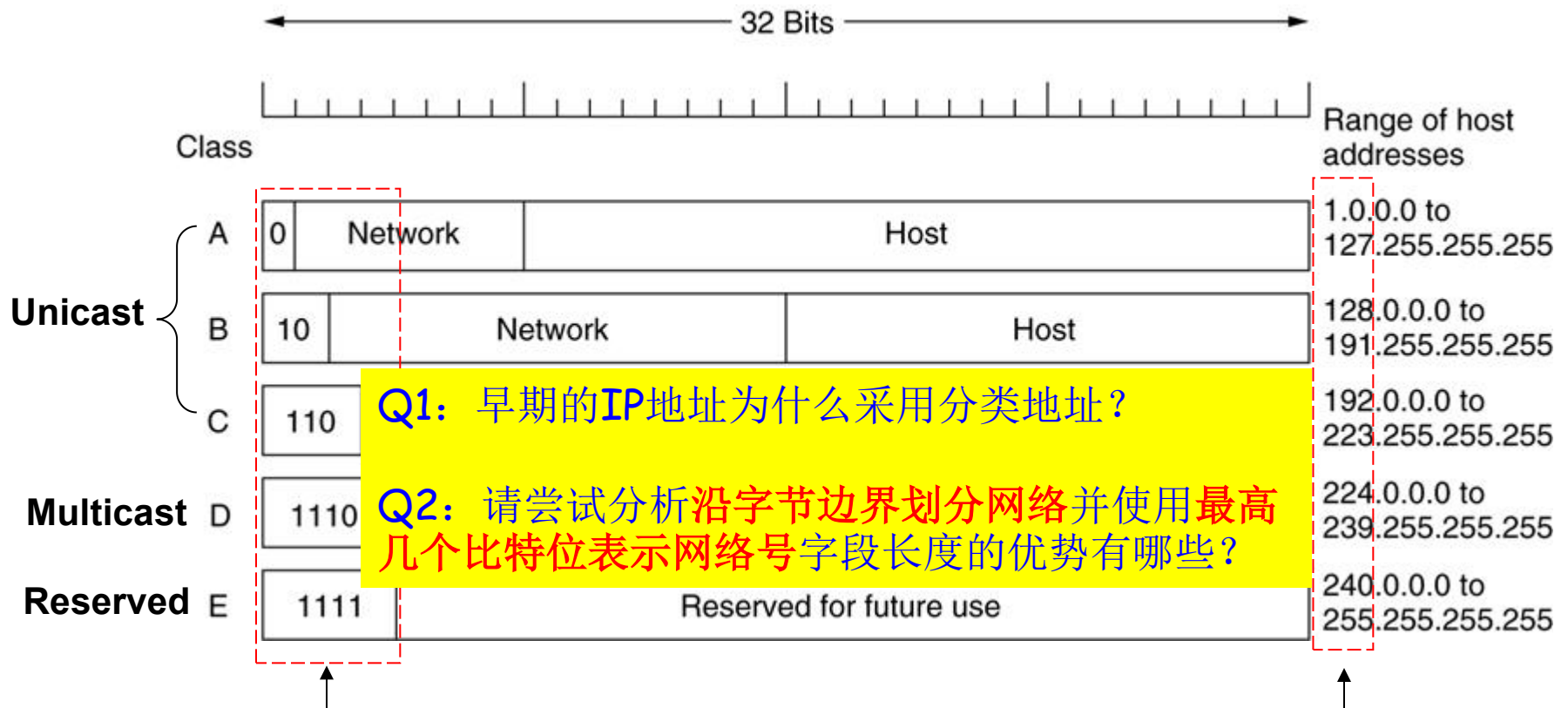
- **Unified addressing**
- Unique for each **network interface**, irrelative with its physical address, which universally defines **1 connection** to Internet.
- If a host/router has 2 interfaces connecting to Internet, then it has 2 IP addresses
- Two types of IP addressing
 - ◆ **Classful** vs. **Classless**



(2) Classful IP Addressing(分类地址)

- IP has a two-level (at least, traditionally) hierarchical addressing mechanism in class A, B and C:

Network ID(net-id) + Host ID (host-id)



(3)Network address

- **Network address** is the first address in the block, is the one that is assigned to the organization.

- 表示：网络号部分保留，主机号部分为0

Network ID + Host ID (全0) 

- 计算：Applying the **default mask**(默认/缺省掩码) to any of the IP addresses in the network

- ◆ IP address bit AND with default mask

eg. 123.50.16.90 AND 255.0.0.0 -> 123.0.0.0

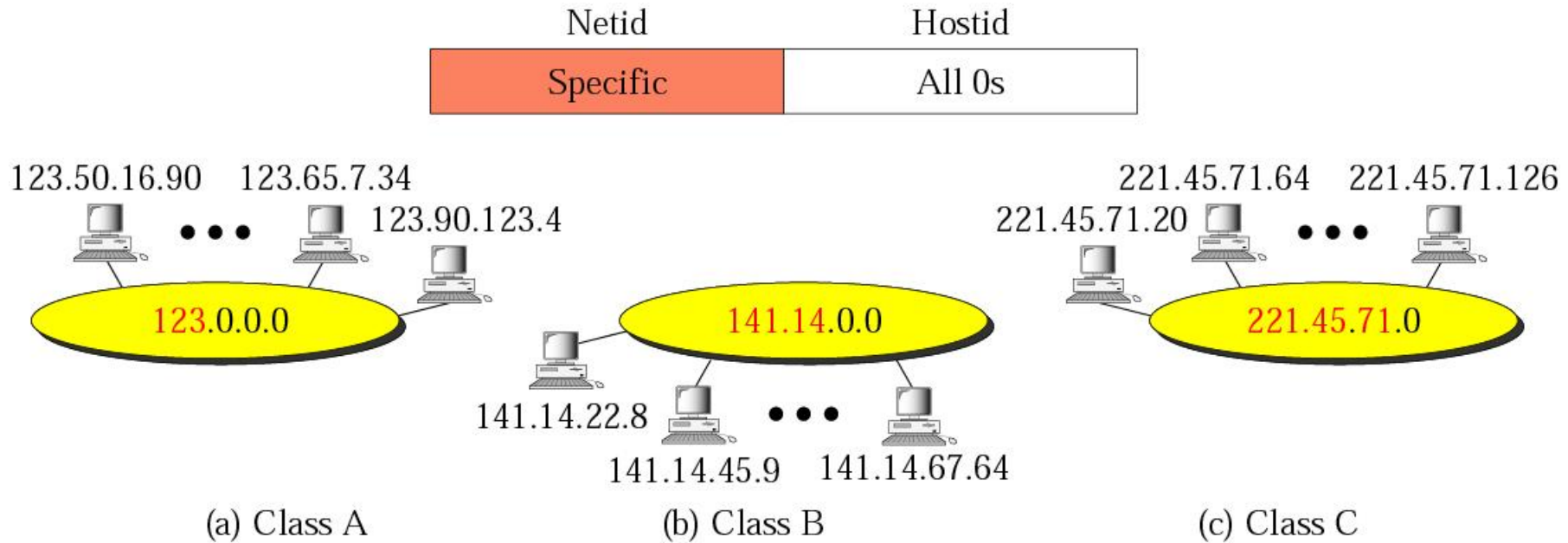
- Default masks

- ◆ Class A: 255.0.0.0

- ◆ Class B: 255.255.0.0

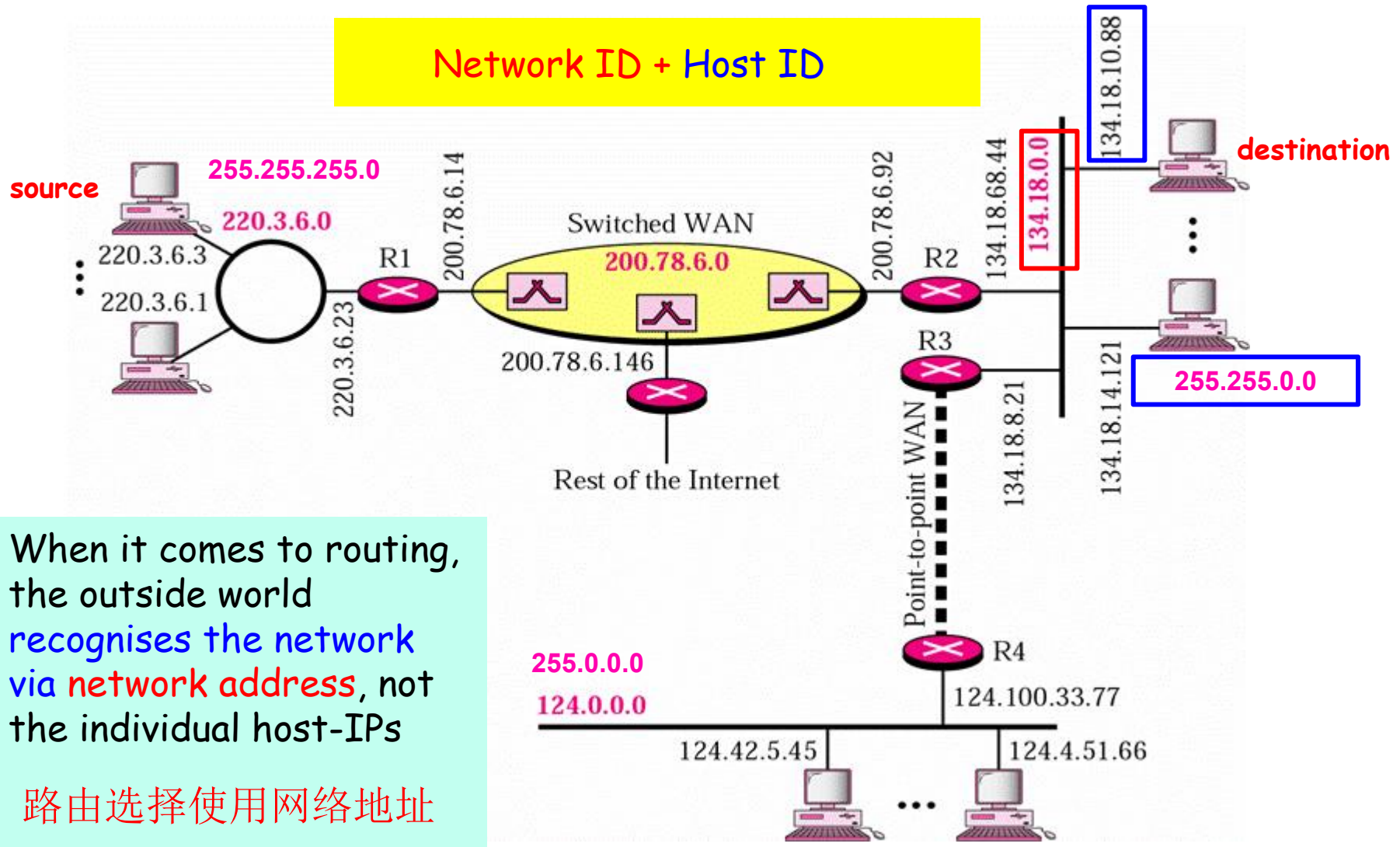
- ◆ Class C: 255.255.255.0

Network address: example



Network Addresses: Usage

Network ID + Host ID



When it comes to routing,
the outside world
recognises the network
via network address, not
the individual host-IPs

路由选择使用网络地址



[Forouzan]

(4) Special IP addresses

0 0	This host
0 0 ... 0 0 Host	A host on this network
1 1	Broadcast on the local network
Network 1 1 1 1 ... 1 1 1 1	Broadcast on a distant network
127 (Anything)	Loopback 环回地址（测试地址）

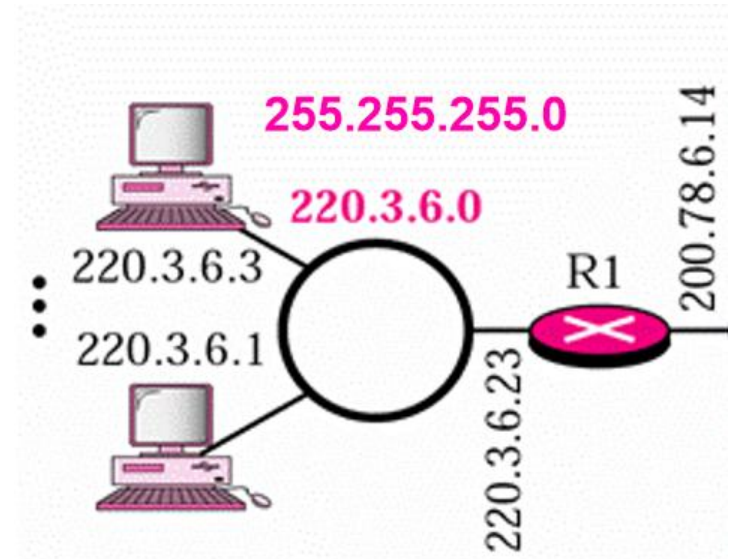
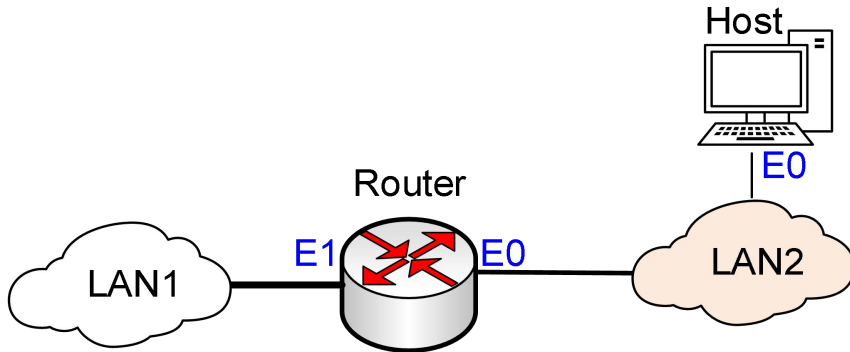
未指定地址(This host): 只能作为IP数据报的源地址不能作为目的地址

广播/组播地址(broadcast address/multicast address): 只能作为IP数据报的目的地址不能作为源地址

测试地址(loop back): 只能做为IP数据报的目的地址不能作为源地址

补：网内的地址数与可分配(可用)地址

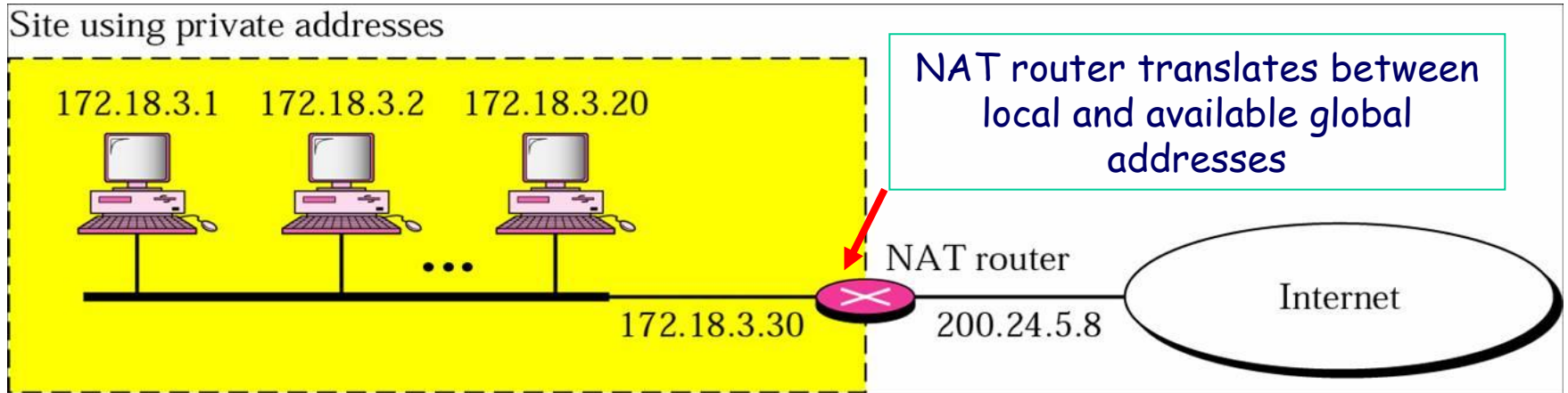
- 网络地址： $x.x.x.x/m$
- 网络号 m 位，主机号= $32-m=n$ 位
- 该网络内共有 2^n 个地址
 - ◆ 第一个地址（主机号为全0）为网络地址
 - ◆ 最后一个地址（主机号为全1）为广播地址
- 可分配（可用）地址数： 2^n-2



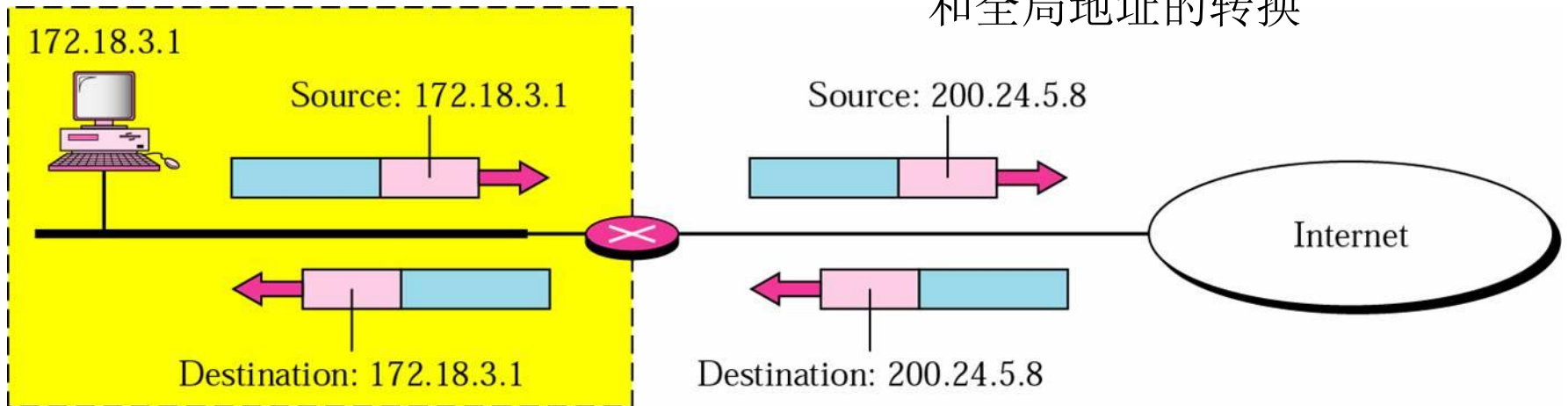
(5)NAT: Network Address Translation

- Problem: IP addresses are scarce
- Solution: using reserved Private IP addresses(私有/专用地址)
 - ◆ 10.0.0.0 - 10.255.255.255/8
 - ◆ 172.16.0.0 - 172.31.255.255/12 (16: 0001 0000)
 - ◆ 192.168.0.0 - 192.168.255.255/16
- NAT router converts internal IP source address to another global IP address
- NAT router is often combined with a firewall in a single device
- Source TCP/UDP port is also converted by NAT box

NAT Sample



NAT路由器负责私有地址
和全局地址的转换



[Forouzan]

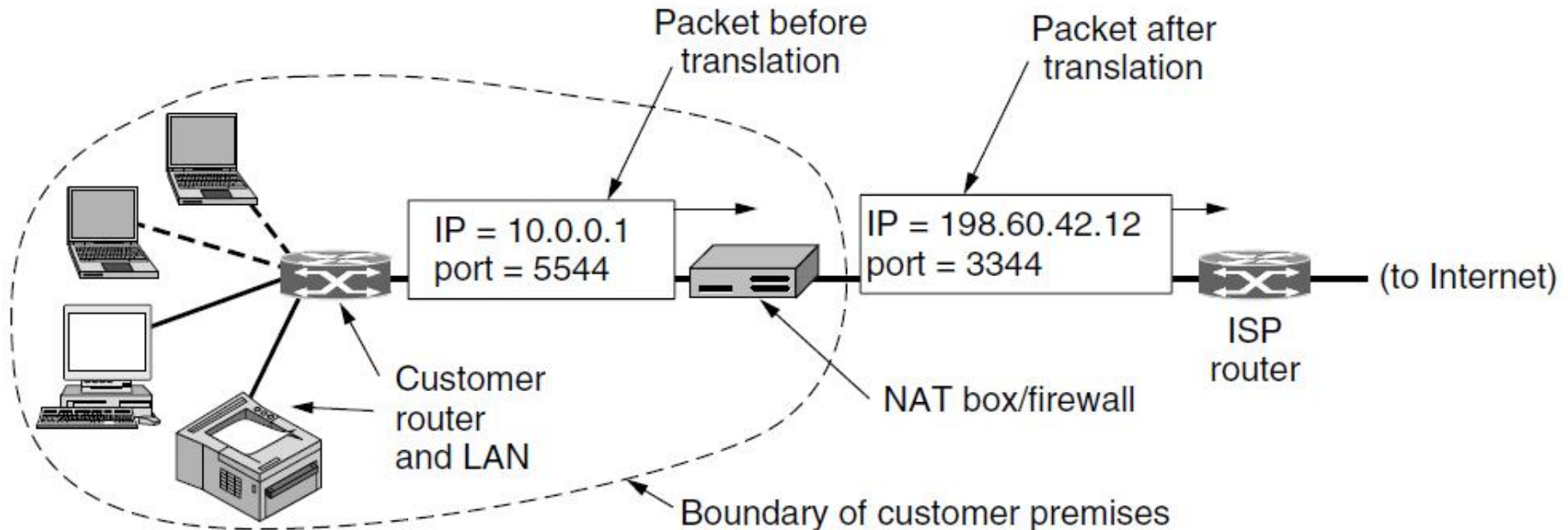
NAT (Network Address Translation)

■ Outgoing packet

- ◆ 10.x.y.z **source address** and TCP **Source port** are replaced
- ◆ Both the IP and TCP checksums are recomputed

■ Incoming packet

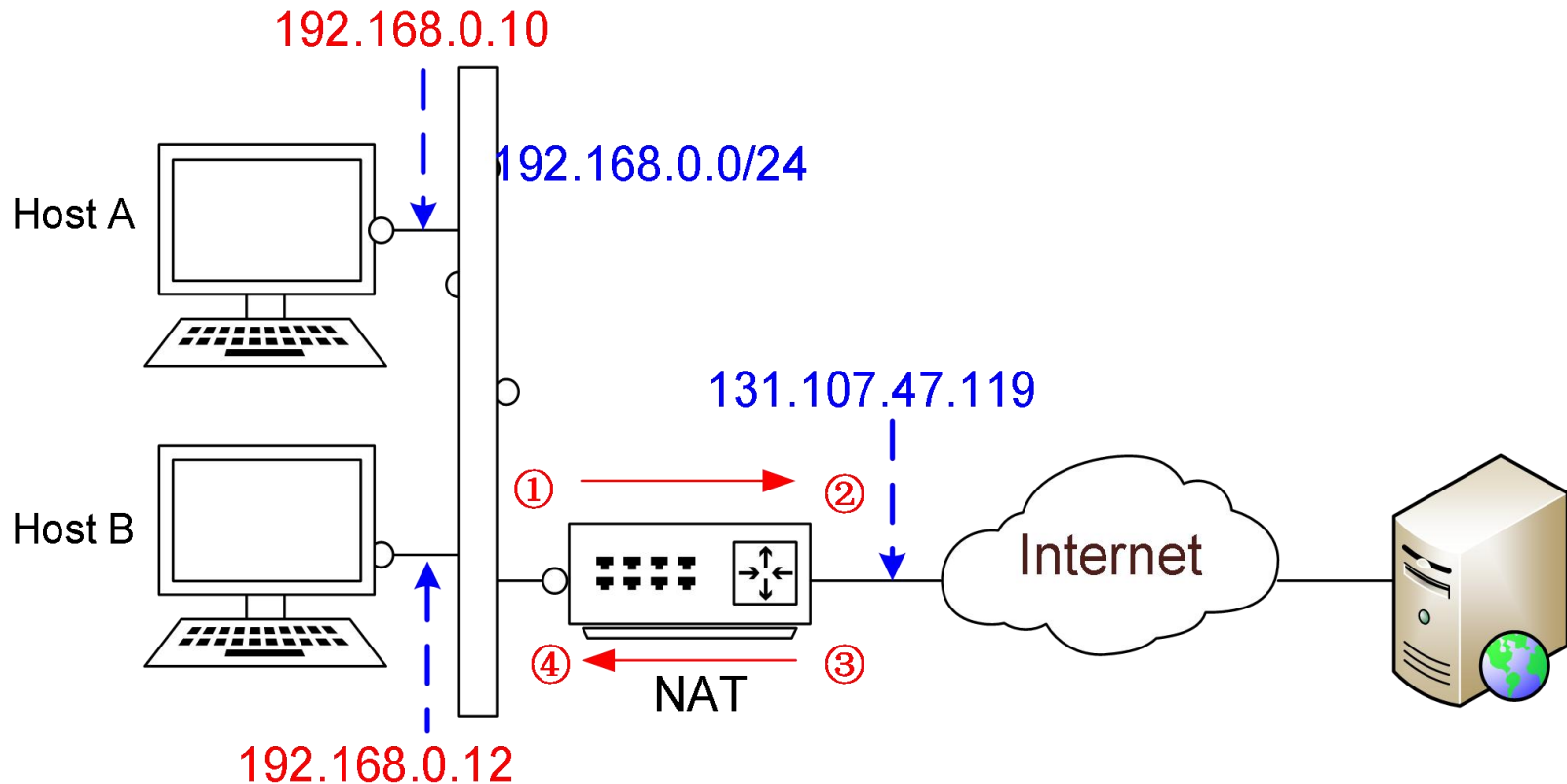
- ◆ Destination port => internal IP address and original TCP Source port, Recompute IP and TCP checksums
- ◆ Deliver the packet using the 10.x.y.z address



What if 2 local hosts talking to same destination?

NAT Table

Local-IP	Local port	Global IP	TCP port
192.168.0.10	1025	131.107.47.119	1025
192.168.0.12	1025	131.107.47.119	1025



What if 2 local hosts talking to same destination?

- Using port number: Layering discipline is violated

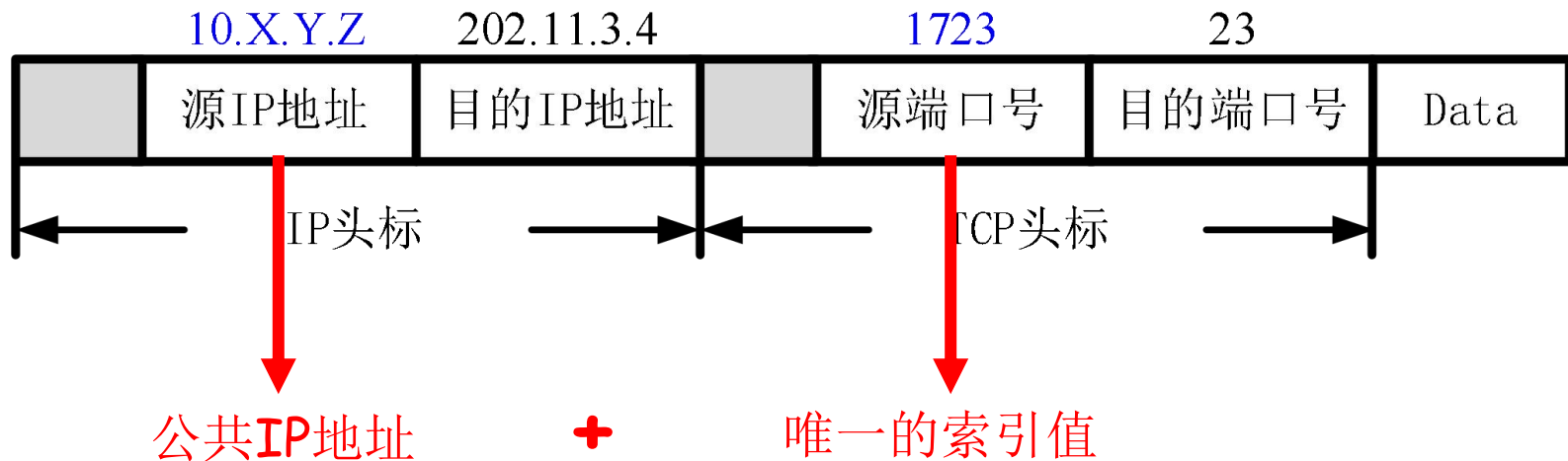


Table in NAT router

Peer	Local: IP+port		Global IP	Global port
202.11.3.4:23	10.1.1.2	1723	198.60.42.12	1723
202.11.3.4:23	10.1.1.3	1723	198.60.42.12	1924
202.11.3.4:23	10.1.1.4	1723	198.60.42.12	1492

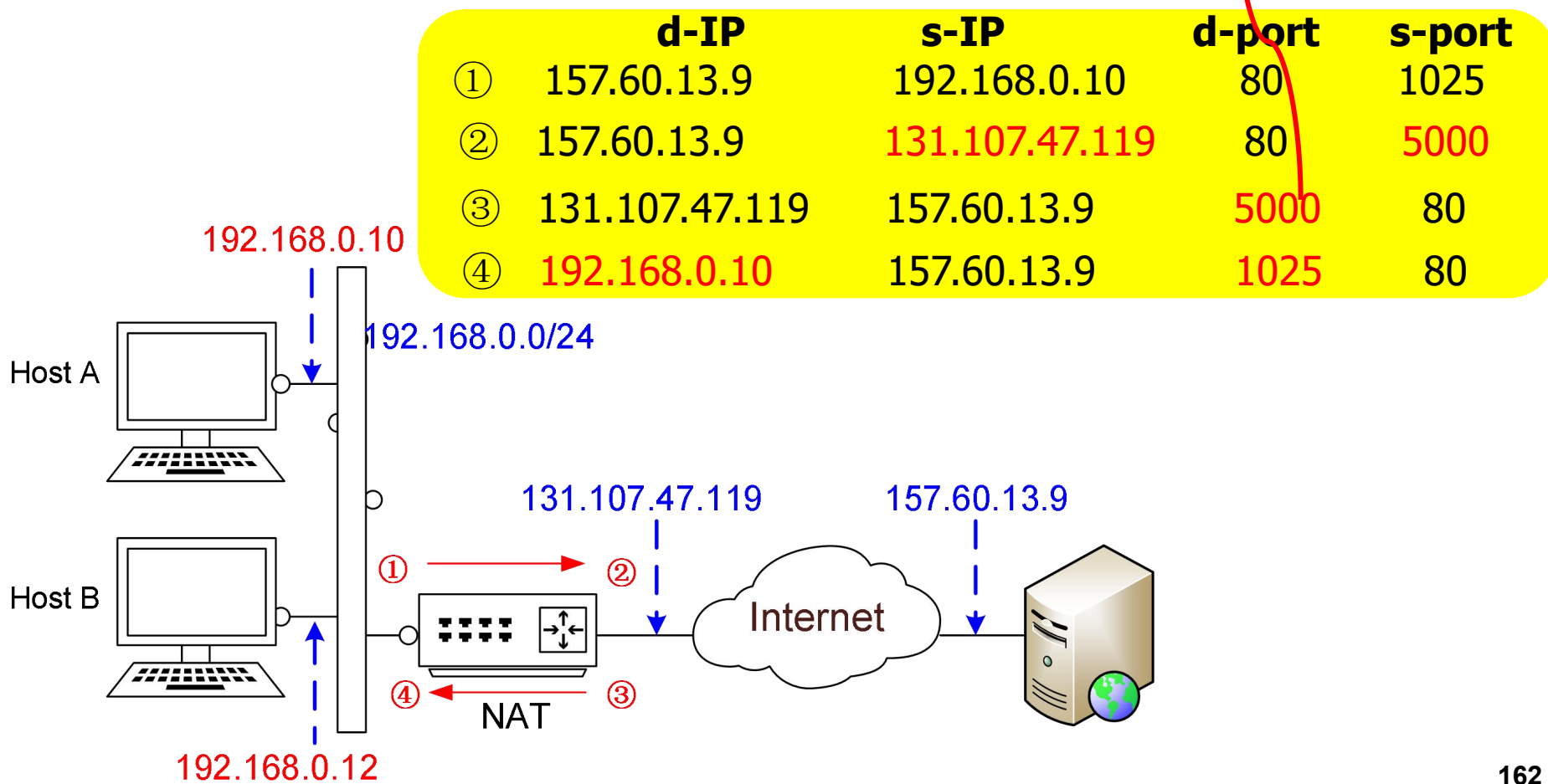
入境IP: 198.60.42.12:1492 -> 10.1.1.4:1723

NAT example

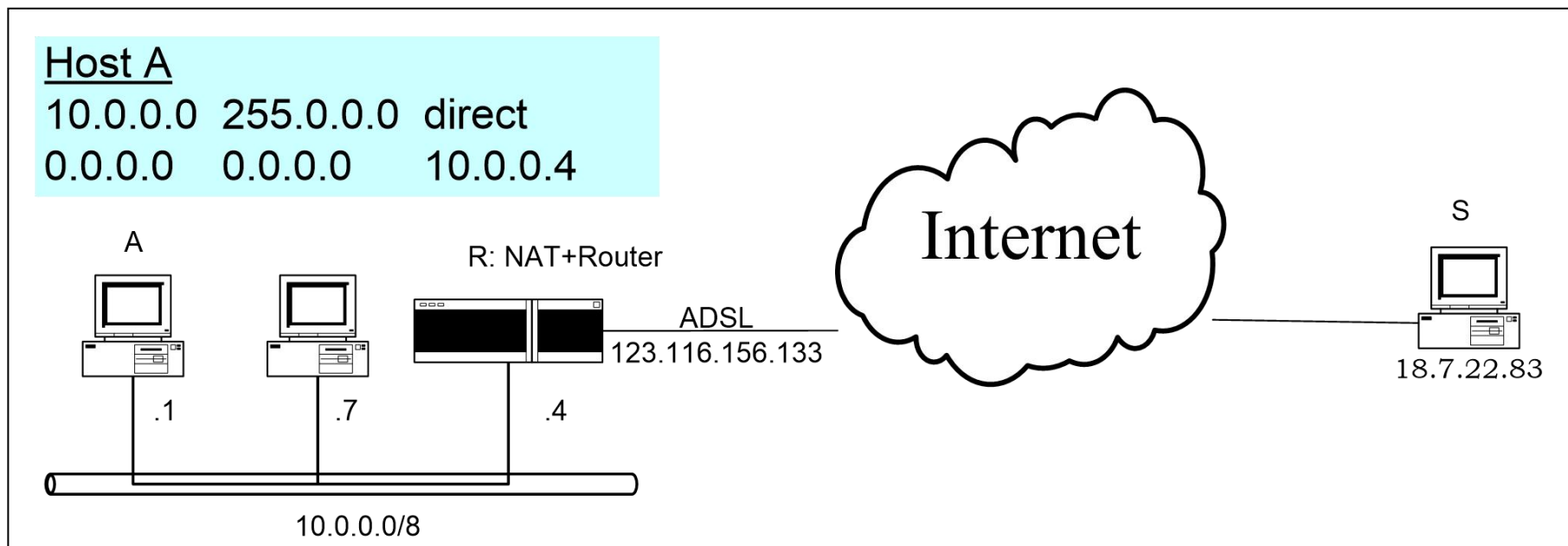
- 场景：Host A访问外网地址为157.60.13.9的web服务器

NAT Table

192.168.0.10 TCP 1025	131.107.47.119 TCP 5000
192.168.0.12 TCP 1025	131.107.47.119 TCP 5001



举例：动态NAT表项



■ A: 发送TCP报 10.0.0.1:3723->18.7.22.83:80

■ R: 路由器动态创建NAT表项:

(Local)10.0.0.1:3723—(Global)123.116.156.133:33120—(Peer)18.7.22.83:80

改TCP报为123.116.156.133:33120->18.7.22.83:80

■ S: 服务器S回数据18.7.22.83:80->123.116.156.133:33120

■ R: 查NAT表: 改TCP报为为18.7.22.83:80->10.0.0.1:3723

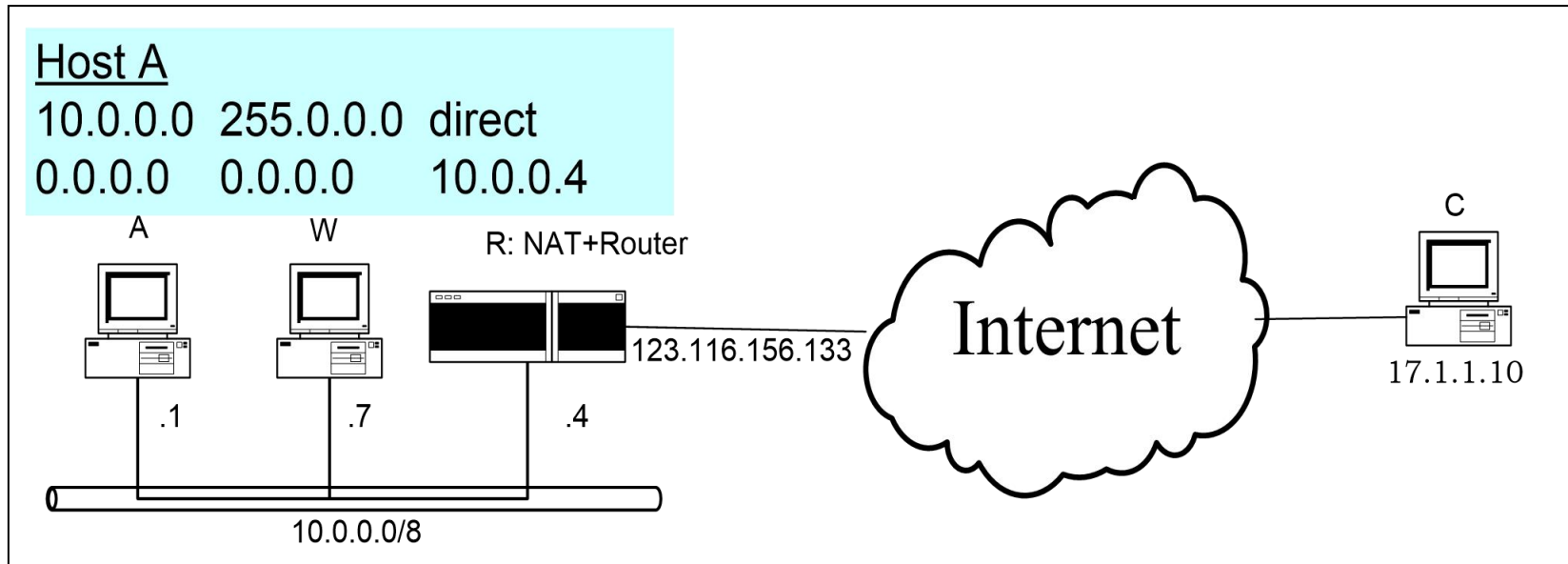
■ A: 收到数据包18.7.22.83:80->10.0.0.1:3723

(连接过程中使用该NAT表项, 当连接关闭时或长久未用的NAT表项, 删除)

OUT: (SRC)10.0.0.1:3723—(DST)18.7.22.83:80=>(SRC)123.116.156.133:33120

IN: (SRC)18.7.22.83:80—(DST)123.116.156.133:33120=>(DST)10.0.0.1:3723

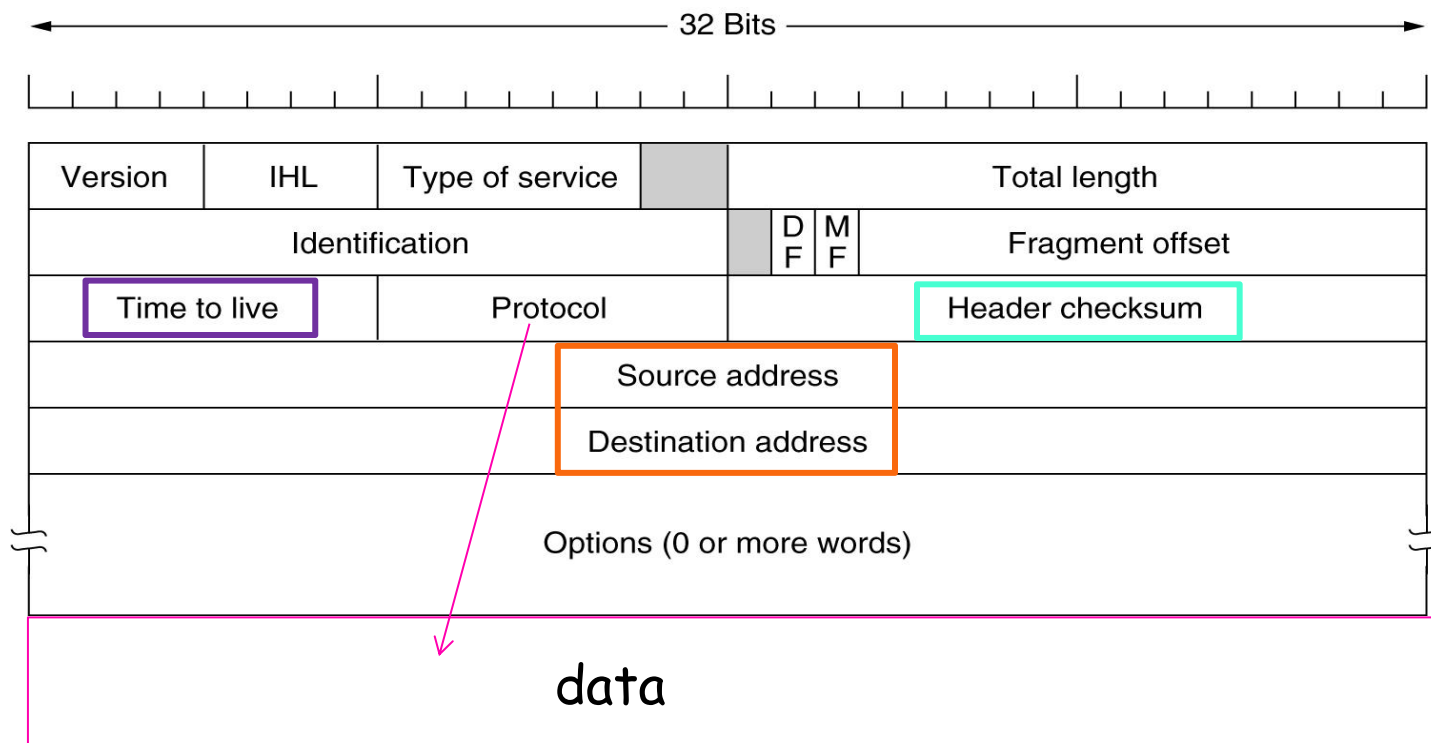
举例：静态NAT表项



- R: 设10.0.0.7为Web服务器: **建立静态NAT表**
(Local)10.0.0.7:80—(Global)123.116.156.133:**80** —(Peer)*:*
- C: 因特网上的客户端C发送TCP数据包:
17.1.1.10:3598->123.116.156.133:80
- R: 查 NAT表, 改为17.1.1.10:3598->10.0.0.7:80
- W: Web服务器回数据10.0.0.7:80->17.1.1.10:3598
- R: 查 NAT表, 改为123.116.156.133:80->17.1.1.10:3598
- C: 收到返回TCP数据报123.116.156.133:80->17.1.1.10:3598

Review(1)

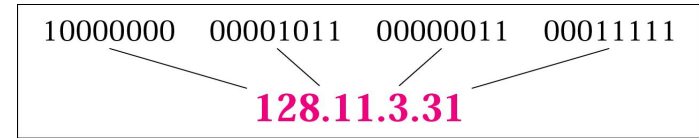
■ IPv4 PDU



Review(2)

■ IP address

- ◆ 32-bit, Dotted-decimal notation
- ◆ Network ID(net-id) + Host ID (host-id)



■ Classful Addressing scheme

- ◆ Unicast address: class A, class B, class C

■ Special IP addresses

- ◆ 0.0.0.0(this host)
- ◆ 127.x.x.x(loopback)
- ◆ 255.255.255.255(broadcast)

0 0	This host
0 0 ... 0 0 Host	A host on this network
1 1	Broadcast on the local network
Network 1 1 1 1 ... 1 1 1 1	Broadcast on a distant network
127 (Anything)	Loopback

■ network address: network ID+all '0' in host ID

- ◆ default mask
- ◆ IP address AND default mask = network address (会计算)
- ◆ number of IP address : 2^n
 - 可分配/可用的IP地址: 2^n-2 (网络地址、广播地址)

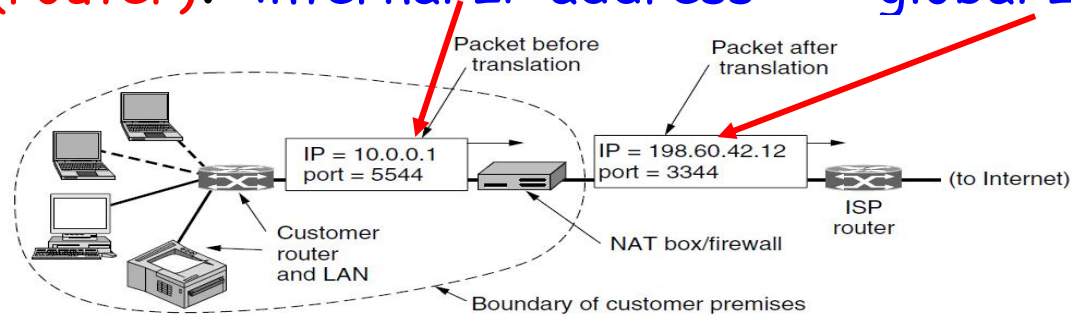
Review(3)

■ NAT

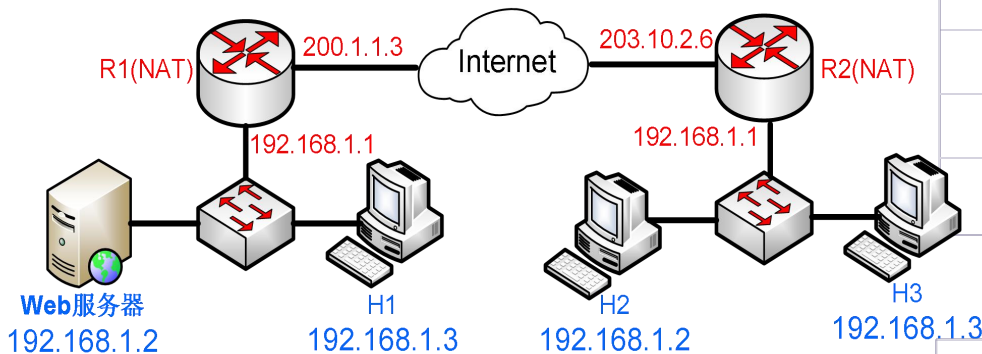
◆ Private IP addresses:

- 10.0.0.0 - 10.255.255.255/8
- 172.16.0.0 - 172.31.255.255/12
- 192.168.0.0 - 192.168.255.255/16

◆ NAT box(router): internal IP address <--> global IP address



◆ NAT table: dynamic & static



R2 NAT table

外网		内网	
IP地址	端口号	IP地址	端口号
203.10.2.6	5001	192.168.1.2	2000
203.10.2.6	5002	192.168.1.3	2000

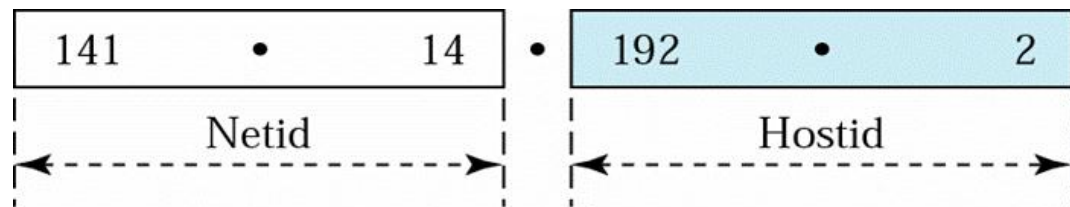
R1 NAT table

外网		内网	
IP地址	端口号	IP地址	端口号
200.1.1.3	80	192.168.1.2	80

(6) Subnetting(子网划分)

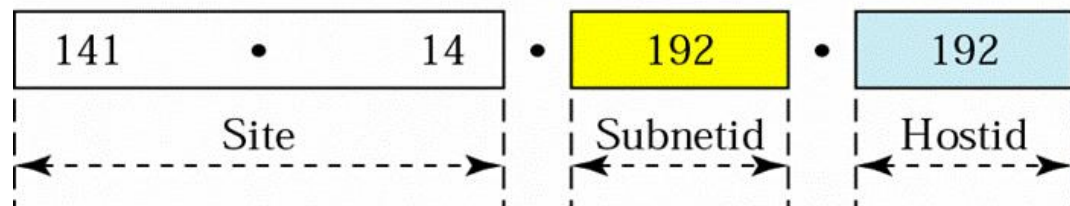
- To manage IP addresses and routing packets more efficiently
- A network is split into several parts for internal use, but still act like a single network to outside world identified by network address
 - ◆ Subnetting is invisible to outside world 子网划分对于外部不可见
- Subnet is identified by using part of **host portion** to identify physical network

2-level
hierarchy



a. Without subnetting

3-level
hierarchy

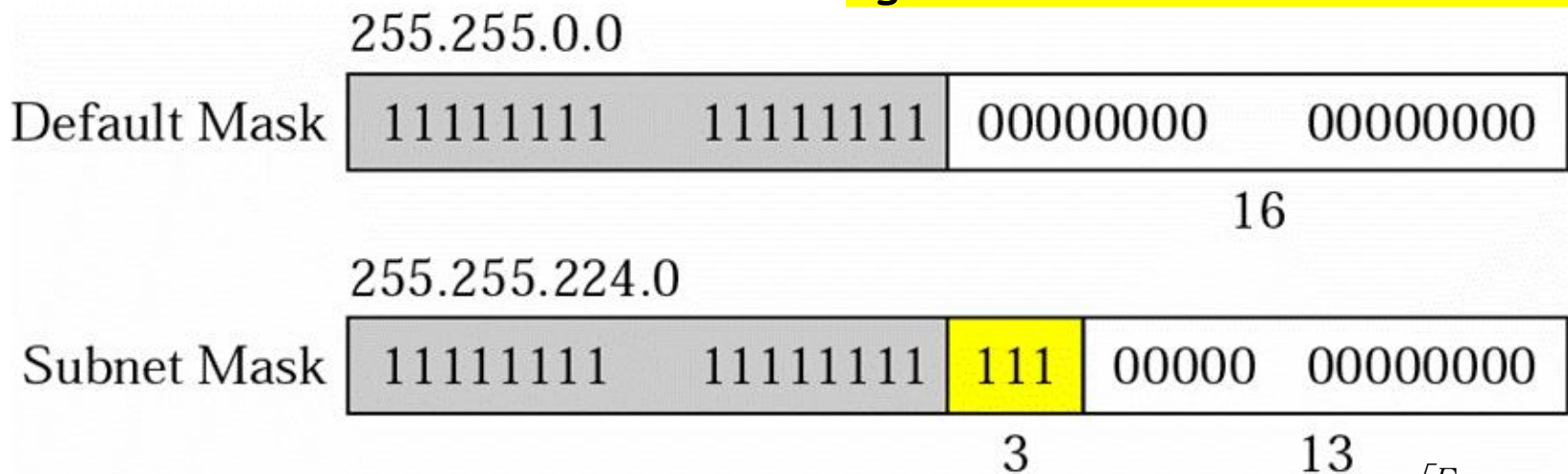


b. With subnetting

[Forouzan]

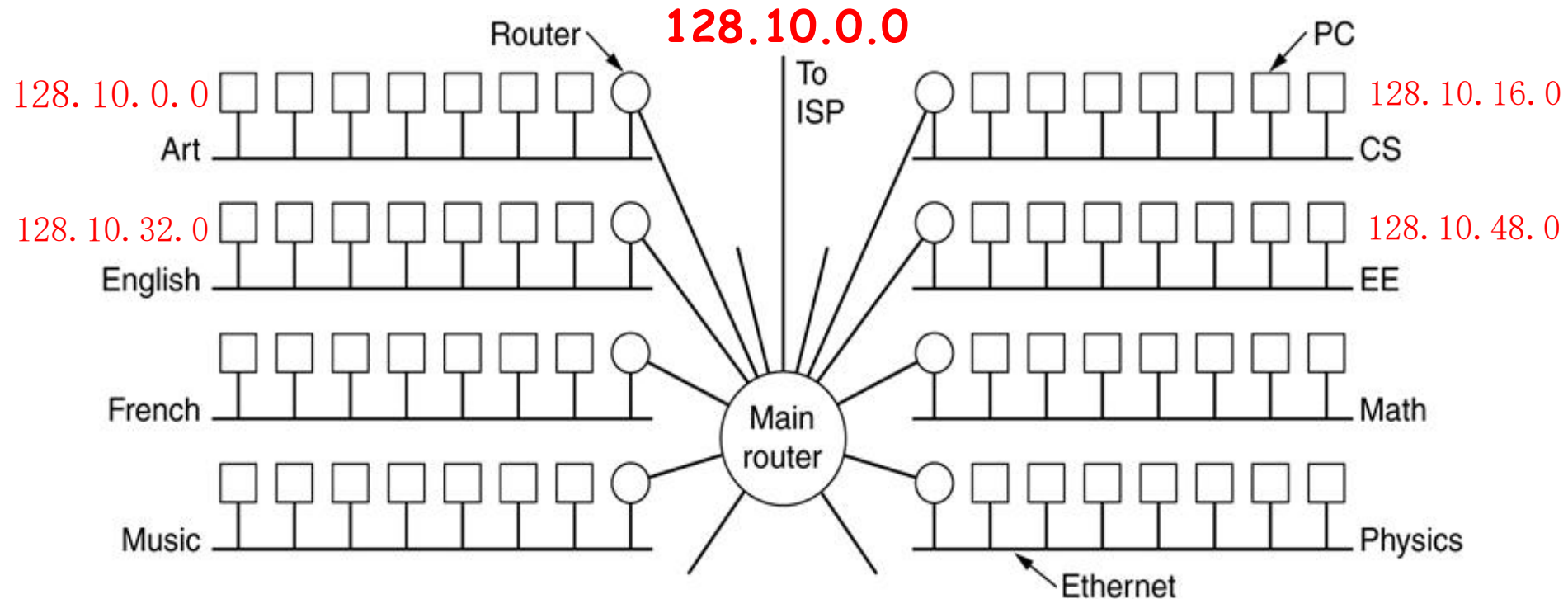
Subnet Mask(子网掩码)

- How a router knows the network address?
 - ◆ Using the **subnet mask**.
- **Boolean ANDed** (按位“与”) with destination address when routing
- If IP addresses of source and destination, and subnet mask are known, then we can find the network address
 - ◆ IP packet is sent to a host in **same subnet**, or
 - ◆ A host in **another network** eg. HA=190.2.3.3, HB=190.2.128.3



Example of Subnetting

A campus network consisting of LANs for various departments

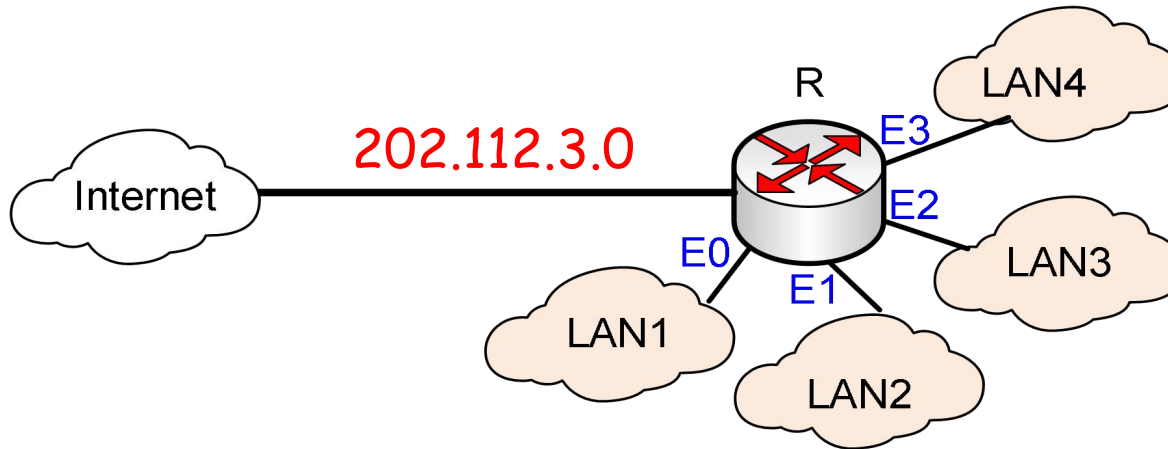


Subnet mask: 255.255.240(11110000).0

What can we get from **subnet mask**?

- Number of IP addresses in the subnet.

Example of Subnetting (2)



Subnet mask: 255.255.255.192

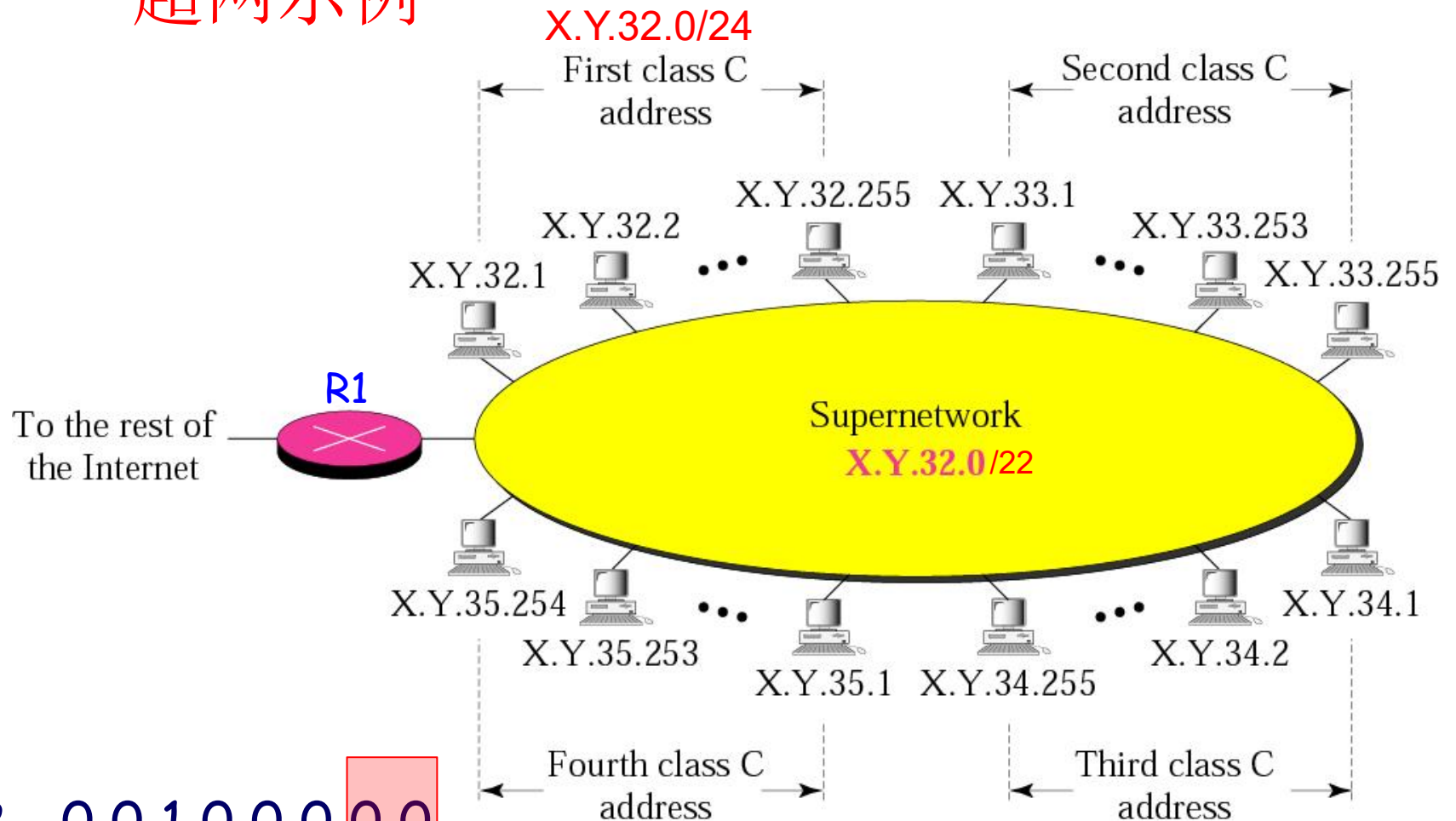
Number of IP addresses in the subnet: 2^6 $2^6-2=62$

LANs	Network address	first IP address assigned	last IP address assigned
LAN1	202.112.3.0	202.112.3.1	202.112.3.62
LAN2	202.112.3.64		
LAN3	202.112.3.128	202.112.3.129	202.112.3.190
LAN4	202.112.3.192		

(7) Supernetting(超网划分)

- Problem of Class C addresses: the size of a block is too small to satisfy the needs of an organisation
- Solution: supernetting
 - ◆ To combine several continuous class C blocks to create a large range of addresses (aggregation)
 - ◆ i.e. several networks are combined to create a supernetwork (超网)
- The number of class C blocks must be a power of 2
- The blocks must be contiguous in the address space (no gaps between the blocks).

超网示例



32:	0	0	1	0	0	0	0
33:	0	0	1	0	0	0	1
34:	0	0	1	0	0	1	0
35:	0	0	1	0	0	1	1

Supernet mask: 255.255.252.0

Network ID: X.Y.32.0/22

[Forouzan]

随堂练习题解析

- 求三个网络：200.2.48.0/23、200.2.52.0/22和200.2.56.0/22聚合后的网络地址。

A. 200.2.48.0/20

B. 200.2.48.0/21

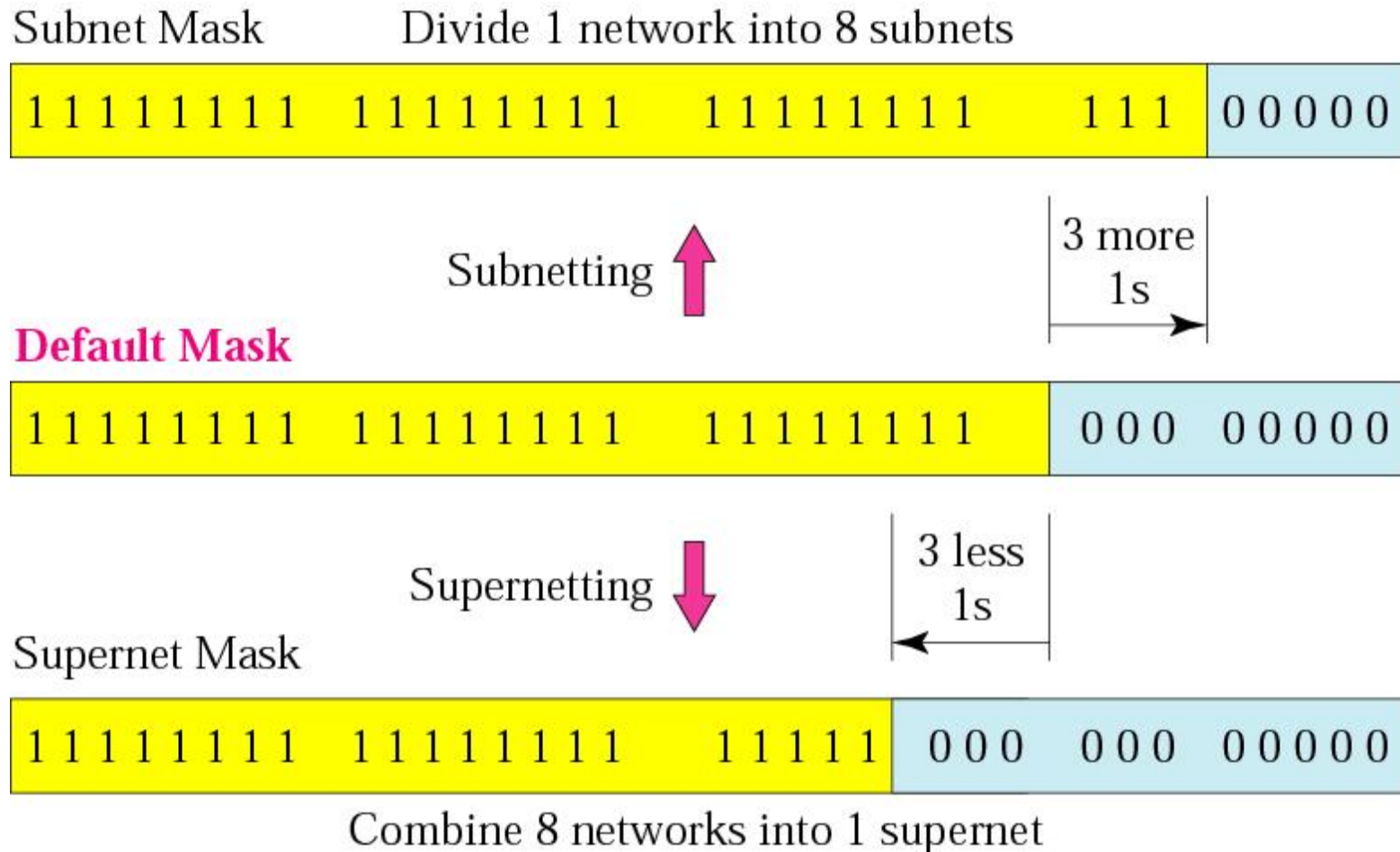
C. 200.2.32.0/20

D. 200.2.32.0/21

48: 0 0 1 1 0 0 0 0
52: 0 0 1 1 0 1 0 0
56: 0 0 1 1 1 0 0 0

200.2.00110000.0/20

Comparison of subnet, default, and supernet masks



[Forouzan]

(8) Classless Addressing(无类别地址)

- Until mid-1990s, a range of addresses meant a block of addresses in class A, B or C.
- Problems of Classful addressing
 - ◆ Continued exponential Internet growth causes address space exhaustion
 - ◆ Explosion in the size of the Internet routing tables
 - ◆ Inefficient use of address space

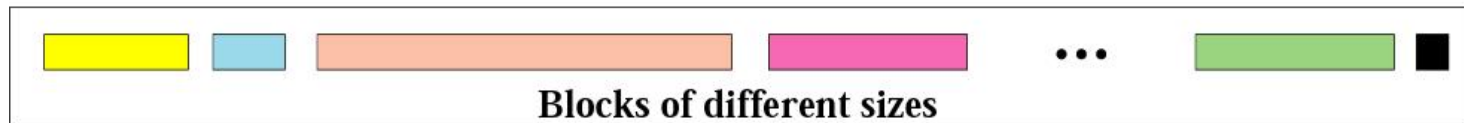
an organization needs 1000 addresses

- a single class C address not enough /24
- a class B address is allocated /16 (a huge waste)
- permits an arbitrary size: /22

■ Solution: Classless addressing (from 1996)

- ◆ VLSM(Variable Length Subnet Mask)可变长子网掩码
 - The idea is to have variable-length blocks that belongs to no class

Address Space



举例. 某C类网192.168.1.0/24可通过VLSM划分为/25(126台)、/26(62台)、/27(30台)、/28(14台)..等不同大小的子网, 适配不同规模的子网

Classless InterDomain Routing (CIDR, 无类别域间选路)

■ CIDR Address Format: slash notation

a.b.c.d/x, where x is # bits in network portion of address



■ CIDR编址中，前缀指明了网络，长度1-32之间

■ Rules of addressing

- ◆ The number of addresses in a network(subnet) **must be a power of 2**

$N=2^{32-n}$, 其中, n 是前缀长度, N 是地址块的地址数

- ◆ 首地址=(任意地址) AND (网络掩码) (即网络号)

- ◆ 末地址=(任意地址) OR [NOT(网络掩码)]

=首地址+地址块中地址数 N

CIDR Prefixes: Dotted Decimal Equivalents

CIDR Notation	Dotted Decimal	CIDR Notation	Dotted Decimal
/1	128.0.0.0	/17	255.255.128.0
/2	192.0.0.0	/18	255.255.192.0
/3	224.0.0.0	/19	255.255.224.0
/4	240.0.0.0	/20	255.255.240.0
/5	248.0.0.0	/21	255.255.248.0
/6	252.0.0.0	/22	255.255.252.0
/7	254.0.0.0	/23	255.255.254.0
/8	255.0.0.0	/24	255.255.255.0
/9	255.128.0.0	/25	255.255.255.128
/10	255.192.0.0	/26	255.255.255.192
/11	255.224.0.0	/27	255.255.255.224
/12	255.240.0.0	/28	255.255.255.240
/13	255.248.0.0	/29	255.255.255.248
/14	255.252.0.0	/30	255.255.255.252
/15	255.254.0.0	/31	255.255.255.254
/16	255.255.0.0	/32	255.255.255.255

问题：采用**CIDR**后，网络地址空间是否可以任意大小？

否，网络大小只能是2的幂次尺寸

一般采取最小够用原则划分网络

■ 某公司（网络地址为**114.23.5.0/24**）有**A、B和C**三个部门，其主机（及路由器）的网络接口总数分别为**64、30和50**个，请给这三个部门根据最小够用原则 顺序划分网络地址，写出每个部门的网络地址。

步骤1：确定各地址块大小 **66(64+2); 32(30+2); 52(50+2)**

步骤2：从主机号最高位开始 **114.23.5. X X X X X X X X**

66: 114.23.5. N X X X X X X X

66: 114.23.5. 0 X X X X X X X / 25 114.23.5. 0 / 25

32: 114.23.5. N N N X X X X X / 27

32: 114.23.5. 1 0 0 X X X X X / 27 114.23.5. 128 / 27

52: 114.23.5. 1 1 X X X X X X / 26 114.23.5. 192 / 26

注意：地址块之间不能重叠，可以有空隙

Subnet

将IP地址空间192.168.1.0/24按网络号递增顺序分配给2个局域网LAN1～LAN2

子网	分配地址数	网络地址	子网掩码
LAN1	95	192.168.1.0/25	255.255.255.128
LAN2	50	192.168.1.128/26	255.255.255.192

CIDR: 路由聚合/汇聚

- CIDR permits route aggregation
- ◆ reduce the Internet routing table

192.24.000	000000.X	/21
192.24.000	100000.X	/20
192.24.000	010000.X	/22

192.24.000000000.X /19

New York



192.24.0.0/19

(1 aggregate prefix)

London



(3 prefixes)

192.24.0.0/21



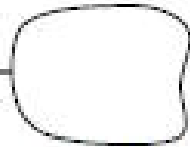
Cambridge

192.24.16.0/20



Oxford

192.24.8.0/22

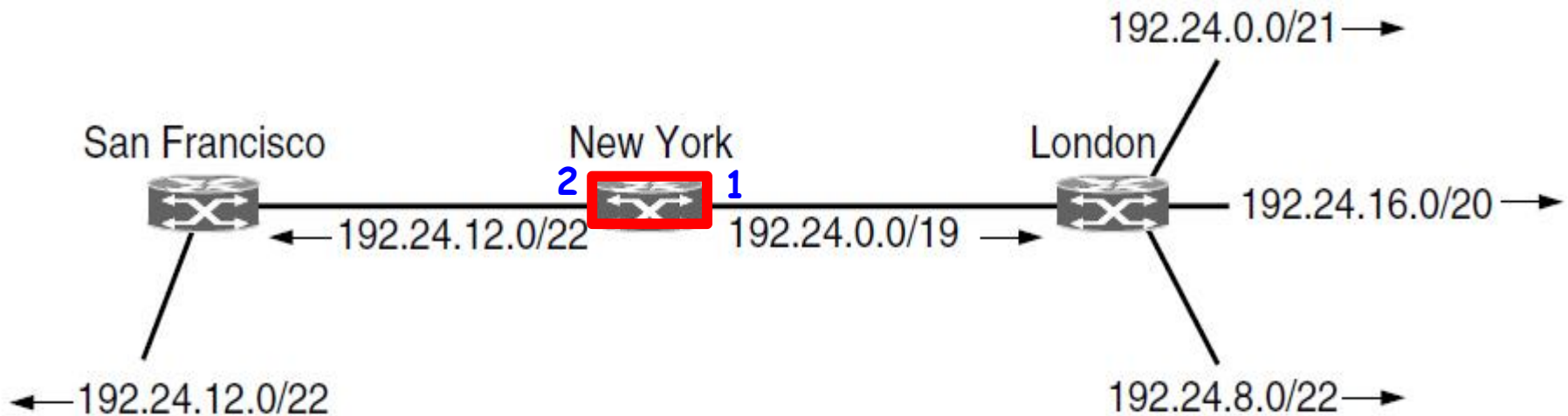


Edinburgh

192.24.12(00001100).0/22

问题：地址空间被放大

Example of routing aggregation



192.24.16.3

192.24.12.5

IP address	outgoing
192.24.0.0/19	Port 1
192.24.12.0/22	Port 2

解决方法：最长匹配原则(路由前缀允许重叠，按最具体路由方向转发)

(9)Routing Table

■ Routing table

- ◆ An array of (IP address, netmask, outgoing line, Next hop)

■ Scan the routing table

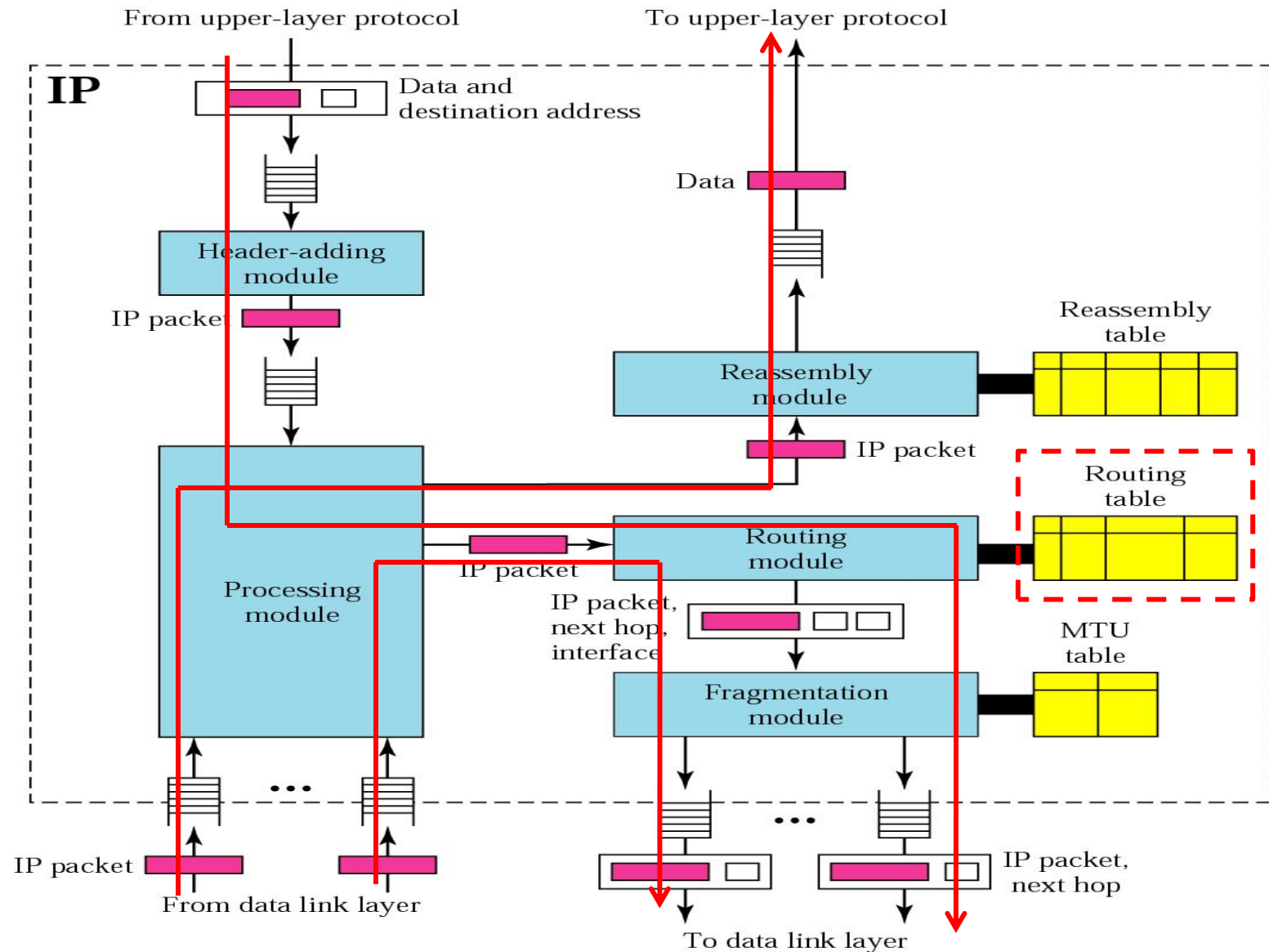
- ◆ Mask the destination address and comparing it to the table entry looking for a match
- ◆ It is possible that multiple entries (with different netmask lengths) match, in which case the longest mask is used
 - If there is a match for a /20 mask and a /24 mask, the /24 entry is used.

■ Implementation

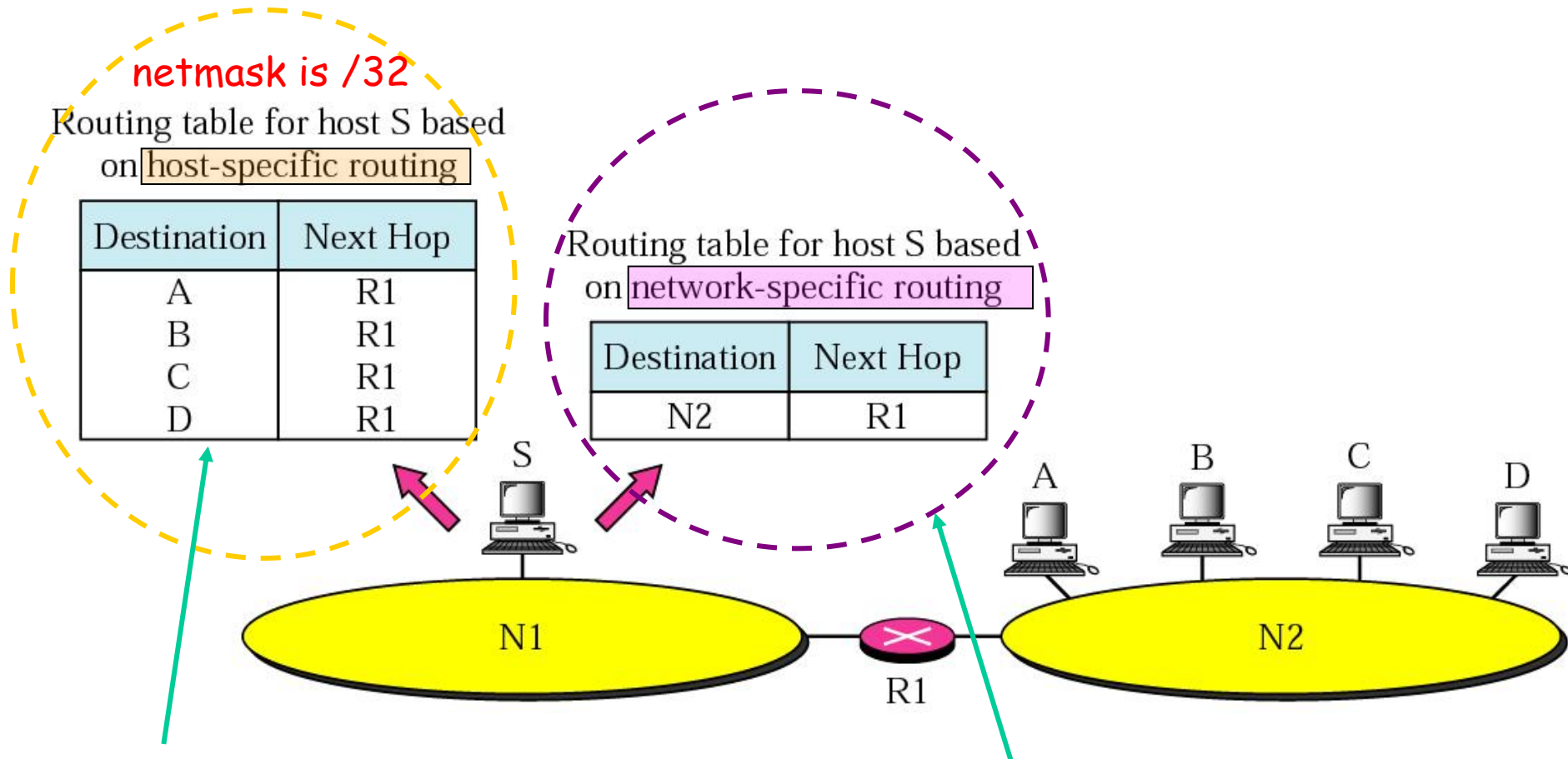
- ◆ Complex algorithms have been devised to speed up the address matching process
- ◆ commercial routers use custom VLSI (Very Large Scale Integration超大规模集成电路) chips with these algorithms embedded in hardware

IP components

Both **source** and **routers** consider routing



Network-specific & host-specific routing



The **destination host address**(**netmask is /32**) is given in the routing table; to have greater control over routing.

Only **one entry** is needed to defined the address of the **network** itself. All host connected to the same network as one single entity.

Default routing(默认/缺省路由)

direct/indirect delivery(直接/间接交付)

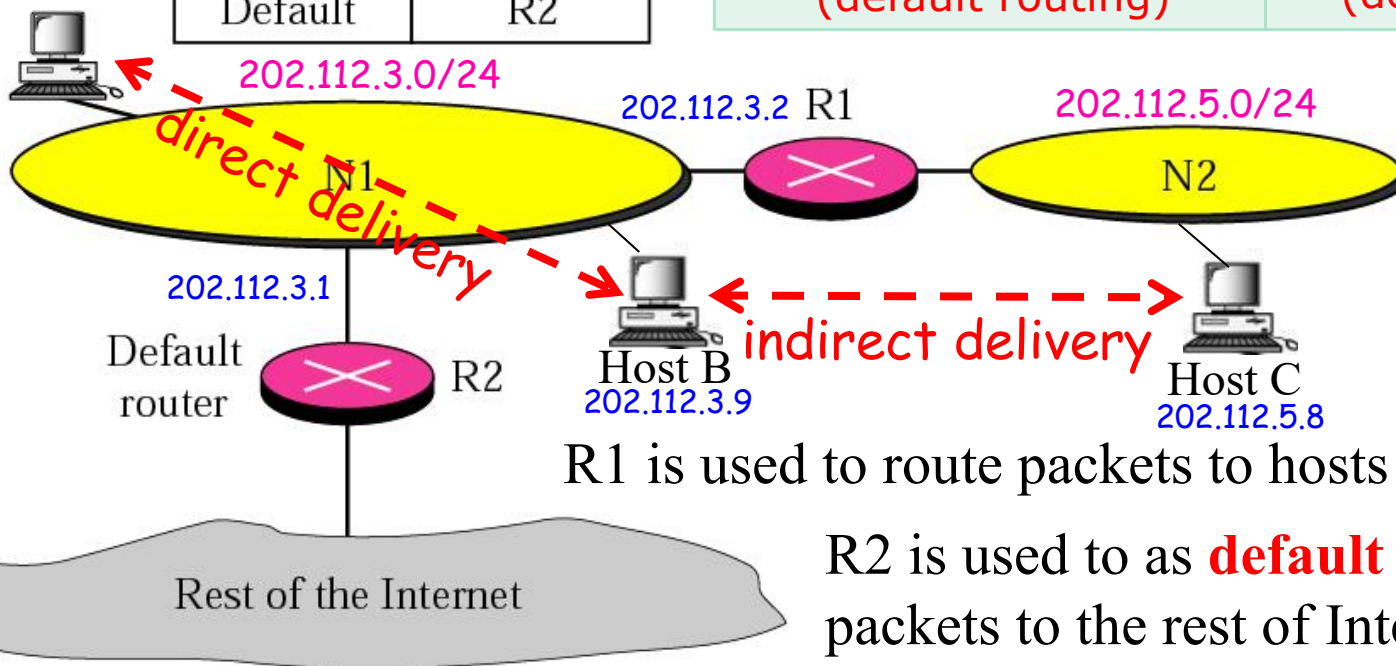
Host A routing table

Destination/netmask	Next Hop
202.112.3.0/24	直连/ -
202.112.5.0/24	202.112.3.2
0.0.0.0/0 (default routing)	202.112.3.1 (default router)

Routing table for host A

Destination	Next Hop
N2	R1
.....	
Default	R2

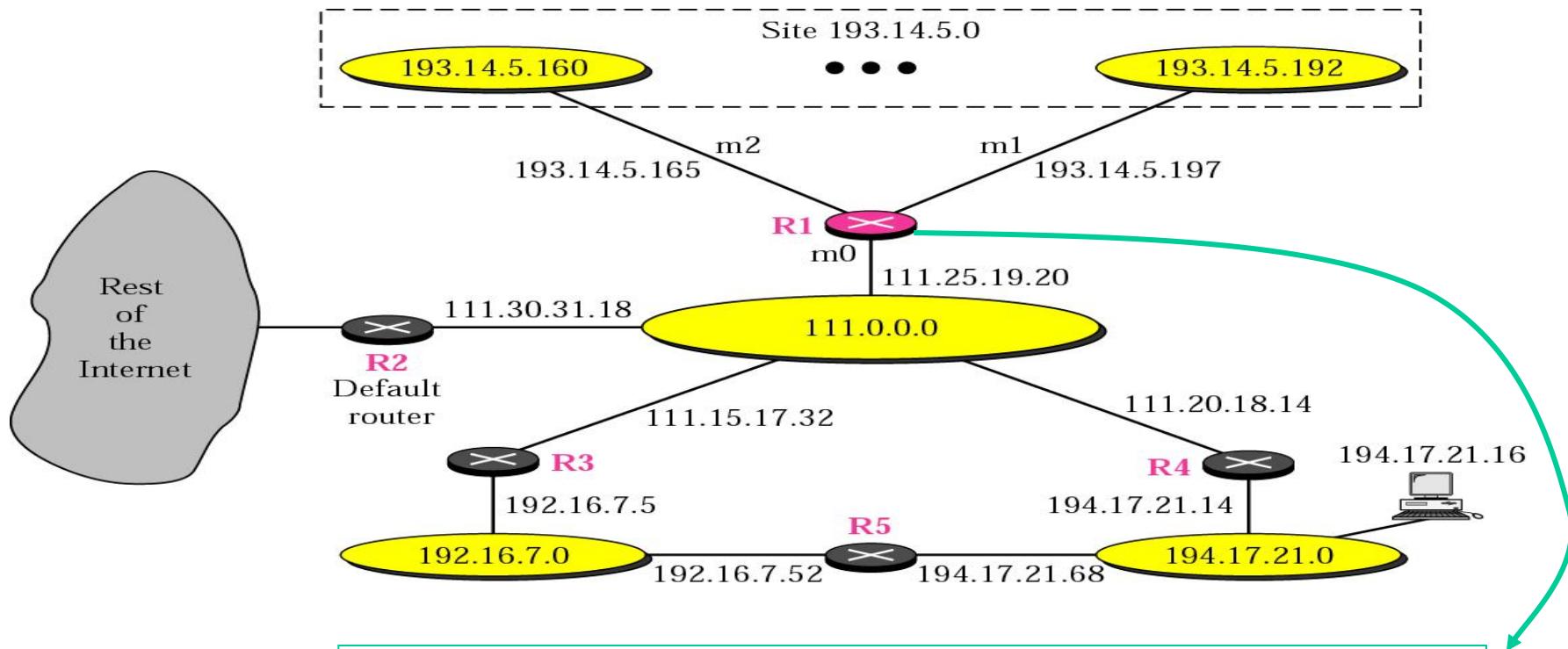
202.112.3.10
Host A



R1 is used to route packets to hosts connected to N2

R2 is used to as **default** to route other packets to the rest of Internet without listing all the networks involved.

Configuration for routing example



Direct delivery

Host-specific

Network-specific

default

Mask	Dest	Next Hop	Int.
255.0.0.0	111.0.0.0	- (直连)	m0
255.255.255.224	193.14.5.160	-	m2
255.255.255.224	193.14.5.192	-	m1
255.255.255.255	194.17.21.16	111.20.18.14	m0
255.255.255.0	192.16.7.0	111.15.17.32	m0
255.255.255.0	194.17.21.0	111.20.18.14	m0
0.0.0.0	0.0.0.0	111.30.31.18	m0

Review

■ classless addressing: Classless InterDomain Routing

◆ address format <IP address/prefix>, netmask

Address Space



◆ advantages

- efficient use of address space: variable-length blocks (会划分)
- reduces the size of routing tables: address blocks aggregation (会聚合)

■ Routing

◆ Routing table

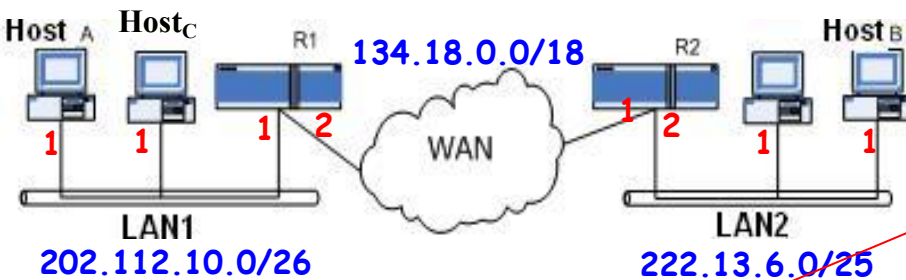
	<u>Mask</u>	<u>Dest</u>	<u>Next Hop</u>	<u>Int.</u>
direct delivery	255.0.0.0	111.0.0.0	-	m0
	255.255.255.224	193.14.5.160	-	m2
	255.255.255.224	193.14.5.192	-	m1
Host-specific	255.255.255.255	194.17.21.16	111.20.18.14	m0
indirect delivery	255.255.255.0	192.16.7.0	111.15.17.32	m0
	255.255.255.0	194.17.21.0	111.20.18.14	m0
default	0.0.0.0	0.0.0.0	111.30.31.18	m0

How does Source make routing?

- Source host applies the **mask** to the destination address of the packet in order to find the corresponding **destination network address**.
 - ◆ If it is **local network**, send directly to **destination**;
 - ◆ Otherwise, send to the **local router(next hop)**

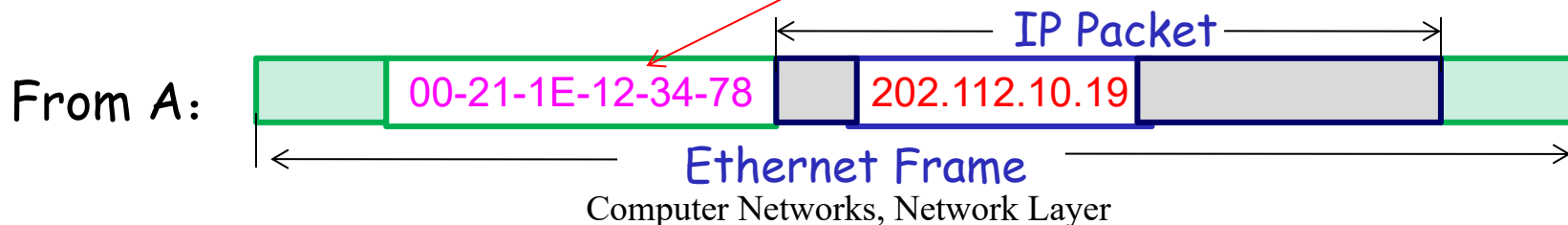
Host A's routing table:

Des IP	mask	port	next hop
202.112.10.0	/26	1	--
0.0.0.0	/0	1	202.112.10.18

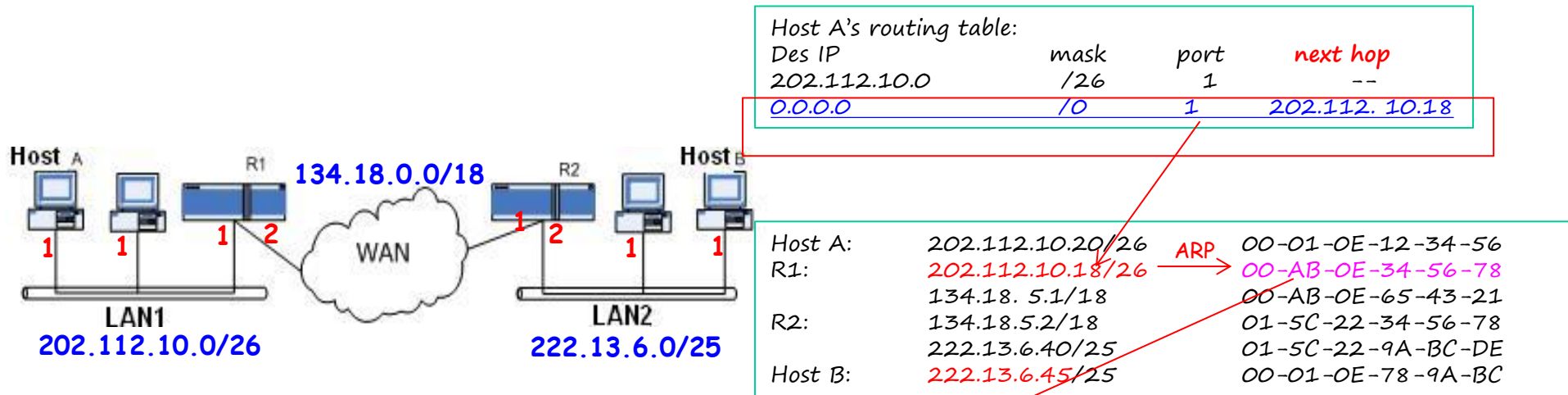


Host A:	202.112.10.20/26	ARP	00-01-0E-12-34-56
Host C:	202.112.10.19/26		00-21-1E-12-34-78
R1:port 1	202.112.10.18/26		00-AB-0E-34-56-78
port 2	134.18.5.1/18		00-AB-0E-65-43-21
R2:port 1	134.18.5.2/18		01-5C-22-34-56-78
port 2	222.13.6.40/25		01-5C-22-9A-BC-DE
Host B:	222.13.6.45/25		00-01-0E-78-9A-BC

Case 1: Des IP: 202.112.10.19

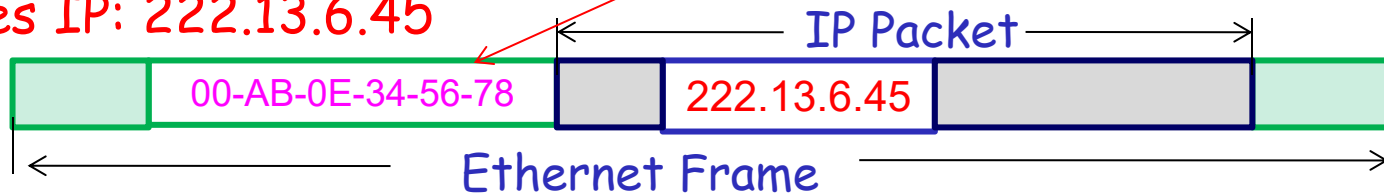


How does Source make routing?



Case 2: Des IP: 222.13.6.45

From A:



How do routers make routing?

- Generally, a routing table needs a minimum of 4 columns: **mask**, **destination network address**, **next hop address** and **outgoing interface**.
- When a packet arrives, the router applies the **mask** to the destination address of the packet in order to find the corresponding **destination network address**.
 - If found **matching entry** (include default routing) in routing table, the packet is **sent out** from the corresponding **interface** in the table.
 - If not found, the packet is **discard** and the router should send a **ICMP error report message** to the source host.

Example 1

Router R1 receives 500 packets for destination **192.16.7.14**; the algorithm applies the masks row by row to the destination address until a match is found:

Rule of thumb: Apply the individual mask (from Routing table) to the received destination address (row-by-row) and see if its matches any of the DEST address stated in its routing table.

Solution

Direct delivery

192.16.7.14	&	255.0.0.0	→	192.0.0.0	no match	to	111.0.0.0
192.16.7.14	&	255.255.255.224	→	192.16.7.0	no match	to	193.14.5.160
192.16.7.14	&	255.255.255.224	→	192.16.7.	no match	to	193.14.5.192

Host-specific

192.16.7.14	&	255.255.255.255	→	192.16.7.14	no match	to	194.17.21.16
-------------	---	-----------------	---	-------------	----------	----	--------------

Network-specific

192.16.7.14	&	255.255.255.0	→	192.16.7.0	match	to	192.16.7.0
-------------	---	---------------	---	------------	--------------	----	------------

Default

192.16.7.14	&	0.0.0.0	→	0.0.0.0	match	to	0.0.0.0
-------------	---	---------	---	---------	--------------	----	---------

Example 1

Router R1 receives 500 packets for destination **192.16.7.14**; the algorithm applies the masks row by row to the destination address until a match is found:

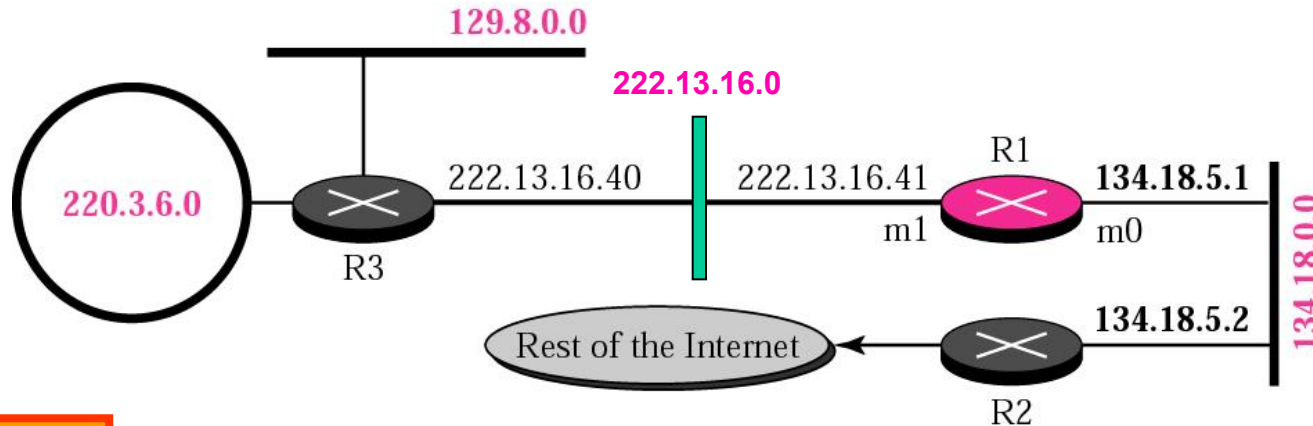
Rule of thumb: Apply the individual mask (from Routing table) to the received destination address (row-by-row) and see if its matches any of the DEST address stated in its routing table.

Solution

	<u>Mask</u>	<u>Dest</u>	<u>Next Hop</u>	<u>Int.</u>
Direct delivery	255.0.0.0	111.0.0.0	---	m0
	255.255.255.224	193.14.5.160	---	m2
Host-specific	255.255.255.224	193.14.5.192	---	m1
	255.255.255.255	194.17.21.16	111.20.18.14	m0
Network-specific	255.255.255.0	192.16.7.0	111.15.17.32	m0
	255.255.255.0	194.17.21.0	111.20.18.14	m0
	0.0.0.0	0.0.0.0	111.30.31.18	m0

Example 2

Make the routing table for router R1 in figure below



Solution

Mask	Destination	Next Hop	Int.
255.255.0.0	134.18.0.0	--	m0
255.255.255.0	222.13.16.0	--	m1
255.255.0.0	129.8.0.0	222.13.16.40	m1
255.255.255.0	220.3.6.0	222.13.16.40	m1
0.0.0.0	0.0.0.0	134.18.5.2	m0

Routing Tables in IP with CIDR

- **Longest-Prefix Matching(最长前缀匹配)**: Longest mask is used for multiple matches
- For each entry in the routing table:
 - ◆ MaskedAddress := EntryMask (bitAND) IPDatagramDestinationAddress;
 - ◆ if (MaskedAddress == EntryDestination), **Mark** the entry

e.g. Destination: **128.125(0111 1101).1.3**, what is next hop?

Mask	Destination	Next Hop
/12	128.96.0.0	145.12.56.29
<u>/17</u>	<u>128.125.0.0</u>	<u>153.202.12.128</u>
<u>/12</u>	<u>128.112.0.0</u>	<u>153.202.14.1</u>
/26	128.105.14.64	153.2.45.101
/32	128.105.14.66	153.2.45.101

总结：路由表查找转发过程

■ 路由表结构

目的网络IP地址	子网掩码	接口	下一跳IP地址
----------	------	----	---------

■ 扫描路由表，查找匹配的网络地址(最长匹配原则)

- ◆ 0项：丢弃IP数据包并报错(ICMP-目的地不可达)
- ◆ 1项：根据Next hop和outgoing转发
- ◆ 多项：选择掩码最长的一项的Next hop和outgoing转发
 - 首先: host-specific routing(/32)
 - 其次: longest network prefix match
 - 最后: default routing(/0)

The network layer in the Internet

- 5.6.1 IPv4 Packet format
- 5.6.2 IP Addressing
- 5.6.3 IPv6
- 5.6.4 ARP, DHCP and ICMP
- 5.6.6&5.6.7 Routing Protocols: OSPF and BGP

IPv6: Motivation

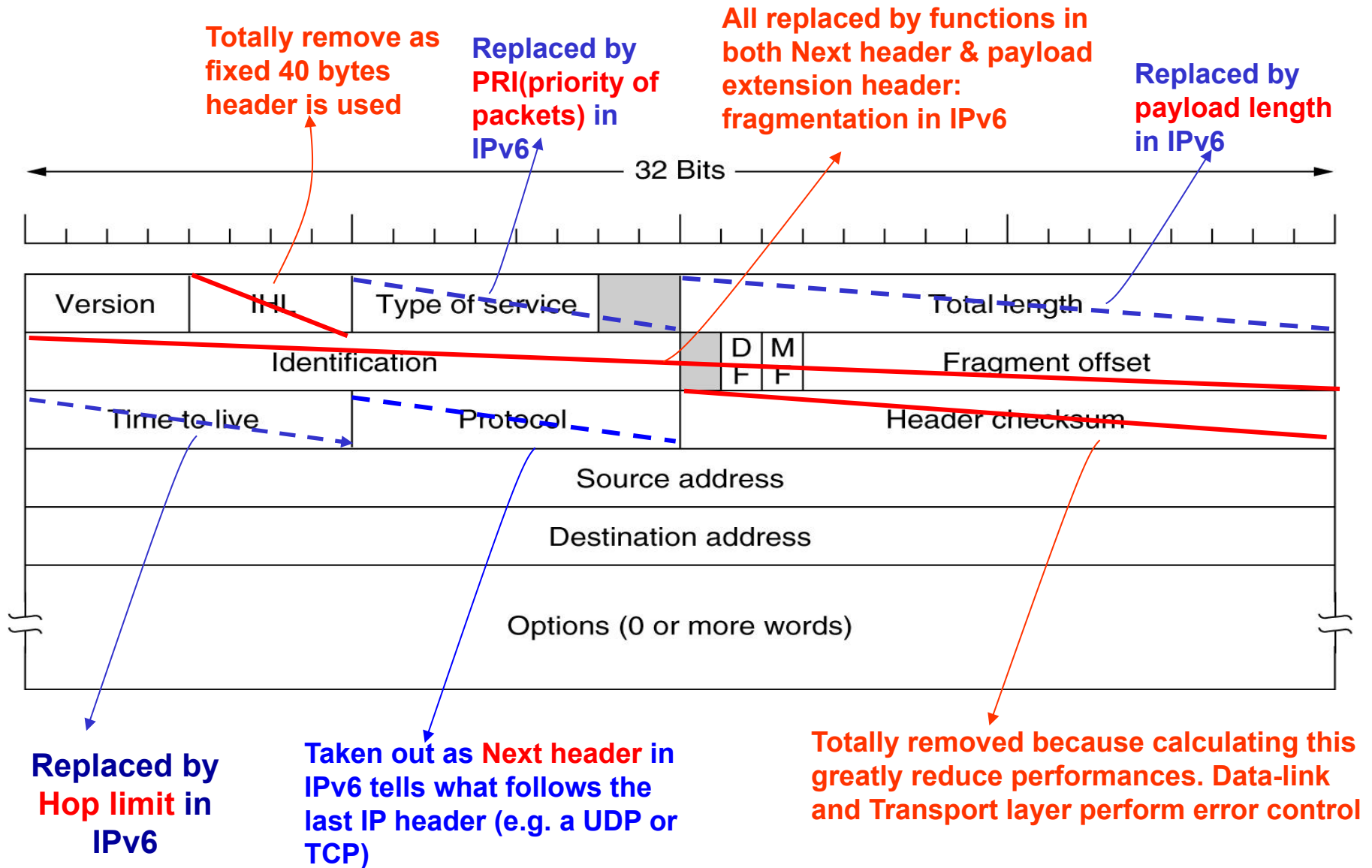
- IPv4 is becoming insufficient for the fast growing Internet and the needs:
 - ◆ The use of **addressing space** is becoming insufficient for conventional netid & hostid two-level addressing structure in Classful addressing
 - ◆ **Real-time application** i.e. video transmission requires **minimum delay** strategies and **reservation of resource** is not provided
 - ◆ Encryption and authentication of data and **security mechanism** is not provided
 - ◆ Limitation of **flexibility** and **mobility** supported

IPv6: Major Goals

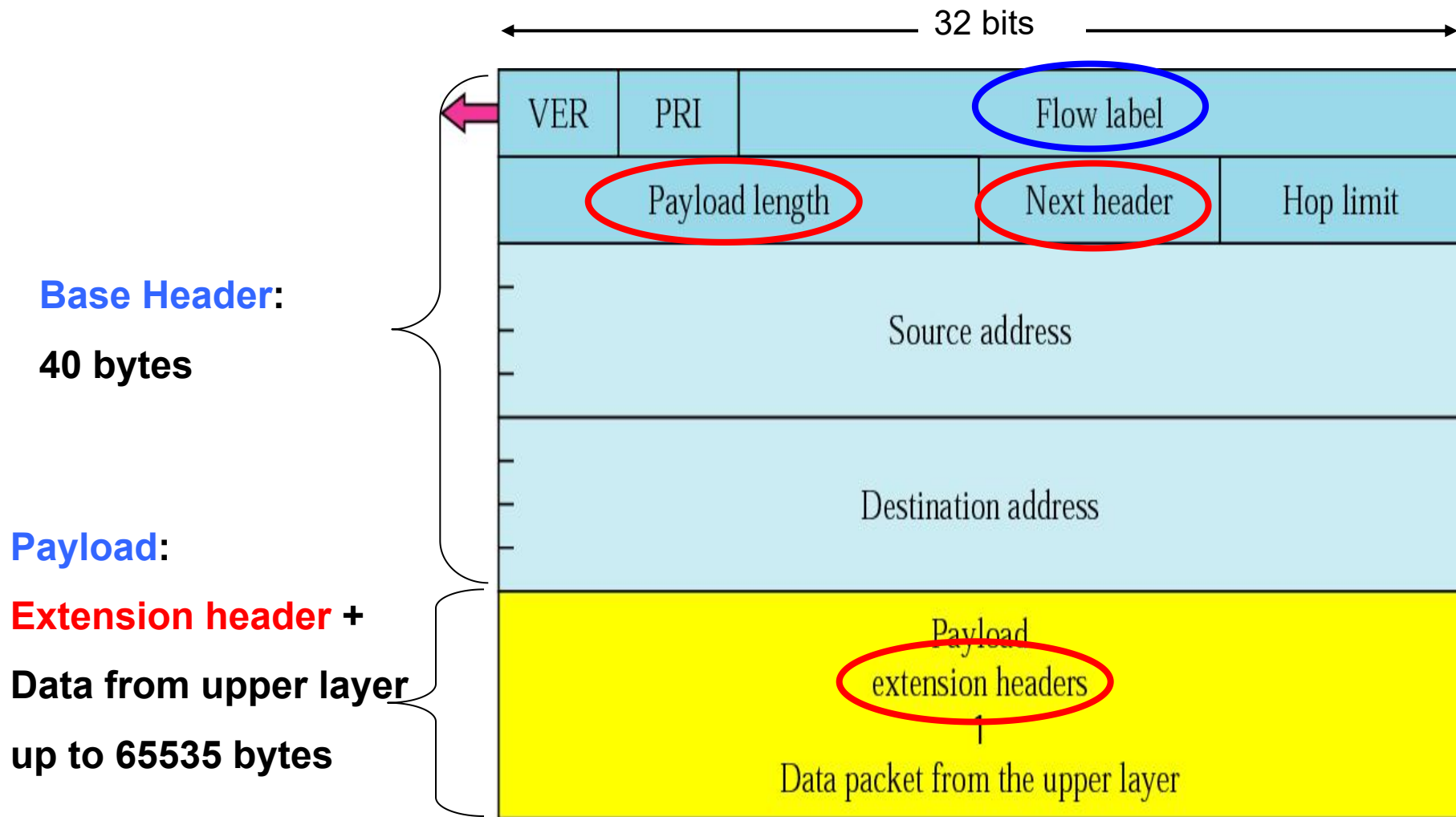
- Address space NAT
- Reduce the size of the routing tables CIDR
- Efficiency: Simplify the protocol, to allow routers to process packets faster
- Security
- QoS: Pay more attention to type of service, particularly for real-time data
- Multicasting
- Mobility : Make it possible for a host to roam without changing its address
- Scalability: Allow the protocol to evolve in the future
- Permit the old and new protocols to coexist for years

} Multimedia,
stream media

IPv6: Revised IPv4 Header



IPv6 Datagram



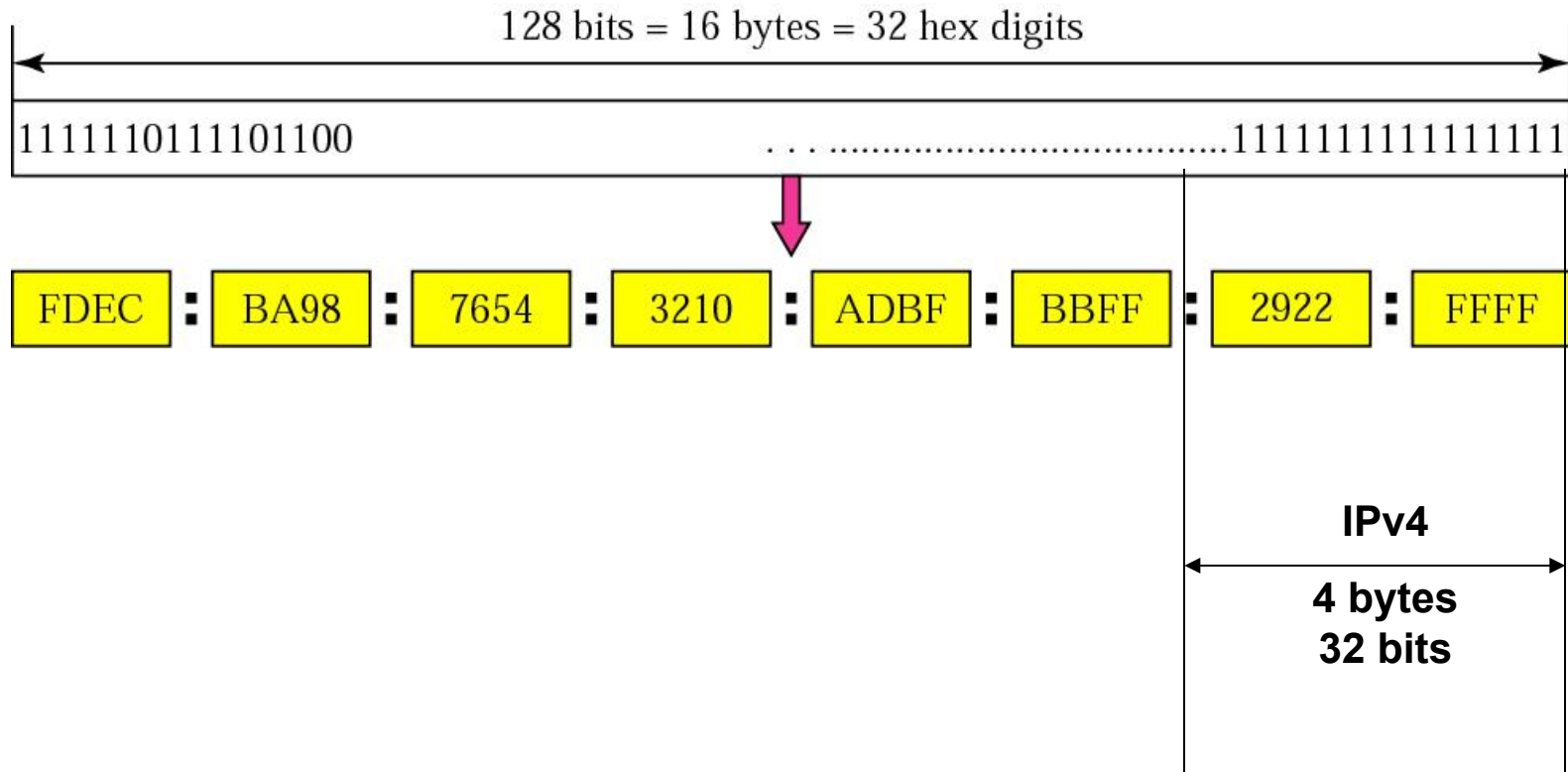
IPv6 Header: Fields

- **VER:** IP version; 4 for IPv4 and 6 for IPv6
- **PRI:** priority of packets; 0-7: for packets that can slow down in event of congestion; 8-15 for real-time traffic with constant sending rate - Audio & Video.
- **Flow label:** pseudo-connection between source and destination. for reserved bandwidth and stringent delay requirement. (Still experimental).
- **Payload length:** tells how many payload bytes follow the 40-bytes header
- **Next header:** the subsequent headers that follow the base header; either **optional extension headers** used by IP or **header** for an upper-layer protocol, e.g. UDP & TCP.
- **Hop limit:** serves the same purpose as the TTL in IPv4.
- **Source and Destination address field:** both 128 bits
- **Extension Headers:** 6 extension headers which is optional in IPv4.

IPv6: Improvement

- **Longer addresses** : 128 bits long > 32 bits currently in IPv4. That means a huge increase of addressing space.
- **Simplification in Header**: contains only 8 vital fields < 13 in IPv4. (speeds up routing process). – 5 fields becomes options.
- **New options**: to support additional functionalities
- **Security**: Encryption and authentication are key features of new IP
- **More Services**: to handle special/real-time multimedia traffic.

IPv6 Address(冒分十六进制)



[Forouzan]

Abbreviated Address

Unabbreviated

FDEC : BA98 : 0074 : 3210 : 000F : BBFF : 0000 : FFFF



FDEC : BA98 : 74 : 3210 : F : BBFF : 0 : FFFF

Abbreviated

Abbreviated

FDEC : 0 : 0 : 0 : 0 : BBFF : 0 : FFFF



FDEC :: BBFF : 0 : FFFF

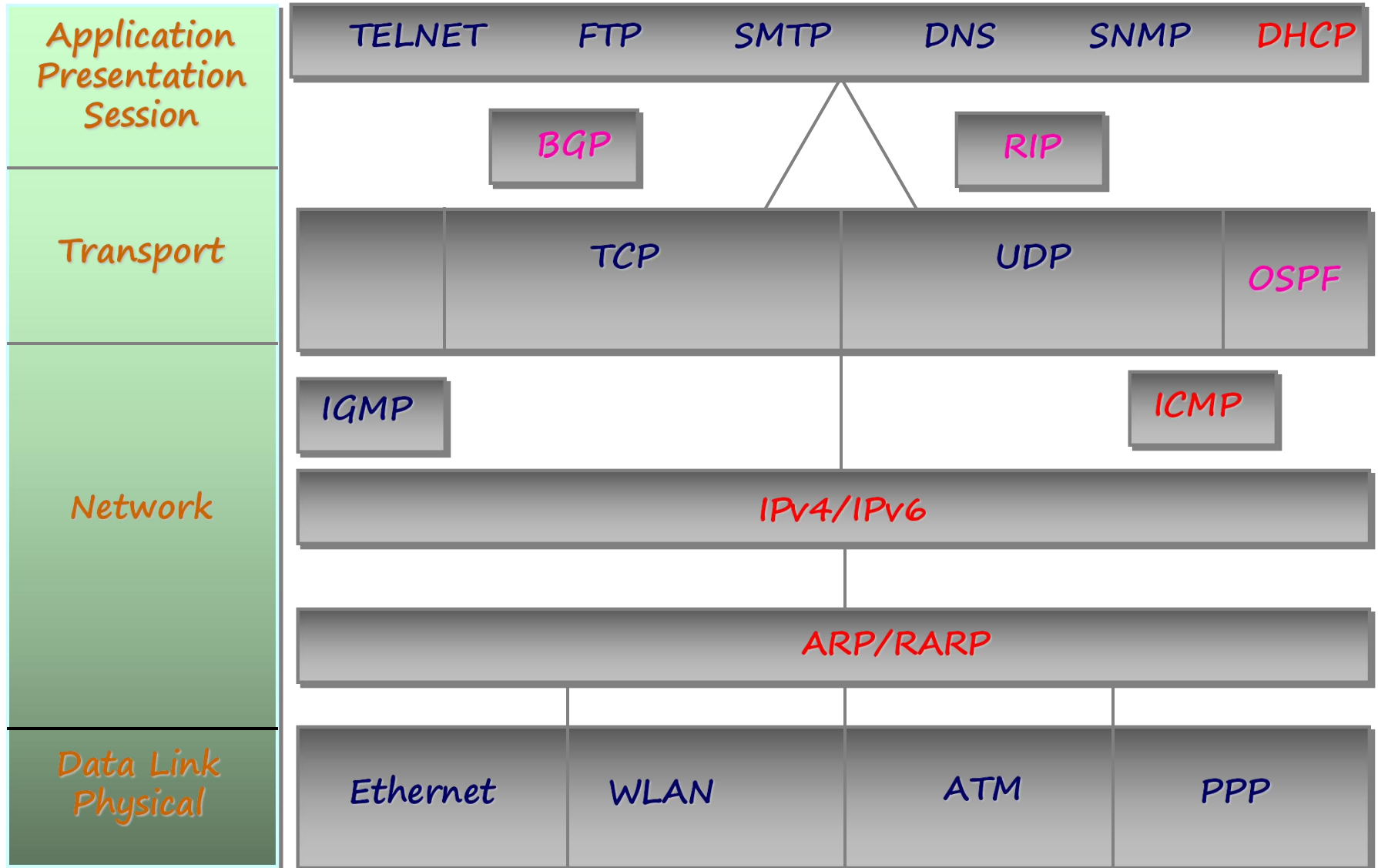
More Abbreviated

The network layer in the Internet

- 5.6.1 IPv4 Packet format
- 5.6.2 IP Addressing
- 5.6.3 IPv6
- 5.6.4 ARP, DHCP and ICMP
- 5.6.6&5.6.7 Routing Protocols: OSPF and BGP

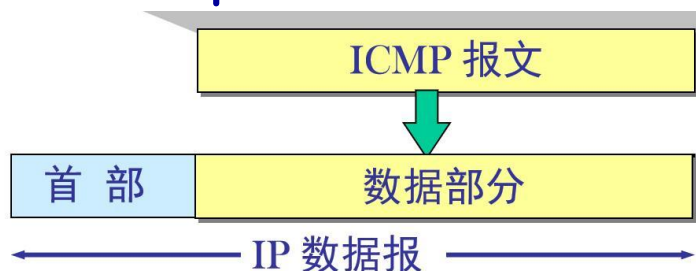
OSI

TCP/IP

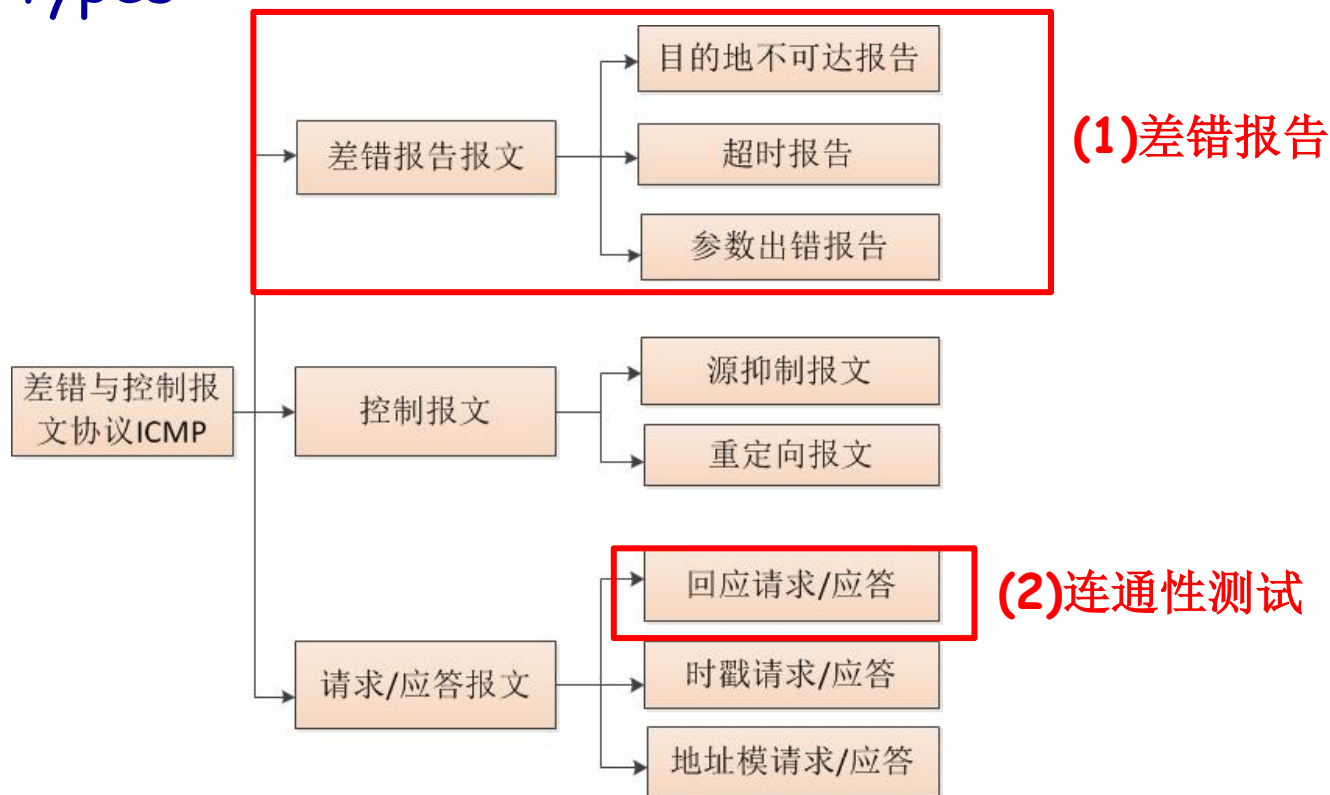


(1) ICMP(因特网控制报文协议)Messages

- An ICMP message is encapsulated in an IP packet.



- Message types



ICMP(因特网控制报文协议)Messages

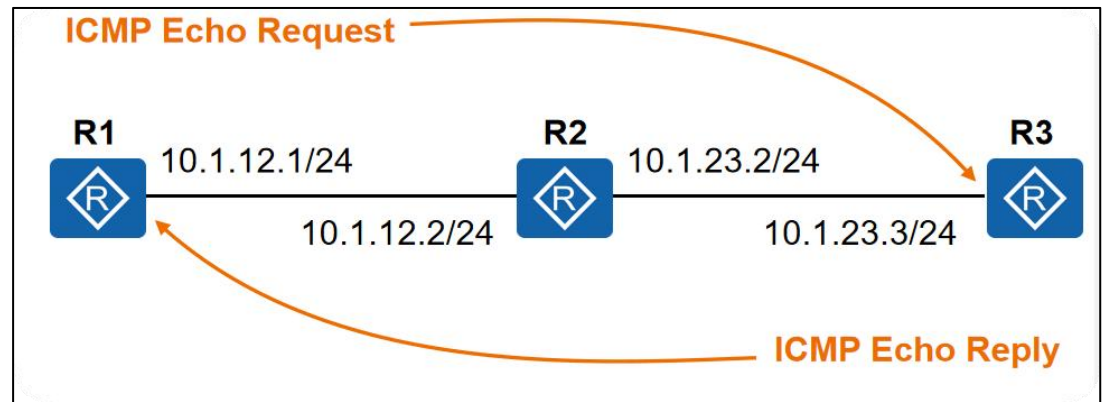
- **Destination Unreachable:** packet could not be delivered
 - ◆ (1) a router cannot locate the destination
 - ◆ (2) a packet with the **DF** bit cannot be delivered because a 'small-packet' network stands in the way.

- **Time Exceeded**
 - ◆ (1) time to live(TTL) exceeded in transit (分组传输超时)
 - ◆ (2) fragment reassembly time exceeded (分片重组超时)
 - ◆ Application: Tracert(Traceroute)

- **Parameter Problem**
 - ◆ A **illegal value** has been detected **in header field**.
 - ◆ Indicate a bug in the sending host's IP software or possibly in the software of a router transited.

ICMP(因特网控制报文协议)Messages

- ◆ Source Quench(源抑制): Congestion control
 - ◆ When a host received this message, it was expected to slow down
- Redirect: used by the router to tell the sending host to update to a better route.
- **ECHO Request and ECHO Reply**
 - ◆ Sent by hosts to see if a given destination **is reachable and currently alive.**
 - ◆ Application: **Ping**

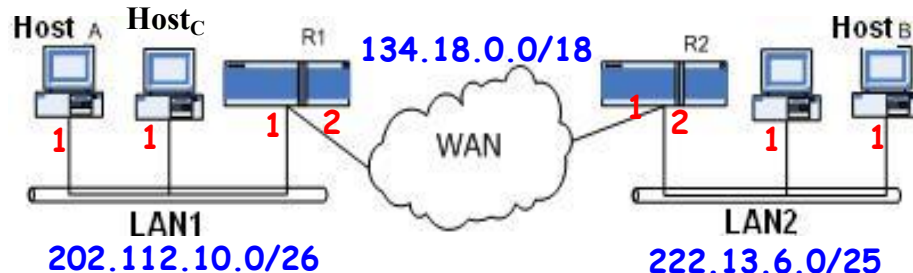


- Time Stamp Request and Reply
- Router advertisement/solicitation: find a nearby router

(2) ARP: Address Resolution Protocol (地址解析协议)

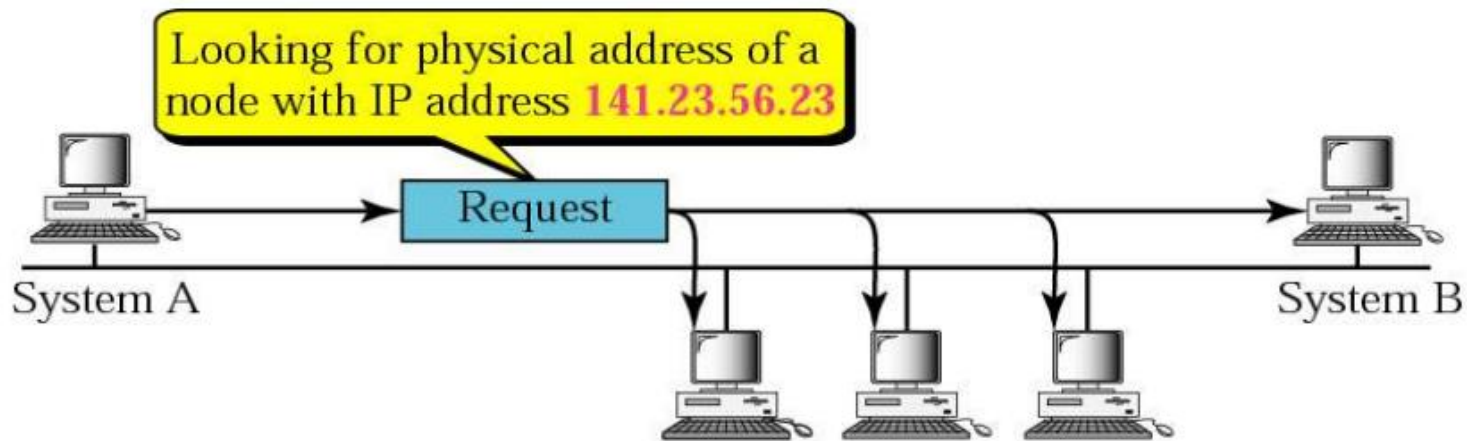
■ How do IP addresses get mapped onto data link layer addresses, such as Ethernet?

- ◆ One solution is to have a **configuration file** somewhere in the system that maps IP addresses onto Ethernet addresses
- ◆ A better solution is to output a **broadcast** packet onto the Ethernet asking: Who owns IP address 192.31.65.5?

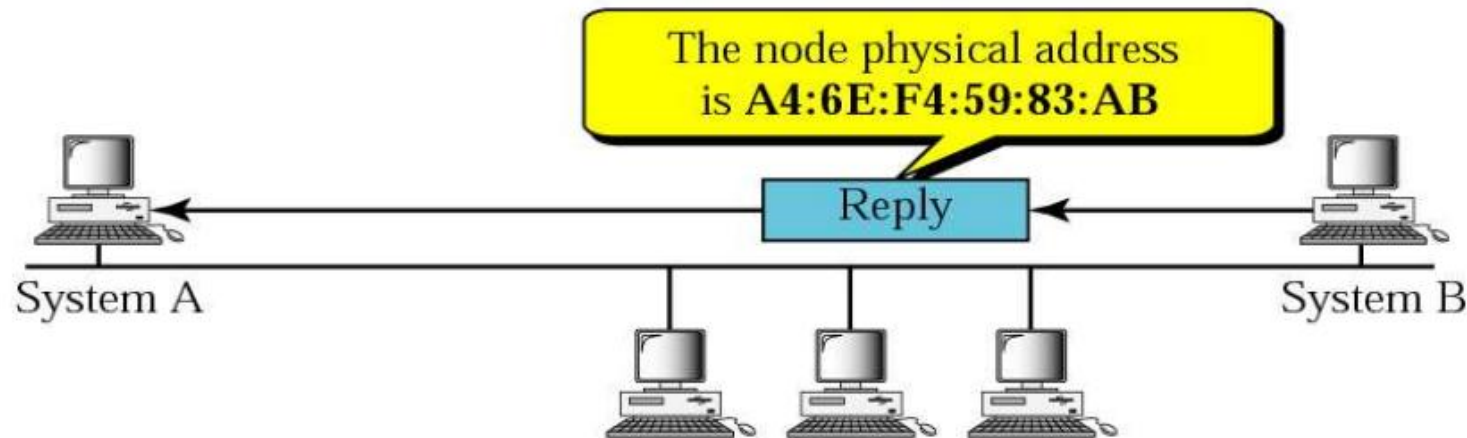


Host A:	202.112.10.20/26	00-01-0E-12-34-56
Host C:	202.112.10.19/26	00-21-1E-12-34-78
R1:port 1	202.112.10.18/26	00-AB-0E-34-56-78
port 2	134.18. 5.1/18	00-AB-0E-65-43-21
R2:port 1	134.18.5.2/18	01-5C-22-34-56-78
port 2	222.13.6.40/25	01-5C-22-9A-BC-DE
Host B:	222.13.6.45/25	00-01-0E-78-9A-BC

(2) ARP Operation

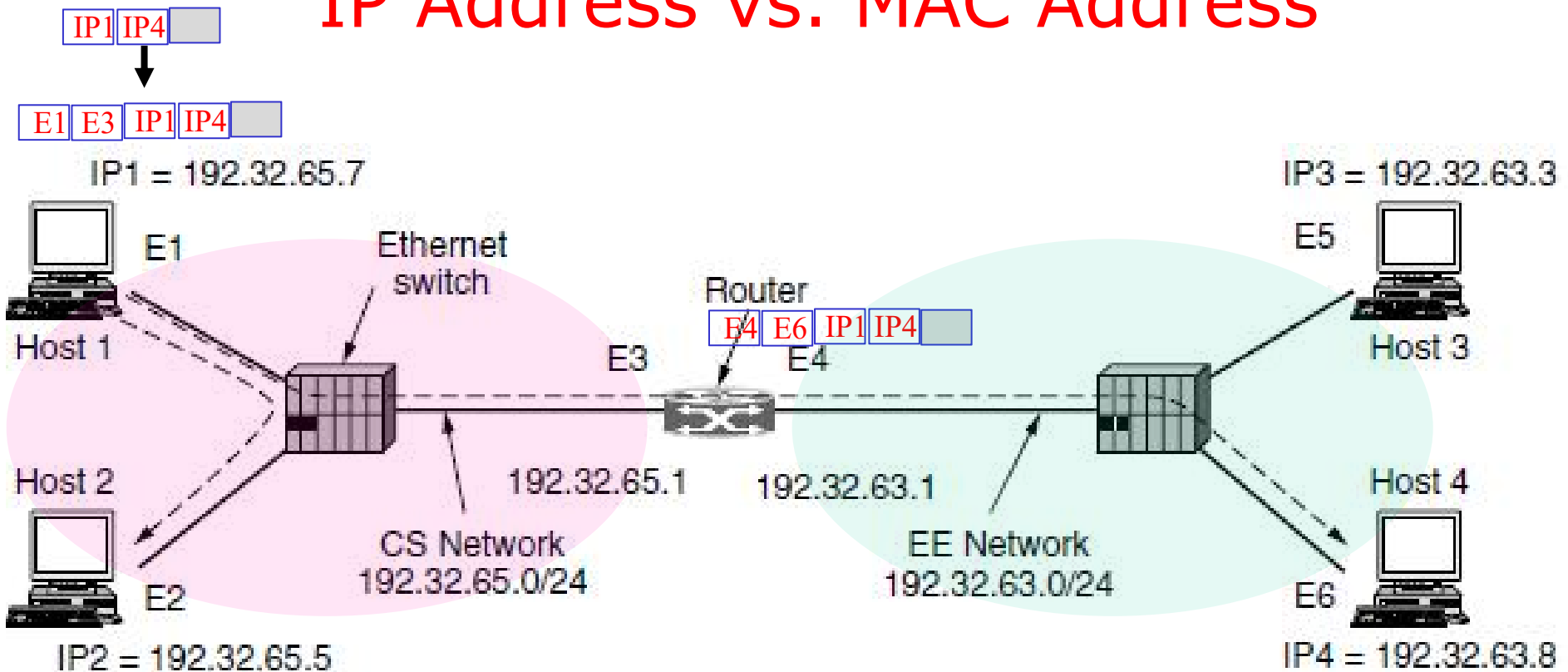


a. ARP request is broadcast



b. ARP reply is unicast

IP Address vs. MAC Address



Frame	Source IP	Source Eth.	Destination IP	Destination Eth.
Host 1 to 2, on CS net	IP1	E1	IP2	E2
Host 1 to 4, on CS net	IP1	E1	IP4	E3
Host 1 to 4, on EE net	IP1	E4	IP4	E6

IP1 IP4

IP地址全程有效，**MAC**只在一个子网内有效

(3) RARP(逆向ARP)

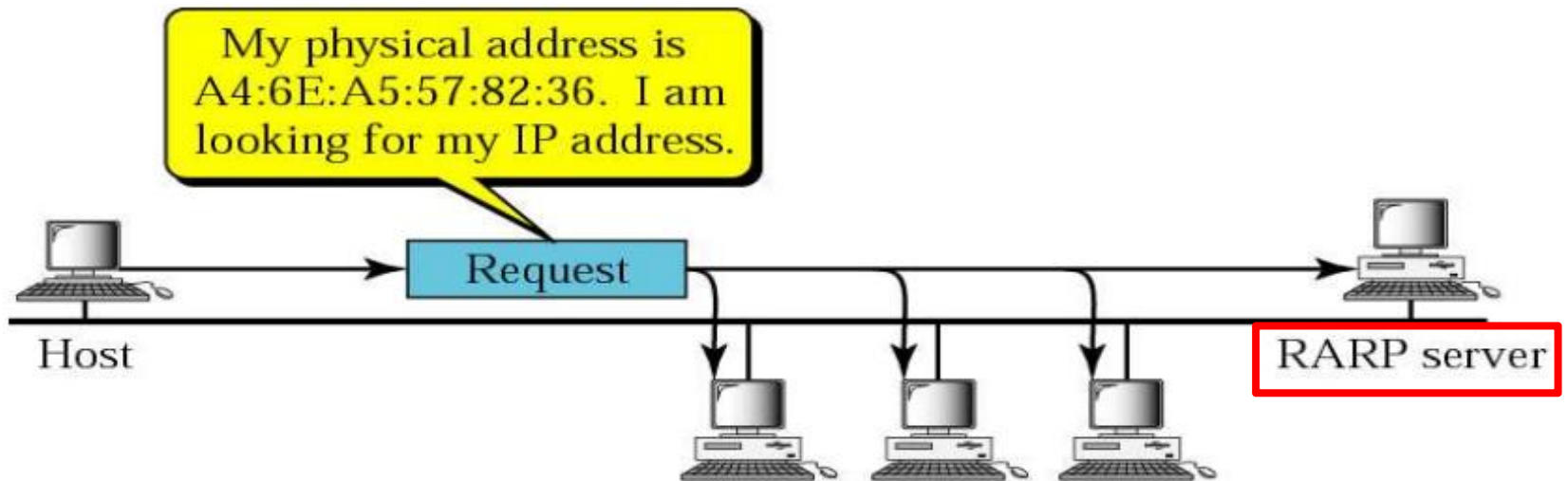
■ Requirement

- ◆ Given an Ethernet address, what is the corresponding IP address?
- ◆ Example: when a diskless workstation is booted
- ◆ Network layer protocol

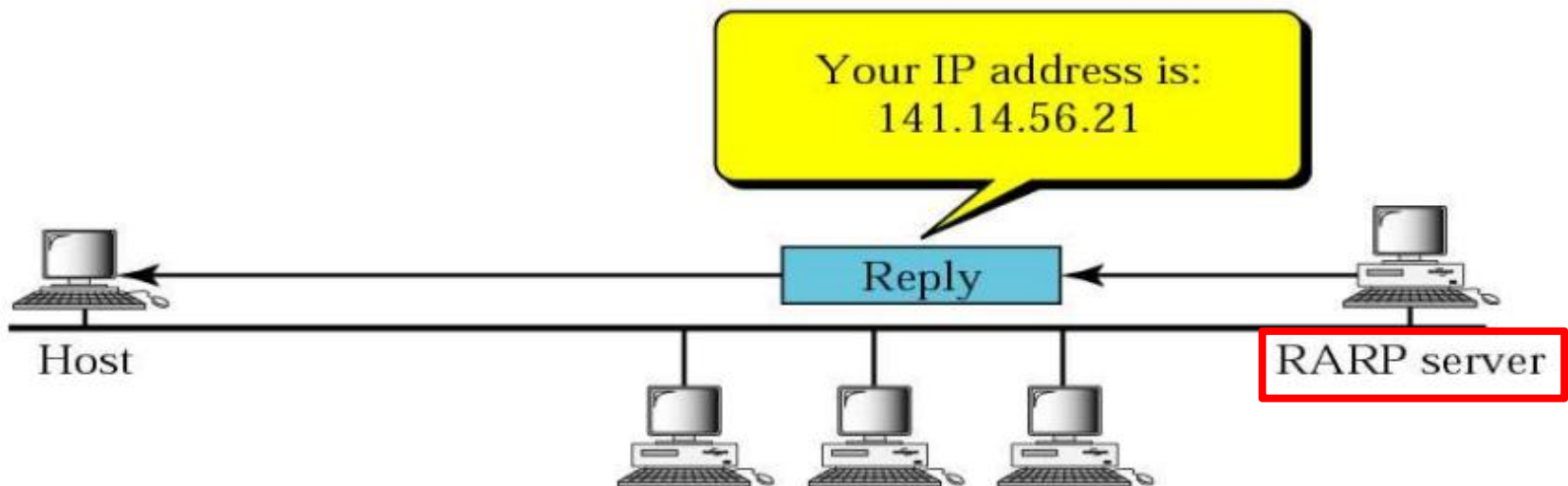
■ RARP (Reverse Address Resolution Protocol)

- ◆ Allows a workstation to broadcast on Ethernet: My Ethernet address is 14-04-05-18-01-25. Does anyone out there know my IP address?

RARP Operation



a. RARP request is broadcast



b. RARP reply is unicast

(4) DHCP(动态主机配置协议)

■ Dynamic Host Configuration Protocol

- ◆ Provides **automatic configuration** of remote hosts including
 - IP address, Router's address, netmask
 - DNS server's address, file server...
- ◆ An application layer protocol, following **client-server paradigm**, using **UDP** to carry its messages
- ◆ **DHCP packets**: Discover/Offer, Request/Ack

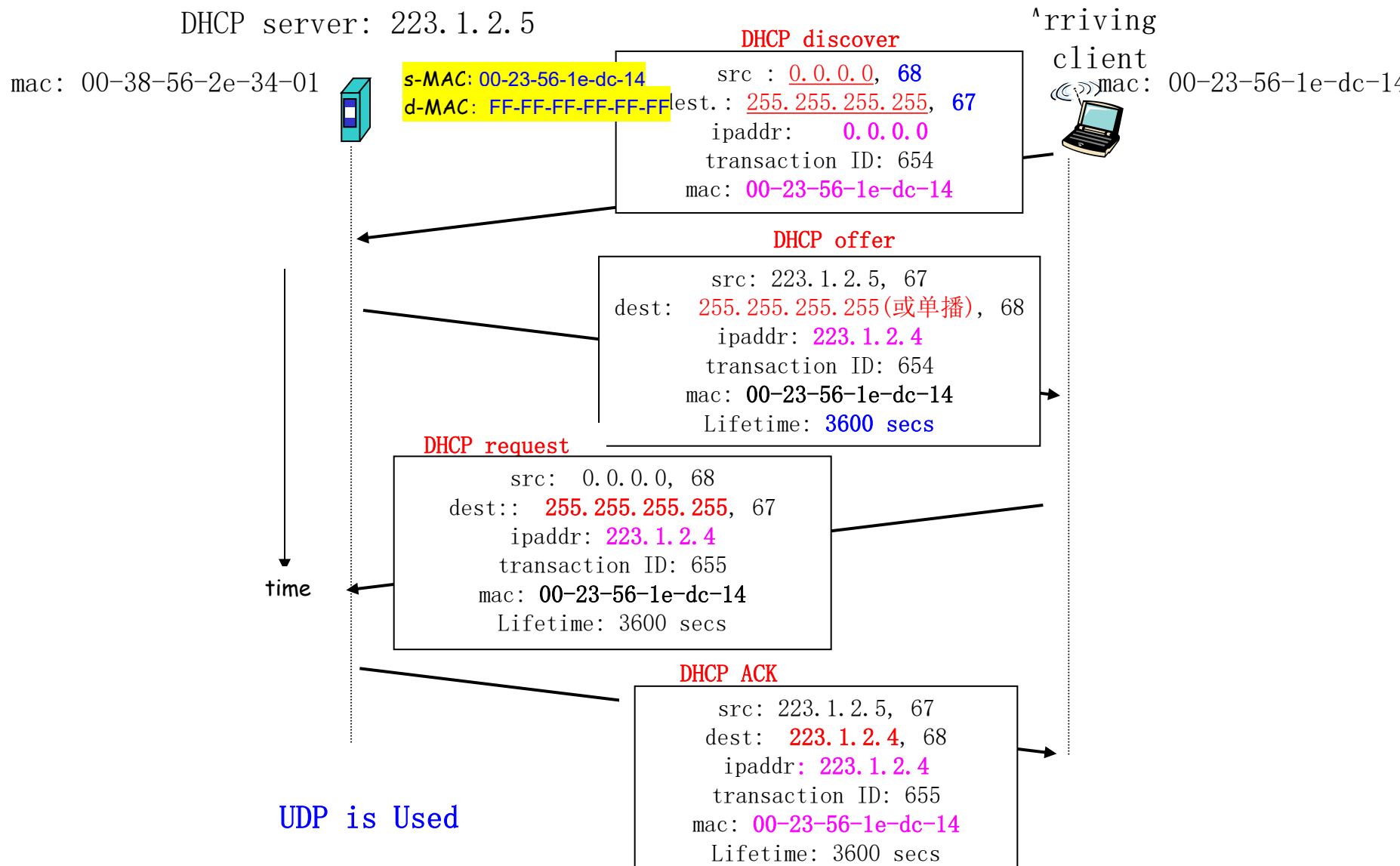
■ An extension to the BOOTP protocol

- ◆ It has largely replaced RARP and BOOTP

■ Leasing & renewal(租赁及续租)

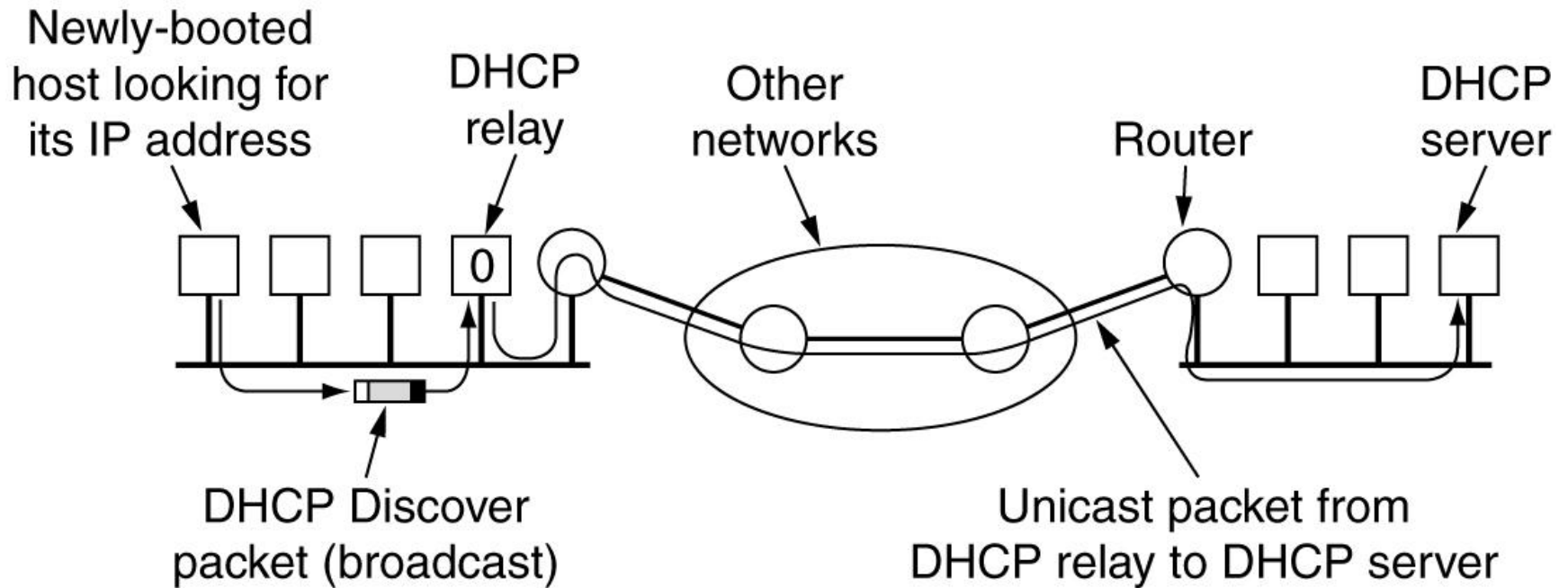
■ What's the difference of RARP and DHCP?

DHCP Client-Server Scenario



DHCP Relay Agent(DHCP中继代理)

- When there is no DHCP server in the LAN, DHCP relay agent relays DHCP messages between DHCP clients and DHCP server



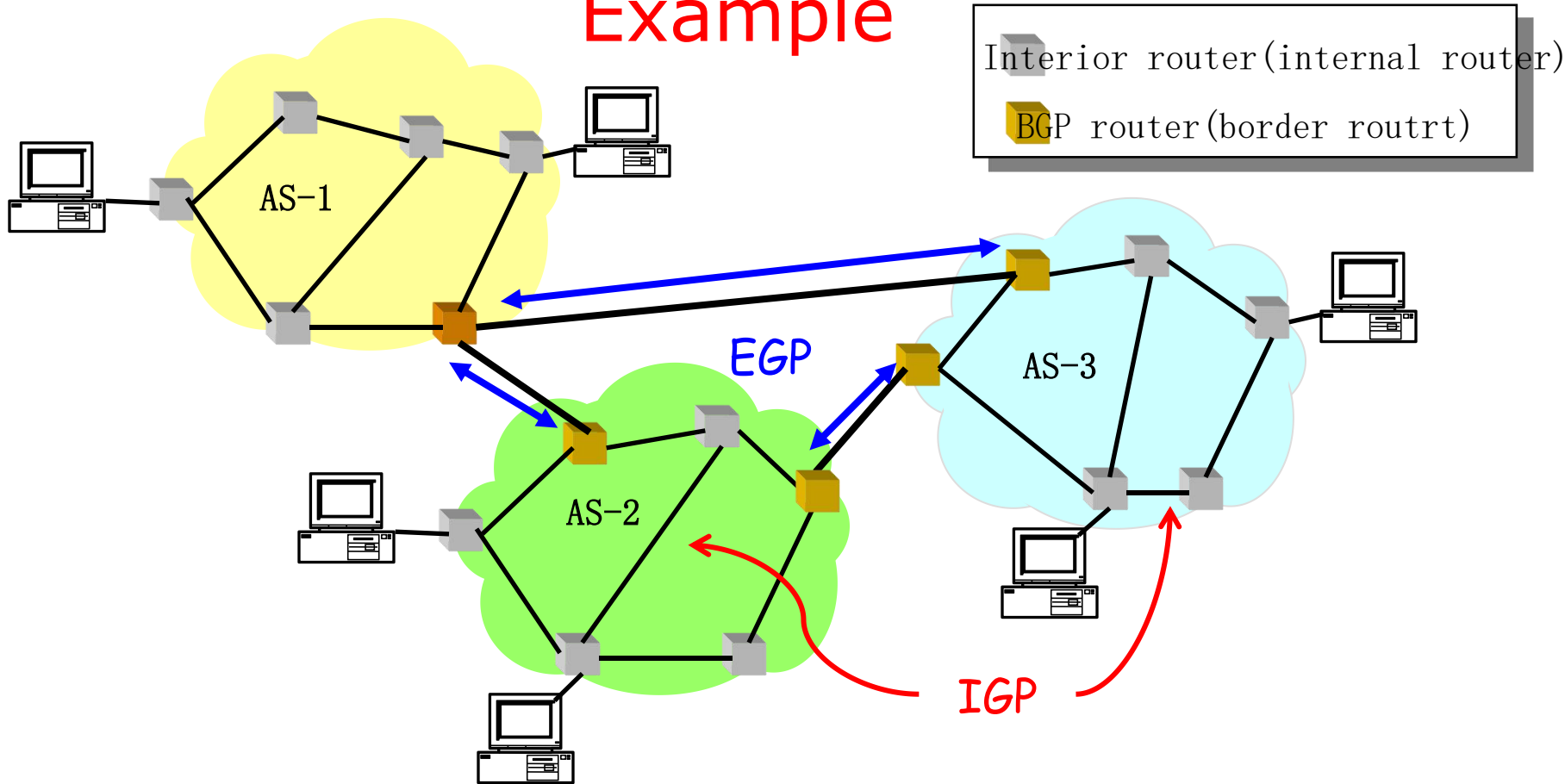
The network layer in the Internet

- 5.6.1 IPv4 Packet format
- 5.6.2 IP Addressing
- 5.6.3 IPv6
- 5.6.4 ARP, DHCP and ICMP
- 5.6.6&5.6.7 Routing Protocols: OSPF and BGP

(1) Routing Architecture in the Internet

- Internet is not a single network
 - ◆ The Internet contains **a large number of diverse networks**
 - ◆ Different networks controlled by different administrations
- Internet is divided in **Autonomous Systems (AS, 自治系统)** that are independently administered by individual organizations.
 - ◆ An AS is a network under single administrative control
 - ◆ Every AS has **a unique AS number**
- An AS is a routing domain
 - ◆ Within an AS
 - Trust other routers,
 - Run a **unified routing protocol** (eg. LSR)
 - ◆ Between ASes
 - Crossing trust boundaries
 - **Lack of information** about other AS's network

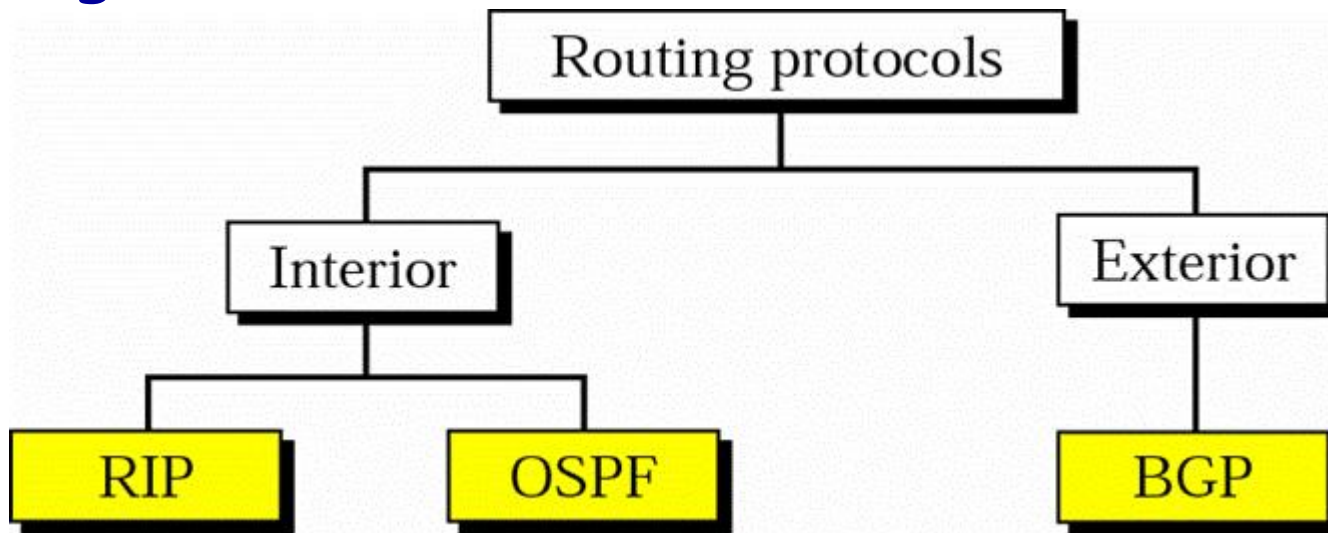
Example



- Each administrative authority use its own routing protocol within the AS, i.e. **Interior Routing Protocol (Interior Gateway Protocol, IGP)**
- One routing protocol is used to exchange routing information among AS, i.e. **Exterior Routing Protocol (Exterior Gateway Protocol, EGP)**

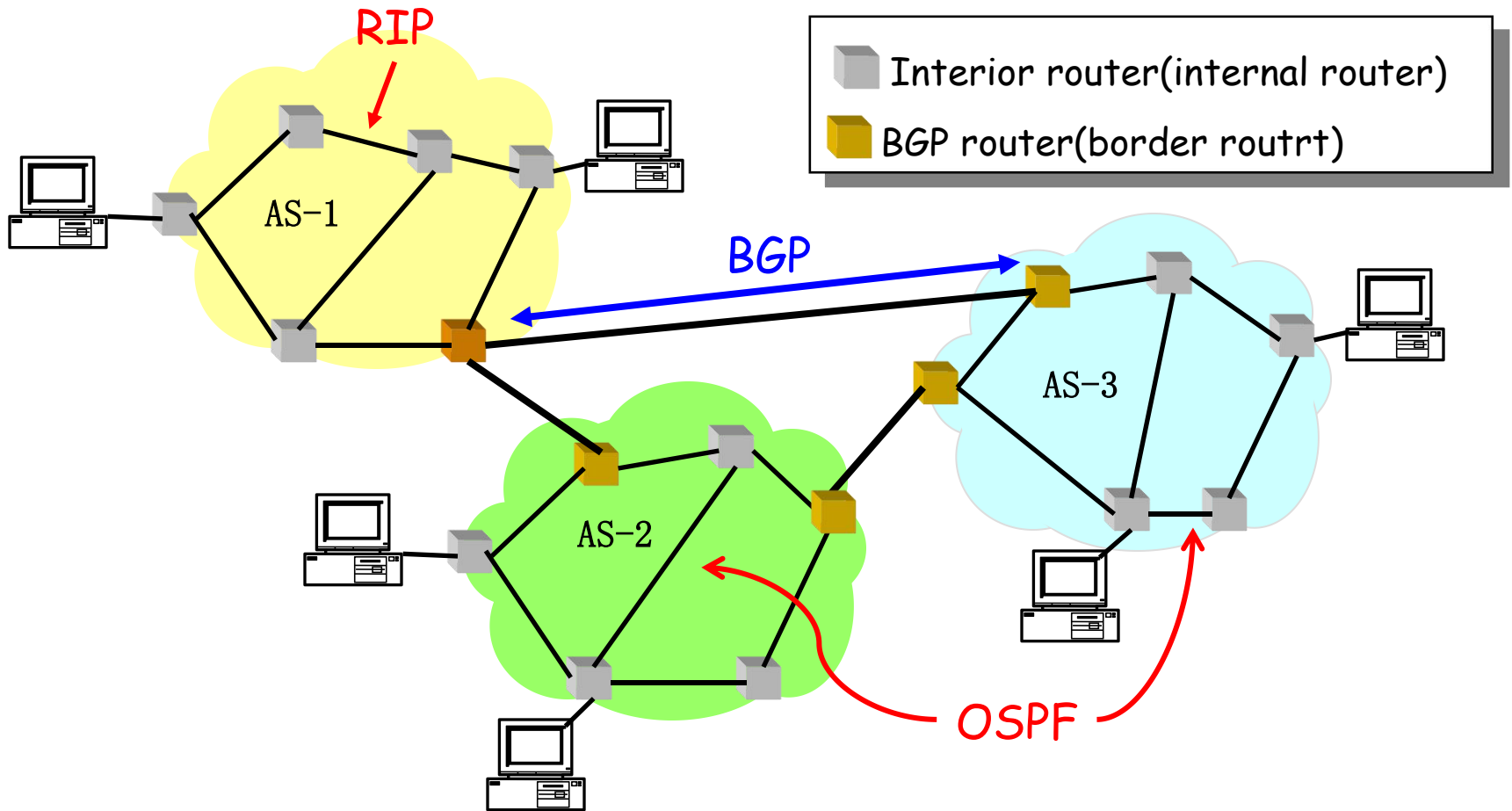
(2) Popular Routing Protocols in Internet

- **RIP**: Distance Vector Routing
- **OSPF, IS-IS**: Link State Routing
- **BGP**: Revised Distance Vector, Path Vector Routing



[Forouzan]

Example



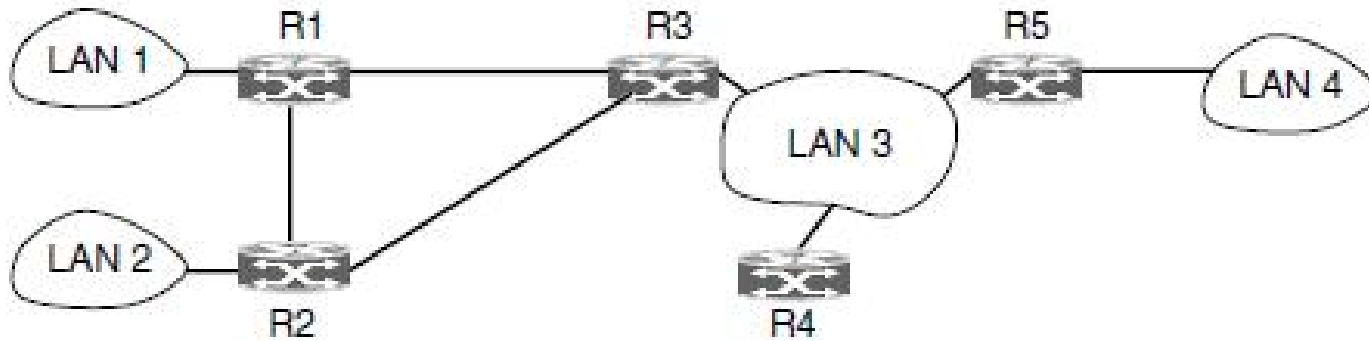
(3) RIP: Features

- Routing Information Protocol (路由信息协议)
- Direct implementation of Distance Vector routing
- Path costs are based on number of hops
 - ◆ RIP treats all network equals; the cost of passing thru a network is the same: one hop count per network
- Number of hops is limited to 16, lack of Scalability
- The update of complete routing table are sent every 30 seconds
- RIP messages are transmitted in UDP packets

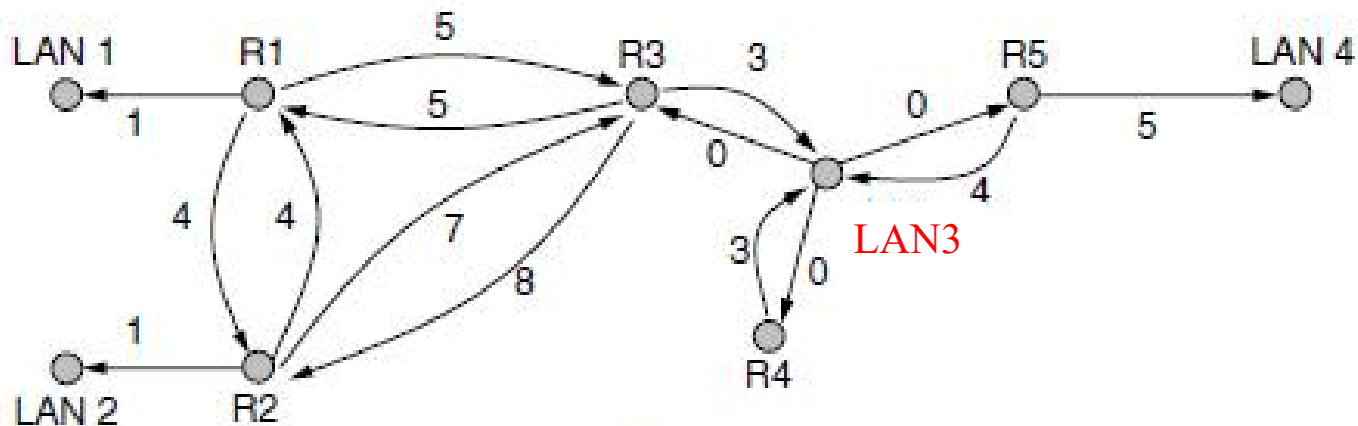
(4) OSPF: Features

- Open Shortest Path First (开放最短路径优先)
- Based on Link State Routing
- OSPF allows the administrator to (only) assign a cost for passing through a network based on the type of service required.
- Each router should have the exact topology of the internet at every moment.
- OSPF standard supports novel concepts such as type of service routing, load balancing and authentication.
- OSPF messages are transported directly in IP packets

OSPF: Graph Representation



(a)



(b)

RIP vs. OSPF

Features	RIPv1	RIPv2	OSPF
RFC	RFC1058	RFC1723	RFC2178
Scheme	Distance-Vector		Link-state
Range of Metrics	16		65,535
Update Frequency	30s		Changes or 30mins
Dead interval	180s		Variable setting
Authentication	No	Yes	Yes
Convergence	Slow	slow	Fast
Support CIDR	No	Yes	Yes
Type of Service	No	No	Yes
Multiple paths	No	No	Yes
Ease of Use	Yes		Complex Setup

(5) BGP

- Ases need a common protocol for communication
- Border Gateway Protocol (边界网关协议)
- Path Vector Routing 路径矢量选路
- Distance vector algorithm with extra information
 - ◆ Every route advertisement contains the entire AS path
 - ◆ No extra computation, just extra storage
- Advantages
 - ◆ Enables router to perform policy routing
 - E.g. avoid path to avoid transiting particular AS
 - E.g. link speed, capacity, tendency to become congested, and overall quality of operation, security
 - E.g. minimizing number of transit ASs
 - ◆ Can avoid loops at an AS level
- BGP using TCP to carry its messages, reliable delivery!

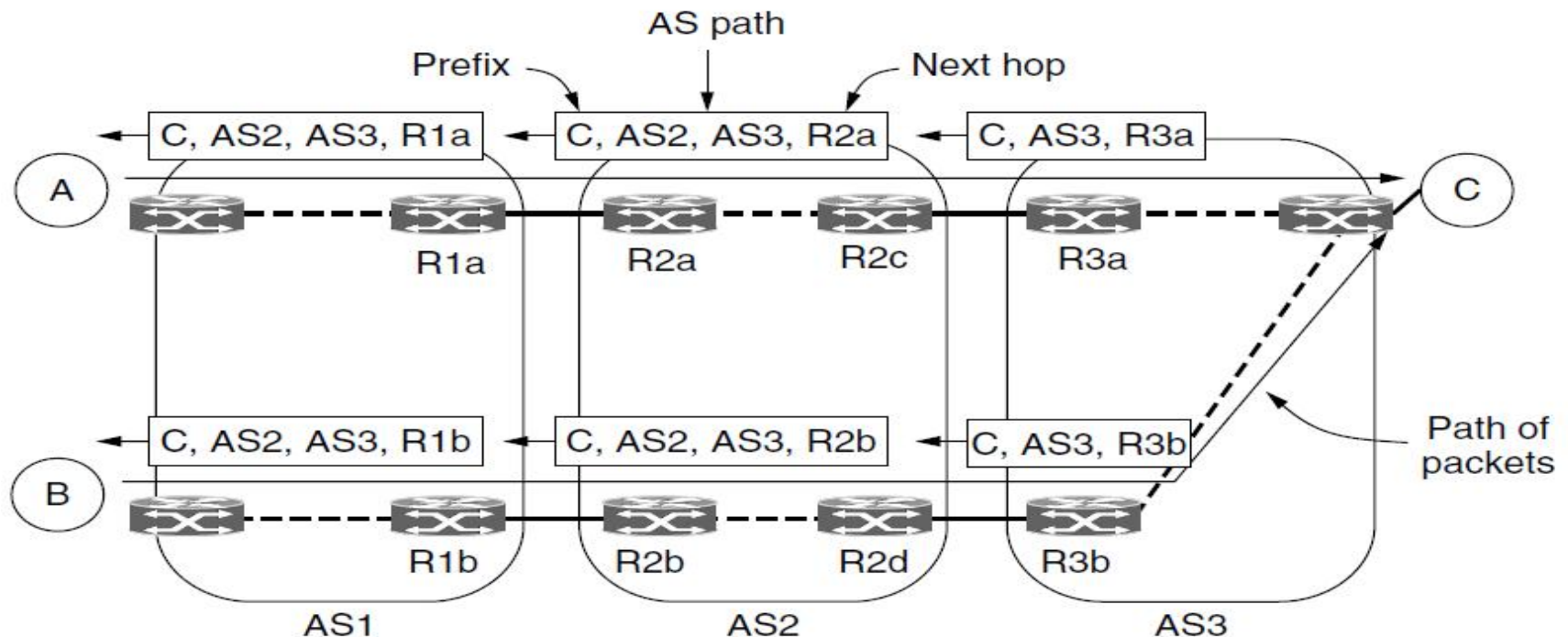
BGP: Routing Table Example

Network	Next Router	Path
N01	R01	AS14, AS23, AS67
N02	R05	AS22, AS67, AS05, AS89
N03	R06	AS67, AS89, AS09, AS34
N04	R12	AS62, AS02, AS09

[Forouzan]

How BGP Routers Choose Route

- Detect and **break** routing **loop**
- Routes via peered networks are chosen in preference to routes via transit providers
- Shorter **AS paths** are better
- Prefer the route that has the lowest cost within AS



因特网的选路协议

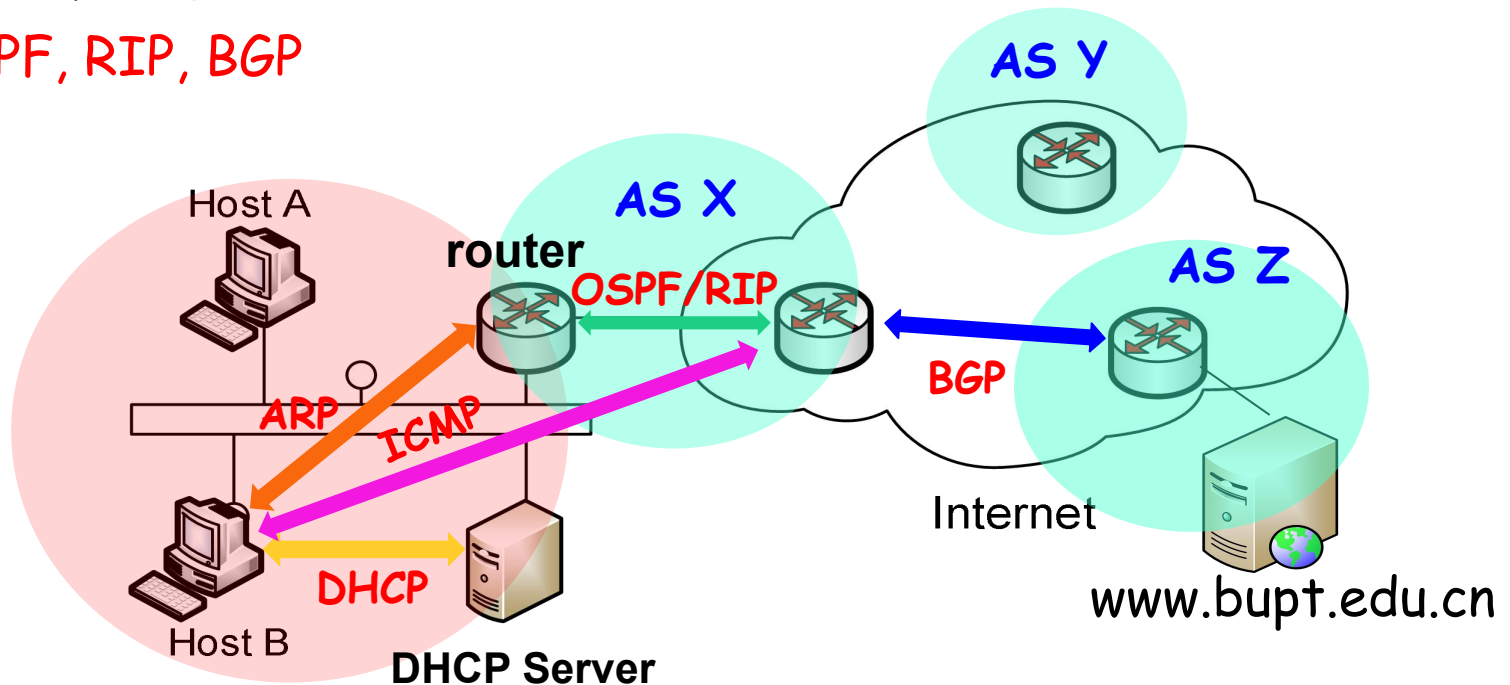
	RIP	OSPF	BGP
应用范围	AS内部(IGP)	AS内部(IGP)	AS之间(EGP)
选路算法	距离矢量选路 DVR	链路状态选路 LSR	路径矢量选路 PVR
路由器之间 交换的信息	到各网络的跳数 (hops-distance)	网络的拓扑结构 (Topology)	到各网络的路径 (Path)
无穷计数 (路径环路)	有	无	无
承载协议	UDP	IP	TCP(高可靠)

Further reading:

[1] A SURVEY ON WIRED, WIRELESS, AND INTERNET OF THINGS ROUTING PROTOCOLS

Review

- Autoconfiguration: **DHCP**, (**RARP**) IP address & routing table
- IP address → MAC address: **ARP** Neighbor table
- Error report: **ICMP**
- Routing table
 - ◆ **IPv4 & IPv6**
 - ◆ **OSPF, RIP, BGP**



Key Points(1)

- Introduction to Network Layer
 - ◆ Position and functions of network layer
 - ◆ Virtual Circuit Switching vs. Datagram Routing
- Routing Algorithms
 - ◆ Static routing vs. Dynamic routing
 - ◆ Sink tree as benchmark
 - ◆ Flooding
 - ◆ Dijkstra algorithm
 - ◆ Distance Vector Routing (DVR): features, how to calculate routing table, drawbacks
 - ◆ Link State Routing (LSR): features, LSP vs. topology
 - ◆ Hierarchical Routing: idea

Key points(2)

■ Congestion Control:

- ◆ Virtual circuit network: Admission control
- ◆ Datagram network: find, notification and decrease sending rate
 - Choke packets, ECN, RED

■ QoS Issues: Token Bucket, Packet Scheduling

■ IP Addressing

- ◆ Classful address: class A, B and C
- ◆ Hierarchical address: Network-id + host-id
- ◆ Network address used in routing
- ◆ Subnetting
- ◆ Supernetting
- ◆ CIDR: slash notation
- ◆ How to calculate subnet mask

Key Points(3)

- IP Packet format
 - ◆ Function of each field
 - ◆ Checksum
 - ◆ Fragmentation
- Routing Protocols: RIP, OSPF and BGP
 - ◆ Features
- Other relative protocols
 - ◆ IPv6: features, addressing, header conversion
 - ◆ ICMP: error report and test
 - ◆ ARP: IP address → MAC address
 - ◆ DHCP: allocate IP address dynamically
 - ◆ IGMP: managing group members
 - ◆ Mobile Routing: idea

Terms

缩写	英文全称	中文
	adaptive algorithms	自适应算法
	addressing	寻址
	address resolution	地址解析
ARP	Address Resolution Protocol	地址解析协议
AS	Autonomous Systems	自治系统
	Admission control	接纳控制
BGP	Border Gateway Protocol	边界网关协议
	backward learning	逆向学习
	broadcast storm	广播风暴
	bridge	网桥
	broadcasting	广播
	checksum	校验和/检查和
	classful addressing	分类地址
	classless addressing	无类别地址
CIDR	Classless InterDomain Routing	无类别域间选路

缩写	英文全称	中文
	Choke Packet	抑制分组
	circuit switching	电路交换
	count-to-infinity	无穷计数
	congestion control	拥塞控制
	connectionless service	无连接服务
	connection-oriented service	面向连接的服务
	cut-through switching	直通交换
	datagram	数据报
	datagram network	数据报网络
DHCP	Dynamic Host Configuration Protocol	动态主机配置协议
	default mask	默认掩码
DVR	Distance Vector Routing	距离矢量选路
ECN	Explicit Congestion Notification	显式拥塞通知

缩写	英文全称	中文
	Ethernet	以太网
	Fair Queue	公平队列
	FIFO Queuing	先进先出队列
	Fixed Routing	固定查表选路
	flooding	洪泛
	flow control	流量控制
	forwarding	转发
	fragmentation	分段/分片
	fragmenting	分段/分片
	gateway	网关
	Hop-by-Hop	逐跳
	Hierarchical Routing	分级选路
ICMP	Internet Control Message Protocol	因特网控制报文协议
IGMP	Internet Group Management Protocol	因特网组管理协议

缩写	英文全称	中文
	implicit notification	隐式通知
	incoming interface	输入接口
	infinite loop	无限循环
	internet	互联网
	Internet	因特网
	internetworking	互通
IP	Internet Protocol	网际协议
	jitter	时延抖动
LAN	Local Area Network	局域网
	Leaky Buckets	漏桶
LSP	Link State Packet	链路状态包
LSR	Link State Routing	连路状态选路
	Longest-Prefix Matching	最长前缀匹配
MTU	Maximum Transmission Unit	最大传输单元
	multicasting	多播/组播

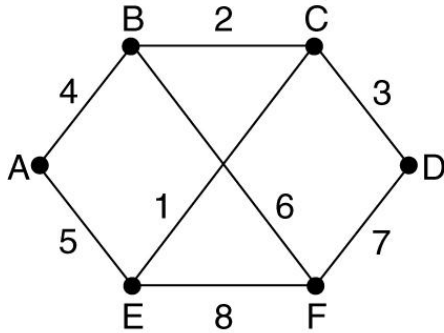
缩写	英文全称	中文
	nonadaptive algorithms	非自适应算法
	network address	网络地址
NAT	Network Address Translation	网络地址翻译
	next hop	下一跳
	outgoing interface	输出接口
OSPF	Open Shortest Path First	开放的最短路径优先
	packet	分组/包
	packetizing	打包
	Packet Scheduling	分组调度
	packet switching	分组交换
	prefix	前缀
	Priority Queuing	优先级队列
PVR	Path Vector Routing	路径矢量选路
QoS	Quality of Service	服务质量
	re-assembly	重装

缩写	英文全称	中文
RARP	Reverse Address Resolution Protocol	逆向地址解析协议
RED	Random Early Detection	随机早期检测
	Random Routing	随机选路
	Reverse Path Forwarding	逆向路径转发
	round robin	循环
	round-trip time	环回时间
	Route Aggregation	路由聚合/会聚
RIP	Routing Information Protocol	选路信息协议
	router	路由器
	routing	选路
	routing algorithm	选路算法
	routing table	路由表
	Resource Reservation	资源预留
	Sink Tree	宿树/汇集树
	spanning tree	生成树)
	static routing	静态选路

缩写	英文全称	中文
	store-and-forward	存储转发
	subnet	子网
	subnet mask	子网掩码
	subnetting	子网划分
	supernetting	超网划分
	supernetwork	超网
	switch	交换机
	Token Buckets	令牌桶
	Traffic Shaping	业务量(流量)整形
TTL	Time-to-Live	生命期
	Tunneling	隧道
VC	Virtual Circuit	虚电路
VCI	Virtual Circuit Identifier	虚电路标识符
	virtual circuit network	虚电路网络
WAN	Wide Area Network	广域网
WFQ	Weighted Fair Queuing	公平加权队列

Self-Study

LSP: Example



(a)

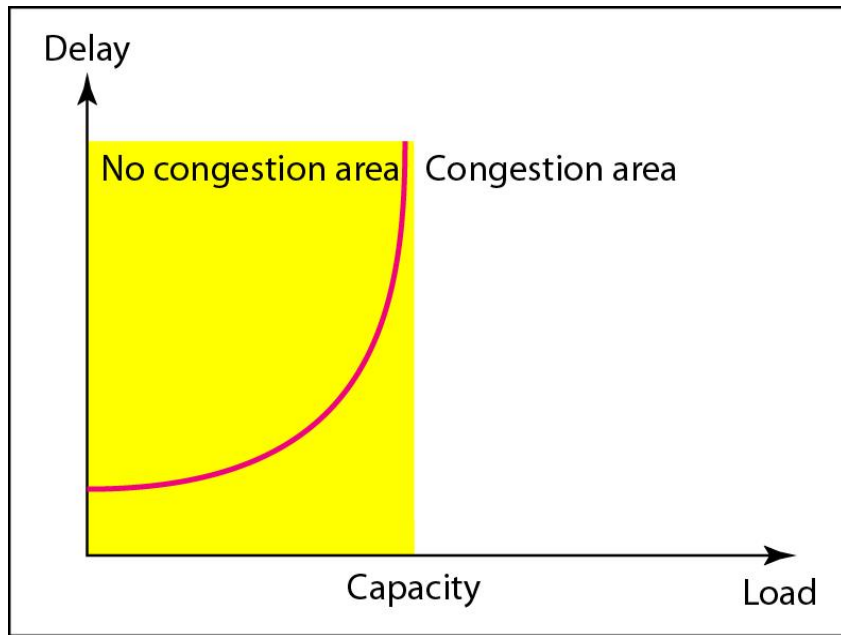
Link		State		Packets	
A		B		C	
Seq.		Seq.		Seq.	
Age		Age		Age	
B 4		A 4		B 2	
E 5		C 2		D 3	
		F 6		F 7	

(b)

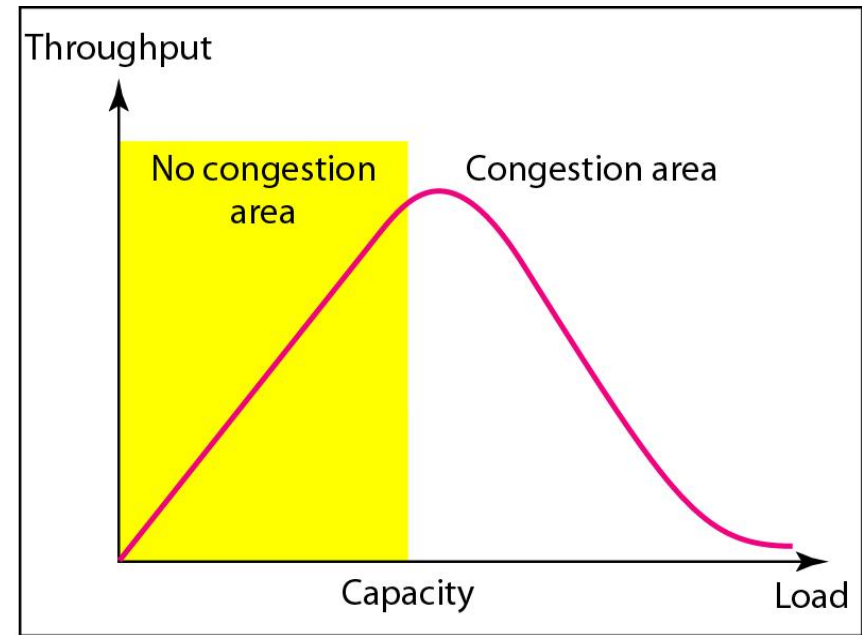
Source	Seq.	Age	Send flags			ACK flags			Data
			A	C	F	A	C	F	
A	21	60	0	1	1	1	0	0	
F	21	60	1	1	0	0	0	1	
E	21	59	0	1	0	1	0	1	
C	20	60	1	0	1	0	1	0	
D	21	59	1	0	0	0	1	1	

Record of LSPs received by B

Network performance: delay(时延) and throughput(吞吐量)



a. Delay as a function of load



b. Throughput as a function of load

■ Congestion Control involves two factors that measure the performance of a network:

- ◆ **Delay**: queuing due to processing delay and propagation delay
- ◆ **Throughput**: number of packets passing through in a unit of time

5.3.1 Network provisioning (网络提供)

- 规划、调整网络，以适应业务量(负载)
- To build a network that is well matched to the traffic that it carries
- Resources are added dynamically when there is serious congestion
- Links and routers that are regularly heavily utilized are upgraded at the earliest opportunity
- 成本高，时间长(约几个月的时间)

5.3.2 Traffic-aware Routing

■ The normal routing schemes

- ◆ Used **fixed link weights**, adapted to changes in topology, but **not** to changes in load

■ Traffic-aware Routing

- ◆ Set **the link weight** to be a function of the **link bandwidth** and **propagation delay** plus the **measured load** or **average queuing delay**
- ◆ **Danger**: **routing tables may oscillate wildly**, leading to erratic routing and many potential problems
- ◆ If load is ignored and only bandwidth and propagation delay are considered, this problem does not occur
- ◆ In **the Internet routing** protocols do **not** generally adjust their routes **depending on the load**

5.3.3 Admission control(接纳/准入控制)

- Widely used in virtual circuit networks
- Idea: do not set up a new virtual circuit unless the network can carry the added traffic without becoming congested. 没有足够资源，就不允许建立VC
- 关键问题：如何描述虚电路的流量特性
- Armed with traffic descriptions(by leaky bucket 漏桶 or token bucket 令牌桶), the network can decide whether to admit the new virtual circuit.
- Combined with traffic-aware routing: considering routes around traffic hotspots as part of the setup procedure

Techniques to improve QoS

- 5.4.2 Traffic shaping
- 5.4.3 Packet Scheduling
- *5.4.4 Admission control(自阉)*
- 5.4.5 Integrated Services(不讲)
- 5.4.6 Differentiated Services(不讲)

Resource Reservation(资源预留) & Admission Control(接纳控制)

■ Resource Reservation

- ◆ Bandwidth.
- ◆ Buffer space.
- ◆ CPU cycles.

Flow: 同一对源/目的主机之间的多个数据包

■ Flow negotiation

- ◆ Sender, receiver, and all routers along the path are involved in the flow negotiation

■ Flow specification (流描述)

- ◆ 描述一个流对网络的要求
- ◆ 源主机产生
- ◆ 沿途路由器和目的主机可修改

Parameter	Unit
Token bucket rate	Bytes/sec
Token bucket size	Bytes
Peak data rate	Bytes/sec
Minimum packet size	Bytes
Maximum packet size	Bytes

Relating flow specifications to router resources

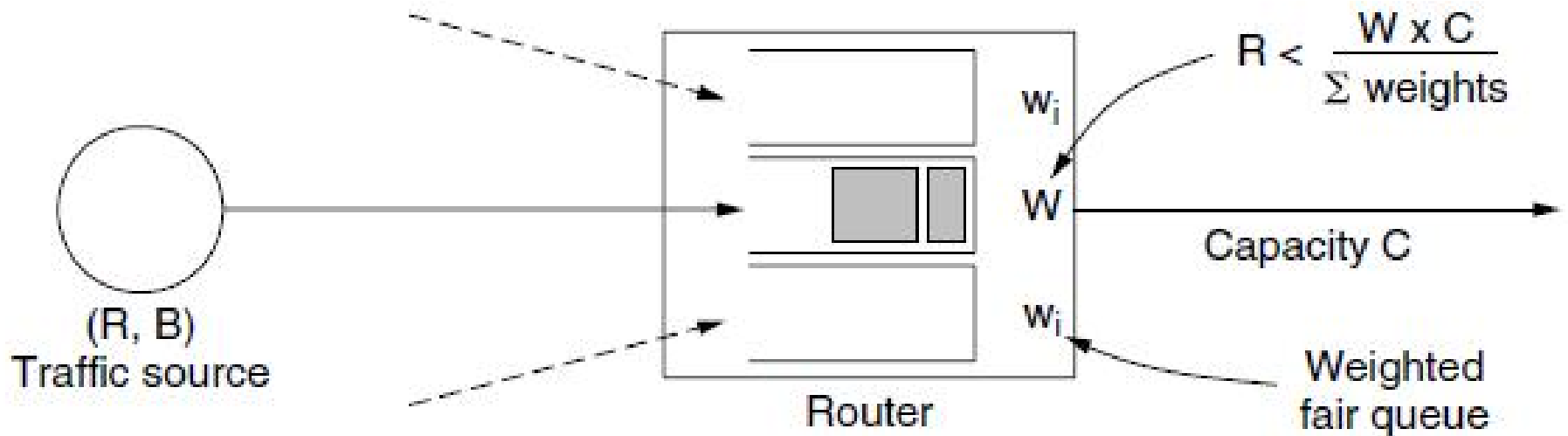
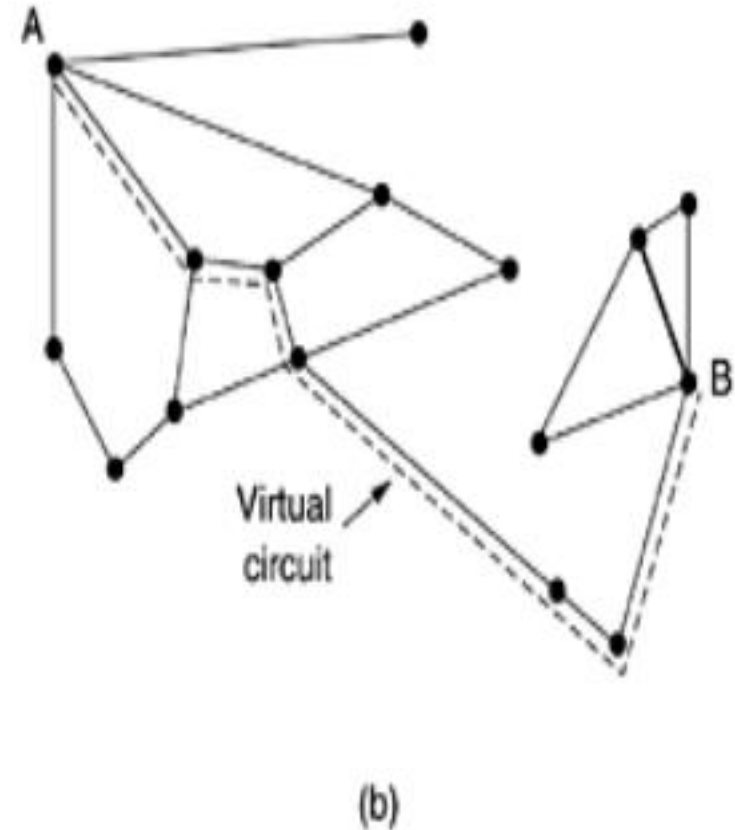
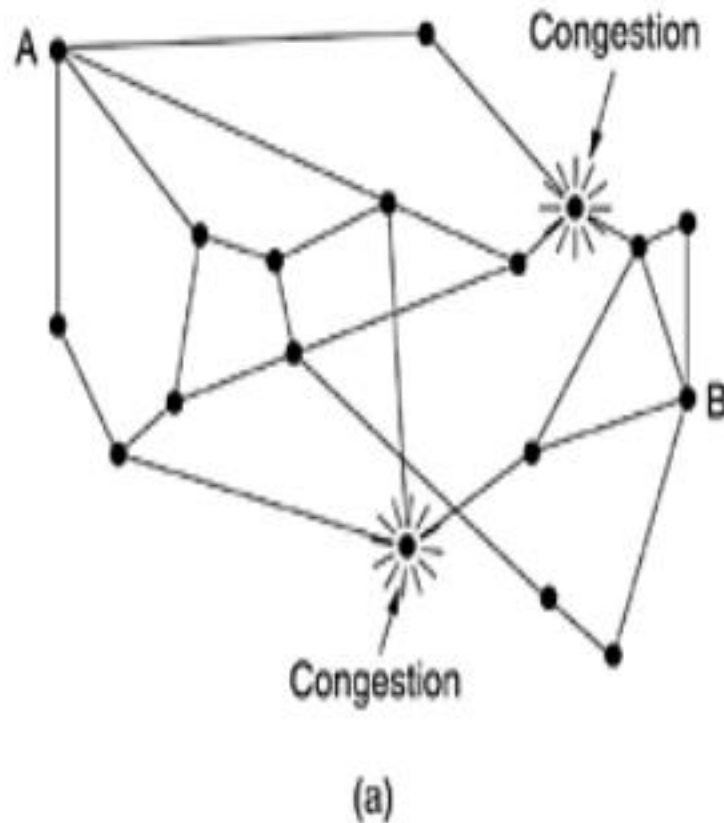


Figure 5-33. Bandwidth and delay guarantees with token buckets and WFQ.

Route new VC around Congestion Areas



The IPv6 Extension Headers

Extension header	Description
Hop-by-hop options	Miscellaneous information for routers
Destination options	Additional information for the destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of the sender's identity
Encrypted security payload	Information about the encrypted contents

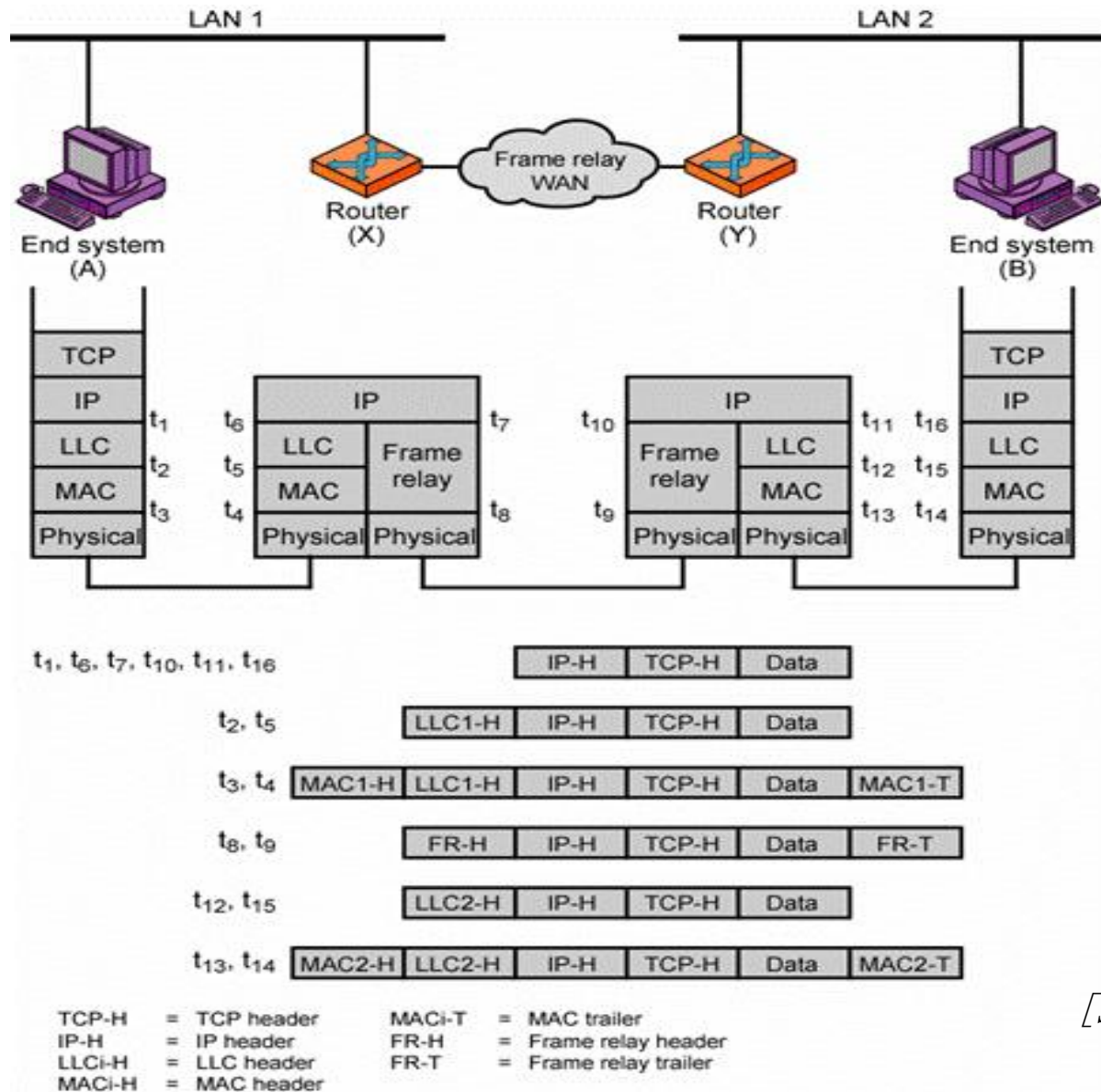
Design Issues of IP

- Design requirements: accommodate differences in constituting networks:
 - ◆ Different addressing schemes
 - ◆ Different maximum packet size
 - ◆ Different network access mechanism
 - ◆ Different timeouts
 - ◆ Different transmission modes (connection-oriented, connectionless)
 - ◆ Error control
 - ◆ Flow control

Different Network Access Mechanism

- Solved by abstracting networking functions in the Networking layer, and relying on Data Link Layer for networking access
- Routers handle the difference in network access mechanism

Operations of IP



[Stallings]

Different Timeout

- Problem: different networks use different timeout mechanisms.
- Solution: Introduce the concept of **Datagram Lifetime**
- Datagrams could loop indefinitely
 - ◆ Consumes resources
 - ◆ Transport protocol may need upper bound on datagram life
- Datagram marked with lifetime
 - ◆ **Time To Live (TTL, 生命期)** field in IP datagram
 - ◆ Once lifetime expires, datagram is discarded (not forwarded)
 - ◆ Hop count

Different Transmission Modes

- Connection-oriented and connectionless modes
- A **connectionless** architecture is adopted by IP

Error Control

- Minimal error control, error check only for the header
- Router should attempt to inform source if packet discarded (using **ICMP**, Internet Control Message Protocol, 因特网控制报文协议)
 - ◆ ICMP messages are encapsulated in IP
 - ◆ Error messages from a router or destination host
 - Destination host is unreachable,
 - TTL reached 0
 - Parameter error
 - ICMP redirect, which tells the source of a better way to a destination, ...
- No guaranteed delivery
- May inform higher layer protocol

Flow Control

- Allows routers and/or destination to limit rate of incoming data
- Limited in connectionless systems
- Send flow control packets
 - ◆ Requesting reduced flow
- Some **ICMP** messages are used for flow control
 - ◆ Source quench messages(源抑制报文, choke packet)

BOOTP(启动协议)

■ BOOTP

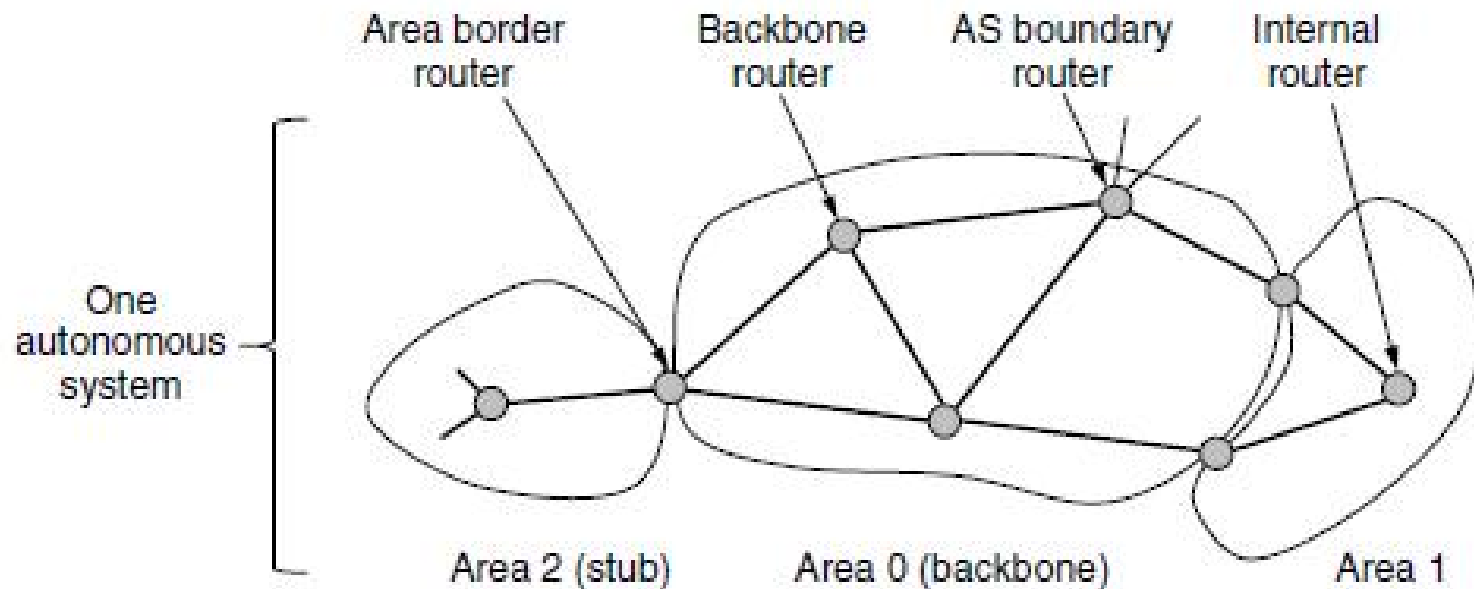
- ◆ Used to configure diskless workstations
- ◆ Unlike RARP, **BOOTP uses UDP messages**, which are forwarded over routers
- ◆ It also provides the **IP address of the file server**, the **IP address of the default router**, and the **subnet mask** to use

■ Problem

- ◆ It requires **manual** configuration of tables mapping IP address to Ethernet address

OSPF: Areas

- OSPF divides an AS(自治系统) into **areas(区域)**, a collection of network, routers all contained within an AS.
- Backbone Router, Boundary Router, Border Router and Internal Router



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